Lecture notes. Lecture 12

Parametrised quantum circuits
variational ansätze, quantum data, probability models, optimisation and classifiers

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These notes introduce parametrised quantum circuits for bachelor students of physics, mathematics and computer science, as the main near-term route to using quantum computers as machine-learning models. A circuit with adjustable gate angles prepares a state on a register of qubits, the state is measured, and a classical computer changes the angles to lower a cost estimated from the measured bit strings. The circuits were first proposed as variational ansätze for ground states of interacting spins, whose description needs a number of amplitudes that grows exponentially with the number of spins. We follow this variational quantum eigensolver from the variational principle through the measurement of energies and gradients to the choice of the circuit and its performance on spin chains. The ground states prepared in this way are quantum data, and circuits trained on them compress the states in a quantum autoencoder, map and classify phases of matter, learn an unknown Hamiltonian from its dynamics and compress circuits through their Choi states. The same circuits then model probability distributions as Born machines, solve binary optimisation problems with the quantum approximate optimisation algorithm, and classify data through quantum kernels and data re-uploading. The final section explains, through the properties of random quantum states, why deep and expressive circuits have vanishing gradients. All numerical results come from exact state-vector simulations or from simulated measurement records with fixed random seeds, and each result states which of the two it is. Results reproduced from the literature are marked as such.

Prerequisites

Qubits and registers of qubits as unit vectors of complex amplitudes, with qubit 0 as the most significant bit of the index. Pauli matrices, Pauli strings and rotations $e^{-i\theta\hat{P}/2}$, the Hadamard and controlled-$\hat{Z}$ gates. The Born rule, measurements in different bases and estimates from a finite number of shots. Hamiltonians as Hermitian matrices, their eigenvalues and ground states, the Schmidt decomposition and the entanglement entropy. Gradient descent and elementary probability. Further physical notions, such as phase transitions, are explained at their first appearance.

The same objects carry different names in physics, mathematics and computer science. Table 1 collects the names used in these notes.

physics	mathematics	computer science
ansatz $\lvert\psi(\boldsymbol{\theta})\rangle$	parametrised family of unit vectors	model with trainable weights
parametrised circuit $\hat{U}(\boldsymbol{\theta})$	product of matrix exponentials $e^{-i\theta_k \hat{P}_k/2}$	program with continuous parameters
gate angle $\theta = \Omega t$	real parameter	weight
energy $\langle\psi\lvert\hat{\mathcal{H}}\rvert\psi\rangle$	quadratic form, Rayleigh quotient	cost or loss function
variational principle	minimum of the Rayleigh quotient	upper bound of the optimum
measurement setting	common eigenbasis of commuting matrices	readout configuration
shots	independent samples	mini-batch of samples
parameter-shift rule	exact difference formula for trigonometric functions	gradient from two function calls
Born machine	distribution $\lvert\psi_{\mathbf{b}}\rvert^2$	generative model over bit strings
trash qubits of an autoencoder	discarded tensor factors	bottleneck of an autoencoder
cost Hamiltonian $\hat{\mathcal{H}}_C$	diagonal matrix of a quadratic form	objective function of a QUBO
ground state of $\hat{\mathcal{H}}_C$	minimiser of a quadratic binary form	optimal solution
state overlap $\lvert\langle\phi(\mathbf{x})\rvert\phi(\mathbf{x}')\rangle\rvert^2$	inner product of feature vectors	kernel
label state	target point on the Bloch sphere	class label
Haar-random state	uniformly distributed unit vector in $\mathbb{C}^d$	random point on a high-dimensional sphere

Table 1: Physics, mathematics and computer-science names for the same objects.

1 Quantum computers as learning models

1.1 Data and processing devices

Deep learning, as developed so far in the course, trains neural networks on a classical computer with data stored in its memory. A quantum computer can enter this picture in two places, as the device that processes the data and as the source of the data. Combining the two possibilities gives four kinds of learning (Figure 1), a classification introduced in Ref. [1] and extended in Ref. [2]. The first letter of each label names the data and the second the algorithm. In the CC case classical data, such as images, texts or the output of a classical simulation, are processed by a classical algorithm, which is the setting of standard deep learning. In the CQ case classical data are processed by a quantum device, for example when the bit strings of a quantum register model a data set or when a circuit classifies points of a plane. In the QC case the data come from a quantum system, as measurement records of a quantum experiment, and a classical algorithm, often a neural network, learns from them. In the QQ case both the data and the processing are quantum, for example when a circuit prepares or compresses the ground state of a quantum many-body system on the register itself.

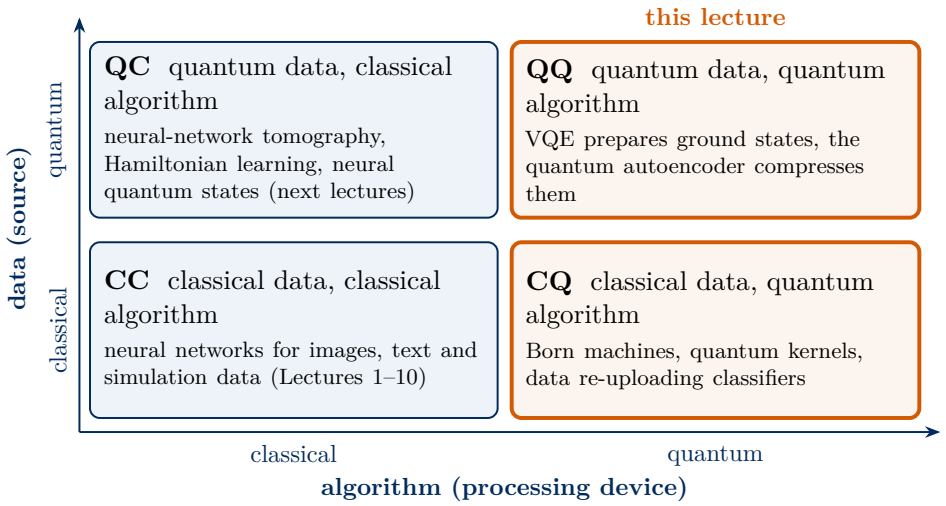


Figure 1: Four kinds of learning, sorted by the source of the data (classical or quantum, vertical) and the device that processes them (classical or quantum, horizontal). The examples of these notes lie in the right column, with a quantum processing device.

Every example in these notes processes data on a quantum device. The variational quantum eigensolver, the quantum autoencoder, the quantum convolutional and graph recurrent neural networks and the circuit autoencoder act on quantum states and belong to QQ, whereas the Born machine, the quantum approximate optimisation algorithm and the classifiers act on classical data and belong to CQ.

1.2 Directions of quantum machine learning

Quantum machine learning has developed along several lines [3], several of which concern these notes and the subsequent ones. The line followed in these notes uses parametrised quantum circuits, sequences of gates whose angles are trained by a classical optimiser in the same way as the weights of a neural network. Such a circuit serves as a trial wave function for a quantum state, as a source of random bit strings that model a probability distribution, or as a classifier of data. A related line uses quantum kernels. A circuit maps every data point to a quantum state, and the overlaps of these states measure the similarity of data points, which a classical support-vector machine then uses to separate classes [4, 5].

Quantum reservoir computing and quantum extreme learning machines leave the quantum part untrained altogether [6, 7]. A fixed quantum system, such as the many-body dynamics of interacting

spins or a disordered ensemble of spins, maps every input into a large number of measured features, and only a linear readout of these features is trained, a convex least-squares problem. A reservoir whose state keeps a fading memory of earlier inputs processes time series, which is reservoir computing, and a version in which each input passes the system once without memory is an extreme learning machine [8]. The fixed scrambler of the quantum scrambling Born machine in Section 12.8 and the fixed quantum kernels of Section 14.4 rest on the same idea of an untrained quantum map followed by a trained classical or single-qubit stage.

Learning from quantum data addresses experiments that produce quantum states. A device that stores several copies of such a state in a quantum memory and processes them together can learn some of its properties with exponentially fewer repetitions of the experiment than any strategy that measures the copies one by one [9]. Classical machine learning, finally, serves quantum physics directly. Neural networks reconstruct quantum states from measurements, learn Hamiltonians from data and represent wave functions as variational ansätze [10].

1.3 Organisation of these notes

We follow the historical order, in which parametrised circuits were proposed as variational ansätze for quantum many-body states before they were used as machine-learning models. Section 2 explains why the ground state of interacting spins is hard to compute classically and states the variational principle. Sections 3–5 build the parametrised circuit, the estimate of its energy from measured bit strings and the gradient of this energy. Sections 6 and 7 compare circuit ansätze in the variational quantum eigensolver for spin chains and show that the symmetry of an ansatz decides which ground states it reaches. Sections 8–11 treat the prepared ground states as quantum data and learn from them on the register. A quantum autoencoder compresses ground states into fewer qubits and, frozen after training, maps phase diagrams without labels. A quantum convolutional neural network classifies a topological phase from labelled ground states, a quantum graph recurrent neural network learns an unknown Hamiltonian from the states it produces, and a circuit autoencoder compresses quantum circuits. Section 12 uses the measured bit strings as samples of a learned probability distribution, and Section 13 uses them as candidate solutions of binary optimisation problems with the quantum approximate optimisation algorithm. Sections 14 and 15 turn circuits into classifiers of classical data, through quantum kernels and through repeated loading of the data into a single qubit. Section 16 uses the properties of random quantum states to explain why deep and expressive circuits become untrainable, and Section 17 summarises the notes.

2 The many-body problem and its classical approximations

2.1 The exponential cost of a many-body state

A chain of N interacting spins-1/2, or equivalently a register of N qubits, is described by a state vector with one complex amplitude for every bit string $\mathbf{b} = b_0 b_1 \dots b_{N-1}$,

$$|\psi\rangle = \sum_{\mathbf{b}\in\{0,1\}^N} \psi_{\mathbf{b}}\, |\mathbf{b}\rangle, \qquad \sum_{\mathbf{b}} |\psi_{\mathbf{b}}|^2 = 1. \tag{2.1}$$

The amplitude of $b_0 \dots b_{N-1}$ is stored at the index $i = \sum_q b_q 2^{N-1-q}$, so qubit 0 is the most significant bit. In double precision each amplitude takes 16 bytes and the vector needs $16 \cdot 2^N$ bytes. Ten spins fit into 16 kilobytes, thirty spins need 17 gigabytes, the memory of a large workstation, and fifty spins need about 18 petabytes (Figure 2). The Hamiltonian $\hat{\mathcal{H}}$, the Hermitian matrix of the energy, has 4^N entries, although for local interactions most of them vanish. Finding the ground state, the eigenvector of the lowest eigenvalue of $\hat{\mathcal{H}}$, or following the evolution $e^{-it\hat{\mathcal{H}}}|\psi\rangle$ in time requires repeated products of $\hat{\mathcal{H}}$ with such vectors. The cost of an exact calculation therefore doubles with every added spin, and exact diagonalisation is limited to a few dozen spins.

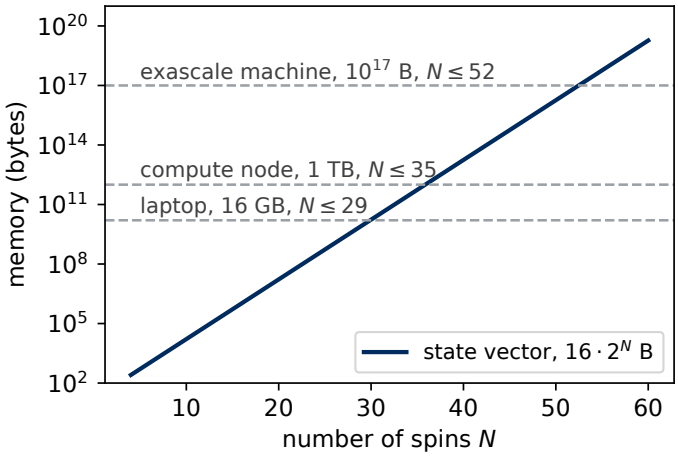


Figure 2: Memory needed to store the state vector of N spins in double precision, $16 \cdot 2^N$ bytes, against N, with the memory of a laptop, of a compute node and of an exascale machine for comparison. Every added spin doubles the memory.

Two spin chains serve as running models in these notes. The transverse-field Ising model (TFIM) on an open chain is

$$\hat{\mathcal{H}}_{\mathrm{TFIM}} = -J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1} - h \sum_{i=0}^{N-1} \hat{X}_i, \tag{2.2}$$

where $\hat{X}_i$ and $\hat{Z}_i$ are the Pauli operators of spin i. The coupling $J > 0$ favours neighbours aligned along z, both in $|0\rangle$ or both in $|1\rangle$, and the transverse field h favours every spin in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, the eigenstate of $\hat{X}$ with eigenvalue $+1$. The two terms do not commute, so the ground state is a superposition of many bit strings. In an infinite chain their competition produces a quantum phase transition at $h = J$ between an ordered phase for $h < J$, in which the spins tend to align along z, and a paramagnetic phase for $h > J$, in which they align with the field. The second model is the XXZ chain,

$$\hat{\mathcal{H}}_{\mathrm{XXZ}} = J \sum_{i=0}^{N-2} (\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \Delta\, \hat{Z}_i \hat{Z}_{i+1}), \tag{2.3}$$

with the anisotropy Δ. At $\Delta = 1$ it is the antiferromagnetic Heisenberg chain. We use $J = 1$ throughout and measure energies in units of J.

2.2 The variational principle

Every method that avoids the full state vector restricts the search to a family of trial states $|\psi(\boldsymbol{\theta})\rangle$ described by a few parameters $\boldsymbol{\theta}$, called an *ansatz*, and chooses the parameters by minimising the energy. To justify this choice, let $E_0 \le E_1 \le E_2 \le \dots$ be the eigenvalues of $\hat{\mathcal{H}}$ and $|E_m\rangle$ its orthonormal eigenvectors. Any normalised state can be expanded as $|\psi\rangle = \sum_m c_m |E_m\rangle$ with $\sum_m |c_m|^2 = 1$, and its mean energy is

$$\langle\psi|\hat{\mathcal{H}}|\psi\rangle = \sum_m |c_m|^2 E_m \;\ge\; E_0 \sum_m |c_m|^2 = E_0, \tag{2.4}$$

because every E_m in the sum is at least E_0. Equality holds only when all the weight lies on ground states. Every trial state therefore gives an upper bound on the ground-state energy (Figure 3), and minimising the energy over an ansatz gives the best bound that the family allows, which is the Rayleigh–Ritz variational method. The minimising state is the best approximation of the ground state within the family in the sense of the energy, and its quality in other respects must be checked separately.

Two product states of the TFIM give bounds without any computation. In $|0\dots0\rangle$ every $\hat{Z}_i\hat{Z}_{i+1}$ has the value $+1$ and every $\langle\hat{X}_i\rangle$ vanishes, so the energy is $-J(N-1)$. In $|+\cdots+\rangle$ every $\langle\hat{X}_i\rangle = 1$

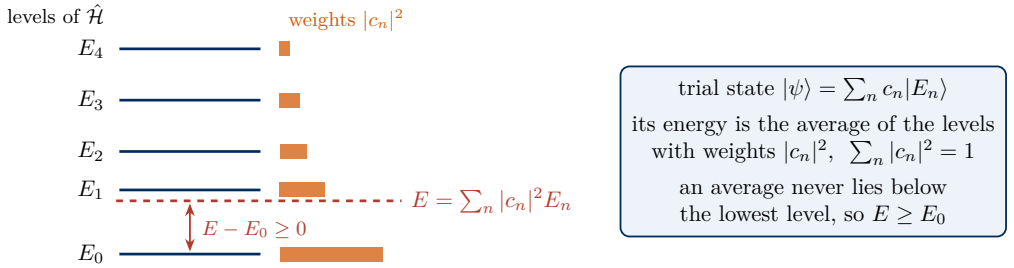


Figure 3: The variational principle (2.4). The energy levels $E_0 \leq E_1 \leq \dots$ of $\hat{\mathcal{H}}$, and the energy of any trial state as a weighted average of them, which lies at or above E_0. Minimising over an ansatz lowers this upper bound.

and every $\langle \hat{Z}_i \hat{Z}_{i+1} \rangle = 0$, so the energy is $-hN$. For $N = 6$ and $h = J$ these bounds are -5 and -6, both above the exact ground-state energy $E_0 = -7.2962$.

2.3 Mean field for the Ising chain

Mean-field theory, the oldest family of ansätze for interacting particles, replaces the ground state by a product state without entanglement, in which each particle feels the average field of all others. For bosons in a trap it leads to the Gross–Pitaevskii equation of Bose–Einstein condensates [11, 12]. For the spin chain every spin occupies the same single-qubit state,

$$|\psi_{\mathrm{MF}}(\theta)\rangle = |\phi(\theta)\rangle^{\otimes N}, \qquad |\phi(\theta)\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle = \hat{R}_y(\theta)|0\rangle, \tag{2.5}$$

where $\hat{R}_y(\theta) = e^{-i\theta\hat{Y}/2}$ is a rotation about the y axis of the Bloch sphere. The Bloch vector of every spin lies in the xz plane at the polar angle θ, so $\langle \hat{Z}_i \rangle = \cos\theta$ and $\langle \hat{X}_i \rangle = \sin\theta$. A $\hat{Y}$ component cannot lower the energy of (2.2) and is left out. The state is stored as one angle in place of 2^N amplitudes. In a product state the correlator factorises, $\langle \hat{Z}_i \hat{Z}_{i+1} \rangle = \langle \hat{Z}_i \rangle \langle \hat{Z}_{i+1} \rangle = \cos^2\theta$, and the energy of the open chain becomes

$$E_{\mathrm{MF}}(\theta) = -J(N-1)\cos^2\theta - hN\sin\theta. \tag{2.6}$$

The open chain has $N-1$ bonds and N spins, and the ratio $(N-1)/N$ tends to one for a long chain, where the energy per spin is $E_{\mathrm{MF}}/N = -J\cos^2\theta - h\sin\theta$. The best angle makes the derivative vanish,

$$\frac{dE_{\mathrm{MF}}}{d\theta} = \cos\theta\,\big[2J(N-1)\sin\theta - hN\big] = 0. \tag{2.7}$$

This stationarity condition is the self-consistent equation of mean-field theory, as the magnetisation $\zeta = \langle \hat{Z}_i \rangle = \cos\theta$ shows. Each spin feels the transverse field h along x and, from its neighbours, the average longitudinal field $2J(N-1)\zeta/N$ along z, where $2(N-1)/N$ is the average number of neighbours on the open chain. This average field of the neighbours is the molecular field that Weiss introduced for ferromagnets [13]. The Bloch vector points along the total field, so its z component is

$$\zeta = \frac{2J(N-1)\zeta/N}{\sqrt{\big(2J(N-1)\zeta/N\big)^2 + h^2}}, \tag{2.8}$$

which is (2.7) for $\cos\theta \neq 0$. The magnetisation appears on both sides, since the field that orients a spin is produced by the orientation of the others. Equation (2.8) always has the solution $\zeta = 0$, and for small h a second solution with $\zeta \neq 0$. Solving for $\sin\theta$ gives the minimum of (2.6),

$$\begin{aligned} \sin\theta^* &= \frac{h}{h_c^{\mathrm{MF}}}, & E_{\mathrm{MF}} &= -J(N-1) - \frac{h^2 N^2}{4J(N-1)} & &\text{for } h < h_c^{\mathrm{MF}} = \frac{2J(N-1)}{N}, \\ \theta^* &= \frac{\pi}{2}, & E_{\mathrm{MF}} &= -hN & &\text{for } h \geq h_c^{\mathrm{MF}}. \end{aligned} \tag{2.9}$$

Figure 4 shows how the minimum of (2.6) moves with the field. Below h_c^{MF} the spins tilt away from z and keep the magnetisation $\zeta = \pm\cos\theta^*$, with two degenerate solutions related by the spin flip $\hat{Z} \to -\hat{Z}$. Above it all spins point along x. In a long chain the mean-field transition lies at $h_c^{\mathrm{MF}} = 2J$, twice the exact critical field $h_c = J$, and for $N = 6$ it lies at $1.667J$ (the ratio $(N-1)/N$ lowers it). The product states $|0\ldots0\rangle$ and $|+\cdots+\rangle$ of Section 2.2 are the members $\theta = 0$ and $\theta = \pi/2$ of this family.

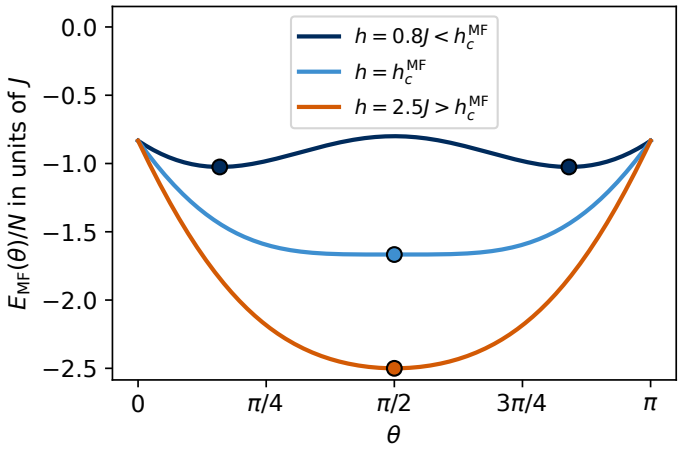


Figure 4: Mean-field energy (2.6) of the six-spin TFIM as a function of the angle θ for several fields h. Below h_c^{MF} the minimum lies at $\sin\theta^* = h/h_c^{\mathrm{MF}}$, above it at $\theta^* = \pi/2$. Look at how the minimum moves with h.

The infinite chain allows an exact comparison, because the TFIM maps onto free fermions and its exact ground-state energy per spin is known [14],

$$e_\infty(h) = -\frac{1}{\pi}\int_0^\pi dk\,\sqrt{J^2 + h^2 - 2Jh\cos k}. \tag{2.10}$$

At $h = J$ the integrand is $2J\sin(k/2)$ and $e_\infty = -4J/\pi \approx -1.2732J$, whereas mean field gives $-J - J/4 = -1.2500J$. Figure 5 compares mean field with the exact results for $N = 6$ and for the infinite chain. Every mean-field energy lies above its exact counterpart, as (2.4) requires, and the difference is largest between the exact critical point and the mean-field transition. At $h = J$ the mean-field energy of the six-spin chain is -6.800 against the exact -7.2962. Far from the transition, at small and at large h, the exact ground state is close to a product state and mean field becomes accurate. In the range $J < h < 2J$ mean field predicts a magnetisation where the exact infinite chain has none, so it overestimates the order.

In Figure 5(b) the mean-field correlator $\cos^2\theta^*$ vanishes above h_c^{MF}, whereas the exact correlator decreases smoothly and stays positive, because quantum fluctuations correlate neighbouring spins without any magnetisation. A product state has zero half-chain entanglement entropy by construction and describes correlations only through the magnetisation.

In the language of circuits the mean-field state (2.5) is a circuit with one $\hat{R}_y(\theta)$ on every qubit and no entangling gate (Figure 6).

2.4 Tensor networks

Tensor networks keep entanglement while storing a number of parameters that grows only polynomially with N. A matrix product state (MPS) writes every amplitude as a product of matrices, one per spin,

$$\psi_{b_0 b_1 \ldots b_{N-1}} = A^{[0]}_{b_0} A^{[1]}_{b_1} \cdots A^{[N-1]}_{b_{N-1}}, \tag{2.11}$$

where $A^{[i]}_{b_i}$ is a matrix of size at most $\chi \times \chi$ chosen by the value of bit b_i, the first and last are a row and a column vector, and χ is the bond dimension (Figure 7). The MPS stores about $2N\chi^2$ numbers.

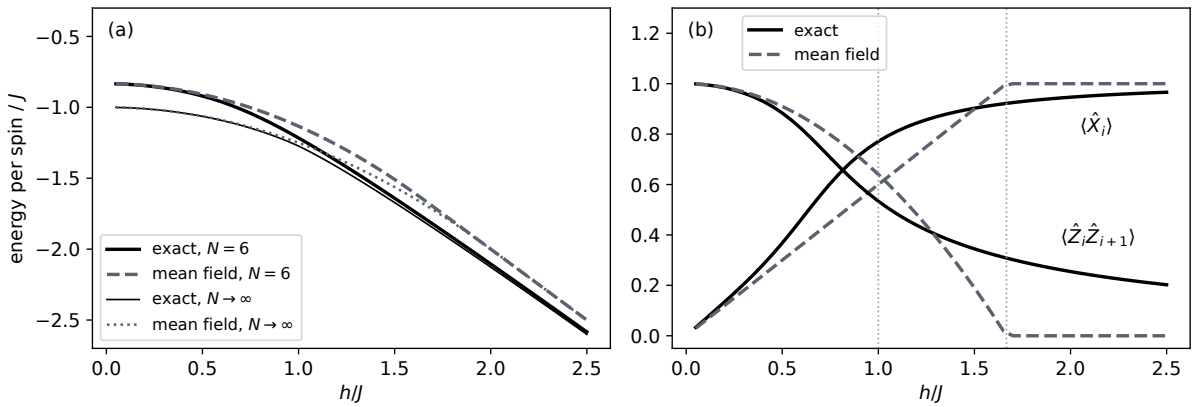


Figure 5: Mean field for the TFIM (2.2). (a) Energy per spin of the mean-field state (2.9) (dashed) and of the exact ground state (solid), for $N = 6$ (thick lines) and for the infinite chain (2.10) (thin lines, mean field dotted). (b) Transverse magnetisation $\langle \hat{X}_i \rangle$ and nearest-neighbour correlator $\langle \hat{Z}_i \hat{Z}_{i+1} \rangle$, averaged over the chain of six spins, exact (solid) and in mean field (dashed). The dotted lines mark the exact critical field $h = J$ and the mean-field transition h_c^{MF} of the six-spin chain. Look at the kink of the mean-field curves at h_c^{MF}, which the exact curves do not have.

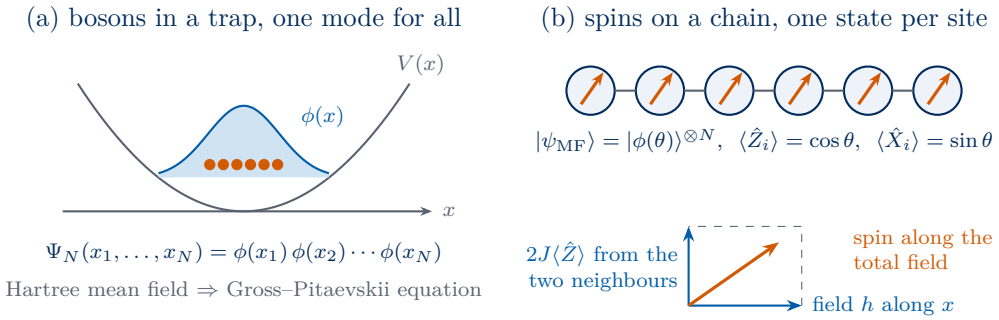
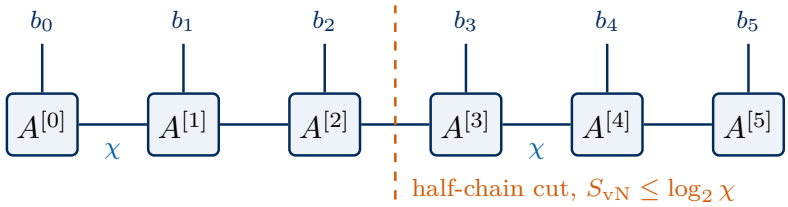


Figure 6: Product ansätze of mean field. (a) Bosons in a trap all occupy one mode $\phi(x)$, the ansatz behind the Gross–Pitaevskii equation. (b) The interacting chain is replaced by independent spins, each in the same state and each feeling the transverse field h and the average field of its neighbours.



$$\psi(b_0, \ldots, b_{N-1}) = A^{[0]}_{b_0} A^{[1]}_{b_1} \cdots A^{[N-1]}_{b_{N-1}}, \quad \text{each } A^{[i]}_{b_i} \text{ a matrix of size at most } \chi \times \chi$$

Figure 7: A matrix product state (2.11). Each spin carries one tensor with a physical index b_i and two bond indices of dimension χ, and contracting the bonds gives the amplitude of a bit string. The bond dimension bounds the entanglement across every cut.

Across any cut of the chain the state has at most χ Schmidt terms, so its entanglement entropy is at most $\log_2 \chi$ bits. A product state is an MPS with $\chi = 1$.

Ground states of gapped one-dimensional Hamiltonians with local interactions obey an area law, with an entanglement entropy across a cut that stays bounded as N grows [15, 16]. A modest χ then describes them accurately, and the density-matrix renormalisation group (DMRG) optimises such states variationally [17, 18]. The required χ grows exponentially with the entanglement that must be represented. Entanglement grows with the width of two-dimensional systems and linearly in time after a sudden change of the Hamiltonian, which limits tensor networks in both situations.

2.5 Monte Carlo methods

Monte Carlo methods estimate expectation values from random samples and never store the full state (Figure 8). Quantum Monte Carlo maps the quantum problem onto the sampling of classical configurations with non-negative weights. For fermions and frustrated magnets the weights take both signs, and the statistical error then grows exponentially with the system size and the inverse temperature. This sign problem is NP-hard in general, so an efficient general solution would solve every problem of the complexity class NP efficiently [19]. Variational Monte Carlo samples bit strings from $|\psi_{\mathbf{b}}|^2$ of a trial wave function whose amplitudes are computed by a classical formula, and a neural network can serve as this formula, with its weights trained to minimise the energy [10].

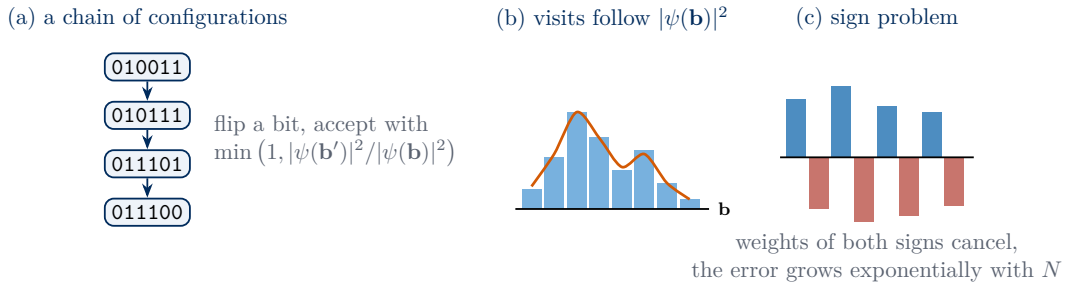


Figure 8: Monte Carlo methods. Configurations are sampled with probabilities given by non-negative weights, and expectation values are estimated as averages over the samples without storing the state. When the weights take both signs, the average is a small difference of large positive and negative contributions, the sign problem.

Each classical method restricts the states it can represent. Mean field excludes entanglement, tensor networks bound it, and Monte Carlo methods are limited by the sign structure of the amplitudes or by the form of the trial wave function.

3 Parametrised circuits as quantum ansätze

3.1 Trial states prepared by a quantum device

Over the last two decades experiments with trapped ions, neutral atoms in optical tweezers and superconducting circuits have reached individual control of tens to hundreds of qubits [20]. Each qubit is prepared in $|0\rangle$, rotated by laser or microwave pulses, coupled to its neighbours by a controlled interaction and measured in the computational basis. A register of N such qubits holds the 2^N amplitudes of (2.1) physically. Such a register can carry a variational ansatz built from the operations of the hardware, with pulse areas and interaction times as parameters optimised by a classical computer and a trial state that is never stored classically. A quantum processor can thus prepare the trial state of a variational calculation itself [21], and the resulting hybrid quantum–classical scheme is the *variational quantum eigensolver* (VQE) [21, 22].

The register starts in $|0\ldots0\rangle$ and a circuit with n adjustable angles $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ acts on it,

$$|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})\,|0\ldots0\rangle, \qquad \hat{U}(\boldsymbol{\theta}) = \hat{W}_n\, e^{-i\theta_n \hat{P}_n/2} \cdots \hat{W}_2\, e^{-i\theta_2 \hat{P}_2/2}\, \hat{W}_1\, e^{-i\theta_1 \hat{P}_1/2}, \tag{3.1}$$

where each $\hat{P}_k$ is a Pauli string, a tensor product of $\hat{I}$, $\hat{X}$, $\hat{Y}$ and $\hat{Z}$ on the qubits, and each $\hat{W}_k$ is a fixed gate without an adjustable angle, such as a controlled-$\hat{Z}$ gate $\widehat{\mathrm{CZ}}$, or the identity. Every parametrised gate is a rotation $\hat{R}_P(\theta) = e^{-i\theta\hat{P}/2}$, and since $\hat{P}^2 = \hat{I}$ its exponential series sums to

$$\hat{R}_P(\theta) = e^{-i\theta\hat{P}/2} = \cos\frac{\theta}{2}\,\hat{I} - i\sin\frac{\theta}{2}\,\hat{P}. \tag{3.2}$$

In the hardware a single-qubit rotation $\hat{R}_x(\theta)$ or $\hat{R}_y(\theta)$ is a resonant laser or microwave pulse of Rabi frequency Ω and duration t, with the pulse area $\theta = \Omega t$. A two-qubit rotation $\hat{R}_{zz}(\theta) = e^{-i\theta\hat{Z}_i\hat{Z}_j/2}$

is an Ising interaction $\frac{J}{2}\hat{Z}_i\hat{Z}_j$ switched on for the time t, with $\theta = Jt$. Changing an angle therefore means changing the length of a pulse or of an interaction.

The ansatz (3.1) is fully quantum, since the trial state exists only on the register, its 2^N amplitudes are never written down and the device returns only bit strings. Different circuit layouts define different ansätze, just as mean field, matrix product states and neural-network wave functions define different classical ansätze. The mean-field state (2.5) is the member with one $\hat{R}_y$ per qubit, and entangling gates such as $\widehat{\mathrm{CZ}}$ or $\hat{R}_{zz}$ take the ansatz beyond product states. Table 1 relates the names used in physics, mathematics and computer science.

3.2 The hybrid loop

The energy $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{\mathcal{H}}|\psi(\boldsymbol{\theta})\rangle$ is a function of n real numbers, and by (2.4) it never falls below E_0. The calculation alternates between the quantum device and a classical computer (Figure 9). The device prepares the trial state and measures it many times, and the classical computer turns the measured bit strings into an estimate of the energy and of its gradient and proposes new angles.

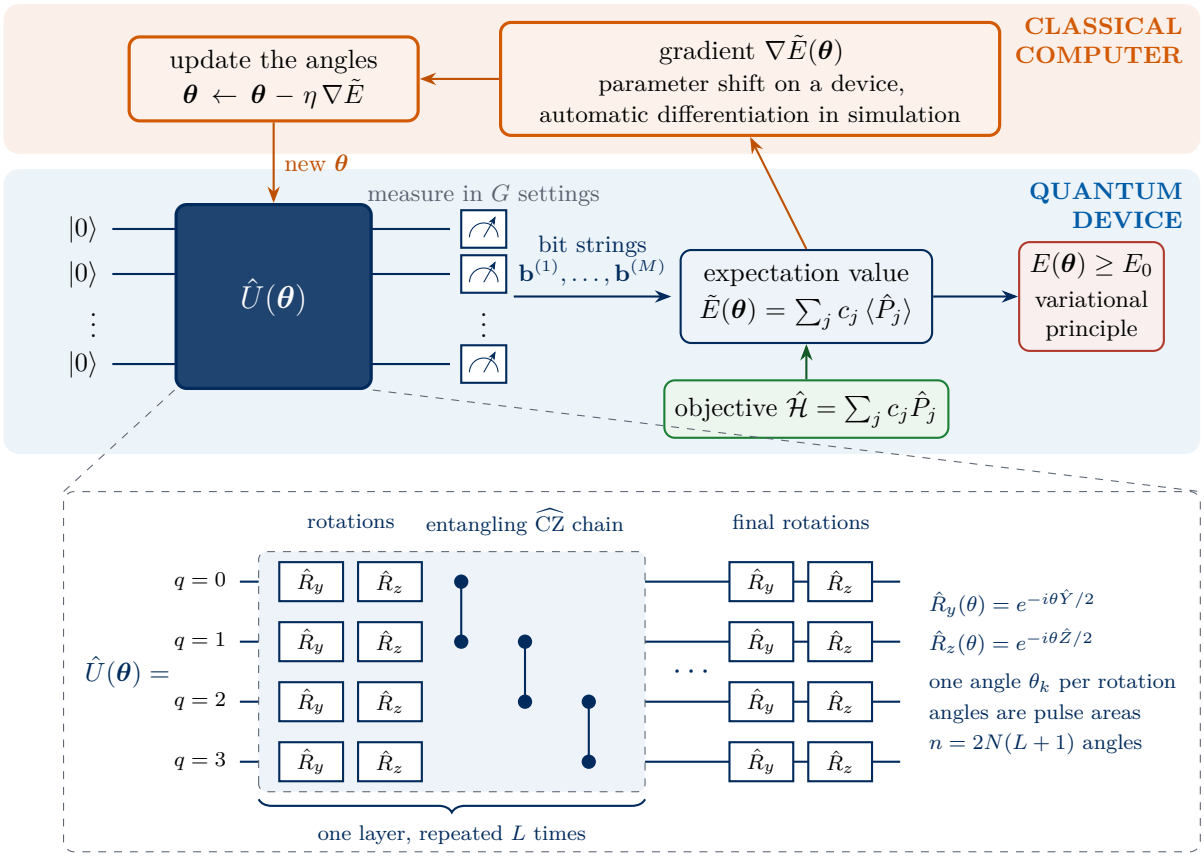


Figure 9: The variational loop. The quantum device prepares $|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle$, measures it in G settings and returns bit strings. The classical computer estimates the expectation value of the objective $\hat{\mathcal{H}} = \sum_j c_j\hat{P}_j$ and its gradient and updates the angles $\boldsymbol{\theta}$. The dashed frame shows the hardware-efficient ansatz inside $\hat{U}(\boldsymbol{\theta})$, layers of single-qubit rotations with independent angles followed by a chain of $\widehat{\mathrm{CZ}}$ gates.

Variational quantum eigensolver

Given a Hamiltonian $\hat{\mathcal{H}} = \sum_j c_j\hat{P}_j$, a circuit $\hat{U}(\boldsymbol{\theta})$ with n angles, a number M of shots per measurement setting and a step size η, the classical computer chooses initial angles $\boldsymbol{\theta}$, and the algorithm repeats three operations.

1. The device prepares $|\psi(\boldsymbol{\theta})\rangle$, rotates the qubits into each measurement setting of $\hat{\mathcal{H}}$ and records M bit strings per setting.

2. The classical computer forms an estimate $\tilde{E}(\boldsymbol{\theta})$ of the energy from the bit strings and, from further runs at shifted angles, an estimate $\tilde{\mathbf{g}}$ of its gradient.
3. It updates the angles, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\tilde{\mathbf{g}}$ or by a variant of gradient descent, and sends them to the device, until the energy stops decreasing.

The output is the last energy estimate, an approximate upper bound on E_0 within its statistical error, and the angles, which prepare the trial state again on demand.

3.3 The accuracy of a variational state

The trained state is also used to compute properties other than the energy, such as magnetisations and correlations, so its distance from the ground state matters as well. The *fidelity* $F = |\langle E_0|\psi\rangle|^2 = |c_0|^2$ is the probability that the trial state passes a test for being the ground state, and $1-F$ is the infidelity. Subtracting E_0 from (2.4) gives

$$E - E_0 = \sum_{m\geq 1} |c_m|^2 (E_m - E_0) \;\geq\; \Delta E \sum_{m\geq 1} |c_m|^2 = \Delta E\,(1-F), \tag{3.3}$$

where $\Delta E = E_1 - E_0$ is the gap between the ground state and the first excited state, assuming a non-degenerate ground state. The energy error therefore bounds the infidelity (Figure 10), $1 - F \leq (E - E_0)/\Delta E$. A small energy error guarantees a fidelity close to one only when it is small compared with the gap. For the critical TFIM with $N = 6$ the gap is $\Delta E = 0.482J$, and an energy error of $10^{-3}J$ guarantees an infidelity of at most about 2×10^{-3}. Near a phase transition, or in an ordered phase where the two lowest levels almost coincide, the gap is small and the bound becomes weak. We use $E - E_0$ or the relative error $\delta E = (E - E_0)/|E_0|$, the fidelity F, and the half-chain entanglement entropy S_{vN} to judge trained states. The fidelity and the entropy require the exact ground state and the full trial state, which only a simulation of a small register provides.

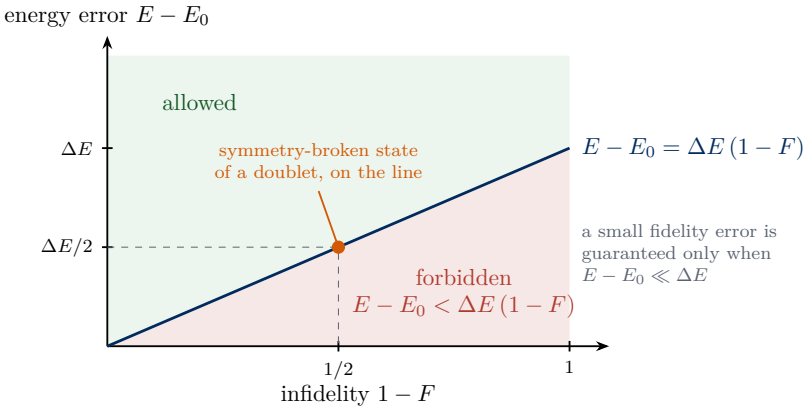


Figure 10: The fidelity bound (3.3). Every unit of weight $1 - F$ outside the ground state raises the energy by at least the gap ΔE, so the energy error bounds the infidelity. A small gap weakens the bound.

4 The energy from bit strings

4.1 Pauli strings and measurement settings

A device returns bit strings, and the energy has to be assembled from them. Every Hamiltonian of N qubits is a real combination of Pauli strings,

$$\hat{\mathcal{H}} = \sum_j c_j \hat{P}_j, \qquad E(\boldsymbol{\theta}) = \sum_j c_j \,\langle \hat{P}_j \rangle_{\boldsymbol{\theta}}, \tag{4.1}$$

where $\langle \hat{P}_j \rangle_{\boldsymbol{\theta}} = \langle \psi(\boldsymbol{\theta}) | \hat{P}_j | \psi(\boldsymbol{\theta}) \rangle$. The TFIM (2.2) has $2N-1$ strings, $N-1$ strings $\hat{Z}_i \hat{Z}_{i+1}$ with coefficient $-J$ and N strings $\hat{X}_i$ with coefficient $-h$, and the XXZ chain (2.3) has $3(N-1)$ strings.

A string of $\hat{Z}$ operators is diagonal in the computational basis. When every qubit is measured and the bit string $\mathbf{b}$ is recorded, $\hat{Z}_i$ takes the value $(-1)^{b_i}$, because $\hat{Z}|0\rangle = |0\rangle$ and $\hat{Z}|1\rangle = -|1\rangle$, and the product $\hat{Z}_i \hat{Z}_{i+1}$ takes the value $(-1)^{b_i + b_{i+1}}$, the parity of the two bits. In general a Pauli string with support S, the set of qubits on which it acts nontrivially, takes the value

$$P(\mathbf{b}) = (-1)^{\sum_{q \in S} b_q} \tag{4.2}$$

after each of its factors has been rotated into $\hat{Z}$. The rotation uses $\hat{H}\hat{X}\hat{H} = \hat{Z}$ for the Hadamard gate $\hat{H}$, so a qubit is measured along x by applying $\hat{H}$ before the readout, and along y by applying $\hat{S}^\dagger$ and then $\hat{H}$, where $\hat{S} = \mathrm{diag}(1, i)$. (The letter $\hat{H}$ denotes the Hadamard gate and $\hat{\mathcal{H}}$ the Hamiltonian.)

One choice of readout basis for every qubit is a *measurement setting*, and one setting measures all strings that carry on each qubit either the identity or the Pauli matrix of that qubit's basis. The TFIM therefore needs $G = 2$ settings, all qubits along z for the bond terms and all qubits along x for the field terms. The XXZ chain needs $G = 3$ settings, all along x, all along y and all along z. Each shot of a setting g gives one value of the part of the energy that this setting measures. For the TFIM

$$e_Z(\mathbf{b}) = -J \sum_{i=0}^{N-2} (-1)^{b_i + b_{i+1}}, \qquad e_X(\mathbf{b}) = -h \sum_{i=0}^{N-1} (-1)^{b_i}, \tag{4.3}$$

where $\mathbf{b}$ in e_X is recorded after a Hadamard gate on every qubit. The energy is the sum of the two averages over the Born distributions of the settings,

$$E(\boldsymbol{\theta}) = \sum_{\mathbf{b}} p_Z(\mathbf{b})\, e_Z(\mathbf{b}) + \sum_{\mathbf{b}} p_X(\mathbf{b})\, e_X(\mathbf{b}), \qquad p_Z(\mathbf{b}) = |\langle \mathbf{b} | \psi \rangle|^2, \quad p_X(\mathbf{b}) = |\langle \mathbf{b} | \hat{H}^{\otimes N} | \psi \rangle|^2. \tag{4.4}$$

4.2 The estimator and its standard error

A device returns M bit strings $\mathbf{b}^{(1)}, \ldots, \mathbf{b}^{(M)}$ in the z setting and M bit strings $\mathbf{b}'^{(1)}, \ldots, \mathbf{b}'^{(M)}$ in the x setting. The estimate of the energy replaces each average in (4.4) by the sample mean (Figure 11),

$$\tilde{E} = \frac{1}{M} \sum_{s=1}^{M} e_Z(\mathbf{b}^{(s)}) + \frac{1}{M} \sum_{s=1}^{M} e_X(\mathbf{b}'^{(s)}). \tag{4.5}$$

Its mean over many repetitions of the experiment is the exact energy. The shots are independent, so the variance of each sample mean is σ_g^2 / M, where

$$\sigma_g^2 = \sum_{\mathbf{b}} p_g(\mathbf{b})\, e_g(\mathbf{b})^2 - \Big(\sum_{\mathbf{b}} p_g(\mathbf{b})\, e_g(\mathbf{b}) \Big)^2 \tag{4.6}$$

is the variance of a single shot of setting $g \in \{Z, X\}$. For a single Pauli string with values ± 1 this is $1 - \langle \hat{P} \rangle^2$, which vanishes only when the state is an eigenstate of $\hat{P}$. The settings are measured independently, so their variances add and the standard error of the energy is

$$\sigma_{\tilde{E}} = \sqrt{\frac{1}{M} \sum_{g=1}^{G} \sigma_g^2}. \tag{4.7}$$

In an experiment σ_g^2 is unknown and is replaced by the sample variance of the recorded values.

As an example we take a six-qubit *hardware-efficient* circuit, named so because it uses only gates that devices apply natively. Each of its four layers rotates every qubit by $\hat{R}_y$ and $\hat{R}_z$ and couples neighbouring qubits by a chain of $\widehat{\mathrm{CZ}}$ gates, which gives $n = 60$ angles. With the critical TFIM as the Hamiltonian and random angles, $M = 1000$ shots per setting give the estimate -0.172 ± 0.107 of the exact energy -0.186. Repeating the whole measurement many times (Figure 12) confirms the scaling

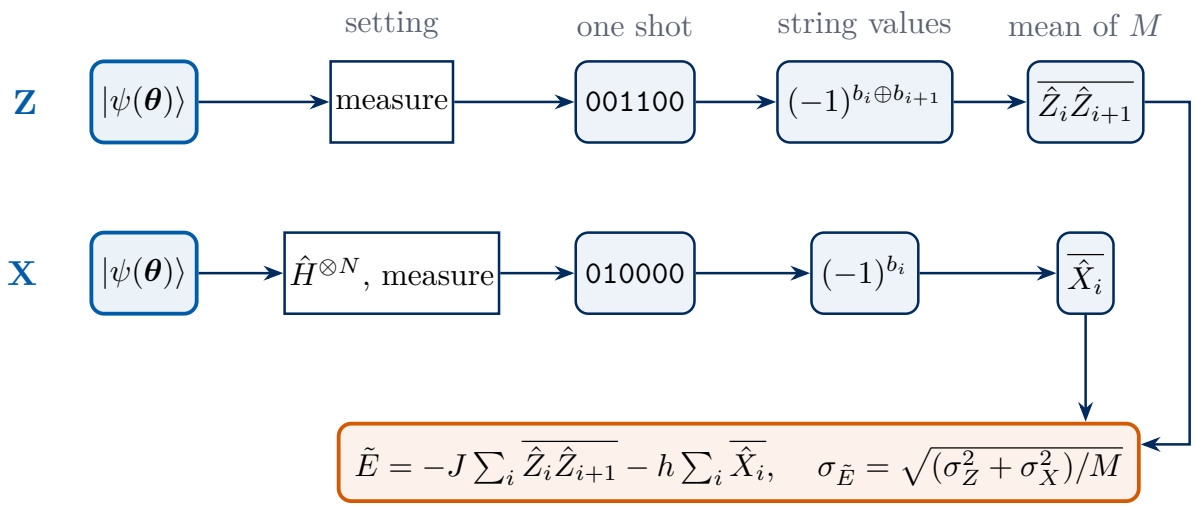


Figure 11: Energy estimate from bit strings for the TFIM. Shots in the z setting give the bond terms as parities of neighbouring bits, shots in the x setting (Hadamard gates before readout) give the field terms, and the two sample means add to $\tilde{E}$ (4.5) with the standard error (4.7).

$M^{-1/2}$ of (4.7) over three orders of magnitude. Every tenfold reduction of the error bar therefore costs a hundred times more shots, and a standard error of $0.01J$ at these angles needs about 1.1×10^5 shots per setting. The single-shot variances depend on the state, so the same target needs a different number of shots at other angles.

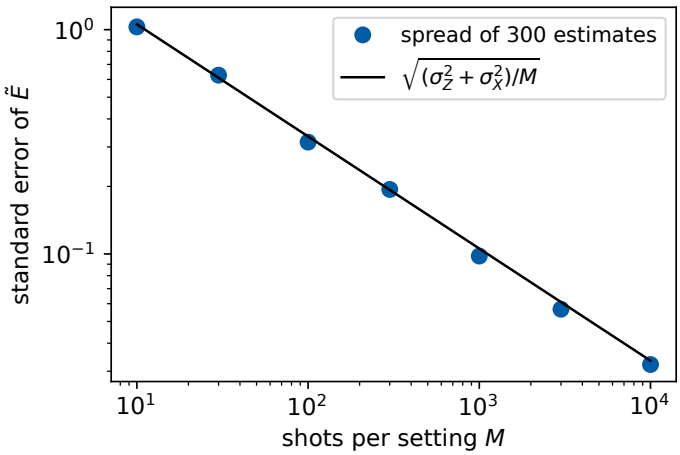


Figure 12: Standard deviation of repeated energy estimates (4.5) against the number M of shots per setting, from simulated measurement records of a six-qubit trial state at fixed random angles (markers), and the prediction (4.7) (line). The slope $-1/2$ on the logarithmic axes is the shot-noise scaling.

The shot budget is the main practical cost of variational algorithms on hardware. Molecular Hamiltonians contain many Pauli strings that fall into many settings, and every setting adds its own record of M shots. Grouping commuting strings into few settings reduces the cost, and the TFIM and XXZ chains with two and three settings are among the cheapest cases.

5 Gradients

5.1 The energy along one angle

Gradient descent needs the derivatives $\partial E/\partial \theta_k$ of the energy with respect to every angle. On a device they have to come from measurements of the same circuit, and their form follows from the dependence

of the energy on a single angle. We split the circuit (3.1) at the rotation that carries θ_k,

$$\hat{U}(\boldsymbol{\theta}) = \hat{A}\, e^{-i\theta_k \hat{P}_k/2}\, \hat{B}, \tag{5.1}$$

where $\hat{B}$ contains all gates before the rotation and $\hat{A}$ all gates after it, and neither depends on θ_k. With $|\phi\rangle = \hat{B}|0\ldots0\rangle$ and the operator $\hat{O} = \hat{A}^\dagger \hat{\mathcal{H}} \hat{A}$, the energy as a function of θ_k alone is

$$E(\theta_k) = \langle\phi|\, e^{i\theta_k \hat{P}_k/2}\, \hat{O}\, e^{-i\theta_k \hat{P}_k/2}\, |\phi\rangle. \tag{5.2}$$

Inserting the rotation (3.2) and multiplying out the four products gives

$$E(\theta_k) = \cos^2\frac{\theta_k}{2}\,\langle\hat{O}\rangle + \sin^2\frac{\theta_k}{2}\,\langle\hat{P}_k\hat{O}\hat{P}_k\rangle + i\sin\frac{\theta_k}{2}\cos\frac{\theta_k}{2}\,\langle[\hat{P}_k,\hat{O}]\rangle, \tag{5.3}$$

with all expectation values in $|\phi\rangle$ and the commutator $[\hat{P}_k,\hat{O}] = \hat{P}_k\hat{O} - \hat{O}\hat{P}_k$. The half-angle identities $\cos^2(\theta/2) = (1+\cos\theta)/2$, $\sin^2(\theta/2) = (1-\cos\theta)/2$ and $2\sin(\theta/2)\cos(\theta/2) = \sin\theta$ turn this into a shifted cosine,

$$E(\theta_k) = a + u\cos\theta_k + v\sin\theta_k = a + b\cos(\theta_k - c), \tag{5.4}$$

where

$$a = \tfrac{1}{2}(\langle\hat{O}\rangle + \langle\hat{P}_k\hat{O}\hat{P}_k\rangle), \qquad u = \tfrac{1}{2}(\langle\hat{O}\rangle - \langle\hat{P}_k\hat{O}\hat{P}_k\rangle), \qquad v = \tfrac{i}{2}\langle[\hat{P}_k,\hat{O}]\rangle, \tag{5.5}$$

the amplitude is $b = \sqrt{u^2+v^2}$ and the phase c satisfies $\cos c = u/b$ and $\sin c = v/b$. All three constants are real, because $i[\hat{P}_k,\hat{O}]$ is Hermitian. The derivation uses only $\hat{P}_k^2 = \hat{I}$, so it holds for every rotation of the circuit and for every Hamiltonian.

Along any single angle the landscape is therefore a cosine of period 2π with one minimum and one maximum, and three energies fix it, $E(0) = a+u$ and $E(\pm\pi/2) = a \pm v$. Figure 13 confirms this for one angle of the six-qubit hardware-efficient circuit of Section 4.2, where the cosine through three points coincides with the energy at all other angles to numerical precision. The fitted constants are $a = -0.582$, $b = 0.563$ and $c = 1.875$. The difficulty of the optimisation comes from the n angles together, since the constants of one angle depend on the values of all others.

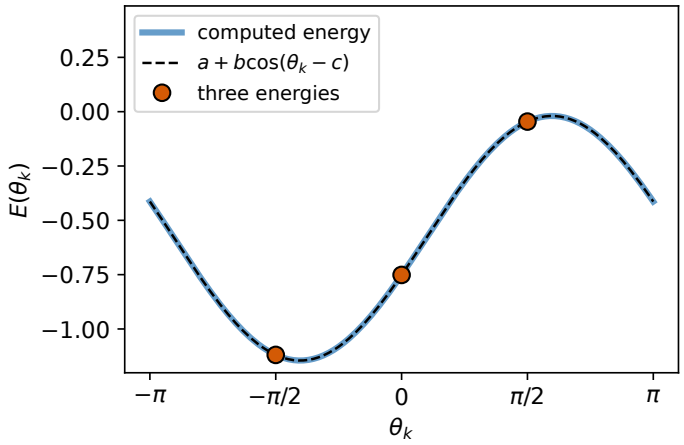


Figure 13: The energy of a six-qubit trial state of the critical TFIM as a function of one angle, with all other angles fixed (solid line), and the cosine (5.4) through the three energies at $\theta_k = 0$ and $\pm\pi/2$ (dots, dashed line). Exact state-vector values.

5.2 The parameter-shift rule

Differentiating (5.4) gives $\partial E/\partial\theta_k = -u\sin\theta_k + v\cos\theta_k$. The same combination appears in the difference of two energies at angles shifted by $\pm\pi/2$,

$$\begin{aligned} E(\theta_k + \tfrac{\pi}{2}) - E(\theta_k - \tfrac{\pi}{2}) &= u\big[\cos(\theta_k + \tfrac{\pi}{2}) - \cos(\theta_k - \tfrac{\pi}{2})\big] + v\big[\sin(\theta_k + \tfrac{\pi}{2}) - \sin(\theta_k - \tfrac{\pi}{2})\big] \\ &= -2u\sin\theta_k + 2v\cos\theta_k, \end{aligned} \tag{5.6}$$

where $\cos(\theta \pm \pi/2) = \mp \sin\theta$ and $\sin(\theta \pm \pi/2) = \pm \cos\theta$. Hence

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\boldsymbol{\theta} + \frac{\pi}{2}\mathbf{e}_k) - E(\boldsymbol{\theta} - \frac{\pi}{2}\mathbf{e}_k)}{2}, \tag{5.7}$$

where $\mathbf{e}_k$ is the unit vector along angle k. This relation is the parameter-shift rule [23, 24] (Figure 14). The shift is finite, the formula is exact, and both energies are measured with the same circuit on the same device, with one angle changed. For a rotation implemented as a pulse of Rabi frequency Ω and duration $t_k = \theta_k/\Omega$, the two shifted circuits differ from the original only in the pulse duration, $t_k \pm \pi/(2\Omega)$.

Many ansätze use one angle in several gates. When an angle φ enters gates $g = 1, 2, \ldots$ with rotation angles $\theta_g = c_g \varphi$ for constants c_g, the energy depends on φ through all of them, and the chain rule gives

$$\frac{\partial E}{\partial \varphi} = \sum_g c_g \frac{\partial E}{\partial \theta_g}. \tag{5.8}$$

Each term is evaluated with the two-point rule (5.7) by shifting the angle of gate g alone and keeping all other occurrences of φ fixed. Shifting φ itself by $\pm\pi/2$ shifts every gate at once and does not give the derivative, because the energy is then a sum of cosines of different frequencies in φ. The mean-field state (2.5) is an example, with θ in all N gates $\hat{R}_y(\theta)$ and $c_g = 1$, and (5.8) reproduces the derivative (2.7).

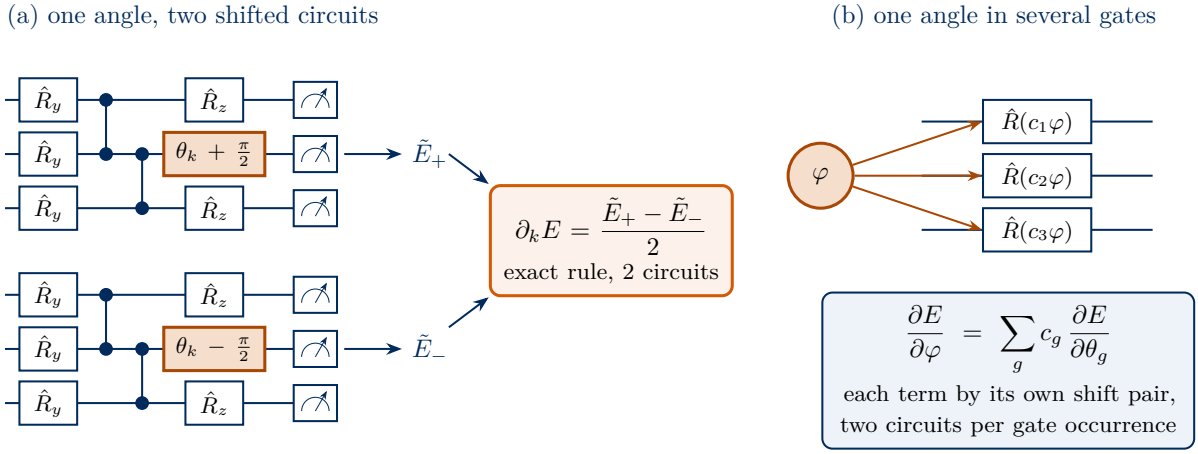


Figure 14: The parameter-shift rule (5.7). Along one angle the energy is a cosine, and the difference of its values at $\theta_k \pm \pi/2$ divided by two is the exact slope at θ_k. For an angle shared by several gates, each gate is shifted separately and the results are summed with the factors c_g of the chain rule (5.8).

5.3 The cost of a gradient on a device

On a device each energy in (5.7) is itself an estimate (4.5). Let n_{rot} be the number of parametrised Pauli rotations of the circuit, the gate occurrences of the angles, which equals n when every angle enters one gate and exceeds n when angles are shared. One gradient then requires $2n_{\text{rot}}$ circuits, each measured in G settings with M shots,

$$\text{shots per gradient} = 2\, n_{\text{rot}}\, G\, M. \tag{5.9}$$

For the six-qubit hardware-efficient circuit of Section 4.2, with $n_{\text{rot}} = n = 60$ and $G = 2$, the gradient needs 120 distinct circuits, and $M = 1000$ shots per setting give 240000 shots for a single gradient. A training run of 600 gradient steps then needs about 1.4×10^8 shots, and the number grows linearly with the number of rotations and with the number of settings.

The statistical error of a gradient component follows from (4.7). The component $\tilde{g}_k = (\tilde{E}_{k,+} - \tilde{E}_{k,-})/2$ combines two independent estimates, so

$$\sigma_{\tilde{g}_k} = \frac{1}{2}\sqrt{\sigma^2_{\tilde{E}_{k,+}} + \sigma^2_{\tilde{E}_{k,-}}} \approx \sqrt{\frac{\sigma^2}{2M}}, \tag{5.10}$$

where $\sigma^2 = \sum_g \sigma_g^2$ is the single-shot variance of the energy, close to its value at the unshifted angles. At the random angles of Section 4.2 with $M = 1000$ this error is about 0.073. Figure 15 compares the estimated components with the exact ones. They scatter within their error bars, and components whose exact value is smaller than about twice the error bar cannot be distinguished from zero with this number of shots. Halving the error bar requires four times as many shots, so small gradients are expensive on hardware.

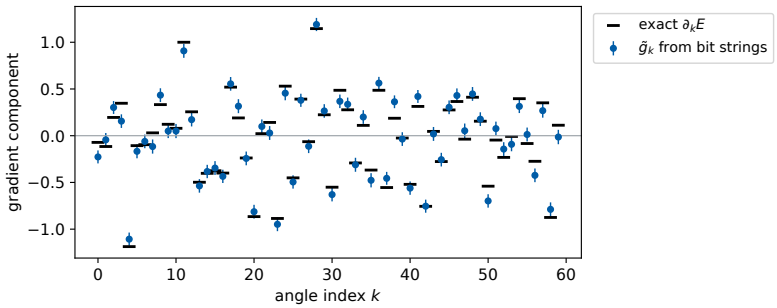


Figure 15: All 60 components of the energy gradient of a six-qubit trial state, estimated with the parameter-shift rule from simulated measurement records with $M = 1000$ shots per setting and shifted circuit (dots with error bars (5.10)), and the exact components (black dashes). Components whose exact value lies within about two error bars of zero are not resolved.

Of the other common gradient estimates, central finite differences $[E(\boldsymbol{\theta} + \epsilon \mathbf{e}_k) - E(\boldsymbol{\theta} - \epsilon \mathbf{e}_k)]/(2\epsilon)$ need the same $2n$ circuits, but the shot noise of the two energies is divided by the small step ϵ, so their error grows as $1/\epsilon$. Simultaneous-perturbation stochastic approximation (SPSA) [25] shifts all angles at once along a random direction with entries ± 1 and needs two energies per step for any n. Its estimate, averaged over the directions, equals the gradient up to corrections of second order in the size of the shift, but a single estimate is noisy and training needs more updates. Appendix A derives the estimator, combines it with Adam and compares the number of circuit configurations that SPSA and the parameter-shift rule need to reach the same energy.

5.4 Automatic differentiation in simulation

A simulation holds the state vector and can differentiate the program that computes the energy. Automatic differentiation applies the chain rule to every elementary operation of that program. Its reverse mode [26] runs the program forwards, stores the intermediate states and propagates derivatives backwards through the gates, so that the whole gradient costs a few energy evaluations for any n. We use the implementation in the JAX library [27]. For the circuit with 60 angles, the reverse-mode gradient and the parameter-shift gradient (5.7), evaluated from exact energies, agree to numerical precision in all components. In a simulation the parameter-shift rule costs $2n$ circuit evaluations, so reverse mode is preferred there. A device has no access to intermediate states, and there the parameter-shift rule is the exact method. All training runs in these notes use exact gradients from reverse-mode differentiation, which correspond to the limit $M \to \infty$ of a device and show what an ansatz can reach.

5.5 Training

With the gradient available, the angles are updated by gradient descent,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla E(\boldsymbol{\theta}), \tag{5.11}$$

with a step size η. We use Adam [28], a variant of gradient descent that scales the step of each angle with running averages of its past gradients and of their squares. Figure 16 shows a training run of the six-qubit hardware-efficient circuit on the critical TFIM, started from small angles near the product state $|0\dots0\rangle$. The energy starts at -4.45, close to the value -5 of $|0\dots0\rangle$, falls below the first excited level E_1 within the first 17 steps and then approaches E_0 slowly. After 600 steps the relative energy error is of order 1×10^{-3} and the fidelity with the exact ground state is 0.998, consistent with the bound (3.3). On a device the same run would cost the shots counted in (5.9).

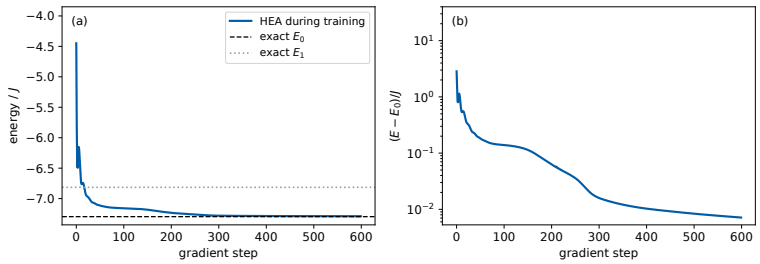


Figure 16: Energy of the six-qubit trial state of the critical TFIM during 600 Adam steps with exact gradients. The dashed and dotted lines mark the exact E_0 and E_1, in (a), and (b) shows the energy error $E-E_0$ on a logarithmic scale.

6 Circuit ansätze

6.1 Requirements and symmetries

The circuit layout fixes the family of states that VQE can reach, so the choice of the ansatz decides the quality of the result. A useful ansatz meets four requirements. It contains a good approximation of the target state, a property called *expressibility*, and its angles are trainable by gradient descent from a reasonable starting point. It uses the native gates and the qubit connectivity of the hardware, because every additional gate adds errors, and it respects the symmetries of the Hamiltonian, which keeps the search inside the symmetry sector of the ground state.

Both running models have symmetries. Every term of the TFIM (2.2) commutes with the *parity*

$$\hat{\Pi}=\prod_{i=0}^{N-1}\hat{X}_i, \tag{6.1}$$

which flips every spin, $|b_0\dots b_{N-1}\rangle\to|\bar{b}_0\dots\bar{b}_{N-1}\rangle$ with $\bar{b}=1-b$. This holds because $\hat{X}_i\hat{X}_{i+1}$ anticommutes with each of $\hat{Z}_i$ and $\hat{Z}_{i+1}$ and therefore commutes with their product. Eigenstates of $\hat{\mathcal{H}}_{\mathrm{TFIM}}$ thus have a definite parity ±1. In the XXZ chain (2.3) the operator $\hat{X}\hat{X}+\hat{Y}\hat{Y}=2(|01\rangle\langle10|+|10\rangle\langle01|)$ exchanges a 0 and a 1 between neighbours and $\hat{Z}\hat{Z}$ is diagonal. Neither term changes the number of bits equal to one, so the Hamiltonian conserves the total magnetisation

$$\hat{Z}_{\mathrm{tot}}=\sum_{i=0}^{N-1}\hat{Z}_i. \tag{6.2}$$

For even N the Heisenberg ground state lies in the sector $\hat{Z}_{\mathrm{tot}}=0$, spanned by the bit strings with equal numbers of zeros and ones.

We compare three ansätze that weigh these requirements differently. In quantum chemistry the standard choice is the unitary coupled-cluster ansatz, which builds the trial state from excitations of electrons out of a reference configuration [21, 29].

6.2 The hardware-efficient ansatz

The hardware-efficient ansatz (HEA) is built only from gates that a device applies natively, without reference to the Hamiltonian [30]. It starts from $|0\dots0\rangle$. Each of its L layers applies $\hat{R}_y$ and then $\hat{R}_z$

with independent angles to every qubit, followed by a chain of $\widehat{\mathrm{CZ}}$ gates on the neighbouring pairs $(0,1),(1,2),\ldots,(N-2,N-1)$, and a final rotation block closes the circuit (Figure 17(a)),

$$|\psi_{\mathrm{HEA}}(\boldsymbol{\theta})\rangle = \hat{R}^{(L)}\,\hat{C}_Z\,\hat{R}^{(L-1)}\cdots\hat{C}_Z\,\hat{R}^{(0)}|0\ldots0\rangle, \qquad \hat{R}^{(\ell)} = \bigotimes_{q=0}^{N-1}\hat{R}_z(\theta^z_{\ell q})\,\hat{R}_y(\theta^y_{\ell q}), \tag{6.3}$$

where $\hat{C}_Z = \prod_{q=0}^{N-2}\widehat{\mathrm{CZ}}_{q,q+1}$. The controlled-$\hat{Z}$ gate $\widehat{\mathrm{CZ}} = \mathrm{diag}(1,1,1,-1)$ changes the sign of the amplitude when both qubits are in $|1\rangle$, so all $\widehat{\mathrm{CZ}}$ gates commute and the chain multiplies the amplitude of $\mathbf{b}$ by $(-1)^{\sum_q b_q b_{q+1}}$. The circuit has

$$n = 2N(L+1) \tag{6.4}$$

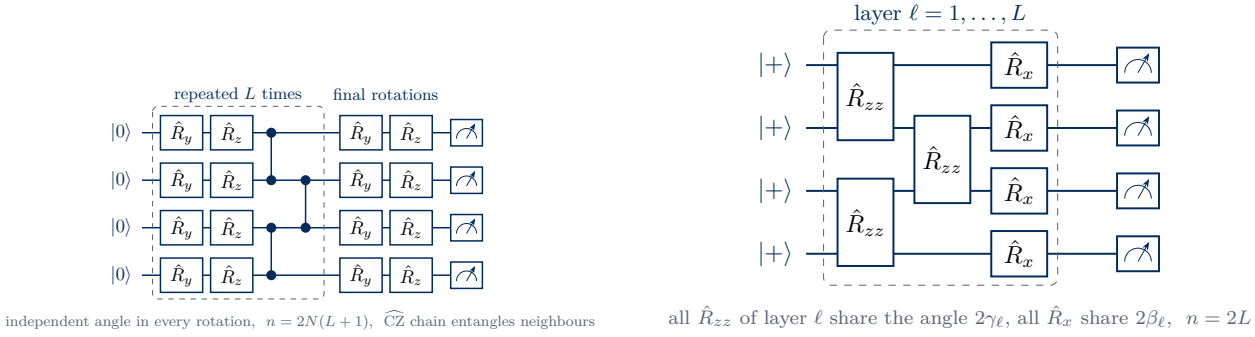
angles and $(N-1)L$ two-qubit gates. One block $\hat{R}_z\hat{R}_y$ takes $|0\rangle$ to any single-qubit state, so the first block alone prepares every product state, including the mean-field state (2.5), and the $\widehat{\mathrm{CZ}}$ layers create entanglement. With growing depth the HEA approaches any state of the register. Its layers carry no information about the Hamiltonian, and its states mix the parity sectors of the TFIM and the magnetisation sectors of the XXZ chain.

6.3 The Hamiltonian-variational ansatz

The Hamiltonian-variational ansatz (HVA) builds every layer from the evolution operators of the terms of the Hamiltonian [31, 32]. For the TFIM it starts in $|+\rangle^{\otimes N}$, the ground state of the field term alone, and applies L layers (Figure 17(b)),

$$|\psi_{\mathrm{HVA}}(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle = \hat{V}_L\cdots\hat{V}_2\hat{V}_1|+\rangle^{\otimes N}, \qquad \hat{V}_\ell = e^{-i\beta_\ell\sum_i\hat{X}_i}\,e^{-i\gamma_\ell\sum_i\hat{Z}_i\hat{Z}_{i+1}}. \tag{6.5}$$

All bonds of a layer share the angle γ_ℓ and all qubits share β_ℓ, so the circuit has only $n = 2L$ angles. In gates, $e^{-i\gamma_\ell\hat{Z}_i\hat{Z}_{i+1}} = \hat{R}_{zz}(2\gamma_\ell)$ on every bond and $e^{-i\beta_\ell\hat{X}_i} = \hat{R}_x(2\beta_\ell)$ on every qubit. A layer therefore contains $N-1$ two-qubit gates, as many as a layer of the HEA, and the circuit has $n_{\mathrm{rot}} = L(2N-1)$ parametrised rotations. Both exponents commute with the parity (6.1), and $|+\rangle^{\otimes N}$ has $\hat{\Pi} = +1$, so every state of the ansatz is even, like the ground state of the TFIM.



(a) hardware-efficient ansatz

(b) Hamiltonian-variational ansatz

Figure 17: (a) The hardware-efficient ansatz (6.3). One layer (dashed box) applies $\hat{R}_y$ and $\hat{R}_z$ with independent angles to every qubit and a chain of $\widehat{\mathrm{CZ}}$ gates on neighbouring pairs, and is repeated L times before a final rotation block. (b) The Hamiltonian-variational ansatz (6.5) for the TFIM. The register starts in $|+\rangle^{\otimes N}$, and each layer applies $\hat{R}_{zz}(2\gamma_\ell)$ on all bonds and $\hat{R}_x(2\beta_\ell)$ on all qubits, with one shared angle per kind of gate and layer. Compare the number of free angles per layer, $2N$ in (a) and two in (b).

The structure of (6.5) follows from approximating the evolution under a sum of non-commuting terms by alternating short evolutions under each term, $e^{-i(\hat{A}+\hat{B})t} \approx (e^{-i\hat{A}t/m}e^{-i\hat{B}t/m})^m$, a Trotter decomposition that becomes exact as $m\to\infty$ [33]. We write the two terms of (2.2) as $\hat{\mathcal{H}}^{ZZ} = -J\sum_i\hat{Z}_i\hat{Z}_{i+1}$ and $\hat{\mathcal{H}}^{X} = -h\sum_i\hat{X}_i$. Each layer $\hat{V}_\ell$ has the form of one such step of evolution under

$s_\ell \hat{\mathcal{H}}^{ZZ} + (1-s_\ell)\hat{\mathcal{H}}^{X}$, a Hamiltonian that equals the field term at $s=0$, the coupling term at $s=1$ and one half of the full model (2.2) at $s=1/2$. The adiabatic theorem states that a system prepared in the ground state of a Hamiltonian stays in the instantaneous ground state when the Hamiltonian changes slowly enough, on a time scale set by the inverse square of the gap to the first excited state [34]. A slow sweep along this path carries $|+\rangle^{\otimes N}$ through the ground states of Ising chains with decreasing field, the target among them, and we start the training from the angles of such a sweep, discretised into layers,

$$\gamma_\ell = -J\tau\, s_\ell, \qquad \beta_\ell = -h\tau\,(1-s_\ell), \qquad s_\ell = \frac{\ell - 1/2}{L}, \tag{6.6}$$

with a time step $\tau = 0.8/J$ and small random offsets, so that $\hat{V}_\ell = e^{-i\tau(1-s_\ell)\hat{\mathcal{H}}^X} e^{-i\tau s_\ell \hat{\mathcal{H}}^{ZZ}}$.

Table 2 compares the two constructions at equal depth. The HEA has an independent angle for every rotation and reaches any state of the register when it has enough layers. The HVA has two angles per layer, starts from the ground state of the field term and prepares only parity-even states adapted to one Hamiltonian.

	HEA (6.3)	HVA (6.5)
gates	$\hat{R}_y$, $\hat{R}_z$, $\widehat{\mathrm{CZ}}$, native to the device	$e^{-i\gamma \hat{Z}_i \hat{Z}_{i+1}}$, $e^{-i\beta \hat{X}_i}$, from the terms of $\hat{\mathcal{H}}$
initial state	$\|0\ldots0\rangle$	$\|+\rangle^{\otimes N}$, ground state of the field term
angles	$n = 2N(L+1)$, one per rotation	$n = 2L$, shared within a layer
symmetry	none, mixes parity sectors	conserves $\hat{\Pi}$, even states only
two-qubit gates per layer	$N-1$	$N-1$

Table 2: The hardware-efficient and the Hamiltonian-variational ansatz for the TFIM at equal depth L.

6.4 A symmetry-preserving ansatz for the XXZ chain

For the Heisenberg chain we use an ansatz that never leaves the sector $\hat{Z}_{\mathrm{tot}} = 0$ of its ground state. It starts from the product of singlets $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ on the pairs $(0,1),(2,3),\ldots$, which has $\hat{Z}_{\mathrm{tot}} = 0$ and is the ground state of (2.3) with every second bond removed. Each layer applies the two-qubit gate

$$\hat{G}(\theta_{xy}, \theta_{zz}) = e^{-i\theta_{xy}(\hat{X}\hat{X}+\hat{Y}\hat{Y})/2}\, e^{-i\theta_{zz}\hat{Z}\hat{Z}/2} \tag{6.7}$$

first on the even bonds $(0,1),(2,3),\ldots$ and then on the odd bonds $(1,2),(3,4),\ldots$, every gate with its own two angles (Figure 18). The two exponents commute, and in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the gate is

$$G = \begin{pmatrix} e^{-i\theta_{zz}/2} & 0 & 0 & 0 \\ 0 & e^{i\theta_{zz}/2}\cos\theta_{xy} & -ie^{i\theta_{zz}/2}\sin\theta_{xy} & 0 \\ 0 & -ie^{i\theta_{zz}/2}\sin\theta_{xy} & e^{i\theta_{zz}/2}\cos\theta_{xy} & 0 \\ 0 & 0 & 0 & e^{-i\theta_{zz}/2} \end{pmatrix}. \tag{6.8}$$

It rotates $|01\rangle$ and $|10\rangle$ into each other and only adds phases to $|00\rangle$ and $|11\rangle$. No gate changes the number of ones, so every state of this symmetry-preserving ansatz (SPA) has $\hat{Z}_{\mathrm{tot}} = 0$. The circuit has $n = 2(N-1)L$ angles and $(N-1)L$ two-qubit gates. Each gate (6.7) is the evolution of two qubits coupled by an anisotropic exchange interaction, and written as rotations it is $\hat{R}_{xx}(\theta_{xy})\hat{R}_{yy}(\theta_{xy})\hat{R}_{zz}(\theta_{zz})$, so $n_{\mathrm{rot}} = 3(N-1)L$. We count it as one two-qubit gate, although a device whose native two-qubit gate is $\widehat{\mathrm{CZ}}$ compiles it into several.

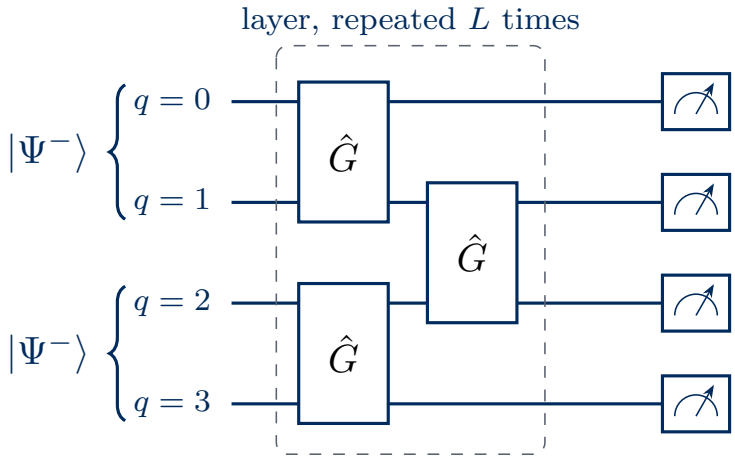


Figure 18: The symmetry-preserving ansatz for the XXZ chain. The register starts in a product of singlets (brackets), and each layer applies the gate $\hat{G}$ of (6.7) on the even and then on the odd bonds, each gate with its own angles $(\theta_{xy}, \theta_{zz})$. No gate changes the number of ones.

6.5 Entanglement capacity

A circuit can create only a limited amount of entanglement, and this bounds the states it reaches. We cut the chain into a left half A and a right half B and recall the Schmidt decomposition from Lecture 11. The 2^N amplitudes are reshaped into a $2^{N/2} \times 2^{N/2}$ matrix whose rows and columns are labelled by the bit strings of the left and the right half. Its singular value decomposition gives $|\psi\rangle = \sum_{k=1}^{\chi} s_k |a_k\rangle_A \otimes |b_k\rangle_B$ with orthonormal $|a_k\rangle$, $|b_k\rangle$ and Schmidt coefficients $s_k > 0$, and the half-chain entanglement entropy in bits is

$$S_{\mathrm{vN}} = -\sum_k s_k^2 \log_2 s_k^2. \tag{6.9}$$

A product state has a single term and $S_{\mathrm{vN}} = 0$, and a Bell pair $(|00\rangle + |11\rangle)/\sqrt{2}$ has $s_1^2 = s_2^2 = \frac{1}{2}$ and carries one bit. A state with χ Schmidt terms across the cut has an entanglement entropy of at most $\log_2 \chi$ bits. Gates inside one half leave the Schmidt coefficients unchanged. A gate across the cut can be written as a sum of r products,

$$\hat{U} = \sum_{k=1}^{r} \hat{A}_k \otimes \hat{B}_k, \tag{6.10}$$

with $\hat{A}_k$ acting on A and $\hat{B}_k$ on B, and applied to a state with χ Schmidt terms it produces at most $r\chi$ terms. For the controlled-$\hat{Z}$ gate $r = 2$, since $\widetilde{\mathrm{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z}$, and the same holds for $\hat{R}_{zz}(\theta) = \cos\frac{\theta}{2}\,\hat{I}\otimes\hat{I} - i\sin\frac{\theta}{2}\,\hat{Z}\otimes\hat{Z}$. One layer of the HEA or of the HVA contains one such gate across the middle cut. After L layers the number of Schmidt terms is therefore at most 2^L, and the half-chain entropy obeys (Figure 19)

$$S_{\mathrm{vN}} \le \min\left(L,\, N/2\right) \text{ bits}, \tag{6.11}$$

where $N/2$ bits is the maximum for $N/2$ qubits on each side. The gate (6.7) has $r = 4$ and adds at most two bits per layer. A ground state with an entropy of one bit across the middle needs at least one layer, and a volume-law state, with an entropy proportional to N, needs a depth that grows with N. The bound is a necessary condition only, since correlations between distant spins also need enough layers for the gates to connect them.

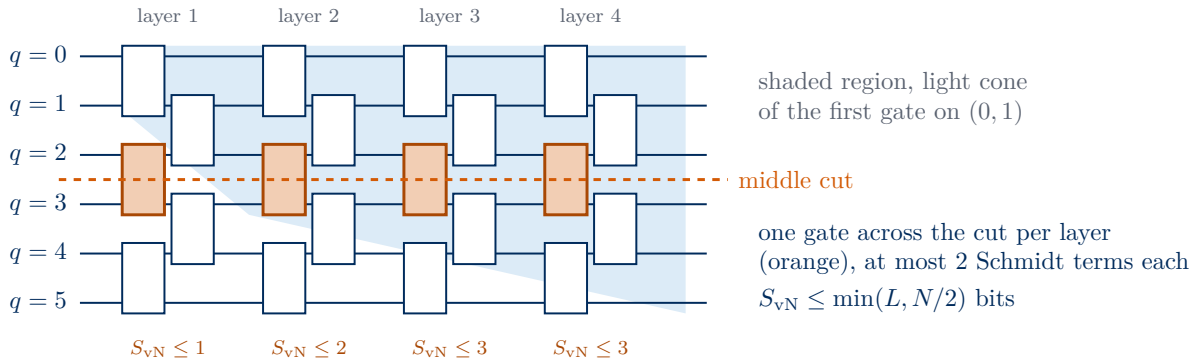


Figure 19: Entanglement capacity of a brickwork circuit. Gates inside one half leave the entanglement across the middle cut unchanged, and each gate across the cut with operator Schmidt rank $r = 2$ at most doubles the number of Schmidt terms, so L layers give at most L bits (6.11).

7 The variational quantum eigensolver for spin chains

7.1 Symmetry sectors in the XYZ chain

An ansatz with a built-in symmetry can miss the ground state completely, as we show for the open XYZ chain of $N = 8$ qubits in a transverse and a longitudinal field,

$$\hat{\mathcal{H}}_{\mathrm{XYZ}} = \sum_{i=0}^{N-2} \left(J_x \hat{X}_i \hat{X}_{i+1} + J_y \hat{Y}_i \hat{Y}_{i+1} + J_z \hat{Z}_i \hat{Z}_{i+1} \right) + \sum_{i=0}^{N-1} \left(h_x \hat{X}_i + h_z \hat{Z}_i \right), \tag{7.1}$$

with $(J_x, J_y, J_z, h_x, h_z) = (1, 0.7, 0.4, 1, 0.5)$. Neighbouring spins interact along all three axes with different strengths. The positive couplings favour antiparallel neighbours, and the fields pull every spin towards $-x$ and $-z$. Neither the magnetisation $\hat{Z}_{\mathrm{tot}}$ (6.2) nor the parity $\hat{\Pi}$ (6.1) commutes with (7.1), so the chain has none of the symmetries for which the ansätze of Section 6 were built. Exact diagonalisation of the 256×256 matrix gives the ground state $|\psi_0\rangle$ with the energy $E_0 = -10.6453$ and the gap $\Delta E = 0.961$.

Besides the energy and the fidelity $F = |\langle \psi_0 | \psi \rangle|^2$, we follow three physically meaningful properties of a trial state $|\psi\rangle$. The transverse magnetisation and the nearest-neighbour correlation along x,

$$m_x = \frac{1}{N} \sum_{i=0}^{N-1} \langle \hat{X}_i \rangle, \qquad C_{xx} = \frac{1}{N-1} \sum_{i=0}^{N-2} \langle \hat{X}_i \hat{X}_{i+1} \rangle, \tag{7.2}$$

are local quantities, which a device estimates from bit strings measured after a Hadamard gate on every qubit. The half-chain entanglement entropy (6.9) follows from the Schmidt coefficients across the middle cut. The exact ground state has $m_x = -0.203$, $C_{xx} = -0.607$ and $S_{\mathrm{vN}} = 1.02$ bits, so a product state cannot represent it.

We compare three ansätze with $L = 4$ layers each. The HEA (6.3) has $n = 80$ angles. The HVA follows the structure of (7.1). Each layer applies the two-qubit bond gate

$$\hat{B}(a, b, c) = e^{-i(a\hat{X}\hat{X} + b\hat{Y}\hat{Y} + c\hat{Z}\hat{Z})} \tag{7.3}$$

first on the even bonds $(0, 1), (2, 3), \ldots$ and then on the odd bonds $(1, 2), (3, 4), \ldots$, and finally the one-qubit gate $e^{-i(\beta\hat{X} + \delta\hat{Z})}$ on every qubit. The three terms in the exponent of (7.3) commute, so $\hat{B}$ is a product of three two-qubit rotations, and it generalises the SPA gate (6.7). In the shared-angle HVA all even bonds of a layer carry one triple (a, b, c), all odd bonds another, and all qubits share (β, δ), which gives eight angles per layer and $n = 32$. The HVA with free angles has the same gates, each with its own angles, and $n = 148$. Both HVA circuits start from the product of the single-qubit ground states of the field term $h_x \hat{X}_i + h_z \hat{Z}_i$. We train every ansatz with exact energies and gradients and Adam [28] for 1200 steps, called epochs, from 3 random starts and keep the start with the lowest final energy.

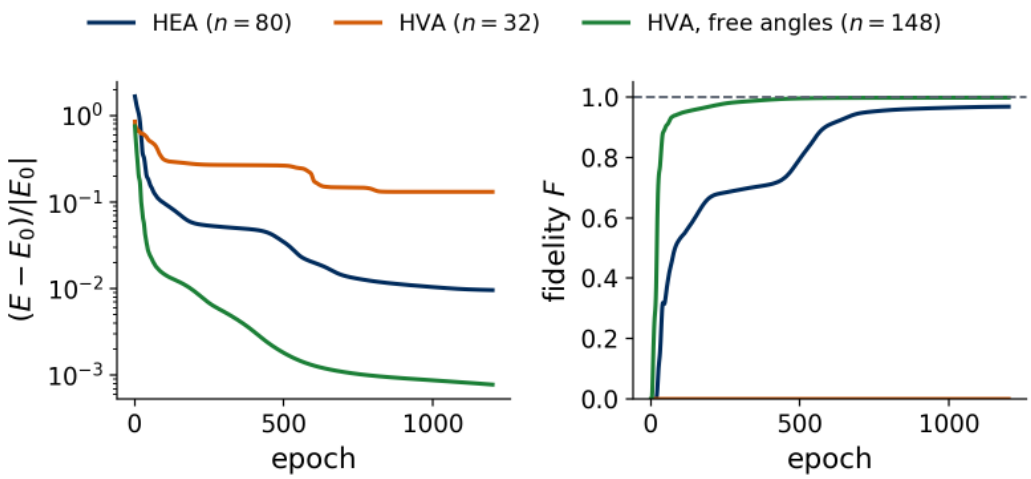


Figure 20: VQE on the XYZ chain (7.1), $N = 8$, with three ansätze of four layers, the HEA with $n = 80$ angles, the shared-angle HVA with $n = 32$ and the HVA with free angles with $n = 148$. Left, the relative energy error $(E - E_0)/|E_0|$ on a logarithmic scale. Right, the fidelity F with the exact ground state, both against the epoch, for the best of three starts. Look at the shared-angle HVA, whose fidelity stays at zero. Exact state-vector simulation with exact gradients.

Figure 20 shows the training. The HVA with free angles reaches a relative energy error of 7.8×10^{-4} and a fidelity of 0.998. The HEA ends with an energy error of 9.6×10^{-3} and a fidelity of 0.968. The shared-angle HVA, with the fewest angles and the most physical structure, stops at an energy error of order ten per cent, and its fidelity is zero at every epoch. A circuit that is too short would give a small but finite fidelity, so an exactly vanishing fidelity points to a symmetry of the ansatz.

The mirror reflection $\hat{\mathcal{R}}$ maps qubit i to qubit $N - 1 - i$ and sends the basis state $|b_0 b_1 \dots b_{N-1}\rangle$ to $|b_{N-1} \dots b_1 b_0\rangle$. Applying it twice gives the identity, so its eigenvalues are $+1$ for mirror-even and -1 for mirror-odd states. All couplings and fields of (7.1) are uniform, so $\hat{\mathcal{R}}\hat{\mathcal{H}}_{\mathrm{XYZ}}\hat{\mathcal{R}} = \hat{\mathcal{H}}_{\mathrm{XYZ}}$, and every non-degenerate eigenstate is either even or odd. The exact ground state is mirror-odd. The gates of the shared-angle HVA have the same symmetry. For even N the reflection maps the set of even bonds onto itself and the set of odd bonds onto itself, the gate (7.3) is unchanged when its two qubits are exchanged, and all bonds of one kind carry the same gate. Every layer $\hat{V}_\ell$ therefore obeys $\hat{\mathcal{R}}\hat{V}_\ell\hat{\mathcal{R}} = \hat{V}_\ell$, and the initial product of identical single-qubit states is mirror-even. For every choice of angles,

$$\hat{\mathcal{R}}|\psi_{\mathrm{HVA}}(\boldsymbol{\theta})\rangle = \hat{\mathcal{R}}\hat{V}_L \cdots \hat{V}_1|\psi_{\mathrm{init}}\rangle = \hat{V}_L \cdots \hat{V}_1\hat{\mathcal{R}}|\psi_{\mathrm{init}}\rangle = |\psi_{\mathrm{HVA}}(\boldsymbol{\theta})\rangle. \tag{7.4}$$

Eigenvectors of $\hat{\mathcal{R}}$ with different eigenvalues are orthogonal, so every HVA state is orthogonal to the mirror-odd ground state and $F = 0$ exactly. A general trial state with $\langle\hat{\mathcal{R}}\rangle = r$ has the weight $(1 - r)/2$ in the odd sector, which gives the bound

$$F \leq \frac{1 - \langle\hat{\mathcal{R}}\rangle}{2} \tag{7.5}$$

for a mirror-odd ground state. The lowest energy that a mirror-even state can reach is the lowest even eigenvalue, which for this chain equals the first excited energy E_1, and four layers of the shared-angle HVA do not reach even that value. The HEA and the free-angle HVA start from mirror-even states as well, but their gates break the symmetry, and training moves both to $\langle\hat{\mathcal{R}}\rangle \approx -1$, as (7.5) requires for a fidelity close to one.

A symmetry built into an ansatz restricts it to one symmetry sector. This saves angles and helps training when the target lies in that sector, and it makes the target unreachable for every depth and every optimiser when the target lies in another one. Before training we therefore determine the quantum numbers of the target, here the reflection eigenvalue, and make sure that the ansatz reaches them, either by starting in the right sector or by breaking the symmetry as the HVA with free angles does.

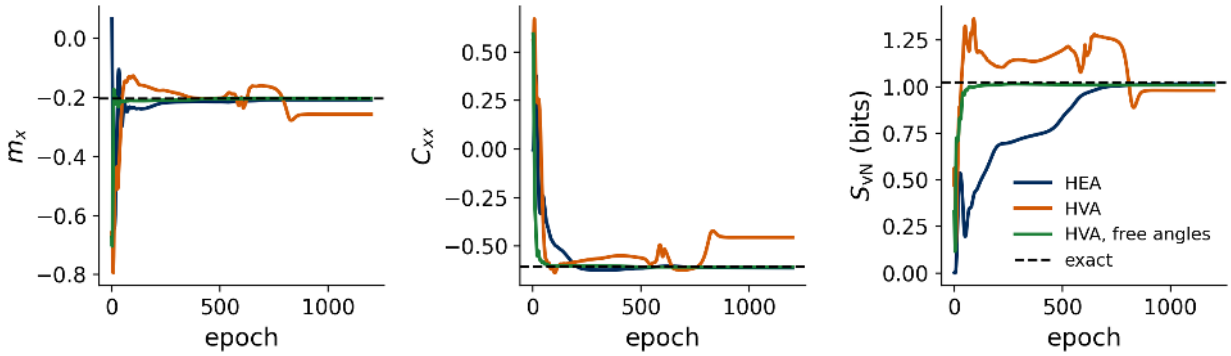


Figure 21: Observables during the training of Figure 20. From left to right, the transverse magnetisation m_x and the nearest-neighbour correlation C_{xx} of (7.2) and the half-chain entropy S_{vN} (6.9) in bits, against the epoch, with the exact values dashed. Compare how fast the local quantities and the entropy reach their exact values. Exact state-vector simulation.

Figure 21 shows which features of the ground state a trial state acquires during training. The HVA with free angles reaches all three exact values early. For the HEA the local observables settle within a few hundred epochs, whereas the entropy reaches the exact value of about one bit only after several hundred more, because the circuit builds the correlations between the two halves gate by gate. The shared-angle HVA ends with almost the exact entropy but with the wrong magnetisation and too weak correlations. A trial state can therefore carry the right amount of entanglement and still be orthogonal to the ground state, so the fidelity has to be checked alongside the energy and the entropy.

7.2 Models, references and training

We compare the ansätze of Section 6 systematically on three chains of $N = 6$ qubits, small enough for exact diagonalisation of the 64×64 Hamiltonians, which gives the exact ground state, the energies E_0 and E_1 and the gap ΔE. The half-chain entropy (6.9) is computed across the cut between qubits 2 and 3, with the 64 amplitudes reshaped into an 8×8 matrix with rows labelled by $b_0 b_1 b_2$ and columns by $b_3 b_4 b_5$. The critical TFIM at $h = J$ has the gap $\Delta E = 0.4821J$ and a ground-state entropy of 0.473 bits. The ordered TFIM at $h = 0.5J$ has a much smaller gap, $0.0235J$, and an entropy of almost one bit, 0.964. The Heisenberg chain, the XXZ chain at $\Delta = 1$, has its ground state in the sector $\hat{Z}_{\mathrm{tot}} = 0$ with an entropy of 1.026 bits. In both TFIM cases the ground state is even and the first excited state odd under $\hat{\Pi}$.

All three ansätze get $L = 6$ layers and therefore the same number $(N-1)L = 30$ of two-qubit gates, with $n = 84$ angles for the HEA, 12 for the HVA and 60 for the SPA. The HEA and the SPA start from small random angles near their reference states $|0 \ldots 0\rangle$ and the singlet product, and the HVA starts from the discretised sweep (6.6). Each run takes 600 Adam steps with exact gradients, and 4 independent starts, called restarts, are trained for every ansatz and model. As a user of VQE would, we keep the restart with the lowest final energy. The fidelity and the entropy are recorded at every step to judge the result, which on a device would require additional measurements.

7.3 The critical Ising chain

Figure 22 follows the training on the critical TFIM. Both ansätze lower the energy error by three to four orders of magnitude, and the best HVA restart, with 12 angles, ends below the best HEA restart, with 84 angles. The restarts of each ansatz end at different energies. The HEA starts from a product state and builds the entanglement gradually, so its entropy approaches the exact value from below. The HVA starts from the sweep (6.6), overshoots the exact entropy during the first tens of steps and then settles near it. For both best states the entropy error $\delta S = |S_{\mathrm{vN}} - S_{\mathrm{vN}}^{\mathrm{exact}}|$ is one to two orders of magnitude larger than the relative energy error (Table 3).

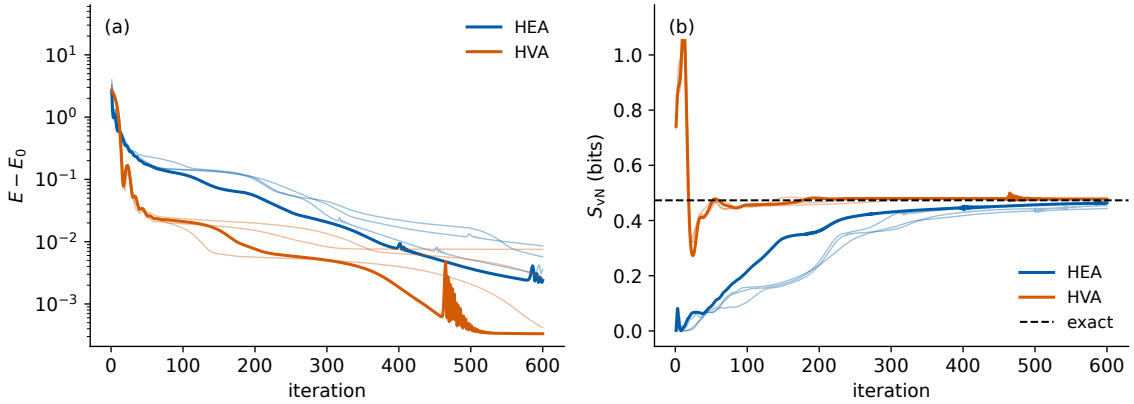


Figure 22: VQE on the critical TFIM, $N = 6$, $h = J$, with the HEA and the HVA at $L = 6$. (a) Energy error $E - E_0$ against the training step on a logarithmic scale. (b) Half-chain entropy S_{vN} against the step, with the exact value dashed. Thick lines are the restart with the lowest final energy, thin lines the other restarts. Exact state-vector simulation with exact gradients.

7.4 Energy errors and entanglement errors

The two errors differ in general, because they depend differently on a small admixture to the ground state. We write a trial state as the ground state with an admixture of amplitude ϵ of a normalised state $|\phi\rangle$ orthogonal to it,

$$|\psi\rangle = \frac{|\psi_0\rangle + \epsilon|\phi\rangle}{\sqrt{1+\epsilon^2}}, \qquad 1 - F = \frac{\epsilon^2}{1+\epsilon^2}, \qquad E - E_0 = \frac{\epsilon^2(\langle\phi|\hat{\mathcal{H}}|\phi\rangle - E_0)}{1+\epsilon^2}. \tag{7.6}$$

The cross terms of the energy, $\epsilon\langle\psi_0|\hat{\mathcal{H}}|\phi\rangle + \text{c.c.}$, vanish because $\hat{\mathcal{H}}|\psi_0\rangle = E_0|\psi_0\rangle$ and $\langle\psi_0|\phi\rangle = 0$, so the energy error is of second order in ϵ, like the infidelity. The ground state is a stationary point of the energy, which is why the variational principle works so well for the energy itself. The entropy has no such stationary point at $|\psi_0\rangle$. The reduced density matrix of the half chain changes at first order in ϵ, through the cross terms $\epsilon(|\psi_0\rangle\langle\phi| + |\phi\rangle\langle\psi_0|)$ traced over the other half, and so does its entropy,

$$S_{\mathrm{vN}}(\psi) - S_{\mathrm{vN}}(\psi_0) = \epsilon\, S_1 + O(\epsilon^2), \tag{7.7}$$

with a coefficient S_1 that depends on $|\phi\rangle$. The entropy error thus scales as $\sqrt{1-F}$, whereas the energy error scales as $1 - F$, and a state with a relative energy error of 10^{-4} can carry a visibly wrong entanglement. Figure 23 confirms both scalings with every state recorded during the training runs of this section, 14400 states in total. All points of panel (a) lie above the bound (3.3) and follow a slope of one, and the entropy errors in panel (b) follow the slope one half of $\sqrt{1-F}$. At small infidelity the entropy is therefore the more sensitive test of the trained state.

7.5 The Heisenberg chain

On the Heisenberg chain we compare the HEA with the SPA (Figure 24). The SPA prepares the ground state to a relative energy error of 1.3×10^{-9}, and its entropy stays at the exact value from the first steps. It starts from the singlet product, which already carries one bit across the middle cut and lies in the sector $\hat{Z}_{\mathrm{tot}} = 0$, so the circuit only redistributes singlet correlations along the chain. The HEA has to build this structure from $|0\ldots0\rangle$ and to remove the weight outside the sector. Its entropy rises to the exact value only after about a hundred steps, and its best state ends at $\delta E = 7.3 \times 10^{-4}$ and $\delta S = 7.6 \times 10^{-3}$ bits.

7.6 The ordered phase and symmetry breaking

At $h = 0.5J$ the exact ground state is close to the even superposition of the two ordered configurations,

$$|\psi_0\rangle \approx \frac{|0\ldots0\rangle + |1\ldots1\rangle}{\sqrt{2}}, \tag{7.8}$$

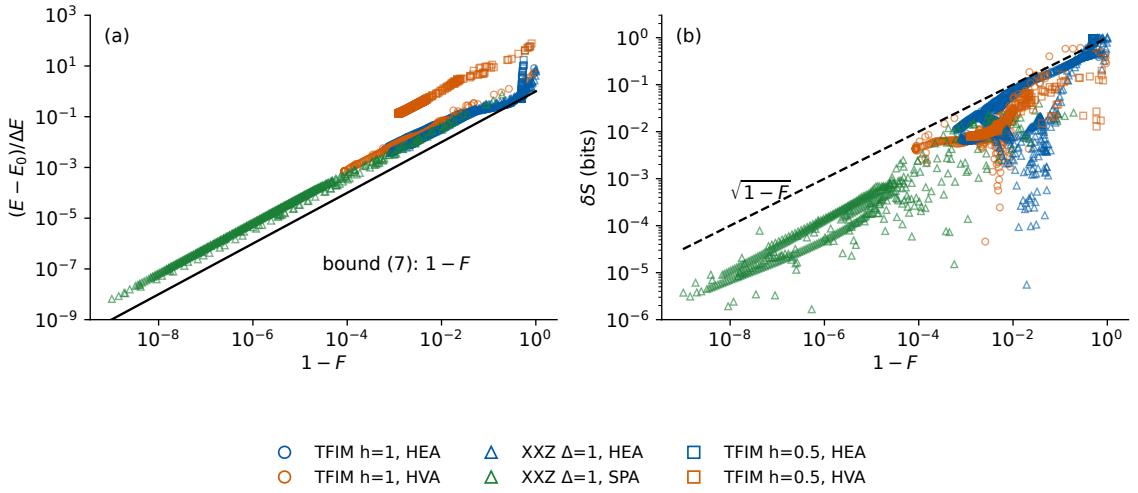


Figure 23: Errors of all states recorded during the VQE runs on the three chains. (a) Energy error in units of the gap, $(E - E_0)/\Delta E$, against the infidelity $1 - F$, with the bound (3.3) as a solid line. (b) Entropy error δS against $1 - F$, with the line $\sqrt{1 - F}$ dashed. Look at the slopes, one in (a) and one half in (b).

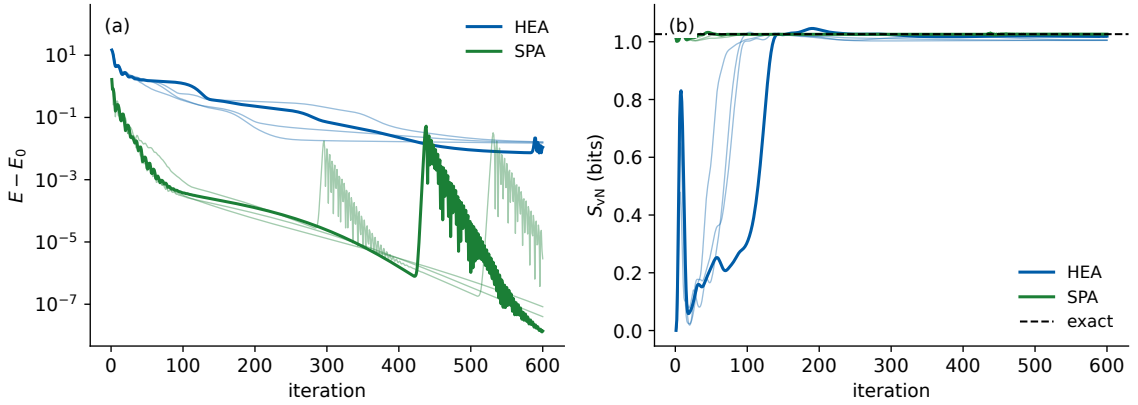


Figure 24: VQE on the Heisenberg chain, $N = 6$, with the HEA and the SPA at $L = 6$, shown as in Figure 22. (a) Energy error. (b) Half-chain entropy with the exact value dashed. The short spikes of the SPA late in training are single large Adam steps, after which the energy returns to its previous level.

dressed by spin flips that the field creates, and the first excited state $|\psi_1\rangle$ is close to the odd superposition. A superposition of two macroscopically different configurations is called a cat state, and across any cut it carries about one bit of entropy. The two levels form a doublet separated by the small gap ΔE, which closes exponentially with N. A trial state within the doublet, $|\psi\rangle = a|\psi_0\rangle + b|\psi_1\rangle$ with $|a|^2 + |b|^2 = 1$, has the parity and the energy

$$\langle\hat{\Pi}\rangle = |a|^2 - |b|^2, \qquad E - E_0 = |b|^2\Delta E = \frac{1 - \langle\hat{\Pi}\rangle}{2}\,\Delta E, \tag{7.9}$$

where the second form uses $|b|^2 = (1 - \langle\hat{\Pi}\rangle)/2$. The symmetry-broken state (Figure 25) $(|\psi_0\rangle + |\psi_1\rangle)/\sqrt{2} \approx |0\ldots0\rangle$, dressed by spin flips, has $\langle\hat{\Pi}\rangle = 0$, fidelity $F = 1/2$ and an energy only $\Delta E/2$ above E_0. It is close to a product state and carries almost no entanglement. A circuit that reaches this state has a relative energy error of order $\Delta E/(2|E_0|)$, small on the scale of the coupling, while its fidelity and entropy are far from the exact values.

Figure 26 shows that every HEA restart descends into the symmetry-broken branch near $|0\ldots0\rangle$, from which it starts, and stays there with $F \approx 1/2$, almost no entanglement and a parity close to zero. Its weight lies almost entirely within the doublet, and its energy error agrees with $\Delta E/2$, as (7.9) predicts (Table 3). Turning this state into the cat state would lower the energy by only $\Delta E/2$ and requires a coherent superposition with $|1\ldots1\rangle$, which differs in all six spins. The HVA keeps the parity at $+1$ at every step and cannot enter the symmetry-broken branch. Its best state reaches a

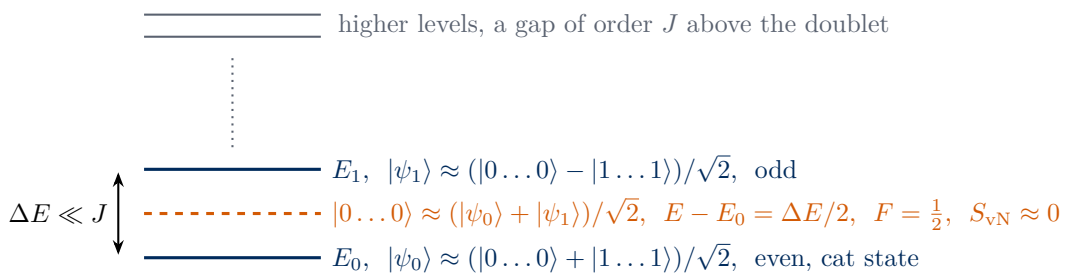


Figure 25: The ground doublet of the ordered TFIM. The even cat state $|\psi_0\rangle$ and the odd state $|\psi_1\rangle$ are split by the small gap ΔE. Their equal superposition is a symmetry-broken branch close to $|0\ldots0\rangle$, with $F = 1/2$, no entanglement and an energy only $\Delta E/2$ above E_0 (7.9).

fidelity close to one and the exact entropy of almost one bit.

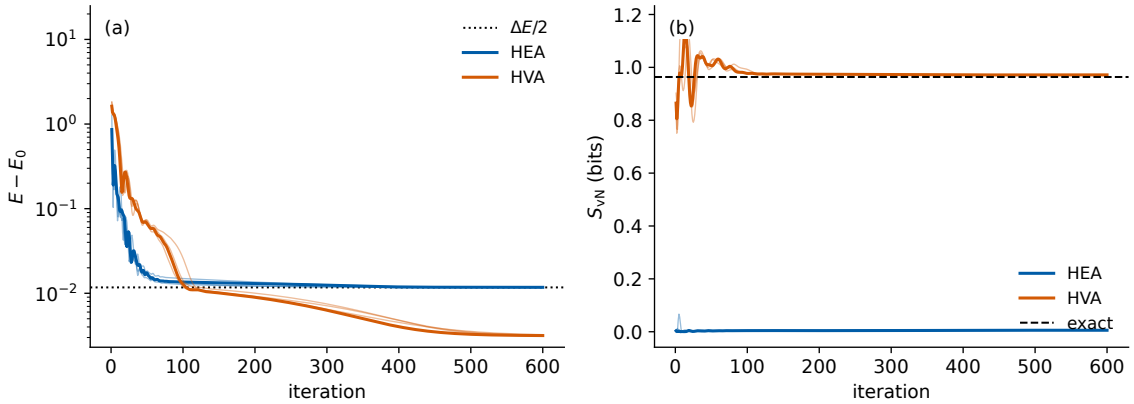


Figure 26: VQE on the ordered TFIM, $N = 6$, $h = 0.5J$, shown as in Figure 22. (a) Energy error, with the dotted line at $\Delta E/2$ of (7.9). (b) Half-chain entropy, with the exact value of almost one bit dashed. The HEA settles at the dotted line with zero entropy.

In the computational basis the exact ground state and the HVA state return 000000 and 111111 in nearly equal shares, about 0.40 each, whereas the HEA state returns 000000 with probability 0.81 and 111111 almost never (Figure 27). Measurements along z alone cannot distinguish the cat state from an equal mixture of the two branches, and the parity $\langle\hat{\Pi}\rangle$, measured in the all-x setting, decides between them.

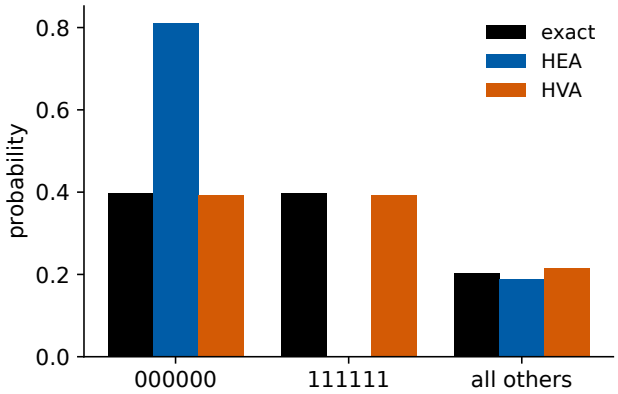


Figure 27: Probabilities of the bit strings 000000 and 111111 and of all other bit strings together in the exact ground state of the ordered TFIM and in the best trained HEA and HVA states, computed from the state vectors.

The HEA fails through two properties together, its missing parity symmetry and its start near $|0\ldots0\rangle$, which lies in one branch. The HVA needs enough depth, because the cat state (7.8) correlates

the two end spins, $N-1$ bonds apart, and each layer of (6.5) extends the region influenced by a spin by one bond in each direction. On the ordered chain the HVA infidelity falls from about 4.8×10^{-1} at one layer to 4.1×10^{-3} at five, while the HEA stays near $1 - F \approx 1/2$ at every depth. The entanglement bound (6.11) allows one bit already at $L = 1$, so the required depth comes from the range of the correlations.

7.7 Excited states

Variational quantum deflation (VQD) finds excited states with the same circuits [35]. After a VQE run has produced an approximate ground state $|\tilde{\psi}_0\rangle$, a second run minimises

$$C(\boldsymbol{\theta}) = E(\boldsymbol{\theta}) + \lambda\, |\langle \tilde{\psi}_0 | \psi(\boldsymbol{\theta}) \rangle|^2, \tag{7.10}$$

where the penalty weight λ raises the cost of any overlap with the ground state. If $|\tilde{\psi}_0\rangle$ were exact, the cost of a trial state $\sum_m c_m |E_m\rangle$ would be $\sum_m |c_m|^2 E_m + \lambda |c_0|^2$, the energy of a Hamiltonian whose ground level is moved up (Figure 28) to $E_0 + \lambda$. For $\lambda > \Delta E$ its minimum is E_1, reached at the first excited state. On a device the overlap is estimated by applying the trial circuit followed by the inverse of the ground-state circuit and counting the outcome $0 \ldots 0$, since

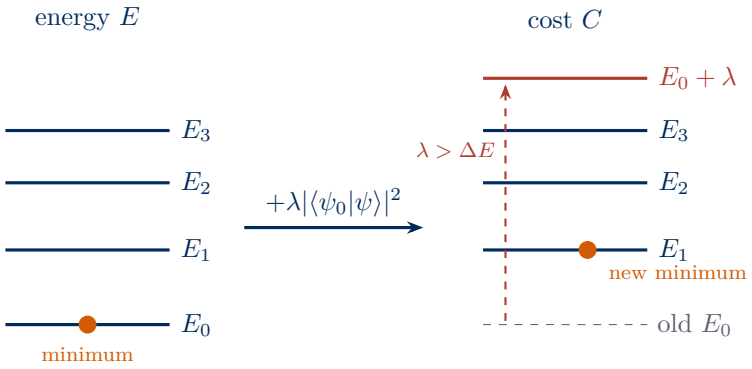


Figure 28: Variational quantum deflation (7.10). The overlap penalty $\lambda|\langle\tilde{\psi}_0|\psi\rangle|^2$ lifts the ground level to $E_0 + \lambda$, so for $\lambda > \Delta E$ the lowest level of the penalised cost is the first excited state E_1.

$$|\langle 0 \ldots 0 | \hat{U}(\tilde{\boldsymbol{\theta}}_0)^\dagger \hat{U}(\boldsymbol{\theta}) | 0 \ldots 0 \rangle|^2 = |\langle \tilde{\psi}_0 | \psi(\boldsymbol{\theta}) \rangle|^2. \tag{7.11}$$

We apply VQD to the critical TFIM with the HEA, its best trained ground state as $|\tilde{\psi}_0\rangle$ and $\lambda = 2J$, larger than the gap. The first excited state is odd under $\hat{\Pi}$, so the HVA, confined to even states, cannot reach it. Figure 29 shows that the energy first drops below E_1 and stays on a plateau where the trial state still overlaps with $|\tilde{\psi}_0\rangle$, since by (2.4) an energy below E_1 requires weight on the ground state. Once this overlap is removed, the energy settles on E_1 with an infidelity below 10^{-3}, and the gap obtained as the difference of the VQD and VQE energies agrees with the exact gap to within $10^{-3}J$. Errors of $|\tilde{\psi}_0\rangle$ propagate into the excited state, because the penalty acts on the approximate ground state and leaves a small admixture of the exact one unpenalised.

7.8 Comparison of the ansätze

Table 3 collects the best restart of every run together with the hardware cost of one gradient by (5.9), $2n_{\mathrm{rot}}$ shifted circuits each measured in G settings, which gives $2n_{\mathrm{rot}}G$ circuit runs of M shots each. In our runs (four restarts, 600 Adam steps), at equal depth and equal two-qubit gate count, the ansätze built for their Hamiltonian need fewer angles and reach better states. The SPA prepares the Heisenberg ground state to numerical precision, and the HVA, with 12 angles, reaches the TFIM ground state at both fields. The HEA, with 84 angles, ends at a lower fidelity on every model and at $F \approx 1/2$ on the ordered chain. The cost of a hardware gradient counts gate occurrences, so the HVA

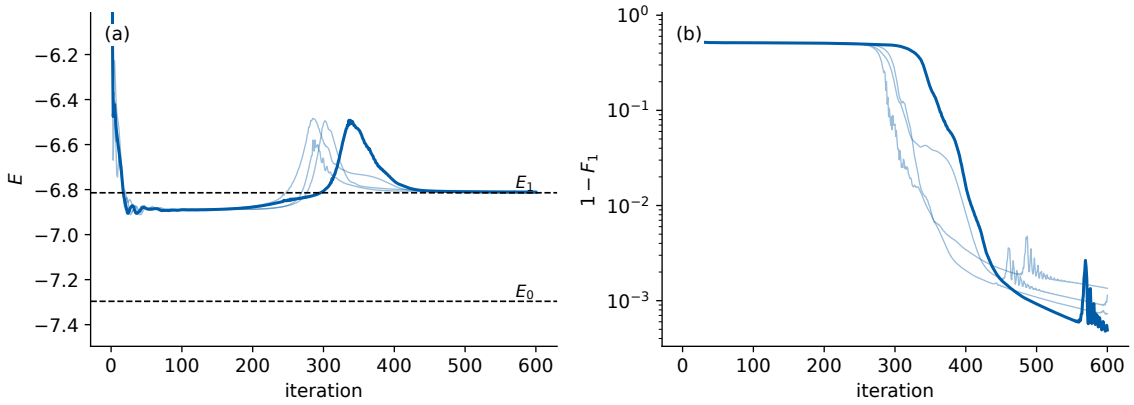


Figure 29: Variational quantum deflation (7.10) for the first excited state of the critical TFIM with the HEA. (a) Energy against the training step, with the exact E_0 and E_1 dashed. (b) Infidelity $1 - F_1$ with the exact first excited state, $F_1 = |\langle\psi_1|\psi\rangle|^2$. Exact state-vector simulation.

with only 12 free angles still needs 264 circuits per gradient. These advantages are tied to the model, since the HVA is defined by one Hamiltonian and the SPA reaches only states with $\hat{Z}_{\text{tot}} = 0$.

model	ansatz	n	n_{rot}	2-qubit gates	circuits	δE	F	δS (bits)
TFIM, $h = J$	HEA	84	84	30	336	3.4×10^{-4}	0.9994	1.1×10^{-2}
TFIM, $h = J$	HVA	12	66	30	264	4.6×10^{-5}	0.9999	4.4×10^{-3}
TFIM, $h = 0.5J$	HEA	84	84	30	336	2.1×10^{-3}	0.5093	9.6×10^{-1}
TFIM, $h = 0.5J$	HVA	12	66	30	264	5.7×10^{-4}	0.9988	8.0×10^{-3}
Heisenberg	HEA	84	84	30	504	7.3×10^{-4}	0.9992	7.6×10^{-3}
Heisenberg	SPA	60	90	30	540	1.3×10^{-9}	1.0000	4.1×10^{-6}

Table 3: Comparison of the ansätze on chains of six qubits with $L = 6$ layers. The columns give the number of angles n, the number of parametrised rotations n_{rot}, the two-qubit gate count, the number $2n_{\text{rot}}G$ of circuit runs (shifted circuits times settings) for one parameter-shift gradient with $G = 2$ settings for the TFIM and $G = 3$ for the Heisenberg chain, and for the best of four restarts the relative energy error δE, the fidelity F and the entropy error δS. Exact state-vector simulation with exact gradients.

7.9 Phase diagram of the Ising chain

A ground state changes character across a phase diagram, and an ansatz that works in one region can fail in another. We train the HEA and the HVA on the TFIM with $N = 8$ qubits at 18 fields between $h = 0.1J$ and $h = 2J$, with a denser grid near the critical field, and compare every trained state with exact diagonalisation. Both ansätze have $L = 8$ layers and 56 two-qubit gates, with $n = 144$ angles for the HEA and 16 for the HVA, and every field is trained independently from the starts used above. Besides the energy per site and the entropy (6.9) we record two magnetisations. The transverse magnetisation m_x of (7.2) measures the alignment with the field. The order along z is harder to measure, because $\hat{\Pi}\hat{Z}_i\hat{\Pi} = -\hat{Z}_i$ implies $\langle\hat{Z}_i\rangle = 0$ in every state of definite parity, so the magnetisation vanishes in the exact ground state of a finite chain on both sides of the transition. We therefore use the parity-invariant order parameter

$$m_z = \sqrt{\Big\langle \Big(\frac{1}{N}\sum_{i=0}^{N-1} \hat{Z}_i\Big)^2 \Big\rangle}, \tag{7.12}$$

which tends to one in the ordered phase and falls to a value of order $1/\sqrt{N}$ in the paramagnet. It takes nearly the same value in the cat state (7.8) and in either of its symmetry-broken branches, since $\big(\sum_i \hat{Z}_i\big)^2$ is diagonal and has no matrix elements between the two branches.

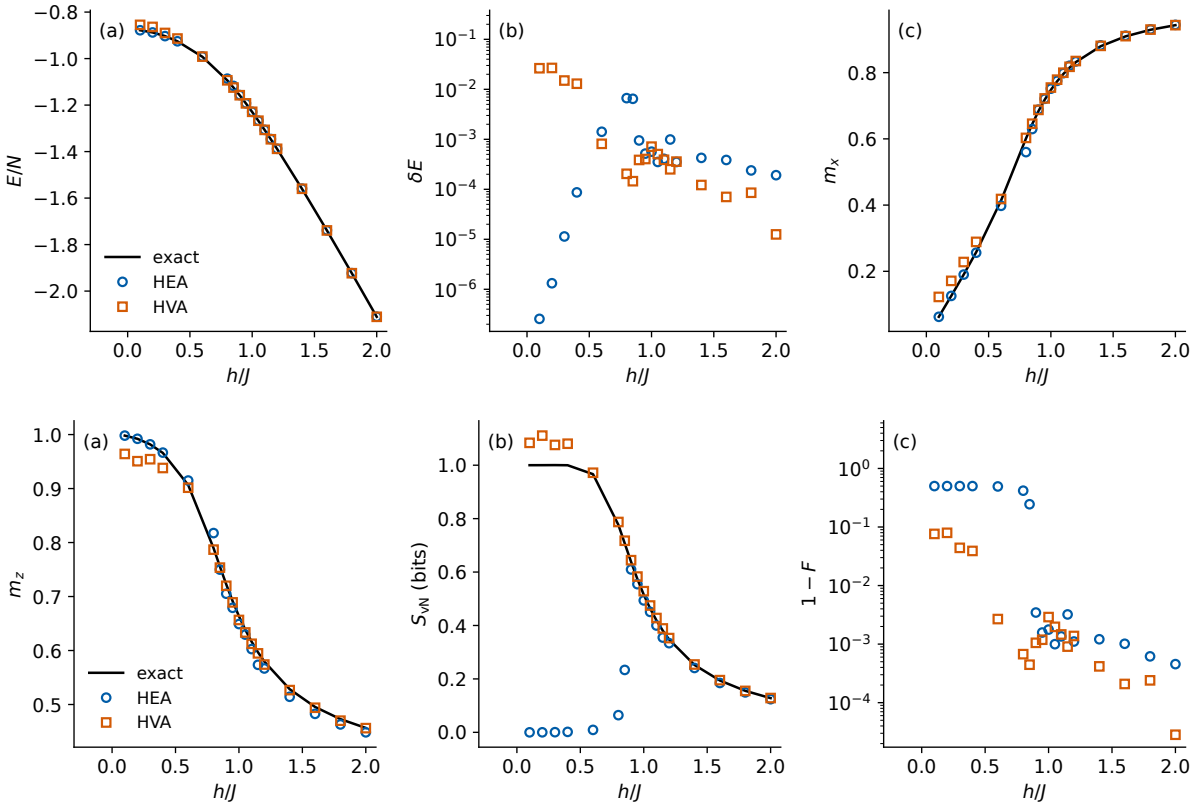


Figure 30: Variational phase diagram of the TFIM with $N = 8$ against the field h/J. Top row, the energy per site, the relative energy error δE and the transverse magnetisation (7.2). Bottom row, the order parameter (7.12), the half-chain entropy and the infidelity $1 - F$. Black lines are exact diagonalisation, blue circles the HEA and orange squares the HVA, each the best of two restarts. Exact state-vector simulation. Compare δE with $1 - F$ in the ordered phase, where the ranking of the two ansätze by energy is the reverse of their ranking by fidelity.

In Figure 30 the energy per site and the transverse magnetisation of both ansätze follow the exact curves on the scale of the figure, and the relative energy error stays below a few per cent at every field. The other panels separate the two ansätze. Deep in the ordered phase the HEA has the smallest energy errors of the whole scan, 1.3×10^{-6} at $h = 0.2J$, while its fidelity is 0.500 and its entropy zero. It sits in a symmetry-broken branch, and its energy error shrinks with h together with the splitting of the doublet, in line with (7.9). Its order parameter agrees with the exact one, because m_z takes nearly the same value in a branch and in the cat state. Its fidelity stays near $1/2$ up to $h = 0.6J$, rises through a crossover, remains below 0.99 up to $h = 0.85J$ and exceeds 0.99 from $h = 0.9J$ upward.

The HVA keeps even parity at every field and follows the exact entropy through the transition. It reaches $F > 0.99$ at every field from $h = 0.6J$ upward. For $h \leq 0.4J$ its fidelity lies between 0.92 and 0.96, its entropy overshoots the exact one bit, and its energy error exceeds that of the trapped HEA. The same eight-layer circuit reaches $F > 0.99$ from $h = 0.6J$, so the deficit is confined to the deep ordered phase, where the circuit must build the long-range correlations of the cat state. Near the critical point the HVA reaches fidelities above 0.99 and the exact entropy, whereas the entropy error of the HEA grows as the field approaches the critical point from the ordered side. A phase diagram drawn from the energy alone would therefore hide the failure of the HEA in the ordered phase, which the entropy and the fidelity reveal.

7.10 The XXZ chain in a transverse field

To test the ansätze on ground states of very different character, we add a magnetic field along x to the XXZ chain,

$$\hat{\mathcal{H}} = J \sum_{i=0}^{N-2} \left(\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \Delta\, \hat{Z}_i \hat{Z}_{i+1} \right) - h_x \sum_{i=0}^{N-1} \hat{X}_i, \tag{7.13}$$

on an open chain of $N = 8$ qubits with $J = 1$. The field term flips single bits and breaks the conservation of $\hat{Z}_{\mathrm{tot}}$, while (7.13) still commutes with the parity (6.1), because $\hat{\Pi}$ commutes with each of $\hat{X}\hat{X}$, $\hat{Y}\hat{Y}$, $\hat{Z}\hat{Z}$ and $\hat{X}$. Every eigenstate therefore has parity ± 1.

On eight sites the phases of the infinite chain appear as smooth crossovers, recognisable in the bit strings that dominate the ground state. For $\Delta < -1$ the coupling favours aligned spins along z, a ferromagnet dominated by 00000000 and 11111111. For $-1 < \Delta < 1$ and small h_x the exchange term dominates and the chain is in the critical XY region, with weight spread over many bit strings with nearly equal numbers of zeros and ones and correlations that decay as powers of the distance. For $\Delta > 1$ the coupling favours anti-aligned neighbours, the Néel antiferromagnet dominated by 01010101 and 10101010. A large h_x turns the spins towards $+x$, almost completely on the ferromagnetic side of the grid and only partly on the antiferromagnetic side. Since $\langle \hat{Z}_i \rangle = 0$ in every parity eigenstate, the two kinds of order are measured by m_z of (7.12) and by the staggered magnetisation

$$m_{\mathrm{stag}} = \sqrt{\left\langle \left(\frac{1}{N} \sum_{i=0}^{N-1} (-1)^i \hat{Z}_i \right)^2 \right\rangle}, \tag{7.14}$$

which equals one in a Néel state and zero in a ferromagnet. Figure 31 summarises the four regimes.

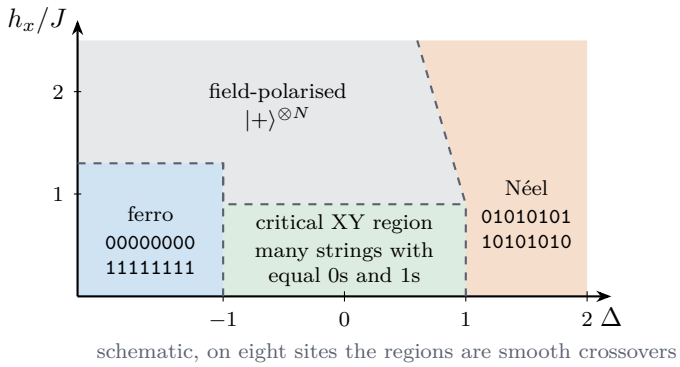


Figure 31: The regimes of the XXZ chain in a transverse field (7.13) with their dominant bit strings, ferromagnet ($\Delta < -1$), critical XY region ($-1 < \Delta < 1$), Néel antiferromagnet ($\Delta > 1$) and the field-polarised state at large h_x. Use it as a legend for the maps of Figure 32.

The exact maps (Figure 32) come from a Lanczos diagonalisation [36] that applies the Hamiltonian to a vector without storing its matrix. In the ferromagnetic corner $m_z \approx 1$ and the entropy is one bit, as for a cat state. The staggered magnetisation grows with Δ from the XY region and stays below one at $\Delta = 2$, because quantum fluctuations reduce the Néel order of eight spins. On the antiferromagnetic side the field changes the ground state in steps, at finite-size crossings between the lowest even and odd levels, so some of the ground states there are odd under $\hat{\Pi}$.

We train two ansätze with $L = 6$ layers and 42 two-qubit gates each on a 20×20 grid. The HEA has $n = 112$ angles. The *HVA with independent angles* applies in each layer the SPA gate (6.7) on the even and then on the odd bonds and a rotation $\hat{R}_x$ on every qubit, each gate with its own angle, $n = 132$ angles in total. Its gates conserve the parity, so we train it from the singlet product (even) and from the same product with $\hat{Z}_0$ applied to its first pair (odd), and keep the lower energy. Both ansätze get two starts per grid point and 300 Adam steps.

The HVA with independent angles reaches $F > 0.99$ at 94% of the grid points (Figure 33), including the odd ground states. It fails in the ferromagnetic corner and along $\Delta \approx -1$, where the

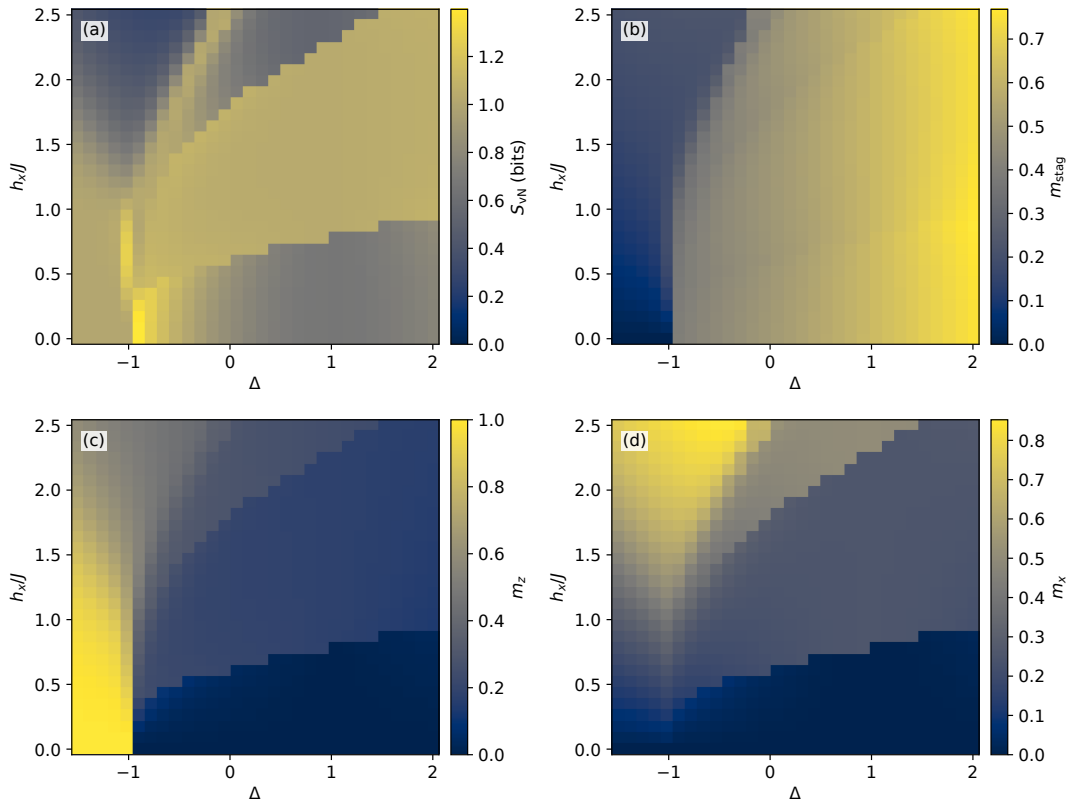


Figure 32: Exact ground state of the XXZ chain in a transverse field (7.13), $N = 8$, on a 30×30 grid of the anisotropy Δ and the field h_x/J. (a) Half-chain entropy, (b) staggered magnetisation (7.14), (c) order parameter m_z (7.12), (d) transverse magnetisation (7.2). Look for the ferromagnet in the lower left corner and for the steps on the antiferromagnetic side.

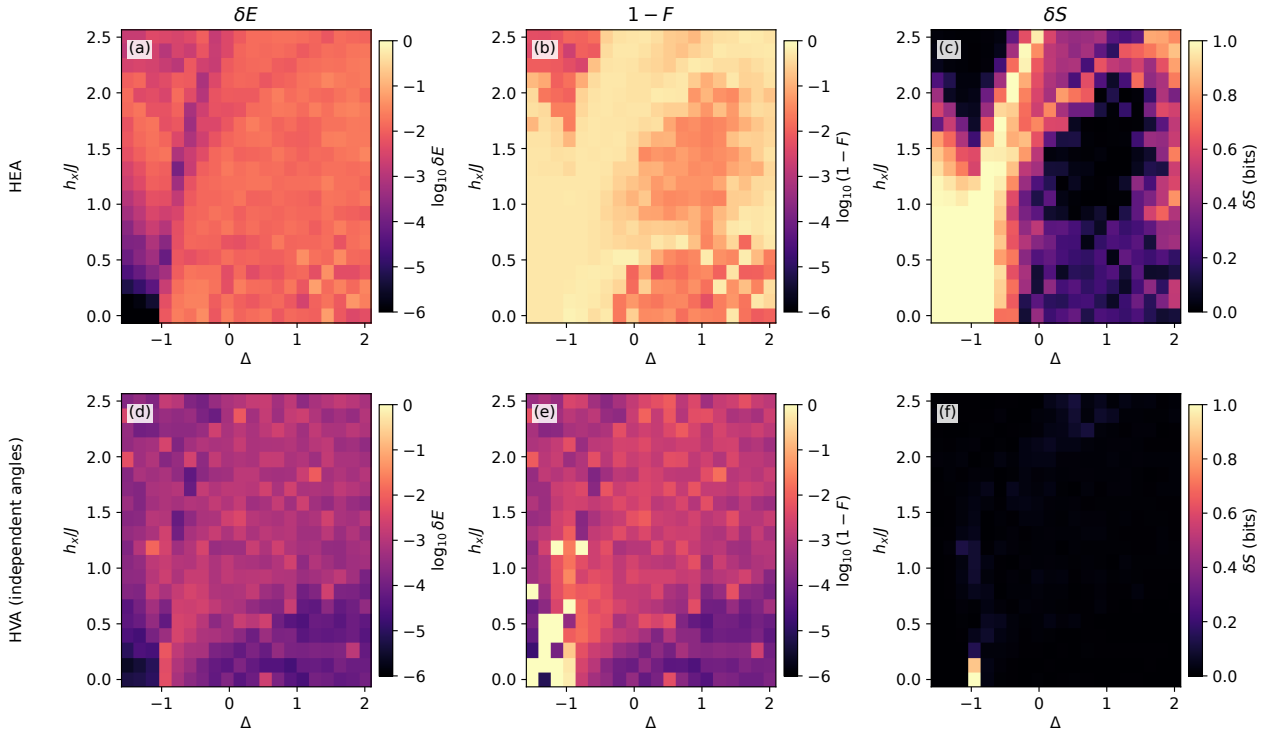


Figure 33: Errors of the trained states on the (Δ, h_x) grid for the HEA (top row) and the HVA with independent angles (bottom row). Left, $\log_{10}$ of the relative energy error δE. Middle, $\log_{10}$ of the infidelity $1 - F$. Right, the entropy error δS in bits. Dark colours mark small errors. Exact state-vector simulation with exact gradients.

even and odd levels are closer than its energy error and the lower trained energy can belong to the wrong sector. The HEA reaches $F > 0.99$ at only 5% of the points. Deep in the ferromagnet it has the smallest energy errors of the map together with $F = 1/2$ and an entropy error of one bit, the symmetry-broken branch of Section 7.6, and elsewhere it builds too little entanglement with this budget. At 34 of the 400 points the HEA has $\delta E < 10^{-3}$ and $F < 0.6$, so a small energy error does not certify the trained state.

8 Compressing quantum data with a quantum autoencoder

8.1 Compressing quantum data

The variational quantum eigensolver prepares a ground state on the register of a quantum device. Such a state is *quantum data*. It exists only on the register, and a classical description of it would require tomography, a reconstruction of all 2^N amplitudes from measurements whose number grows exponentially with N. Storing such states or sending them to another device costs scarce qubits, so it is useful to compress them with a circuit that acts on the states themselves.

A classical autoencoder is a neural network that maps an input vector through a narrow layer, the code, and back to a vector of the input size. It is trained to reproduce its inputs, and it succeeds when the data occupy a low-dimensional part of the input space. The quantum autoencoder of Ref. [37] transfers this idea to quantum states. The input is a state of N qubits, and the narrow layer is a smaller register of $N - k$ *latent* qubits. The remaining k qubits, here the last k of the register, are the *trash* qubits.

Compression task

We are given an ensemble $\{p_i, |\psi_i\rangle\}$ of states of N qubits, each prepared with probability p_i. We look for one circuit $\hat{U}(\boldsymbol{\theta})$ with trainable angles $\boldsymbol{\theta}$ that moves the information of every $|\psi_i\rangle$ into the $N - k$ latent qubits and leaves the k trash qubits in the fixed reference state $|0\ldots0\rangle$, so that the inverse circuit $\hat{U}(\boldsymbol{\theta})^\dagger$ rebuilds every $|\psi_i\rangle$ from the latent qubits and k fresh qubits in $|0\rangle$. Success is measured by the fidelity of the rebuilt states with the inputs.

The encoder $\hat{U}(\boldsymbol{\theta})$ first acts on the input (Figure 34). The trash qubits are then reset to $|0\rangle$, which discards whatever they carried, and the decoder $\hat{U}(\boldsymbol{\theta})^\dagger$, the encoder run backwards, acts on the latent qubits together with the fresh trash qubits and returns the output state. The compression of a state is lossless if and only if the encoder leaves the trash qubits in $|0\ldots0\rangle$,

$$\hat{U}(\boldsymbol{\theta})|\psi_i\rangle = |\psi_i^{\mathrm{c}}\rangle_{\mathrm{latent}} \otimes |0\ldots0\rangle_{\mathrm{trash}} \tag{8.1}$$

with a compressed state $|\psi_i^{\mathrm{c}}\rangle$ of the latent qubits. The reset then changes nothing, and the decoder returns $\hat{U}^\dagger\hat{U}|\psi_i\rangle = |\psi_i\rangle$.

Ref. [37] considers two training costs. The first is the average fidelity of the output with the input, $C_1 = \sum_i p_i F_i$, with the reconstruction fidelity F_i of state i defined in (8.3) below. It measures the quantity of interest, but evaluating it on a device requires the output and a fresh copy of the input side by side. The second cost is the average probability $C_2 = \sum_i p_i\, p_{0,i}$ that all trash qubits of the encoded state are found in 0, which requires measurements of the trash qubits alone. Condition (8.1) holds for every state of the ensemble when $C_2 = 1$, so $C_1 = 1$ if and only if $C_2 = 1$, and the encoder can be trained on the trash qubits alone.

8.2 Following the state through the autoencoder

The reset discards information, so between encoder and decoder the register is in general in a mixed state, which we describe by density matrices. For a pure input $|\psi_{\mathrm{in}}\rangle$ the autoencoder passes through

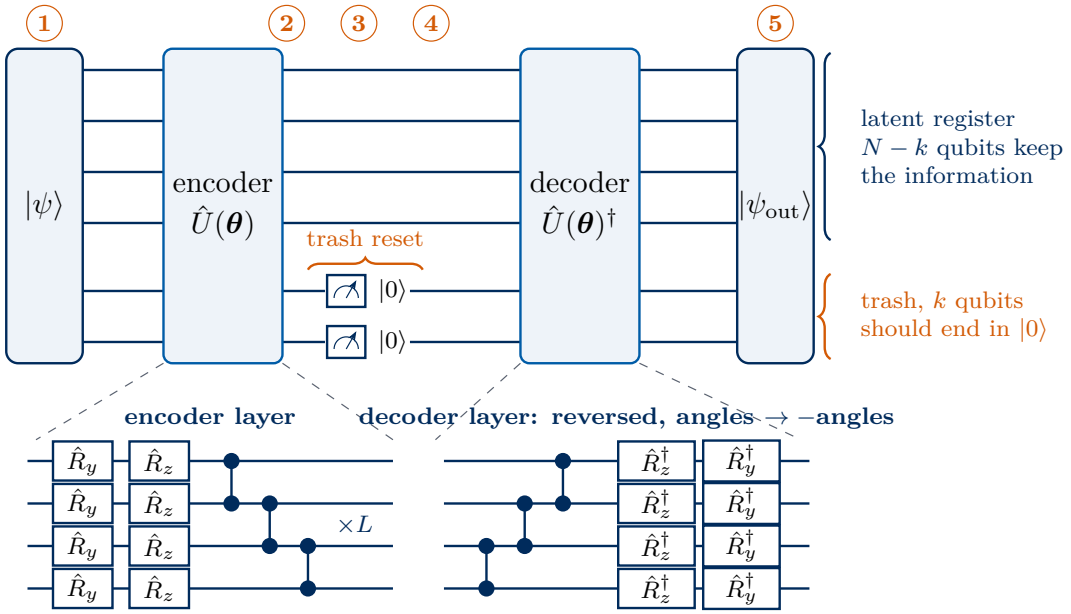


Figure 34: The quantum autoencoder. The encoder $\hat{U}(\boldsymbol{\theta})$ moves the information of an N-qubit state into $N-k$ latent qubits. The k trash qubits are reset to $|0\rangle$, and the decoder $\hat{U}(\boldsymbol{\theta})^\dagger$ rebuilds the state from the latent qubits and the fresh trash qubits.

five stages,

$$\begin{aligned}\hat{\rho}_{\text{in}} = |\psi_{\text{in}}\rangle\langle\psi_{\text{in}}| \xrightarrow{\hat{U}} \hat{\rho}_1 = \hat{U}\hat{\rho}_{\text{in}}\hat{U}^\dagger \xrightarrow{\text{Tr}_{\text{trash}}} \hat{\rho}_{\text{lat}} = \text{Tr}_{\text{trash}}\hat{\rho}_1 \\ \xrightarrow{\text{reset}} \hat{\rho}_2 = \hat{\rho}_{\text{lat}} \otimes |0\ldots0\rangle\langle 0\ldots0| \xrightarrow{\hat{U}^\dagger} \hat{\rho}_{\text{out}} = \hat{U}^\dagger\hat{\rho}_2\hat{U}.\end{aligned} \tag{8.2}$$

The partial trace Tr_{trash} sums the diagonal over the basis states of the trash qubits and leaves the density matrix $\hat{\rho}_{\text{lat}}$ of the latent qubits, which predicts every measurement on the latent qubits alone (Figure 35). The quality of the reconstruction is the fidelity

$$F = \langle\psi_{\text{in}}|\hat{\rho}_{\text{out}}|\psi_{\text{in}}\rangle, \tag{8.3}$$

the probability that the output passes a test for being the input state. If (8.1) holds, $\hat{\rho}_{\text{lat}}$ is the pure state $|\psi^{\text{c}}\rangle\langle\psi^{\text{c}}|$ and $F=1$. Otherwise the latent and trash qubits are entangled after the encoder, $\hat{\rho}_{\text{lat}}$ is mixed, and part of the state is lost at the reset.

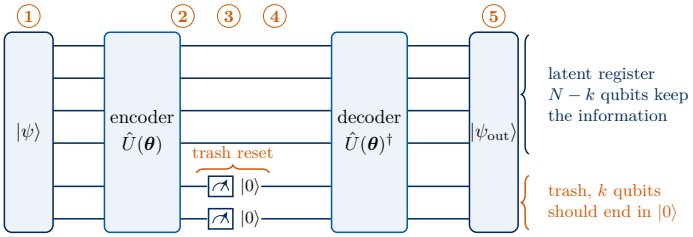


Figure 35: The stages of (8.2), with the circled numbers 1–5 marking $\hat{\rho}_{\text{in}}$, $\hat{\rho}_1$, $\hat{\rho}_{\text{lat}}$, $\hat{\rho}_2$ and $\hat{\rho}_{\text{out}}$. The encoder acts on the input, the trash qubits are traced out and replaced by fresh qubits in $|0\rangle$, and the decoder returns the output, which is compared with the input through the fidelity (8.3).

A formula for F follows from writing the encoded state as a sum over the 2^k bit strings t of the trash qubits,

$$\hat{U}(\boldsymbol{\theta})|\psi_{\text{in}}\rangle = \sum_{t\in\{0,1\}^k} |\alpha_t\rangle_{\text{latent}} \otimes |t\rangle_{\text{trash}}, \qquad \sum_t \langle\alpha_t|\alpha_t\rangle = 1, \tag{8.4}$$

where the latent vectors $|\alpha_t\rangle$ are not normalised. A measurement of the trash qubits returns the bit string t with probability $\langle\alpha_t|\alpha_t\rangle$, and in particular all trash bits read zero with probability

$$p_0 = \langle\alpha_0|\alpha_0\rangle, \tag{8.5}$$

called the *trash fidelity* of the input. The partial trace gives $\hat{\rho}_{\text{lat}} = \sum_t |\alpha_t\rangle\langle\alpha_t|$, and the reset replaces every $|t\rangle$ by $|0\ldots0\rangle$. The overlap of the decoded branch t with the input is $\langle\psi_{\text{in}}|\hat{U}^\dagger(|\alpha_t\rangle \otimes |0\ldots0\rangle) = \langle\alpha_0|\alpha_t\rangle$, because by (8.4) only the trash component $|0\ldots0\rangle$ of $\hat{U}|\psi_{\text{in}}\rangle$ contributes. Inserting this into (8.3) gives

$$F = \sum_t |\langle\alpha_0|\alpha_t\rangle|^2. \tag{8.6}$$

The term $t = 0\ldots0$ equals p_0^2, and the Cauchy–Schwarz inequality $|\langle\alpha_0|\alpha_t\rangle|^2 \le \langle\alpha_0|\alpha_0\rangle\langle\alpha_t|\alpha_t\rangle$ bounds the sum by $p_0 \sum_t \langle\alpha_t|\alpha_t\rangle = p_0$, so

$$p_0^2 \le F \le p_0. \tag{8.7}$$

A trash fidelity close to one therefore guarantees a faithful reconstruction, which is the quantitative form of the equivalence of the two costs C_1 and C_2.

In a three-qubit example, qubit 0 is the latent qubit and qubits 1 and 2 are the trash, $k = 2$. The input

$$|\psi_{\text{in}}\rangle \propto \cos a\,|000\rangle + \sin a\,|111\rangle + \varepsilon\,|100\rangle \tag{8.8}$$

is a weighted cat state of three qubits with an admixture of weight ε. The encoder consists of two $\widehat{\text{CNOT}}$ gates with qubit 0 as control and qubits 1 and 2 as targets. They map $|000\rangle \to |000\rangle$ and $|111\rangle \to |100\rangle$, so for $\varepsilon = 0$ the encoded state is $(\cos a|0\rangle + \sin a|1\rangle) \otimes |00\rangle$ and the compression is lossless. The admixture is mapped to $|1\rangle \otimes |11\rangle$ and lands in the branch $t = 11$, where it is entangled with the trash. Figure 36 shows the five density matrices for $a = \pi/5$ and $\varepsilon = 0.3$. The latent matrix is mixed, with purity $\text{Tr}\hat{\rho}_{\text{lat}}^2 = 0.901$, and the output keeps the corners of the cat state but has lost the admixture, so that $F = 0.868$.

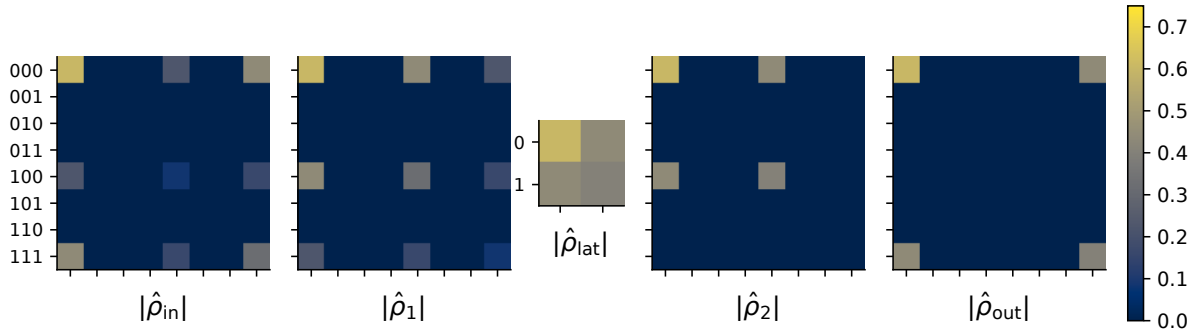


Figure 36: Absolute values of the matrix elements of the five density matrices of (8.2) for the three-qubit input (8.8) with $a = \pi/5$ and $\varepsilon = 0.3$, encoded by two $\widehat{\text{CNOT}}$ gates. Rows and columns follow the bit strings $b_0b_1b_2$, and the latent matrix has the rows 0 and 1 of qubit 0. Follow the weak row of the admixture $|100\rangle$, which moves to row 111 after the encoder and is absent from the output. Exact calculation.

8.3 The trash Hamming distance

A device measures the trash qubits in the computational basis. Each shot gives a trash bit string, and the number of ones in it is its Hamming distance from $0\ldots0$ (Figure 37). We define the *trash Hamming distance* as the mean number of ones per trash qubit,

$$D_H = \frac{1}{k} \sum_{j \in \text{trash}} \langle\hat{n}_j\rangle, \qquad \hat{n}_j = \frac{\hat{I} - \hat{Z}_j}{2}, \tag{8.9}$$

where the expectation value is taken in the encoded state $\hat{U}|\psi_{\text{in}}\rangle$ and the operator $\hat{n}_j$ has the eigenvalue one on $|1\rangle$ and zero on $|0\rangle$ of trash qubit j. Every term involves a single qubit, so D_H is a *local* cost. Its estimate from M shots is the fraction of ones among the kM recorded trash bits. The probability that at least one trash bit reads one, $1 - p_0$, is the corresponding *global* cost, because it depends on all trash qubits together.

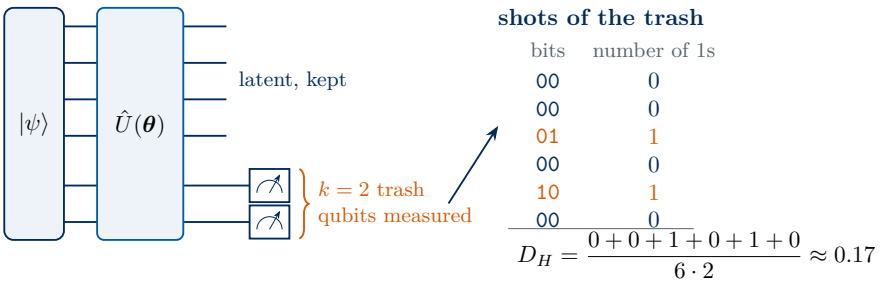


Figure 37: The trash Hamming distance (8.9). Every shot returns a bit string of the k trash qubits, and its number of ones is its Hamming distance from $0 \ldots 0$. Averaged over the shots and divided by k, it gives D_H, which vanishes only when every trash qubit reads zero in every shot.

If $D_H = 0$, every trash qubit reads zero with certainty, so the trash qubits are in the pure state $|0 \ldots 0\rangle$, which gives $p_0 = 1$ and, by (8.7), $F = 1$. Away from this optimum the local cost still controls the reconstruction. A trash string $t \neq 0 \ldots 0$ contains at least one and at most k ones, so $D_H = \frac{1}{k} \sum_t |t| \langle \alpha_t | \alpha_t \rangle$, with $|t|$ the number of ones in t, lies between $(1 - p_0)/k$ and $1 - p_0$. The left inequality is the union bound, by which the probability that some trash bit reads one is at most the sum of the k single-bit probabilities. Together,

$$D_H \leq 1 - p_0 \leq k \, D_H, \tag{8.10}$$

and inserting $p_0 \geq 1 - k D_H$ into (8.7) gives the guarantee

$$F \;\geq\; \max\{0,\, 1 - k D_H\}^2. \tag{8.11}$$

A small measured Hamming distance therefore certifies a good reconstruction, although the device never measures F itself. For the three-qubit example with $\varepsilon = 0.3$ the bound gives $F \geq 0.697$, below the actual value 0.868. Local costs of this kind keep the gradient measurable for shallow circuits on large registers, whereas global costs do not [38]. The bounds (8.10) are linear and hold for ensemble averages directly, and the lower bound (8.11) holds for the average fidelity as well, because the right-hand side is a convex function of D_H. In a simulation we also record the *trash entropy* S_B, the von Neumann entropy in bits of the reduced state of the trash qubits after the encoder, which measures how strongly the trash remains entangled with the latent qubits. Minimising D_H already removes this entanglement, since the purity of the trash state obeys $\mathrm{Tr}\hat{\rho}_B^2 \geq p_0^2 \geq (1 - k D_H)^2$, and $D_H = 0$ leaves the trash in the product state $|0 \ldots 0\rangle$. On a device the entropy is not an observable, whereas the purity, or the Rényi-2 entropy $S_2 = -\log_2 \mathrm{Tr}\hat{\rho}_B^2 \leq S_B$, is measured with a SWAP test on two copies [39] or with randomized measurements [40]. Purity alone is a weaker requirement than $D_H = 0$, since it accepts any fixed pure trash state.

8.4 Schmidt rank and compressibility

Whether a reset loses information depends on the entanglement between the latent and the trash qubits, and the Schmidt decomposition describes this entanglement completely. Every pure state of two parts A and B can be written as

$$|\psi\rangle = \sum_{r=1}^{\chi} s_r \, |a_r\rangle_A \otimes |b_r\rangle_B, \qquad \hat{\rho}_B = \mathrm{Tr}_A |\psi\rangle\langle\psi| = \sum_{r=1}^{\chi} s_r^2 \, |b_r\rangle\langle b_r|, \tag{8.12}$$

with orthonormal states $|a_r\rangle$ of A, orthonormal states $|b_r\rangle$ of B, and Schmidt coefficients $s_1 \geq s_2 \geq \cdots \geq s_\chi > 0$ with $\sum_r s_r^2 = 1$. The number of terms χ is the *Schmidt rank*, which equals the rank of the reduced density matrix $\hat{\rho}_B$. A product state has $\chi = 1$, and the larger χ and the more even the coefficients, the more entangled the two parts are. To compute the decomposition, we reshape the state vector into the matrix ψ_{ab}, with a the bit string of A and b that of B, as for the entropy (6.9). Its singular values are the Schmidt coefficients. In the autoencoder A is the latent and B the trash register, and condition (8.1) requires $\chi = 1$ across this cut together with the trash state $|0 \ldots 0\rangle$.

A unitary $\hat{W}_B$ acting on B alone can rotate the χ states $|b_r\rangle$ into the basis states $|r\rangle \otimes |0 \dots 0\rangle$, in which $r = 0, \dots, \chi - 1$ is written in binary on the first $m = \lceil \log_2 \chi \rceil$ qubits of B and all other qubits of B are in $|0\rangle$. These qubits can then be reset without loss, so the information of B fits into m qubits. Fewer qubits do not suffice, since a unitary on B keeps the rank χ of $\hat{\rho}_B$, whereas a state of $m' < m$ qubits times $|0 \dots 0\rangle$ has rank at most $2^{m'} < \chi$. If only m' qubits are kept, the largest probability of finding the others in 0 is the sum of the $2^{m'}$ largest s_r^2. For a random state of six qubits with $\chi = 3$ across the cut between the first two and the last four qubits, the four qubits of B fit into 2, and keeping a single qubit retains only the weight 0.924.

An encoder acting on the whole register compresses any single known state, even into zero latent qubits, because any normalised $|\psi\rangle$ is the first column of some unitary matrix whose inverse maps $|\psi\rangle$ to $|0 \dots 0\rangle$. A shallow encoder has less freedom, because a gate acting across the cut between latent and trash qubits, written as a sum of r products (6.10), multiplies the Schmidt rank by at most r, and its inverse does the same, so it divides the Schmidt rank by at most r. For the $\widehat{\mathrm{CZ}}$ gate of the HEA $r = 2$, so an encoder that reduces a state of Schmidt rank χ to a product needs at least $\log_2 \chi$ such gates across the cut.

Compression becomes a learning task when one encoder has to serve a family of states $|\psi_1\rangle, \dots, |\psi_{n_\mathrm{f}}\rangle$. The family enters through its average density matrix

$$\bar{\rho} = \frac{1}{n_\mathrm{f}} \sum_{m=1}^{n_\mathrm{f}} |\psi_m\rangle\langle\psi_m|, \tag{8.13}$$

a positive matrix with unit trace whose eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ give the weight of the family along the corresponding eigenvectors. The projector $\hat{P}_0 = \hat{I}_\mathrm{latent} \otimes |0 \dots 0\rangle\langle 0 \dots 0|_\mathrm{trash}$ onto the states with all trash bits zero has rank 2^{N-k}, and the trash fidelity averaged over the family obeys

$$\bar{p}_0 = \mathrm{Tr}\big[\hat{P}_0 \, \hat{U} \bar{\rho} \, \hat{U}^\dagger\big] \leq \sum_{j=1}^{2^{N-k}} \lambda_j. \tag{8.14}$$

The encoder rotates the eigenvectors of $\bar{\rho}$ and keeps its eigenvalues, so the largest weight it can place in the 2^{N-k}-dimensional range of $\hat{P}_0$ is the weight of the 2^{N-k} leading eigenvectors, and an ideal encoder reaches the bound by rotating these eigenvectors into that range. This is the quantum analogue of principal-component analysis, in which data are projected onto the directions of largest variance. A family is therefore compressible into $N - k$ qubits exactly when the weight of $\bar{\rho}$ beyond its 2^{N-k} largest eigenvalues vanishes, and approximately when this weight is small. The one-qubit family $\{|0\rangle, |+\rangle\}$, for example, has $\lambda_1 = \cos^2(\pi/8) \approx 0.854$, which is the best average trash fidelity of any compression into zero qubits. All ground states of the Ising chain are even under the parity (6.1), so they span at most 2^{N-1} dimensions, and one trash qubit can always be freed by an ideal encoder.

8.5 Schmidt spectrum after the encoder

The Schmidt spectrum across the cut between latent and trash qubits shows what a trained encoder achieves. We take ground states of the Ising chain (2.2) with $N = 8$ qubits, an HEA encoder (6.3) with four layers and 80 angles, and $k = 3$ trash qubits. For each field h we train a separate encoder on the single ground state at that field, with the Hamming distance (8.9) as the cost, Adam [28] and the best of four random starts. As a second case we freeze the encoder trained at $h = 2$ and apply it to the ground states at all other fields. The weight p_0 of the product branch $|\alpha_0\rangle \otimes |000\rangle$ cannot exceed the weight s_1^2 of the best product term, so $1 - s_1^2 \leq 1 - p_0$.

With an encoder trained at each field (Figure 38, left), a single Schmidt term carries almost all the weight, $1 - s_1^2 \leq 10^{-3}$ at every field, so the encoded states are nearly products of a latent and a trash state. At $h = 0$ the ground state is the cat state $(|0 \dots 0\rangle + |1 \dots 1\rangle)/\sqrt{2}$, which a ladder of $\widehat{\mathrm{CNOT}}$ gates disentangles completely, and its spectrum has a single coefficient. The encoder trained at $h = 2$ and frozen (Figure 38, right) leaves the spectrum spread in the ordered phase, with $s_1^2 \approx 0.5$ at $h = 0$, so the latent and trash qubits stay entangled and the reset loses information.

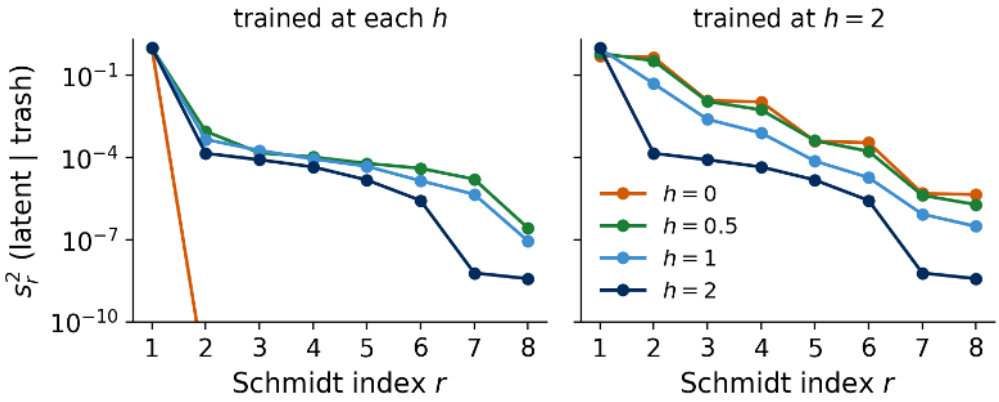


Figure 38: Squared Schmidt coefficients s_r^2 across the cut between the five latent and the three trash qubits after the encoder, for ground states of the Ising chain with $N = 8$ at four fields, on a logarithmic scale. Left, an encoder trained on the ground state at each field. Right, the encoder trained at $h = 2$ and frozen. Compare the weight beyond the first coefficient in the two panels. Exact state-vector simulation.

8.6 Compressing families of Ising ground states

We compress whole families of exact ground states of the Ising chain (2.2) with $N = 6$ qubits. The encoder is the HEA with $L = 3$ layers and $n = 48$ angles, and we train with Adam on the average of a cost over eight ground states with fields $h \in [1.5, 2]$, in the paramagnetic phase. Four starts are trained for 1500 steps each with exact gradients, and the start with the lowest final cost is kept.

Training with the global cost $1 - \bar{p}_0$ and with the local cost $\bar{D}_H$ gives nearly the same result for $k = 3$ (Figure 39). Both runs reduce the average trash infidelity $1 - \bar{p}_0$ by more than two orders of magnitude within a few hundred steps and end near 1.5×10^{-3} and 1.7×10^{-3}. The advantage of the local cost appears only on larger registers. The remaining infidelity is a limit of the three-layer circuit, since by (8.14) an ideal encoder compresses this family almost perfectly for every $k \leq 5$.

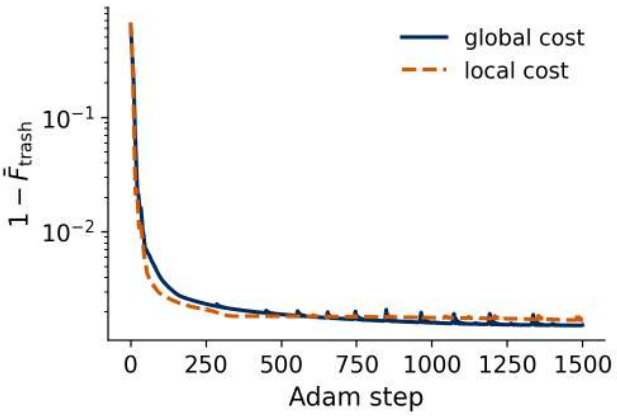


Figure 39: Training of the quantum autoencoder on eight paramagnetic ground states of the Ising chain, $N = 6$, $k = 3$ trash qubits, hardware-efficient encoder with 48 angles. The average trash infidelity $1 - \bar{F}_{\text{trash}} = 1 - \bar{p}_0$ against the Adam step, for training with the global cost $1 - \bar{p}_0$ (solid) and with the local cost, the trash Hamming distance $\bar{D}_H$ (8.9) (dashed). Exact state-vector simulation.

Figure 40 compares two families as the number k of trash qubits grows. The trained encoder compresses the paramagnetic family into a single latent qubit, $k = 5$, with an average infidelity of 3.2×10^{-3}, because its ground states differ only by small corrections to $|+\rangle^{\otimes N}$. A second family of 21 ground states spread over $h \in [0, 2]$ crosses the transition, and its trained infidelity is one to two orders of magnitude higher, rising from 2×10^{-2} at $k = 1$ to 1.7×10^{-1} at $k = 5$. An ideal encoder would compress this family almost perfectly up to $k = 4$ and loses 6.0×10^{-3} at $k = 5$. The gap between the trained and the ideal values is therefore a limit of the circuit, since the states themselves are compressible. A shallow circuit disentangles the nearly product states of the paramagnet easily,

whereas undoing the cat-like correlations of the ordered states and serving both kinds of state with the same angles requires more layers.

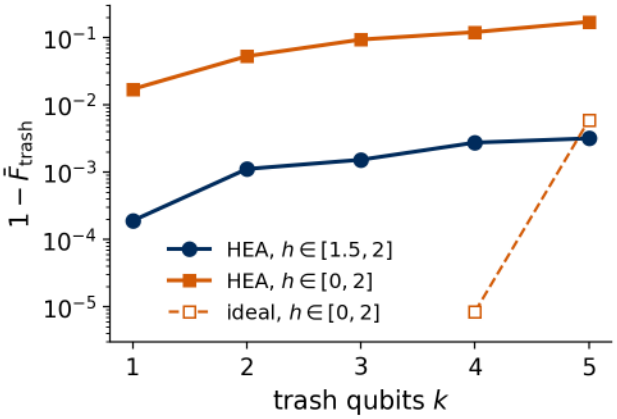


Figure 40: Average trash infidelity after training against the number k of trash qubits, for the paramagnetic family $h \in [1.5, 2]$ (circles) and the family $h \in [0, 2]$ that crosses the transition (squares), with the ideal bound (8.14) for the second family (open squares). Ideal values below 10^{-5} lie off the scale. Exact state-vector simulation.

8.7 Compressibility across two chains

A separate encoder trained on each single ground state measures how compressible that state is for a given circuit. We repeat the training of Section 8.5, $N = 8$, four HEA layers and the Hamming distance as the cost, for $k = 1, 3, 5, 7$ trash qubits, on ground states of the Ising chain at 21 fields $h \in [0, 2]$ and of the XXZ chain (2.3) at 22 values of the anisotropy $\Delta \in [-2, 3]$. The XXZ chain is a ferromagnet for $\Delta < -1$, whose ground state is a cat state of all spins up and all spins down, has a gapless region with power-law correlations for $-1 < \Delta < 1$ and orders antiferromagnetically for $\Delta > 1$. Besides D_H we record the reconstruction fidelity F and the trash entropy S_B.

For the Ising chain every ground state is compressed with $F \geq 0.97$ at every k (Figure 41), including the cat states of the ordered phase, which four layers disentangle completely. The small dips scatter from point to point and from one k to the next, so they reflect the optimisation more than the states. The XXZ chain is compressed perfectly in the ferromagnet. Just above $\Delta = -1$, where the half-chain entropy peaks, the fidelity drops to 0.643 at $k = 5$ and the trash entropy rises, and compression stays imperfect across the gapless region. A larger trash register can never be compressed better than a smaller one by the best encoder, so the better results at $k = 7$ than at $k = 5$ show that four random starts do not always find the best angles of this circuit. The difficulty of compression follows the entanglement of the state, which makes the critical XXZ chain the hardest case for this circuit.

8.8 Detecting a phase boundary

An encoder trained on one family of states compresses similar states well and different states badly. We freeze the six-qubit encoders trained on the paramagnetic family with $k = 2$ and $k = 3$ and apply them to the ground states at 41 fields between $h = 0$ and $h = 2$. The trash infidelity $1 - p_0$ of each state then serves as an *anomaly score*, an idea introduced for mapping phase diagrams in Ref. [41]. No state of the ordered phase and no order parameter enters the training.

In Figure 42(a) the score stays at the level of the training window, a few 10^{-3}, across the paramagnetic side, starts to rise near $h = J$ and reaches about 0.6 at $h = 0$. The half-chain entropy in panel (b) rises over the same range of fields, from 0.13 bits at $h = 2$ to one bit at $h = 0$, the value of the cat state. The frozen encoder thus separates the two regimes of the chain without labels. In a

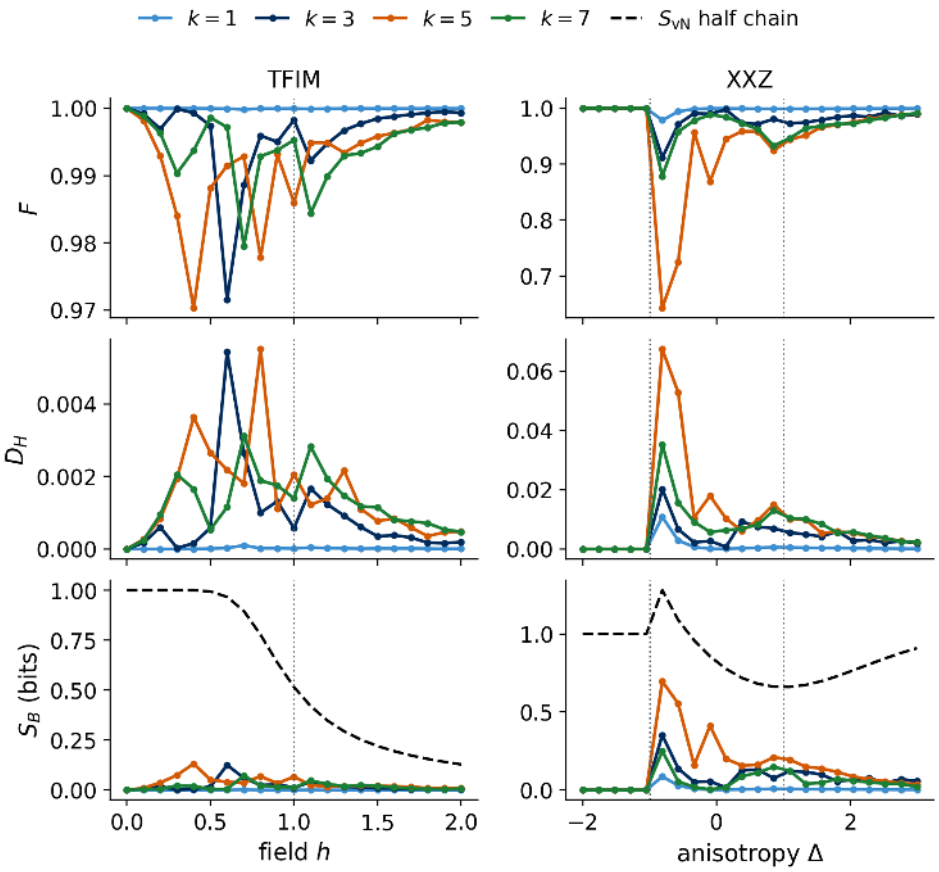


Figure 41: Compressibility of single ground states with $N = 8$, a separately trained four-layer encoder at every point and $k = 1, 3, 5, 7$ trash qubits. Left column, the Ising chain against the field h. Right column, the XXZ chain against the anisotropy Δ. Rows show the reconstruction fidelity F, the trash Hamming distance D_H and the trash entropy S_B, with the exact half-chain entropy dashed in the bottom row. Dotted lines mark $h = J$ and $\Delta = \pm 1$. The two columns have different vertical scales. Exact state-vector simulation.

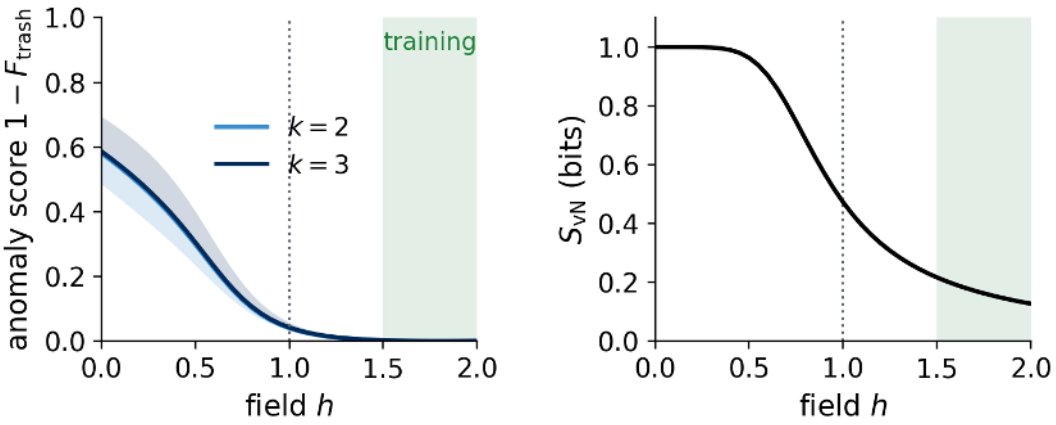


Figure 42: The frozen autoencoder as an anomaly detector for the Ising chain, $N = 6$. (a) Anomaly score $1 - p_0$ against the field for encoders trained in the green window with $k = 2$ and $k = 3$ trash qubits. Lines show the kept start and bands the range over four starts. (b) Exact half-chain entanglement entropy. The dotted line marks the critical field $h = J$ of the infinite chain. Exact state-vector simulation.

chain of six qubits both curves change smoothly, and locating the transition itself would require a comparison of several chain lengths.

The score also depends on the circuit. An encoder that had learned only the rotation taking $|+\rangle$ to $|0\rangle$ on every qubit would give the cat state a score of $1 - 2^{-k}$, since each trash qubit of the cat state then reads zero or one with equal probability. The trained encoders stay below this value, and the four starts, which all compress the training family equally well, give scores at $h = 0$ that differ by about 0.1. The behaviour of an encoder on states it never saw depends on how its circuit acts outside the training family, which the training does not fix. The field at which the score starts to rise is common to all starts and follows the entanglement, whereas the height of the score deep in the ordered phase depends on the encoder.

8.9 Anomaly maps of the XXZ chain in a field

The same idea maps a two-dimensional phase diagram. We take the XXZ chain in a transverse field (7.13) with $N = 8$, whose exact ground-state maps are shown in Figure 32. The sign of the field does not matter here, since the rotation $\prod_i \hat{Z}_i$ reverses $\hat{X}_i$ and $\hat{Y}_i$ on every site and leaves the couplings unchanged. We choose six reference points, one in each region of the diagram, the ferromagnet, the XY region, the Néel antiferromagnet, the canted ferromagnet and canted antiferromagnet at intermediate field, and the field-polarised paramagnet. At each reference point we train an encoder with $k = 3$ on the single ground state, freeze it and evaluate its Hamming distance D_H on the ground states of a grid of 24×25 points in the plane of $\Delta \in [-2, 2]$ and $h_x \in [0, 2.5]$.

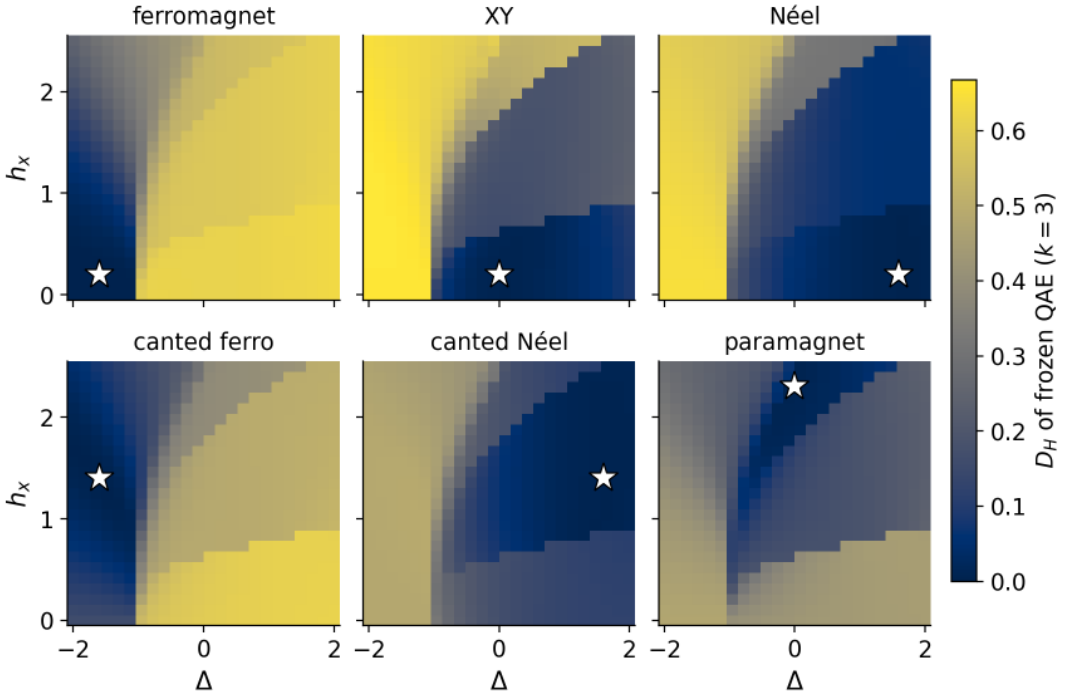


Figure 43: Anomaly maps of the XXZ chain in a transverse field, $N = 8$. Each panel belongs to one encoder with $k = 3$, trained on the ground state at the star and frozen, and shows its trash Hamming distance D_H on the ground states of the whole (Δ, h_x) grid. Dark regions are compressed as well as the training state. Compare the borders of the dark regions with the exact maps of Figure 32. Exact state-vector simulation.

Every map isolates a region around its own reference point (Figure 43). Its borders run along $\Delta = -1$, where the ferromagnet ends, and along the steps that the level crossings of the eight-site chain produce on the antiferromagnetic side, visible in the exact maps as well. Six frozen encoders, each trained on one state, thus divide the plane into regions that agree with the exact order parameters, although no encoder was given an order parameter or a label.

8.10 Variational quantum anomaly detection

Ref. [41] formulated this procedure as variational quantum anomaly detection, which maps a phase diagram entirely on a quantum device (Figure 44). A ground-state preparation such as the VQE prepares the ground state at one point of parameter space, an encoder is trained on it by minimising the trash Hamming distance (8.9), and the frozen encoder is evaluated on the ground states across parameter space, where a large Hamming distance marks a different phase. Retraining at such a point and repeating the procedure finds all phases, and since the ground states never leave the device, no tomography, order parameter or labels are needed. The main example of Ref. [41] is a chain of twelve sites with interacting particles and alternating hopping amplitudes, which has three phases, a charge-density wave (CDW), a Mott insulator (MI) and a topological Mott insulator (TMI) that no local order parameter distinguishes from the MI. Three encoders, each trained on one state of one phase, isolate their own phases (Figure 45), and the encoder trained in the MI separates the TMI from it.

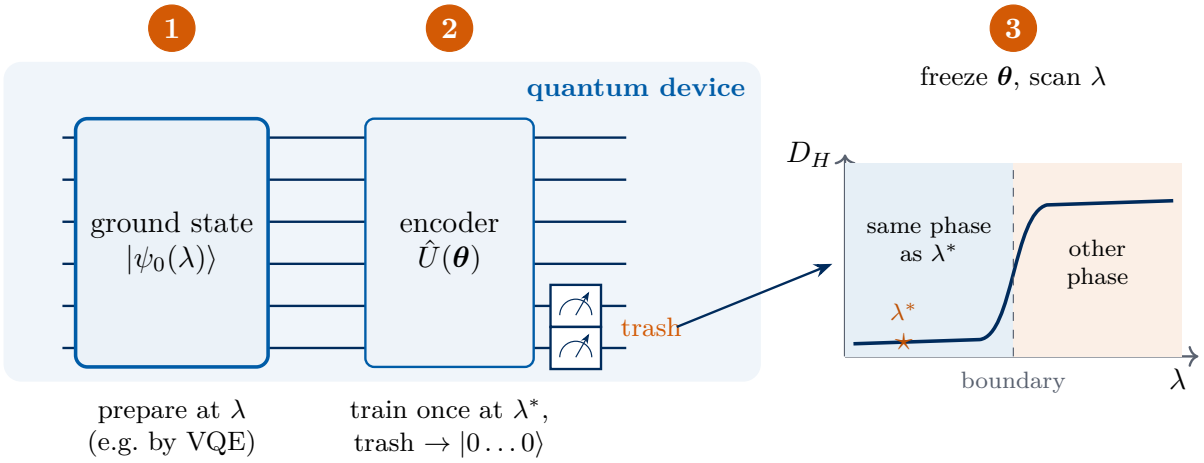


Figure 44: Variational quantum anomaly detection. A ground state is prepared on the device at one point of parameter space, an encoder is trained on it to clear the trash qubits, and the frozen encoder is evaluated on ground states across parameter space. Where it fails to clear the trash, a different phase begins.

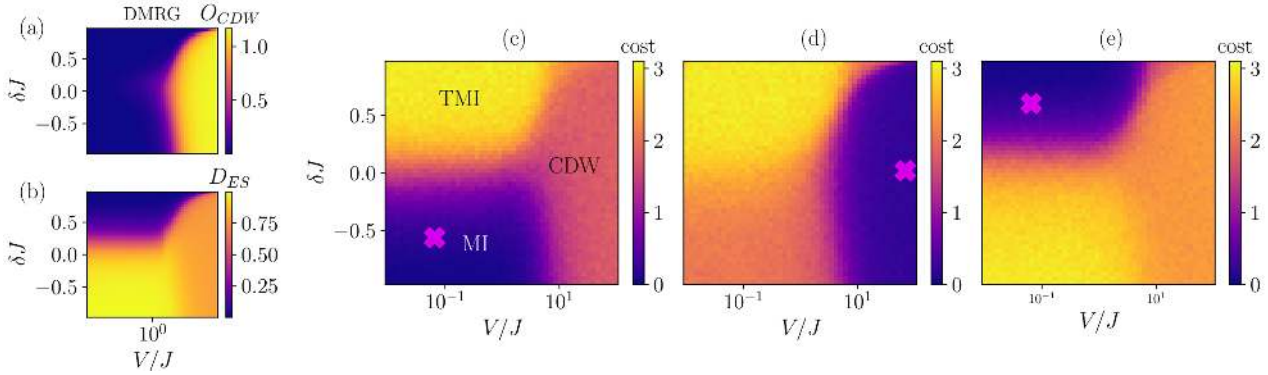


Figure 45: Variational quantum anomaly detection for the chain of Ref. [41], against the repulsion V/J between particles on neighbouring sites and the alternation δJ of the hopping amplitude. (a) CDW order parameter and (b) a marker of the TMI computed from the Schmidt coefficients, both from the density-matrix renormalisation group. (c)–(e) Trash Hamming distance, with six trash qubits, of encoders trained on a single ground state (cross) in the MI, CDW and TMI phase, respectively. Dark regions resemble the training state. Reproduced from Ref. [41] under the Creative Commons Attribution 4.0 licence.

9 Classifying phases with a quantum convolutional neural network

9.1 From anomaly detection to classification

Anomaly detection located changes of phase without labels. When the phase of some ground states is known, a circuit can be trained to return the phase of a new state directly on the device that prepared it. The quantum convolutional neural network (QCNN) of Ref. [42] does this with a circuit modelled on the classical convolutional network. It is the supervised counterpart of anomaly detection, trained on labelled states, and it reads out a single qubit, so the state never has to be reconstructed. We apply it to a phase that no local measurement reveals.

9.2 The cluster-Ising chain

The states to classify are ground states of the cluster-Ising chain of Ref. [42]. We place N qubits on a ring, so that indices are taken modulo N, whereas Ref. [42] uses an open chain,

$$\hat{\mathcal{H}} = -J \sum_{i=0}^{N-1} \hat{K}_i - h_1 \sum_{i=0}^{N-1} \hat{X}_i - h_2 \sum_{i=0}^{N-1} \hat{X}_i \hat{X}_{i+1}, \qquad \hat{K}_i = \hat{Z}_{i-1} \hat{X}_i \hat{Z}_{i+1}, \tag{9.1}$$

with $J = 1$. Each of the three terms prefers a different ground state. The three-qubit operators $\hat{K}_i$ square to the identity and commute with each other, because two of them overlap either not at all or on two sites, where an $\hat{X}$ meets a $\hat{Z}$ twice. At $h_1 = h_2 = 0$ the ground state is their common eigenstate with $\hat{K}_i = +1$ for every i, the *cluster state*. It is strongly entangled, and every single qubit in it is completely mixed, $\langle \hat{X}_i \rangle = \langle \hat{Z}_i \rangle = 0$. A strong field h_1 turns every spin into $|+\rangle$, a paramagnet, and a strong negative coupling $h_2 \lesssim -J$ produces an antiferromagnet with alternating spins along x. Along the line $h_2 = 0$ the model is exactly solvable, since the Jordan–Wigner transformation [43] maps it onto free fermions, and the transition from the cluster phase to the paramagnet lies at $h_1 = J$ [42].

The Hamiltonian has two symmetries, the flips of all even spins and of all odd spins,

$$\hat{X}_{\text{even}} = \prod_{i \text{ even}} \hat{X}_i, \qquad \hat{X}_{\text{odd}} = \prod_{i \text{ odd}} \hat{X}_i. \tag{9.2}$$

Both commute with every term of (9.1), because each $\hat{K}_i$ contains its two $\hat{Z}$ operators on sites of equal parity and each flip changes the sign of both. The group generated by the two flips is called $\mathbb{Z}_2 \times \mathbb{Z}_2$. The phase around the cluster state is a *symmetry-protected topological* (SPT) phase. Its ground state is unique, breaks neither symmetry and has featureless single-qubit expectation values, so no local order parameter distinguishes it from the paramagnet. Nevertheless no path of Hamiltonians that keeps the gap open and respects the symmetries connects the two phases.

9.3 String order

The difference between the SPT phase and the paramagnet lies in a non-local correlation. In the cluster state every $\hat{K}_i$ has the eigenvalue $+1$, so the product of every second one, $\hat{K}_1 \hat{K}_3 \cdots \hat{K}_{2m-1}$, has the eigenvalue $+1$ as well. In this product the $\hat{Z}$ factors between two neighbouring $\hat{K}$ meet on the same site and square to the identity (Figure 46), and the product telescopes into the string operator

$$\hat{S} = \hat{Z}_0 \, \hat{X}_1 \hat{X}_3 \cdots \hat{X}_{2m-1} \, \hat{Z}_{2m}. \tag{9.3}$$

Its expectation value, the string order parameter, therefore equals one in the cluster state for every length of the string. In the paramagnet $|+\rangle^{\otimes N}$ the two end operators $\hat{Z}_0$ and $\hat{Z}_{2m}$ average to zero and $\langle \hat{S} \rangle = 0$, and in the antiferromagnet it vanishes for the same reason. In a long chain $\langle \hat{S} \rangle$ stays finite throughout the SPT phase and vanishes outside it [44]. Table 4 summarises the three phases. We take $2m = N/2$, half of the ring.

We write S_n for the string order parameter $\langle \hat{S} \rangle$ of the ground state of a ring of n sites. We label a point (h_1, h_2) as SPT when the string order parameter S_{12} of the exact ground state of a ring of

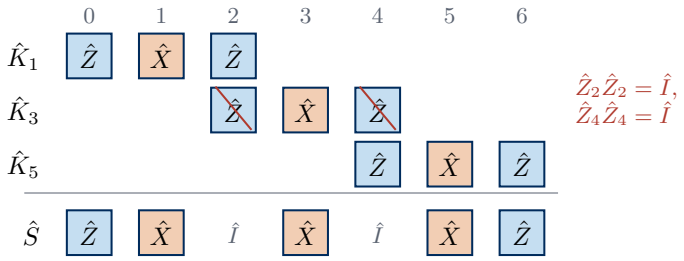


Figure 46: String order of the cluster state. The product $\hat{K}_1\hat{K}_3\hat{K}_5$ of three cluster operators (9.1) has the inner factors $\hat{Z}_2\hat{Z}_2$ and $\hat{Z}_4\hat{Z}_4$ equal to the identity and telescopes into the string $\hat{S} = \hat{Z}_0\hat{X}_1\hat{X}_3\hat{X}_5\hat{Z}_6$, which therefore has $\langle\hat{S}\rangle = 1$ in the cluster state.

	SPT	paramagnet	antiferromagnet
typical state	cluster state	$\lvert{+}{+}\cdots{+}\rangle$	$\lvert{+}{-}{+}{-}\cdots\rangle$
local order	none	$\langle\hat{X}_i\rangle \approx 1$, no symmetry broken	$\langle\hat{X}_i\hat{X}_{i+1}\rangle \approx -1$
string order $\langle\hat{S}\rangle$	≈ 1	≈ 0	≈ 0
open chain	edge modes, fourfold degenerate ground state	unique ground state	two degenerate patterns

Table 4: The three phases of the cluster-Ising chain (9.1). Only the string order parameter (9.3) separates the SPT phase from the paramagnet.

12 sites exceeds one half (Figure 49(a)). The longer ring gives a sharper boundary than the 8-qubit chain that we classify. The line $S_{12} = 1/2$ agrees within about 0.06 with the maxima of the fidelity susceptibility of the same ring, an independent criterion that marks where the ground state changes fastest. At $h_1 = 0.5$ it crosses $h_2 = 0.44$, close to the infinite-chain value 0.423 obtained by the density-matrix renormalisation group in Ref. [42].

9.4 The circuit

The SPT phase shows up only in a correlation along the whole chain, invisible to any single qubit or pair of qubits, so a classifier must combine the information of all qubits. The QCNN does this in the way a convolutional network processes an image. A classical convolutional network has three kinds of layers. A convolution layer slides one small filter, a kernel of a few weights, over the whole image, so that the same local feature is detected at every position with a number of weights independent of the image size. A pooling layer coarse-grains the resulting map, for instance by keeping one of every two neighbouring values. A fully connected, or dense, layer finally maps the few remaining numbers to class probabilities. The QCNN keeps this structure for an input that is a quantum state of N qubits (Figure 47).

A convolution layer applies one two-qubit unitary $\hat{U}_\ell$ to all neighbouring pairs, first to $(q_0, q_1), (q_2, q_3), \ldots$ and then to the shifted pairs. The same unitary at every position is the shared kernel. A pooling layer measures one qubit of every pair and applies a single-qubit unitary $\hat{V}_\ell$ to its partner if the outcome is one. The measured qubit is not used again, so the register halves. In our parametrisation the partner also receives a trainable rotation for the outcome zero, which gives two single-qubit unitaries per pooling. When two qubits remain, a general two-qubit unitary $\hat{F}$ acts on them, the analogue of the dense layer, and one of them is measured. The probability of the outcome one is the output of the network.

Each layer combines neighbouring qubits, and each pooling merges pairs of qubits into one, then pairs of those, so that after $\log_2 N$ poolings the last qubit depends on the whole chain (Figure 48). The effect resembles zooming out of an image, where local details disappear and the overall pattern remains. Training chooses the gates for which this last qubit answers whether the state is topological. Ref. [42] showed that a QCNN with suitable gates removes local perturbations of the cluster state at

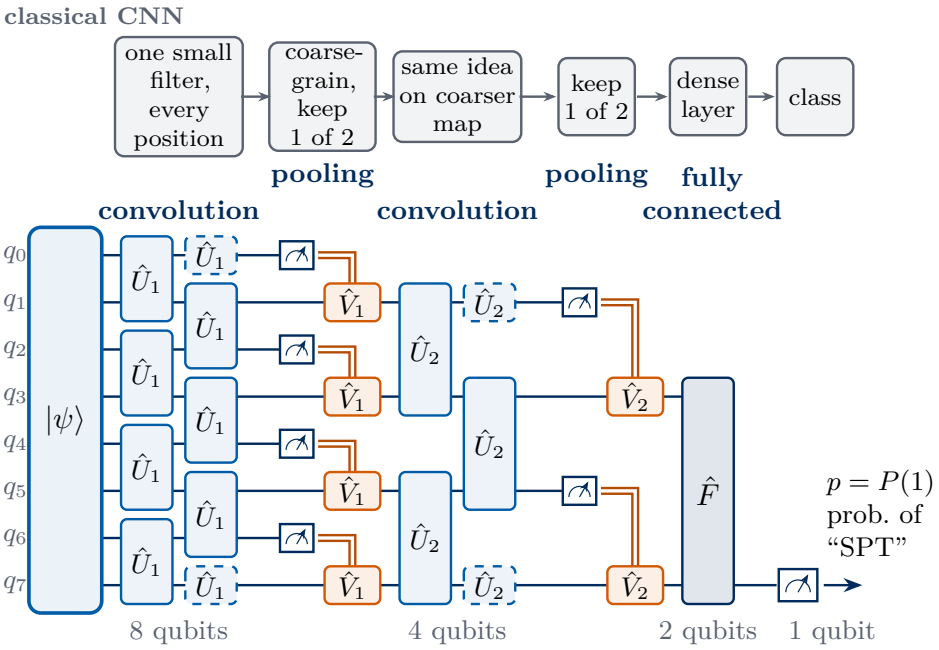


Figure 47: Quantum convolutional neural network on a ring of $N = 8$ qubits. A convolution applies one shared two-qubit unitary $\hat{U}_\ell$ to the even and then to the odd neighbouring pairs, dashed gates acting on the pairs that close the ring. Pooling measures every second qubit and rotates its partner conditioned on the outcome, which halves the register. A fully connected unitary $\hat{F}$ acts on the last two qubits and one of them is measured. The top row lists the corresponding layers of a classical convolutional network.

every scale, in the same way as the syndrome measurements of a quantum error-correcting code.

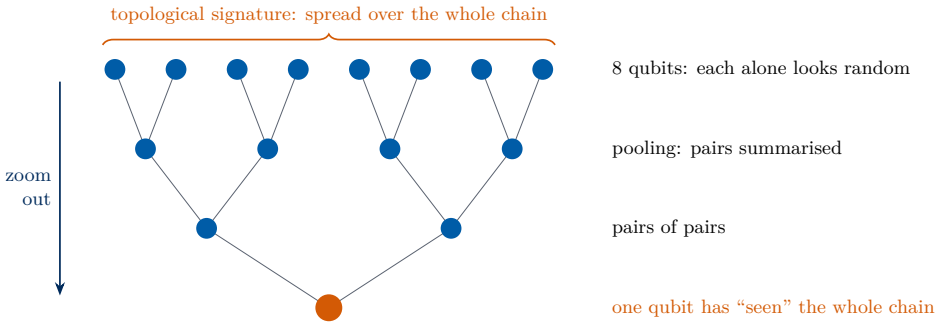


Figure 48: Pooling lets one qubit see the whole chain. Each qubit of the input alone looks random. Every pooling layer summarises pairs, then pairs of pairs, and the last qubit depends on the correlation spread over all qubits.

For a simulation the measurement in a pooling layer can be replaced by the controlled unitary

$$\hat{C} = |0\rangle\langle 0|_a \otimes \hat{V}_b^{(0)} + |1\rangle\langle 1|_a \otimes \hat{V}_b^{(1)} \tag{9.4}$$

on the measured qubit a and its partner b, after which qubit a is ignored. A measurement whose outcome is never used again leaves the other qubits in the same reduced state as no measurement at all, so the whole QCNN can be simulated as one unitary $\hat{W}(\boldsymbol{\theta})$ on the state vector. A general two-qubit unitary has 15 real parameters, which we write as the exponential of a combination of the 15 two-qubit Pauli strings other than the identity, and a single-qubit unitary has three. With one shared unitary per layer, each convolution and pooling unit carries $15 + 6 = 21$ angles, and a register of N qubits, a power of two, needs $\log_2 N - 1$ units and the final $\hat{F}$, so the number of angles

$$n = 21(\log_2 N - 1) + 15 \tag{9.5}$$

grows only logarithmically with N, with $n = 57$ for $N = 8$.

9.5 Training and results

Classification task

The input is the ground state $|\psi(h_1, h_2)\rangle$ of the cluster-Ising ring (9.1) with $N = 8$ qubits, prepared on the device. The output is the probability p of reading one on the output qubit, and the state is classified as SPT when $p > 1/2$. The training data are $M = 40$ ground states at random points (h_1, h_2) of the rectangle $h_1 \in [0.05, 1.6]$, $h_2 \in [-1.6, 1.6]$, labelled by the string order of the 12-site ring. The trained parameters are the 57 angles of the shared convolution, pooling and final unitaries. Success is the accuracy of the predicted phase diagram on a grid of 528 points.

The output of the QCNN for an input state $|\psi\rangle$ is the probability of reading one on the output qubit,

$$p_{\boldsymbol{\theta}}(|\psi\rangle) = \langle\psi|\hat{W}(\boldsymbol{\theta})^\dagger(|1\rangle\langle 1|_{\text{out}} \otimes \hat{I})\hat{W}(\boldsymbol{\theta})|\psi\rangle, \tag{9.6}$$

where $\hat{W}(\boldsymbol{\theta})$ is the whole circuit of convolutions, poolings and $\hat{F}$, and $|1\rangle\langle 1|_{\text{out}}$ projects the output qubit. On a device $p_{\boldsymbol{\theta}}$ is the frequency of the outcome one in repeated shots. The training states $|\psi_\alpha\rangle$ carry the labels $y_\alpha = 1$ for SPT and $y_\alpha = 0$ otherwise, and we minimise the mean squared error between output and label,

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2M}\sum_{\alpha=1}^{M}\left[y_\alpha - p_{\boldsymbol{\theta}}(|\psi_\alpha\rangle)\right]^2, \tag{9.7}$$

with exact gradients and Adam [28], keeping the best of four random starts. On a device the gradient would come from the parameter-shift rule.

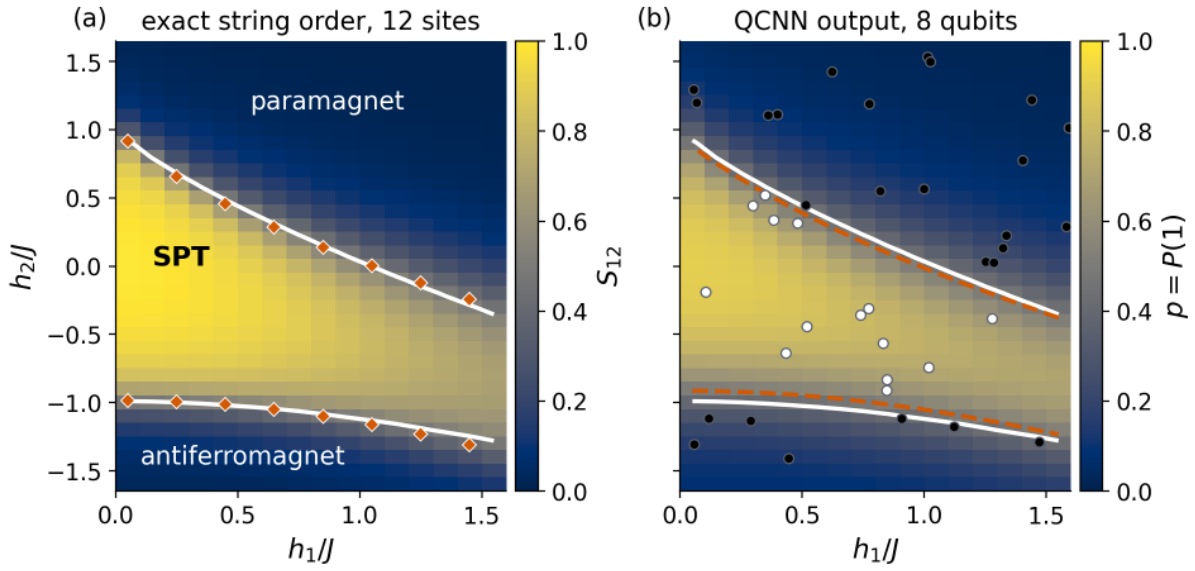


Figure 49: Phase diagram of the cluster-Ising ring (9.1). (a) Exact string order parameter S_{12} of the 12-site ring, with the reference boundary $S_{12} = 1/2$ (white) and the maxima of the fidelity susceptibility (diamonds). (b) Output p of the trained QCNN for the 8-qubit ground states, with the learned boundary $p = 1/2$ (dashed), the reference boundary (white) and the 40 training points (white for SPT, black otherwise). Exact state-vector simulation.

On the grid of 528 points the trained network classifies 97% of the 8-qubit states correctly, and the four starts give between 95% and 98% (Figure 49). The rule $S_8 > 1/2$, which evaluates the string order parameter on the same 8-qubit states and thus uses the knowledge of the right observable, reaches 96%. The network therefore classifies as well as the string order parameter, from 40 labelled

examples and without being told which observable to measure. At $h_1 = 0.5$ its learned boundary lies at $h_2 = 0.39$, against the reference 0.44 of the 12-site ring and 0.423 of the infinite chain. Its errors lie next to the phase boundary, where the 8-qubit states change gradually with the couplings and even the exact 8-qubit string order disagrees with the 12-site reference.

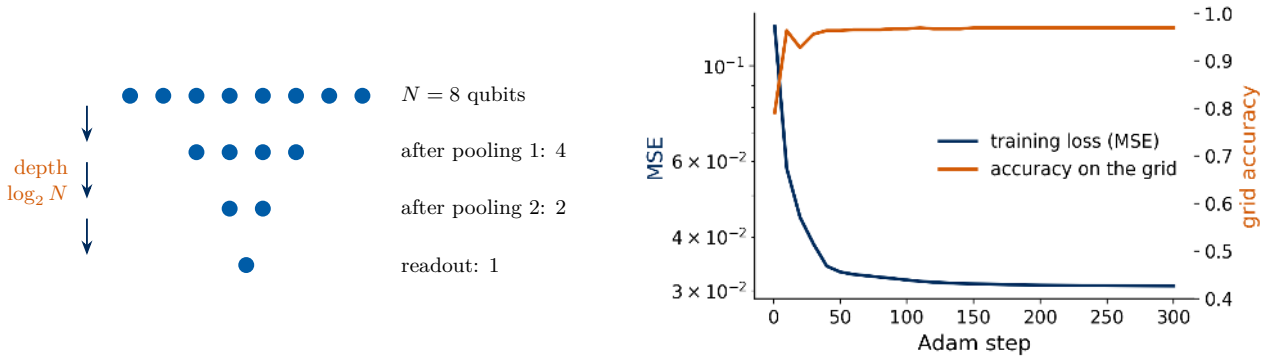


Figure 50: Trainability of the QCNN. Left, every unit halves the register, so the circuit has $\log_2 N$ short layers. Right, the training loss (9.7) on a logarithmic scale and accuracy on the grid against the Adam step for the kept start. Exact state-vector simulation.

The loss levels off within about fifty steps (Figure 50). The circuit trains easily because its depth grows as $\log_2 N$ and its output is a single-qubit measurement. For this architecture the variance of the gradient decreases at most polynomially with N [45], so the QCNN avoids the vanishing gradients of deep unstructured circuits. Ref. [42] simulated a QCNN of this kind for 135 spins and found that near the transition it decides the phase from fewer copies of the input state than a measurement of the string order parameter.

10 Learning a Hamiltonian with a quantum graph recurrent neural network

10.1 Learning from quantum dynamics

A circuit can also be trained to reproduce the dynamics of a physical system. A quantum device or a material evolves under a Hamiltonian $\hat{\mathcal{H}}$ that we do not know, including which qubits interact and how strongly. We can prepare a state, let the system evolve for a time t and keep the result, and these evolved states are quantum data. The quantum graph recurrent neural network (QGRNN) of Ref. [46] is a circuit that evolves states with a trainable Hamiltonian, tuned until it reproduces the data, so that the trained parameters are the learned Hamiltonian.

In the learning loop of Figure 51, the initial state $|\psi_0\rangle$ is a state we know how to prepare, so we can make as many identical copies as we need. One copy goes to the unknown system, which returns the data state. Another copy enters a circuit that evolves it with a trainable Hamiltonian and returns the model state. A SWAP test measures how close the two outputs are, and a classical optimiser adjusts the parameters to bring them together. When the two states agree at many times, the trained parameters are the hidden Hamiltonian.

10.2 A Hamiltonian on a graph

We consider a device of $N = 4$ qubits whose Hamiltonian is an Ising model with a transverse field on a graph with edge set E,

$$\hat{\mathcal{H}} = \underbrace{\sum_{(i,j)\in E} J_{ij}\,\hat{Z}_i\hat{Z}_j}_{\hat{\mathcal{H}}_E} + \underbrace{\sum_{i=0}^{N-1} (h_i\hat{Z}_i + g\,\hat{X}_i)}_{\hat{\mathcal{H}}_V}, \tag{10.1}$$

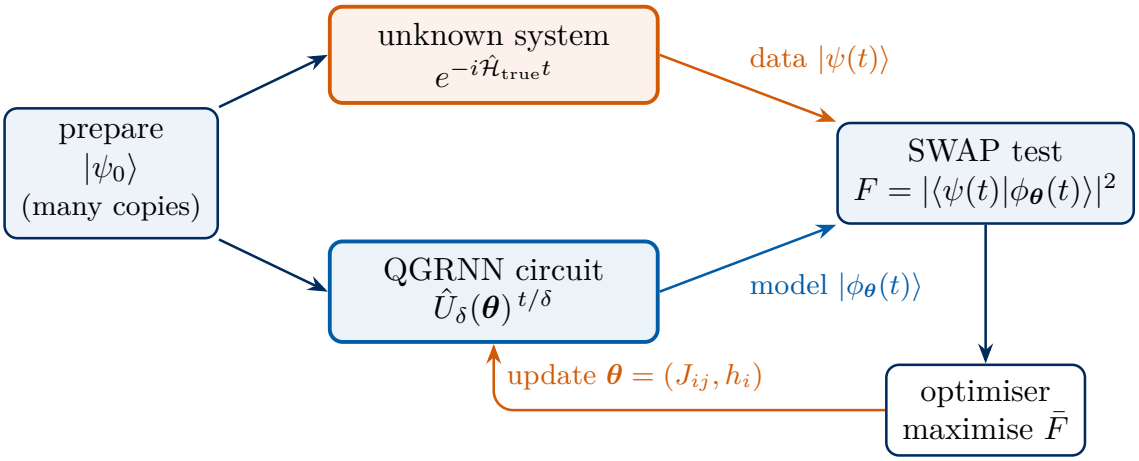


Figure 51: Learning loop of the QGRNN. Copies of the same initial state $|\psi_0\rangle$ go to the unknown system and to the trainable circuit, a SWAP test measures the fidelity of the two outputs, and the optimiser updates the couplings and fields.

with a coupling J_{ij} on every edge, a longitudinal field h_i on every qubit and a transverse field g. Every qubit is a node of the graph and every pair of interacting qubits is an edge (Figure 52), and a pair without an edge has $J_{ij} = 0$. The edge part $\hat{\mathcal{H}}_E$ and the node part $\hat{\mathcal{H}}_V$ do not commute. The couplings, the fields and the edge set are hidden, and only $g = 1$ is known. In our simulation the hidden graph is the ring 0–1–2–3–0 with couplings between -1.1 and 1.2 and fields between -0.6 and 0.7. Learning the couplings of all six pairs of qubits therefore also learns the graph.

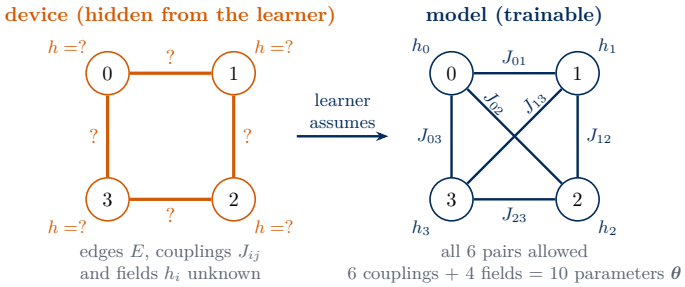


Figure 52: The unknown Hamiltonian (10.1) lives on a graph. Every node is a qubit with a field h_i, and every edge is a coupling $J_{ij} \neq 0$. The hidden graph is a ring of four qubits, and the model allows a coupling on all six pairs.

10.3 Quantum data

The data consist of pairs of an initial state $|\psi_0\rangle$ and the data state

$$|\psi(t)\rangle = e^{-i\hat{\mathcal{H}}t}|\psi_0\rangle, \tag{10.2}$$

what the unknown system makes of $|\psi_0\rangle$ after a time t. An eigenstate of $\hat{\mathcal{H}}$ would be useless as an initial state, because it only acquires a global phase, and every Hamiltonian that shares this eigenvector reproduces its evolution. We therefore model an imperfect preparation of a low-energy state, the exact ground state followed by small random rotations $\hat{R}_y(\varepsilon)$, $\varepsilon \in [0.2, 0.6]$, of every qubit, which leaves between 77 and 87 per cent of the weight in the ground state. Three such initial states, each evolved to five times $t_k \leq 1.5$, give 15 data pairs (Figure 53).

10.4 One layer is one Trotter step

The model has to evolve a state under a Hamiltonian of the form (10.1) with trainable parameters. The exact evolution cannot be split into the evolutions under $\hat{\mathcal{H}}_E$ and $\hat{\mathcal{H}}_V$, because the two parts do

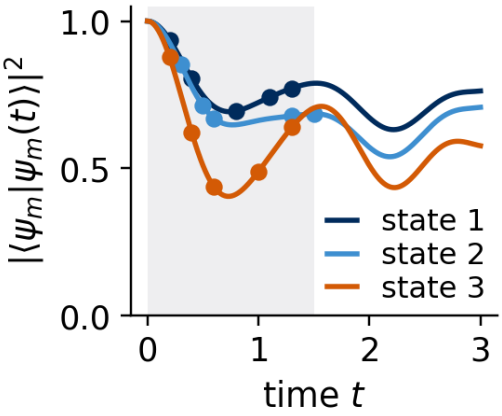


Figure 53: Quantum data of the QGRNN. Overlap $|\langle\psi_m|\psi_m(t)\rangle|^2$ of each of the three initial states $|\psi_m\rangle$, with its evolved state (10.2) against the time t, with dots at the five data times. An exact eigenstate would stay at one. Exact state-vector simulation.

not commute. Over a short time δ the splitting error is small, and repeating the short step approaches the exact evolution [33],

$$e^{-i\hat{\mathcal{H}}t} = \lim_{P\to\infty} \left(e^{-i\delta\hat{\mathcal{H}}_E}\, e^{-i\delta\hat{\mathcal{H}}_V}\right)^P, \qquad \delta = t/P. \tag{10.3}$$

Each factor is a layer of gates. The node factor is a product of single-qubit rotations $e^{-i\delta(h_i\hat{Z}_i+g\hat{X}_i)}$, and the edge factor is a product of two-qubit gates $e^{-i\delta J_{ij}\hat{Z}_i\hat{Z}_j}$, which multiply every computational basis state by a phase. With parameters $\boldsymbol{\theta} = \{J_{ij}, h_i\}$ the QGRNN applies the layer $\hat{U}_\delta(\boldsymbol{\theta}) = e^{-i\delta\hat{\mathcal{H}}_E(\boldsymbol{\theta})}e^{-i\delta\hat{\mathcal{H}}_V(\boldsymbol{\theta})}$ to the initial state $P = t/\delta$ times (Figure 54) and outputs the model state

$$|\phi_{\boldsymbol{\theta}}(t)\rangle = \hat{U}_\delta(\boldsymbol{\theta})^{t/\delta}|\psi_0\rangle. \tag{10.4}$$

The model graph is the complete graph, so $\hat{\mathcal{H}}_E(\boldsymbol{\theta})$ contains a coupling for all six pairs and $\boldsymbol{\theta}$ has ten components. The same parameters enter every layer, as the same weights enter every time step of a recurrent neural network. In the name, *graph* refers to the pattern of interacting qubits, *recurrent* to the identical layer applied again and again, and *neural network* to the trainable parameters. With the hidden parameters and $\delta = 0.05$ the model differs from the exact evolution of the ground state by an infidelity below 4×10^{-3} for $t \le 1.5$.

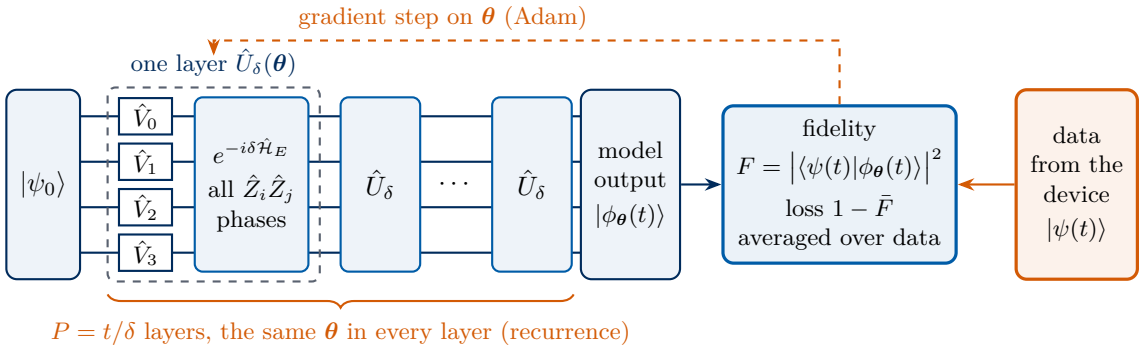


Figure 54: The QGRNN circuit. One layer $\hat{U}_\delta(\boldsymbol{\theta})$ is a Trotter step, single-qubit rotations $\hat{V}_i$ under the node terms followed by the $\hat{Z}_i\hat{Z}_j$ phases of all edges, and the same layer is repeated t/δ times with the same trainable couplings and fields.

10.5 The SWAP test

The model state is compared with the data state through the fidelity $F = |\langle\psi(t)|\phi_{\boldsymbol{\theta}}(t)\rangle|^2$, which equals one when the two states coincide up to a global phase. A device does not output amplitudes, so it

measures the fidelity with the SWAP test [47] (Figure 55). For two states $|a\rangle$ and $|b\rangle$ held in registers of equal size, an ancilla qubit in $|0\rangle$ passes a Hadamard gate $\hat{H}$, controls a swap of the two registers and passes a second Hadamard gate before it is measured. The state of ancilla and registers evolves as

$$\begin{aligned} |0\rangle|a\rangle|b\rangle &\xrightarrow{\hat{H}} \tfrac{1}{\sqrt{2}}\big(|0\rangle+|1\rangle\big)|a\rangle|b\rangle \xrightarrow{\text{controlled SWAP}} \tfrac{1}{\sqrt{2}}\big(|0\rangle|a\rangle|b\rangle+|1\rangle|b\rangle|a\rangle\big) \\ &\xrightarrow{\hat{H}} \tfrac{1}{2}|0\rangle\big(|a\rangle|b\rangle+|b\rangle|a\rangle\big)+\tfrac{1}{2}|1\rangle\big(|a\rangle|b\rangle-|b\rangle|a\rangle\big). \end{aligned} \tag{10.5}$$

The probability of reading 0 is the squared norm of the first branch. With normalised states the two norm terms give one each, and the two cross terms give $\langle a|b\rangle\langle b|a\rangle = |\langle a|b\rangle|^2$ each, so

$$P(0) = \tfrac{1}{4}\big\||a\rangle|b\rangle+|b\rangle|a\rangle\big\|^2 = \tfrac{1}{4}(2+2|\langle a|b\rangle|^2) = \tfrac{1}{2}(1+F), \qquad F = |\langle a|b\rangle|^2. \tag{10.6}$$

Identical states always return 0, and orthogonal states return 0 and 1 with equal probability. From S repetitions with n_0 zeros the estimate is $\tilde{F} = 2n_0/S - 1$, with a statistical error of order $S^{-1/2}$. The test uses only fresh copies of the two states and never their amplitudes, which suits quantum data.

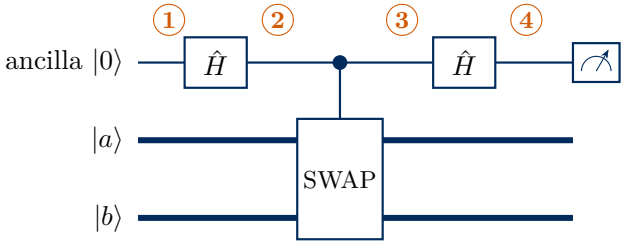


Figure 55: The SWAP test. An ancilla in $|0\rangle$ passes a Hadamard gate, controls the swap of the registers holding $|a\rangle$ and $|b\rangle$ and passes a second Hadamard gate. The circled numbers mark the four states of (10.5). The probability of reading 0 is $(1+|\langle a|b\rangle|^2)/2$ (10.6).

10.6 Training and identifiability

In a training step one register holds the data state $|\psi(t_k)\rangle$ and the other the model state $|\phi_{\boldsymbol{\theta}}(t_k)\rangle$, and the SWAP test returns their fidelity (Figure 56). The loss averages the fidelities over the data pairs, labelled by the initial state m and the time t_{mk},

$$\mathcal{L}(\boldsymbol{\theta}) = 1 - \frac{1}{15}\sum_{m=1}^{3}\sum_{k=1}^{5}\big|\langle\psi_m(t_{mk})|\phi_{m,\boldsymbol{\theta}}(t_{mk})\rangle\big|^2, \tag{10.7}$$

where $|\psi_m(t)\rangle$ and $|\phi_{m,\boldsymbol{\theta}}(t)\rangle$ are the data and model states of initial state m. In our simulation we evaluate (10.7) exactly from the state vectors. The circuit starts with a coupling on every pair of qubits and random values, because the graph of the device is not known in advance.

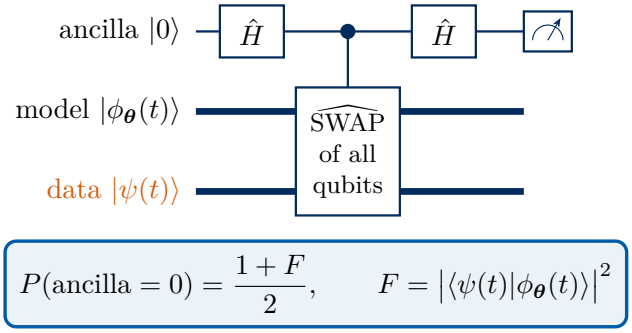


Figure 56: One training step. The data state from the unknown system and the model state from the QGRNN enter the two registers of a SWAP test, whose ancilla statistics give their fidelity.

Agreement of the states identifies the Hamiltonian because of the short-time expansion

$$e^{-i\hat{\mathcal{H}}t}|\psi_0\rangle = |\psi_0\rangle - it\,\hat{\mathcal{H}}|\psi_0\rangle - \frac{t^2}{2}\,\hat{\mathcal{H}}^2|\psi_0\rangle + \dots \tag{10.8}$$

Agreement at many times fixes the vectors $\hat{\mathcal{H}}|\psi_0\rangle$, $\hat{\mathcal{H}}^2|\psi_0\rangle$ and the higher powers, and several initial states fix the action of $\hat{\mathcal{H}}$ on enough of the state space to determine its ten parameters. A single data pair is reproduced almost perfectly, with a loss of 7.5×10^{-4}, yet the learned parameters differ from the hidden ones by up to 0.18, because one fidelity cannot fix ten numbers. With all fifteen pairs the largest error drops to 0.05.

10.7 Learned couplings and graph

We start from couplings and fields drawn uniformly from $[-0.5, 0.5]$, a dense random graph, and train with Adam [28] for 500 steps. The loss falls from 0.49 to close to its final value $1.7e-03$ within about one hundred steps (Figure 58, left). Every learned coupling and field lies within 0.05 of its hidden value (Figure 57). The two pairs without an edge end with couplings of about 0.01, whereas the four edges keep couplings of at least 0.58 in magnitude, so the learned graph is the hidden ring.

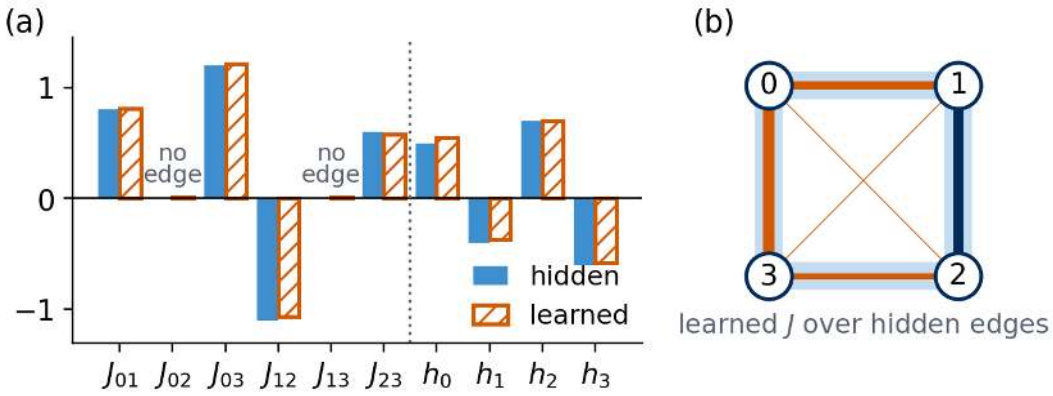


Figure 57: QGRNN trained on fifteen data pairs of a four-qubit Ising Hamiltonian on a ring, with all six pairs allowed in the model. (a) Hidden (filled) and learned (hatched) couplings J_{ij} and fields h_i. (b) Learned couplings drawn over the hidden edges (light blue), with line width proportional to $|J_{ij}|$, orange for $J_{ij} > 0$ and navy for $J_{ij} < 0$. The thin diagonal lines are the couplings learned for the two pairs without an edge. Exact state-vector simulation, Trotter step $\delta = 0.05$.

The final loss lies below the loss of the hidden parameters, $2.3e-03$. The data follow the exact evolution, whereas the model is a Trotterised evolution, so the best fit shifts the parameters slightly to compensate part of the Trotter error, and this shift is the remaining parameter error. Repeating the training with smaller steps shows the loss of the hidden parameters falling as δ^2 and the largest parameter error falling roughly in proportion to δ, from 0.12 at $\delta = 0.1$ to 0.011 at $\delta = 0.0125$ (Figure 58, right), at the cost of t/δ layers and therefore proportionally deeper circuits.

Recovering a Hamiltonian from the dynamics it generates is the task of Hamiltonian learning [48]. Ref. [46] also applies quantum graph networks to the preparation of entangled states for networks of quantum sensors, to the clustering of graphs and to the classification of graph isomorphism.

11 Compressing circuits with a quantum circuit autoencoder

11.1 Circuits as quantum data

The autoencoder of Section 8.1 compresses states, but quantum information is also stored in circuits, for example in a set of circuits that load classical data into a register, or in a device whose action is known only as a black box. The quantum circuit autoencoder of Ref. [49] compresses such a circuit

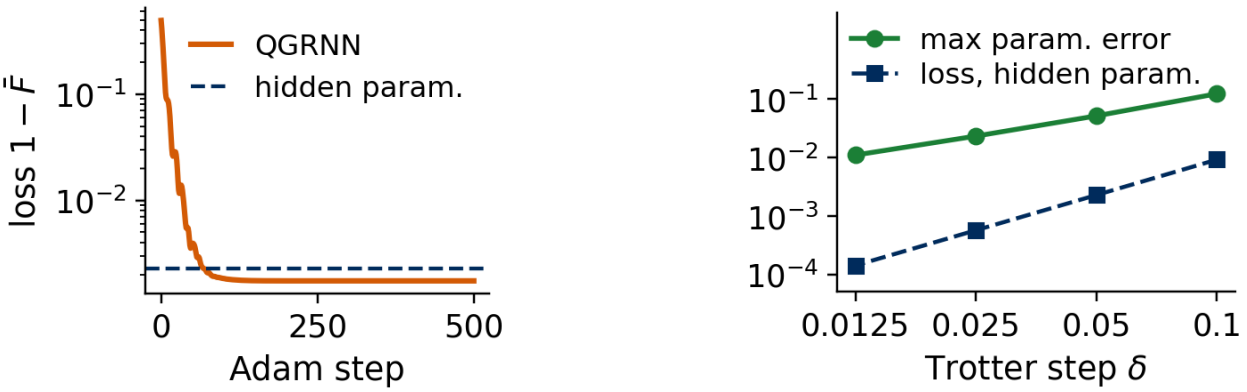


Figure 58: Left, the loss (10.7) of the QGRNN against the Adam step for $\delta = 0.05$. Right, the largest parameter error after training and loss of the hidden parameters against the Trotter step δ. Exact state-vector simulation.

into fewer qubits. A circuit is a map from input states to output states, and to compare or compress maps we first turn them into states.

Circuit compression task

We can run an unknown channel $\mathcal{E}$ on N qubits as a black box. We look for two encoder circuits $\hat{U}$ and $\hat{V}$ and a channel $\mathcal{F}$ on $N-k$ qubits that, wrapped in the encoders, rebuild $\mathcal{E}$. Success is measured by the Choi fidelity of the rebuilt channel with $\mathcal{E}$, defined below.

A circuit of gates acts on a density matrix as $\hat{\rho} \mapsto \hat{W}\hat{\rho}\hat{W}^\dagger$ with a unitary $\hat{W}$. Noise, or a circuit drawn at random from a list, produces a more general *quantum channel*, a linear map that can be written in the Kraus form

$$\mathcal{E}(\hat{\rho}) = \sum_a \hat{K}_a \hat{\rho} \hat{K}_a^\dagger, \qquad \sum_a \hat{K}_a^\dagger \hat{K}_a = \hat{I}. \tag{11.1}$$

A unitary circuit has the single Kraus operator $\hat{W}$, and a circuit $\hat{W}_j$ applied with probability p_j has the Kraus operators $\sqrt{p_j}\hat{W}_j$. We use four one-qubit channels as examples. The identity leaves $\hat{\rho}$ unchanged and the Hadamard gate gives $\hat{H}\hat{\rho}\hat{H}$. Dephasing gives $(1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z}$ and shrinks the off-diagonal elements by $1-2p$. Complete depolarisation returns $\hat{I}/2$ for every input and has the Kraus operators $\hat{I}/2$, $\hat{X}/2$, $\hat{Y}/2$, $\hat{Z}/2$.

11.2 The Choi–Jamiołkowski isomorphism

To capture the action of $\mathcal{E}$ on all inputs at once, we let it act on one half of a maximally entangled state. A second register of the same dimension $D = 2^N$, the reference R, is prepared together with the system S in

$$|\Phi^+\rangle = \frac{1}{\sqrt{D}} \sum_{i=0}^{D-1} |i\rangle_S \otimes |i\rangle_R, \tag{11.2}$$

which is a product of N Bell pairs $(|00\rangle + |11\rangle)/\sqrt{2}$, one for each qubit of the system and its partner in the reference, because the sum over bit strings factorises qubit by qubit. Each Bell pair is prepared from $|00\rangle$ by a Hadamard gate on the system qubit followed by a $\widehat{\text{CNOT}}$ gate with the system qubit as control and its reference partner as target, since $|00\rangle \to (|00\rangle + |10\rangle)/\sqrt{2} \to (|00\rangle + |11\rangle)/\sqrt{2}$. The channel acts on the system while the reference is left untouched (Figure 59(a)), and the result is the *Choi state* [50,51]

$$\hat{J}_\mathcal{E} = (\mathcal{E} \otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{D} \sum_{i,j=0}^{D-1} \mathcal{E}(|i\rangle\langle j|) \otimes |i\rangle\langle j|. \tag{11.3}$$

The second form follows from inserting (11.2) and the linearity of $\mathcal{E}$. It lists the images of all matrix units $|i\rangle\langle j|$, which determine $\mathcal{E}$ on every input. The state $\hat{J}_{\mathcal{E}}$ is a density matrix of $2N$ qubits, so its preparation needs twice as many qubits as the circuit. Tracing out the system leaves $\mathrm{Tr}_S \hat{J}_{\mathcal{E}} = \hat{I}/D$, since a channel acting on the system cannot change the state of the untouched reference.

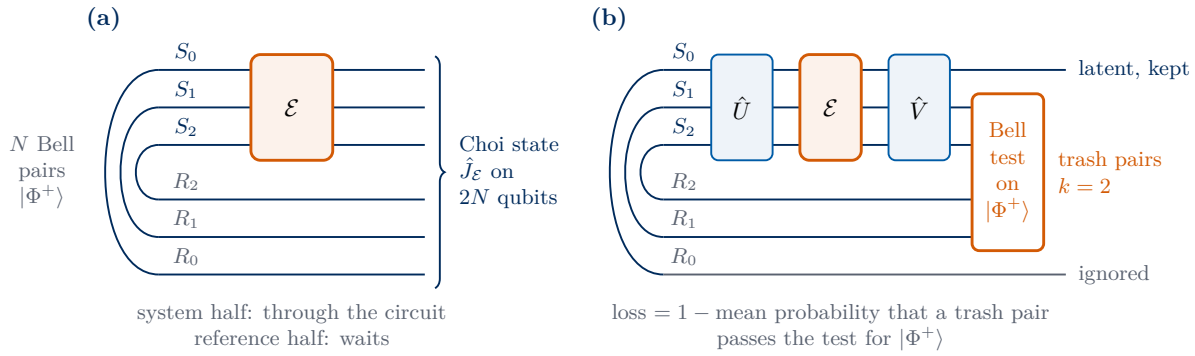


Figure 59: (a) Choi state of a channel $\mathcal{E}$ on $N = 3$ qubits. Each system qubit S_q starts in a Bell pair with a reference qubit R_q, the system half passes through $\mathcal{E}$ and the reference half waits. The reference wires are drawn in reverse order so that the pairs nest. (b) Register for training the circuit autoencoder. The encoders $\hat{U}$ and $\hat{V}$ act before and after $\mathcal{E}$, the latent pair (S_0, R_0) is not measured, and the $k = 2$ trash pairs are tested for the Bell state.

The channel can be read back from its Choi state. Multiplying the reference by the transpose of an input $\hat{\rho}$ and tracing it out gives, for each term of (11.3), the factor $\mathrm{Tr}\big[\hat{\rho}^T|i\rangle\langle j|\big] = \langle j|\hat{\rho}^T|i\rangle = \rho_{ij}$, so

$$D\,\mathrm{Tr}_R\big[(\hat{I}\otimes\hat{\rho}^T)\,\hat{J}_{\mathcal{E}}\big] = \sum_{i,j}\rho_{ij}\,\mathcal{E}\big(|i\rangle\langle j|\big) = \mathcal{E}(\hat{\rho}). \tag{11.4}$$

The map $\mathcal{E} \mapsto \hat{J}_{\mathcal{E}}$ is therefore invertible, and every statement about a channel can be phrased as a statement about a state.

Inserting (11.1) into (11.3) gives $\hat{J}_{\mathcal{E}} = \sum_a |K_a\rangle\!\rangle\langle\!\langle K_a|$ with the unnormalised vectors $|K_a\rangle\!\rangle = (\hat{K}_a \otimes \hat{I})|\Phi^+\rangle$. The rank of $\hat{J}_{\mathcal{E}}$ is therefore the smallest number of Kraus operators that represent $\mathcal{E}$, the *Kraus rank*. Table 5 lists the four examples. The identity gives the Bell state itself and the Hadamard gate another pure state, both of rank one, as for every unitary circuit. Dephasing mixes two pure states with weights $1-p$ and p, and complete depolarisation gives the maximally mixed state $\hat{I}/4$ of rank four. A channel that destroys more information therefore has a more mixed Choi state.

channel	$\mathcal{E}(\hat{\rho})$	eigenvalues of $\hat{J}_{\mathcal{E}}$	rank
identity	$\hat{\rho}$	$1, 0, 0, 0$	1
Hadamard	$\hat{H}\hat{\rho}\hat{H}$	$1, 0, 0, 0$	1
dephasing	$(1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z}$	$1-p,\ p,\ 0,\ 0$	2
complete depolarisation	$\hat{I}/2$	$\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$	4

Table 5: Choi states of four one-qubit channels. The rank equals the smallest number of Kraus operators.

Two channels are compared through the fidelity of their Choi states, $F(\hat{J}_{\mathcal{E}_1}, \hat{J}_{\mathcal{E}_2})$, with $F(\hat{\rho},\hat{\sigma}) = \big(\mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\,\hat{\sigma}\sqrt{\hat{\rho}}}\big)^2$. For two unitary circuits it equals $|\mathrm{Tr}(\hat{W}_1^\dagger \hat{W}_2)|^2/D^2$, and for the identity and dephasing it equals $\langle\Phi^+|\hat{J}_{\mathrm{deph}}|\Phi^+\rangle = 1-p$. The Choi fidelity measures an average over inputs, since for a channel compared with a unitary the output fidelity averaged over all pure inputs equals $(DF+1)/(D+1)$ [52,53]. A Choi fidelity close to one therefore guarantees good outputs on average, although individual inputs can have a lower fidelity.

11.3 Encoders and the lossless condition

As for states, we call the last k qubits trash and the first $N-k$ latent. The encoders are a unitary $\hat{U}$ applied before $\mathcal{E}$ and a unitary $\hat{V}$ applied after it, giving the encoded channel $\Lambda = \hat{V} \circ \mathcal{E} \circ \hat{U}$, read from right to left. The compressed channel on the latent qubits feeds the trash qubits with the maximally mixed state and discards them,

$$\mathcal{F}(\hat{\sigma}) = \mathrm{Tr}_{\mathrm{trash}}\big[\Lambda(\hat{\sigma} \otimes \hat{I}/2^k)\big], \tag{11.5}$$

and its Choi state is $\hat{J}_\Lambda$ with the trash qubits and their reference partners traced out. Decoding places $\mathcal{F}$ next to an untouched trash register and undoes the encoders, $\tilde{\mathcal{E}} = \hat{V}^\dagger \circ (\mathcal{F} \otimes \mathrm{id}_{\mathrm{trash}}) \circ \hat{U}^\dagger$. The reconstruction is exact if and only if $\Lambda = \mathcal{F} \otimes \mathrm{id}_{\mathrm{trash}}$, the lossless condition of Ref. [49], since then $\tilde{\mathcal{E}} = \hat{V}^\dagger \circ \Lambda \circ \hat{U}^\dagger = \mathcal{E}$. The identity $(\hat{U} \otimes \hat{I})|\Phi^+\rangle = (\hat{I} \otimes \hat{U}^T)|\Phi^+\rangle$, checked by comparing amplitudes, gives

$$\hat{J}_{\hat{V} \circ \mathcal{E} \circ \hat{U}} = (\hat{V} \otimes \hat{U}^T)\, \hat{J}_{\mathcal{E}}\, (\hat{V} \otimes \hat{U}^T)^\dagger, \tag{11.6}$$

so $\hat{V}$ rotates the output half and $\hat{U}^T$ the input half, and no choice of encoders changes the eigenvalues of the Choi state.

A single unitary circuit $\hat{W}$ is compressed trivially by $\hat{U} = \hat{I}$ and $\hat{V} = \hat{W}^\dagger$, which give $\Lambda = \mathrm{id}$, in the same way as a single state can always be compressed by its own preparation circuit (Section 8.4). Compression becomes a learning task when a family $\hat{W}_1, \ldots, \hat{W}_{n_{\mathrm{f}}}$ must share one pair of encoders. The family is represented by the channel $\mathcal{E}(\hat{\rho}) = n_{\mathrm{f}}^{-1} \sum_j \hat{W}_j \hat{\rho} \hat{W}_j^\dagger$ that applies one member at random, and its Choi state is the average of the pure Choi states of the members, the analogue of the average density matrix (8.13).

11.4 A bound from the Choi spectrum

By (11.6), $\hat{J}_{\tilde{\mathcal{E}}}$ is a unitary rotation of $\hat{J}_{\mathcal{F}} \otimes |\Phi^+\rangle\langle\Phi^+|_{\mathrm{trash}}$. The first factor is a density matrix of dimension d^2 with $d = 2^{N-k}$, and the second is pure, so $\hat{J}_{\tilde{\mathcal{E}}}$ has rank at most d^2. The fidelity of a state $\hat{\rho}$ with any state $\hat{\sigma}$ of rank r obeys $F(\hat{\rho}, \hat{\sigma}) \leq \mathrm{Tr}(\hat{P}_\sigma \hat{\rho})$ with $\hat{P}_\sigma$ the projector onto the range of $\hat{\sigma}$. By the maximum principle (8.14), this trace is at most the sum of the r largest eigenvalues of $\hat{\rho}$. Hence

$$F(\hat{J}_{\mathcal{E}}, \hat{J}_{\tilde{\mathcal{E}}}) \leq \sum_{i=1}^{d^2} \lambda_i, \tag{11.7}$$

with $\lambda_1 \geq \lambda_2 \geq \ldots$ the eigenvalues of $\hat{J}_{\mathcal{E}}$ (Proposition 3 of Ref. [49]). A channel is compressible into $N-k$ qubits only if its Choi state carries nearly all its weight on 4^{N-k} eigenvectors. Complete depolarisation of N qubits has $\hat{J}_{\mathcal{E}} = \hat{I}/4^N$ and the bound 4^{-k}, so it cannot be compressed at all. The bound limits every pair of encoders, and a given encoder circuit need not reach it, as for the bound (8.14) on families of states.

11.5 Training on the trash qubits

Evaluating the Choi fidelity requires both $2N$-qubit Choi states side by side, whereas the variational algorithm of Ref. [49] measures only the trash. The register of Figure 59(b) is prepared in the Choi state $\hat{J}_\Lambda$ of the encoded channel. The latent pair is left unmeasured, so the latent input is the maximally mixed state of (11.5). Each trash qubit c is tested together with its reference partner $\tilde{c}$ for the Bell state by a controlled-NOT gate, a Hadamard gate and a measurement with outcome 00. The loss is

$$\mathcal{L} = 1 - \frac{1}{k} \sum_{c \in \mathrm{trash}} \langle\Phi^+|\hat{\rho}_{c\tilde{c}}|\Phi^+\rangle, \tag{11.8}$$

a sum of two-qubit terms and hence a local cost in the sense of Ref. [38]. It vanishes if and only if the encoded channel is lossless. If every trash pair leaves as a perfect Bell pair, the trash qubits have passed

through Λ unchanged. A qubit maximally entangled with its partner carries no correlations with any third system, so no information has moved between latent and trash qubits. Then $\Lambda = \mathcal{F} \otimes \mathrm{id}_{\mathrm{trash}}$ (Proposition 1 of Ref. [49]).

11.6 Example

We compress a family of $n_{\mathrm{f}} = 6$ circuits on $N = 3$ qubits, $\hat{W}_j = \hat{\mathcal{S}}(\hat{R}_j \otimes \hat{I} \otimes \hat{I})\hat{\mathcal{S}}^{\dagger}$, where each $\hat{R}_j$ is a random single-qubit unitary and $\hat{\mathcal{S}}$ is a fixed hardware-efficient circuit (6.3) with two layers and random angles. Every member looks like a generic entangling circuit, while in the basis rotated by $\hat{\mathcal{S}}$ all members act on the first qubit alone. The Choi state of the family therefore has rank four, and the bound (11.7) for one latent qubit, $k = 2$ and $d^2 = 4$, equals one (Figure 60). The encoders $\hat{U} = \hat{\mathcal{S}}$, $\hat{V} = \hat{\mathcal{S}}^{\dagger}$ reconstruct the family exactly, whereas the training has no access to $\hat{\mathcal{S}}$. We train hardware-efficient encoders $\hat{U}(\boldsymbol{\theta})$ and $\hat{V}(\boldsymbol{\varphi}) = \hat{U}(\boldsymbol{\varphi})^{\dagger}$ with two layers and 18 angles each on the loss (11.8), from 16 random starts with 400 steps of Adam [28]. The encoders have the same form as $\hat{\mathcal{S}}$, so lossless encoders exist inside the ansatz, which is not guaranteed for an unknown channel.

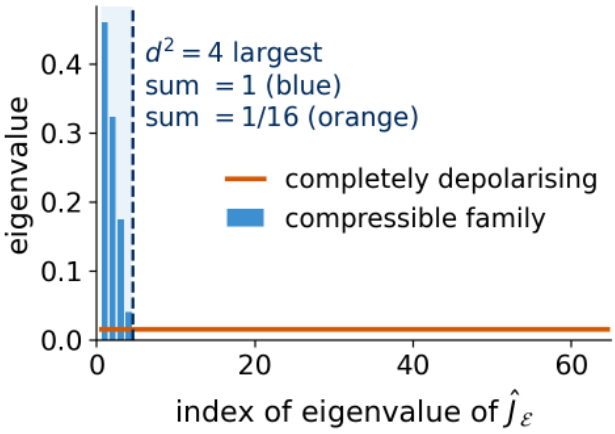


Figure 60: Eigenvalues of the Choi state $\hat{J}_{\mathcal{E}}$ of a three-qubit channel, sorted, for the compressible family of six circuits (bars) and for complete depolarisation (line). The shaded window holds the $d^2 = 4$ largest eigenvalues allowed by one latent qubit. Their sum, the bound (11.7), is 1 for the family and 1/16 for depolarisation. Exact calculation.

Seven of the 16 starts reach a loss below 10^{-3} and a Choi fidelity above 0.999 (Figure 61). The best start ends at a loss of 1.5×10^{-5} and a Choi fidelity of $1 - 1.8 \times 10^{-5}$, close to the bound. Its Choi fidelity, never used in training, rises together with the fall of the trash loss, as the lossless condition predicts. The other nine starts stop on plateaus of the loss near 0.19 and 0.21 with Choi fidelities between 0.34 and 0.46, so the trash loss alone identifies the starts to keep. For complete depolarisation every trash pair leaves in the state $\hat{I}/4$ whatever the encoders, the loss stays at 3/4 with a vanishing gradient, and the Choi fidelity stays at its bound 1/16.

Our example is an exact simulation of three qubits, whereas on a device each pass probability in (11.8) is estimated from a finite number of Bell tests and the preparation of the Choi state doubles the number of qubits. The Choi fidelity certifies the reconstruction only on average over inputs. Ref. [49] also uses trained circuit autoencoders to flag anomalous circuits by their reconstruction fidelity and to reduce depolarising noise.

12 Circuits as probability models

12.1 Generative modelling

Up to this point a circuit prepared a state whose energy was of interest. Measured in the computational basis, the same circuit produces random bit strings whose distribution can itself be the object of

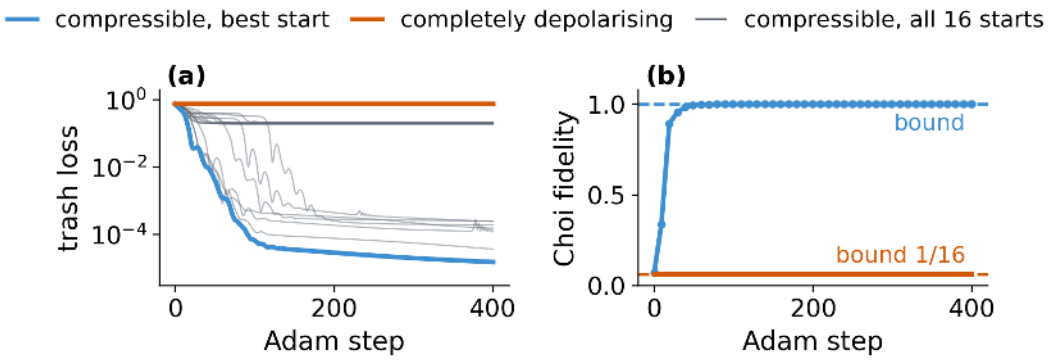


Figure 61: Training of the circuit autoencoder for $N = 3$, $k = 2$. (a) Trash loss (11.8) against the Adam step for 16 random starts on the compressible family (grey, best start blue) and for complete depolarisation (orange). (b) Choi fidelity of the reconstruction for the best start and for depolarisation, with the bounds (11.7) dashed. Exact simulation of density matrices.

interest, which turns the circuit into a machine-learning model.

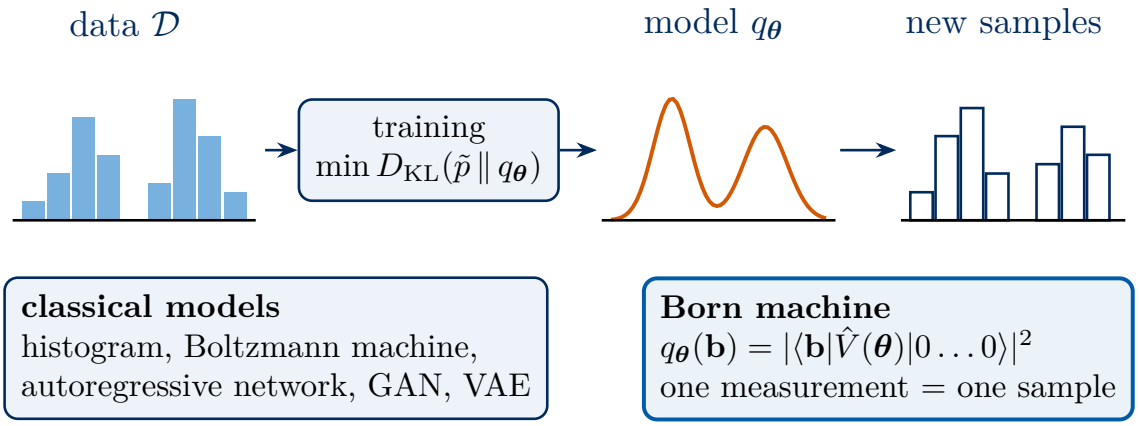


Figure 62: Generative modelling. Samples from an unknown distribution p form the data set $\mathcal{D}$, a model $q_{\boldsymbol{\theta}}$ is trained on them, and the trained model produces new samples that resemble the data.

We consider an unknown distribution p over a space of outcomes and a data set

$$\mathcal{D} = \{x^{(1)}, x^{(2)}, \ldots, x^{(|\mathcal{D}|)}\} \tag{12.1}$$

of $|\mathcal{D}|$ independent samples drawn from it. The learner sees the samples and never the distribution p itself. A *generative model* is a family of distributions $q_{\boldsymbol{\theta}}$ labelled by parameters $\boldsymbol{\theta}$, from which new samples can be drawn (Figure 62). Training adjusts $\boldsymbol{\theta}$ so that $q_{\boldsymbol{\theta}}$ comes close to p in a measure of distance between distributions, and a trained model produces new samples that resemble the data without copying them. Among the classical families, a histogram of the data can be sampled by picking a bin with the stored frequency. A Boltzmann machine assigns the probability $e^{-E_{\boldsymbol{\theta}}(x)}/Z$ through an energy function with trainable couplings and a normalisation Z. An autoregressive network writes the probability of a vector as a product of conditional probabilities, one component after another, and generative adversarial networks and variational autoencoders transform random noise into samples with a neural network. The quantum model of this section addresses the same task and differs in how the probabilities arise.

12.2 A wave function as a probability distribution

A state of N qubits is a superposition (2.1) of all 2^N bit strings with amplitudes $\psi_{\mathbf{b}}$. By the Born rule a measurement of every qubit in the computational basis returns the bit string $\mathbf{b}$ with probability

$$q(\mathbf{b}) = |\psi_{\mathbf{b}}|^2 \geq 0, \qquad \sum_{\mathbf{b}} q(\mathbf{b}) = \langle\psi|\psi\rangle = 1. \tag{12.2}$$

The squared amplitudes therefore form a probability distribution over 2^N outcomes, and its normalisation is guaranteed by the normalisation of the state. Drawing a sample from this distribution requires no computation, since one preparation of the state followed by a measurement of all qubits returns one bit string with these probabilities. A circuit with angles $\boldsymbol{\theta}$ prepares a family of states and with it a family of distributions $q_{\boldsymbol{\theta}}$, and changing the angles moves probability between bit strings. A product state, prepared without entangling gates, gives independent bits, and entangling gates correlate them. In the variational quantum eigensolver the bit strings served only to estimate an energy. In this section their distribution is itself the object that is trained.

12.3 Data as bit strings

A register of N qubits returns bit strings $\mathbf{b} = b_0 b_1 \dots b_{N-1}$, so the data must be written as bit strings before a circuit can model them. Discrete data such as spin configurations or binary images are bit strings already. A continuous variable $x \in [x_{\min}, x_{\max}]$ is first discretised by cutting the interval into 2^N bins of equal width

$$w = \frac{x_{\max} - x_{\min}}{2^N} \tag{12.3}$$

and number them from left to right by the bin index

$$k(x) = \Big\lfloor \frac{x - x_{\min}}{w} \Big\rfloor \in \{0, 1, \dots, 2^N - 1\}, \tag{12.4}$$

where $\lfloor \cdot \rfloor$ rounds down to an integer and the right end $x = x_{\max}$ is assigned to the last bin. The index written in binary,

$$k = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q}, \tag{12.5}$$

is the bit string $\mathbf{b}$, with b_0 the most significant bit, the same ordering as the amplitudes of the register. Every value of x in bin k therefore corresponds to the computational basis state $|\mathbf{b}\rangle$, and every measured bit string corresponds back to a bin. When a value of x is needed, we take the centre of the bin,

$$x_k = x_{\min} + \Big(k + \frac{1}{2}\Big) w. \tag{12.6}$$

For $N = 3$ and the interval $[-4, 4]$ the width is $w = 1$, and there are eight bins labelled $000, 001, \dots, 111$ (Figure 63). The value $x = -0.37$ lies in the bin $[-1, 0)$ with index $k = \lfloor 3.63 \rfloor = 3$, and $3 = 0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1$ is the bit string 011. The value belongs to the basis state $|011\rangle$, and a measurement that returns 011 is read as the bin centre $x_3 = -0.5$. The discretisation loses the position of x inside its bin, an error of at most $w/2$. Each added qubit doubles the number of bins and halves w, so the resolution improves exponentially with the number of qubits.

The data define the empirical distribution over bit strings,

$$\tilde{p}(\mathbf{b}) = \frac{n(\mathbf{b})}{|\mathcal{D}|}, \tag{12.7}$$

where $n(\mathbf{b})$ counts how often the bit string $\mathbf{b}$ occurs in the data set, and the tilde marks an estimate obtained from samples. As a running example we take a mixture of two Gaussian peaks on $x \in [-4, 4]$,

$$p(x) \propto 0.4\, e^{-(x+1.6)^2/(2\sigma^2)} + 0.6\, e^{-(x-1.6)^2/(2\sigma^2)}, \qquad \sigma = 0.6, \tag{12.8}$$

discretised on $N = 6$ qubits into 64 bins of width $w = 0.125$, and a data set of $|\mathcal{D}| = 3000$ values drawn from it (Figure 64). The bin probabilities $p(\mathbf{b})$ are known only because we chose the target, and they serve to judge how well a trained model generalises beyond the finite data. The two peaks make the bits strongly correlated. The left peak lies around bin 19, bit string 010011, and the right peak around bin 44, bit string 101100, so the leading bits of a value near one peak are the complements of those near the other.

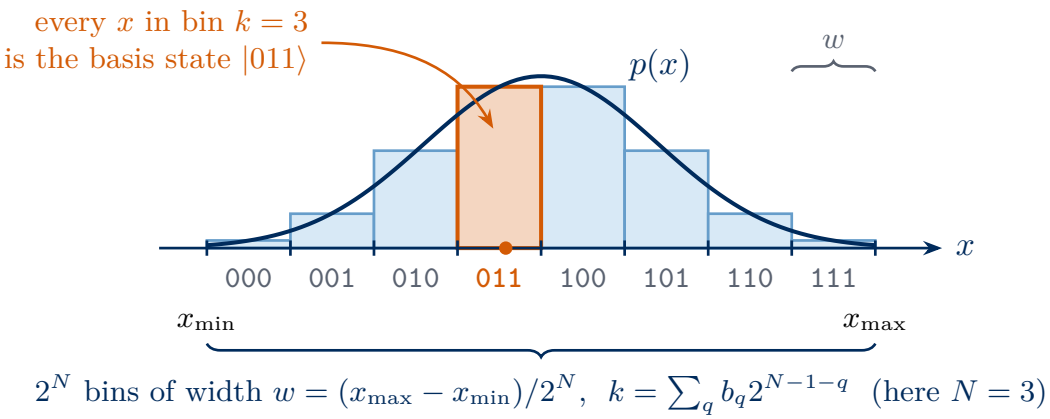


Figure 63: Continuous data as bit strings. The interval $[x_{\min}, x_{\max}]$ is cut into 2^N bins, here $N = 3$, and the binary form (12.5) of the bin index is the bit string of a computational basis state. A value in bin 3 corresponds to $|011\rangle$, and the bin probabilities of the distribution $p(x)$ become a distribution over bit strings.

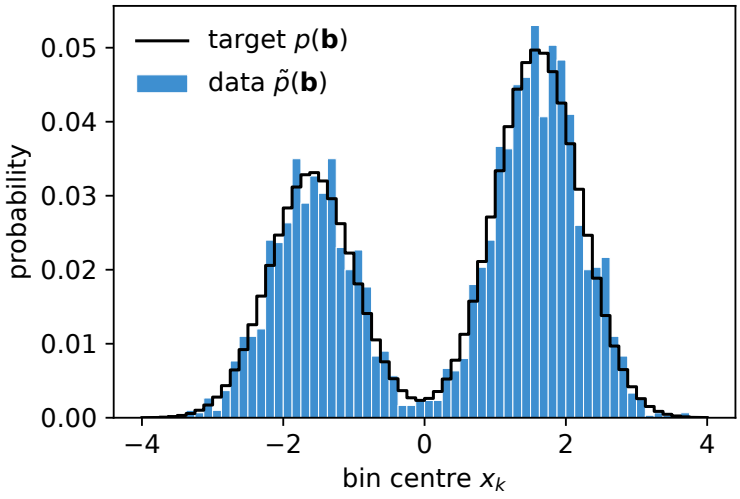


Figure 64: The two-peak target (12.8) on 64 bins (line) and the empirical distribution (12.7) of 3000 samples (bars). Some bins in the tails received no sample. The total variation distance (12.14) between the two is 0.065.

12.4 The Born machine

A circuit $\hat{V}(\boldsymbol{\theta})$ acts on the register prepared in $|0\rangle^{\otimes N}$ and produces the state $|\psi(\boldsymbol{\theta})\rangle = \hat{V}(\boldsymbol{\theta})|0\rangle^{\otimes N}$. A measurement of every qubit in the computational basis returns the bit string $\mathbf{b}$ with the Born-rule probability

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \left|\langle \mathbf{b}|\hat{V}(\boldsymbol{\theta})|0\rangle^{\otimes N}\right|^2. \tag{12.9}$$

These probabilities are non-negative and sum to one for every $\boldsymbol{\theta}$, because the state is normalised, so (12.9) is a family of distributions over the 2^N bit strings (Figure 65). A model whose probabilities are squared amplitudes is called a *Born machine*. The idea was introduced with matrix product states as the source of the amplitudes [54] and compared with Boltzmann machines in Ref. [55]. With a parametrised circuit as the source, the model is the quantum circuit Born machine [56, 57], whose sampling hardness and training were studied in Ref. [58].

Sampling needs no further computation, since one run of the circuit followed by a measurement of all qubits draws one bit string from $q_{\boldsymbol{\theta}}$ and, through (12.6), one value of x. On a device $q_{\boldsymbol{\theta}}(\mathbf{b})$ is accessible only as a frequency $\tilde{q}(\mathbf{b}) = n(\mathbf{b})/M$ over M shots, and a bit string of small probability may not appear at all. A simulation stores all 2^N amplitudes and gives the exact $q_{\boldsymbol{\theta}}$. The circuit is the same object as in VQE, where the bit strings serve only to estimate an energy and the goal is the state that minimises it. In a Born machine the distribution of the bit strings is the goal, and the samples themselves are the output.

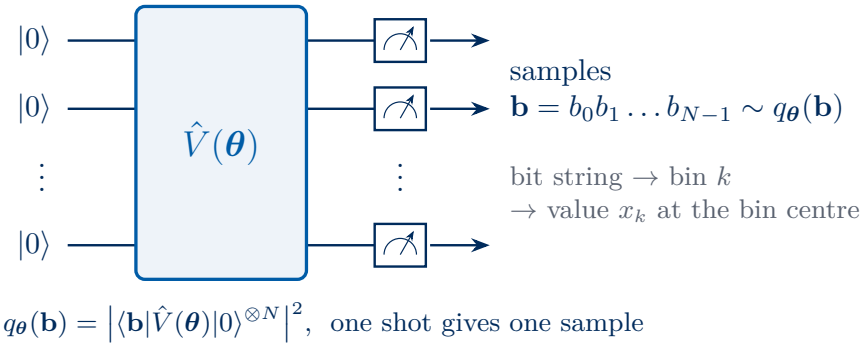


Figure 65: A quantum circuit Born machine. A parametrised circuit $\hat{V}(\boldsymbol{\theta})$ acts on $|0\rangle^{\otimes N}$, and each measurement of all qubits returns one sample $\mathbf{b} \sim q_{\boldsymbol{\theta}}(\mathbf{b})$ of the model (12.9).

12.5 The loss function

Training needs a number that measures how far $q_{\boldsymbol{\theta}}$ is from the data. The Kullback–Leibler (KL) divergence [59] of q from p is

$$D_{\mathrm{KL}}(p \,\|\, q_{\boldsymbol{\theta}}) = \sum_{\mathbf{b}} p(\mathbf{b}) \, \ln \frac{p(\mathbf{b})}{q_{\boldsymbol{\theta}}(\mathbf{b})}, \tag{12.10}$$

with the convention $0 \ln 0 = 0$, measured in nats because of the natural logarithm. It is non-negative and vanishes only for $q_{\boldsymbol{\theta}} = p$, which follows from $\ln u \leq u - 1$ applied to $u = q_{\boldsymbol{\theta}}/p$. It is asymmetric in its arguments, and the weight $p(\mathbf{b})$ makes it most sensitive to the bit strings that are frequent under p.

With the empirical distribution (12.7) in place of p, the KL divergence becomes the negative log-likelihood of the data, the standard loss of statistics. Splitting the logarithm gives

$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_{\boldsymbol{\theta}}) = \underbrace{\sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln \tilde{p}(\mathbf{b})}_{-H(\tilde{p})} \; \underbrace{- \sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln q_{\boldsymbol{\theta}}(\mathbf{b})}_{\text{depends on } \boldsymbol{\theta}} . \tag{12.11}$$

The first sum is minus the Shannon entropy $H(\tilde{p})$ of the data, in nats, which does not depend on $\boldsymbol{\theta}$. In the second sum each bit string occurs with the weight $n(\mathbf{b})/|\mathcal{D}|$, so the sum over bit strings is a sum over the data points,

$$-\sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln q_{\boldsymbol{\theta}}(\mathbf{b}) = -\frac{1}{|\mathcal{D}|} \sum_{\mathbf{b}} n(\mathbf{b}) \ln q_{\boldsymbol{\theta}}(\mathbf{b}) = -\frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \ln q_{\boldsymbol{\theta}}(\mathbf{b}^{(i)}) \equiv \mathcal{L}_{\mathrm{NLL}}(\boldsymbol{\theta}), \tag{12.12}$$

where $\mathbf{b}^{(i)}$ is the bit string of the data point $x^{(i)}$. Therefore

$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_{\boldsymbol{\theta}}) = \mathcal{L}_{\mathrm{NLL}}(\boldsymbol{\theta}) - H(\tilde{p}), \tag{12.13}$$

and minimising the KL divergence to the data and maximising the likelihood of the data are the same task with the same gradient.

The loss (12.12) contains $\ln q_{\boldsymbol{\theta}}(\mathbf{b})$ for every bit string in the data. If the model gives a data bit string a vanishing probability, the loss diverges, so training starts from a model that covers all bit strings, for example with every qubit near $|+\rangle$, where $q_{\boldsymbol{\theta}}$ is close to the uniform distribution 2^{-N}. In a simulation the exact $q_{\boldsymbol{\theta}}$ is available and the loss and its gradient follow by automatic differentiation. On a device the loss is costly, because it needs the probability of each data bit string, which can only be estimated from shot frequencies, and a zero frequency makes the estimate of $\ln q$ infinite. The gradient follows from the parameter-shift rule (5.7), since $q_{\boldsymbol{\theta}}(\mathbf{b})$ is the expectation value of the projector $|\mathbf{b}\rangle\langle\mathbf{b}|$ and depends on each angle as a cosine.

A sampled model is judged by a quantity that stays finite when some bins receive no shot. The total variation distance

$$d_{\mathrm{TV}}(p, q) = \frac{1}{2} \sum_{\mathbf{b}} |p(\mathbf{b}) - q(\mathbf{b})| \tag{12.14}$$

is the largest difference between the probabilities that the two distributions assign to any set of bit strings. It lies between 0 and 1 and can be computed directly from shot frequencies.

12.6 Product circuits and entangling circuits

The circuit we use alternates rotation blocks $\hat{B}_\ell = \bigotimes_q \hat{R}_z(\phi_{\ell q})\hat{R}_y(\vartheta_{\ell q})$ with chains of $\widehat{\mathrm{CZ}}$ gates,

$$\hat{V}(\boldsymbol{\theta}) = \hat{B}_L\,\hat{C}_Z \cdots \hat{B}_1\,\hat{C}_Z\,\hat{B}_0, \tag{12.15}$$

the layout of the HEA (6.3) with $n = 2N(L+1)$ angles. Without entangling gates, $L = 0$, the circuit prepares a product state. Each qubit q ends in $\cos(\vartheta_q/2)|0\rangle + e^{i\phi_q}\sin(\vartheta_q/2)|1\rangle$ up to a global phase, the amplitude of a bit string is a product of one-qubit amplitudes, and the Born probability factorises,

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \prod_{q=0}^{N-1} q_q(b_q), \qquad q_q(0) = \cos^2\frac{\vartheta_q}{2}, \quad q_q(1) = \sin^2\frac{\vartheta_q}{2}. \tag{12.16}$$

Every bit is an independent biased coin, and the phases ϕ_q have no effect on the outcomes. The best product model of a distribution p follows from (12.10), because the logarithm of a product splits into a sum over qubits. Summing $p(\mathbf{b})$ over all bits except b_q gives the marginal $p_q(b_q)$, the probability that bit q takes the value b_q,

$$D_{\mathrm{KL}}\Big(p \,\Big\|\, \prod_q q_q\Big) = -H(p) - \sum_{q=0}^{N-1} \sum_{b_q \in \{0,1\}} p_q(b_q)\ln q_q(b_q). \tag{12.17}$$

Each inner sum is smallest for $q_q = p_q$, by the non-negativity of the KL divergence between two one-bit distributions. The best product model is therefore the product of the marginals, and no training of the angles goes below the value of (12.17) at $q_q = p_q$. For the two-peak target the marginals mix the bits of the two peaks, and their product places probability on bit strings that combine the leading bits of one peak with the trailing bits of the other, between and outside the peaks. For the empirical distribution $\tilde{p}$ of the data set, which is the distribution the training sees, the product bound is 0.500 nats.

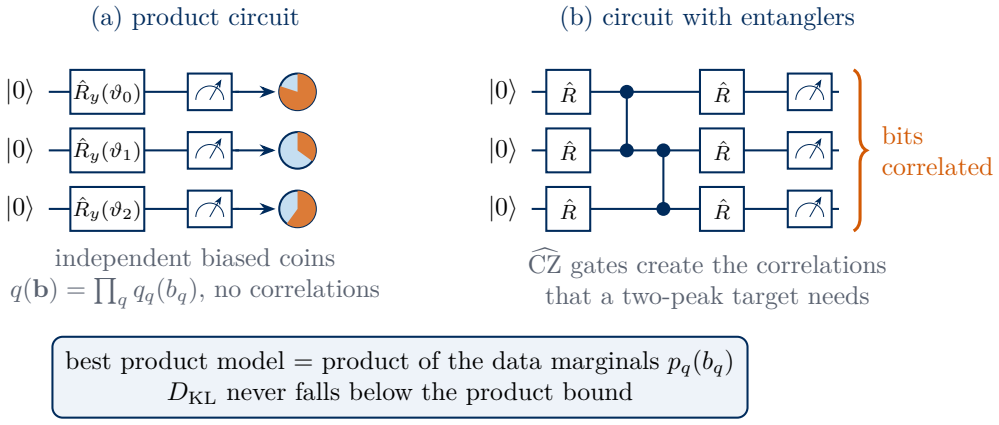


Figure 66: Product and entangling circuits as probability models. Without entangling gates every bit is an independent biased coin and the model is a product of one-bit distributions (12.16). Entangling gates between rotation layers correlate the bits, which a two-peak target requires.

Entangling layers make the amplitudes, and therefore the bits, correlated (Figure 66). We train circuits of depth $L = 0, \ldots, 5$ by minimising the KL divergence to the data with exact probabilities and automatic differentiation, from three random starts near the uniform distribution each, with Adam (Figure 67). The product circuit converges to the bound (12.17). One entangling layer lowers the KL divergence from 0.5 to below 0.1, and both divergences fall quickly up to $L = 2$. Beyond it the loss on the data keeps decreasing by an order of magnitude, while the divergence from the target

stays near 0.0137. The deeper circuits start to reproduce the statistical fluctuations of the 3000 data values, including empty bins, which are not part of p. Fitting such fluctuations is called overfitting, and the divergence from a target that the training never sees detects it. In practice p is unknown, and a part of the data set held back from training plays the role of the target.

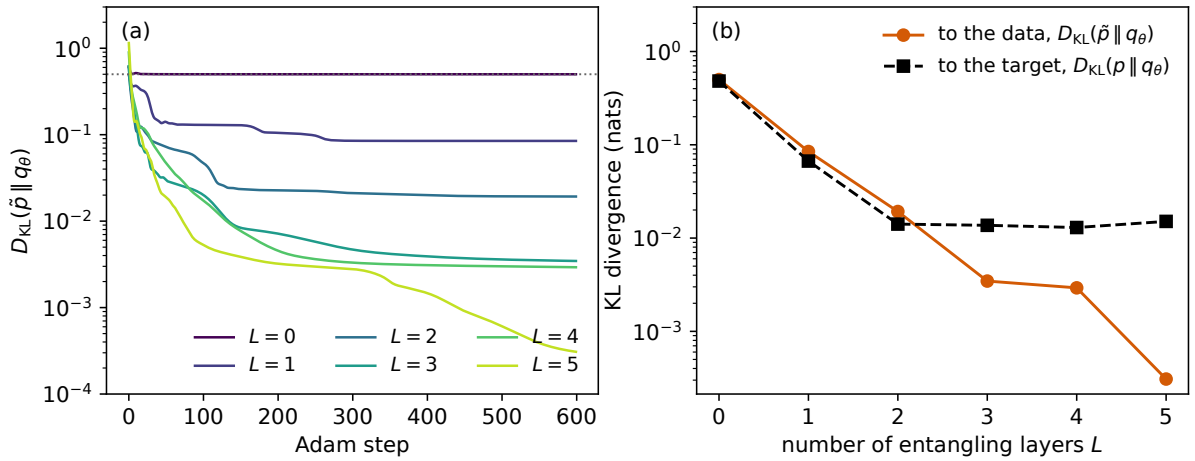


Figure 67: Born machines of depth $L = 0, \ldots, 5$ trained on the two-peak data. (a) KL divergence to the data against the Adam step for the best start at each depth, with the product bound (12.17) dotted. (b) Final KL divergence against the number of entangling layers, to the data (circles) and to the exact target (squares). Exact probabilities from the state vector.

The saturation comes from the finite data set. With a second data set ten times larger, whose empirical distribution lies at $d_{\mathrm{TV}} = 0.015$ from the target, the divergence of the trained circuits from the target drops to 0.0019 at $L = 3$ and 0.0014 at $L = 5$, whereas the $L = 2$ circuit improves less, because its error comes mostly from the limited circuit. In Figure 68 the product model spreads probability over the whole interval and misses both peaks, one entangling layer places both peaks at the right positions with distorted shapes, and three layers follow the target in the peaks and in the tails.

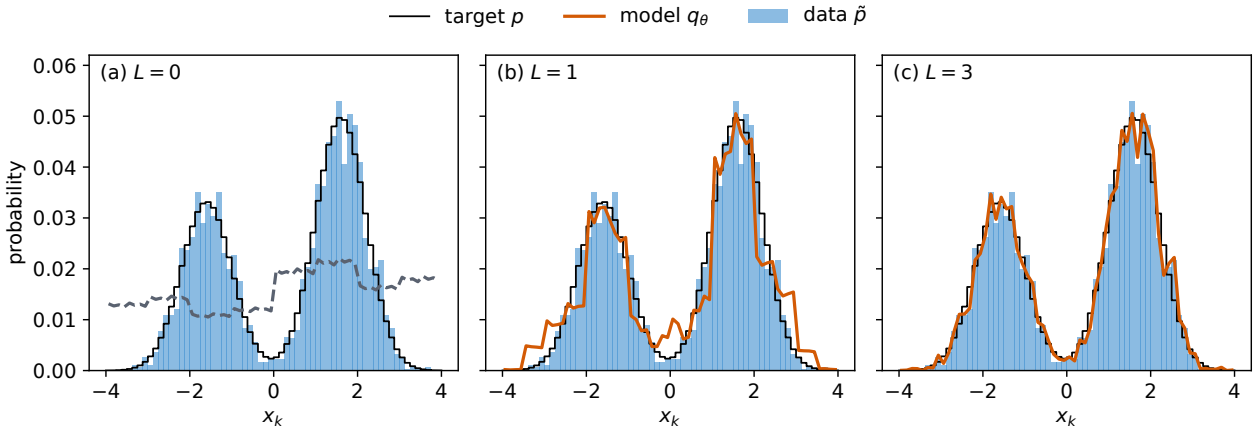


Figure 68: Trained model distributions $q_{\boldsymbol{\theta}}$ (orange lines, grey dashed for the product circuit) against the bin centres x_k, with the exact target p (black steps) and the empirical distribution $\tilde{p}$ of the data (bars), for (a) the product circuit, $L = 0$, (b) one entangling layer and (c) three entangling layers. Exact model probabilities.

On a device the trained model is used and tested through its samples. We measure the $L = 3$ model 10000 times in simulation and convert the bit strings to bin frequencies and values of x (Figure 69). The sampled histogram reproduces the two peaks with their weights and widths, and its total variation distance from the target, 0.051, is close to the value 0.055 of the exact model probabilities. The KL divergence from samples would be infinite, since 2 bins with nonzero target probability received no shot.

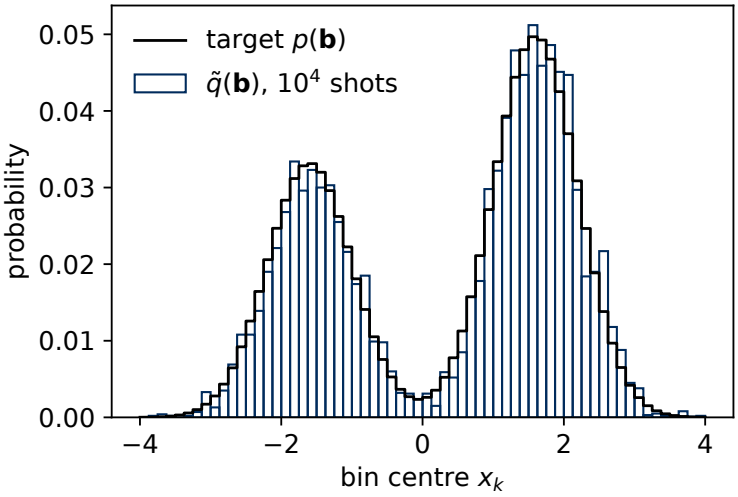


Figure 69: Frequencies of 10000 bit strings sampled from the trained $L = 3$ Born machine, read as bin centres (12.6) (bars), and the exact target (line). Simulated measurement record.

12.7 Trainability and ancilla qubits

The Born machine meets the same balance of expressibility and trainability as VQE. Deep circuits with every angle trainable are expressive, but the gradients of their angles become exponentially small in the number of qubits, a limitation analysed in Section 16. One remedy fixes the entangling part of each layer and trains only single-qubit rotations, which keeps the number of trainable angles small and the gradients of the rotations measurable. Every distribution is the squared amplitude of some pure state, $\psi_{\mathbf{b}} = \sqrt{p(\mathbf{b})}$, but a circuit of fixed depth reaches only a small part of the pure states of N qubits. Ancilla qubits that take part in the circuit but are not measured enlarge the set of distributions reachable at a given depth. Ignoring their outcomes sums the Born probabilities over their bits $\mathbf{a}$,

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \sum_{\mathbf{a}} \left|\langle \mathbf{b}, \mathbf{a}|\psi(\boldsymbol{\theta})\rangle\right|^2 = \langle \mathbf{b}|\hat{\rho}_{\text{meas}}|\mathbf{b}\rangle, \tag{12.18}$$

where $\hat{\rho}_{\text{meas}}$ is the reduced density matrix of the measured qubits (Figure 70). The measured qubits thus follow a mixture of Born distributions, at the cost of extra qubits.

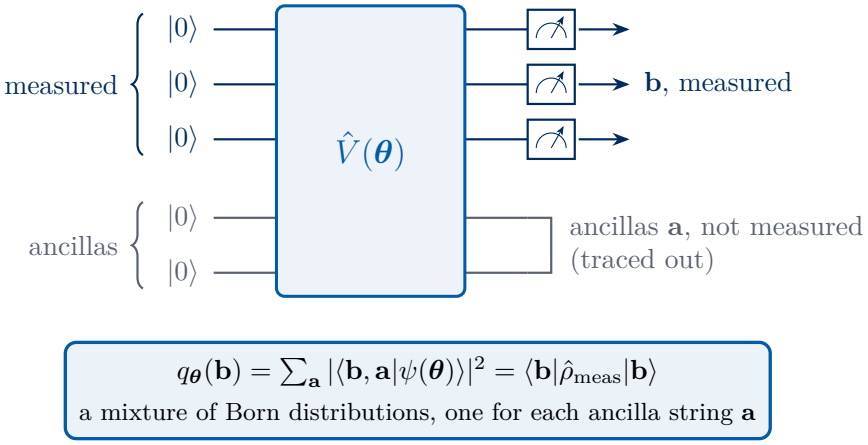


Figure 70: Ancilla qubits in a Born machine. The circuit acts on measured and ancilla qubits, and only the measured qubits are read out. Summing over the unread ancilla bits (12.18) makes the model a mixture of Born distributions.

12.8 The quantum scrambling Born machine

The quantum scrambling Born machine [60] combines the two remedies of the previous subsection. Each of its L layers rotates every qubit by trainable angles, applies a fixed entangling unitary $\hat{U}_{\mathrm{s}}$, called the scrambler, and rotates every qubit once more (Figure 71),

$$\hat{V}(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} \hat{R}_y^{(\ell)}\, \hat{U}_{\mathrm{s}}\, \hat{R}_z^{(\ell)} \hat{R}_x^{(\ell)}, \qquad \hat{R}_a^{(\ell)} = \bigotimes_{q=0}^{N-1} \hat{R}_a(\theta_{\ell q}^{a}), \quad a \in \{x, y, z\}. \tag{12.19}$$

Only the $3NL$ single-qubit angles are trained. The scrambler has no adjustable parameters, so on hardware it is calibrated once and then applied in every layer. Some qubits act as ancillas, which are not measured, so the model is a mixture of Born distributions (12.18). The construction leaves open which properties the fixed entangler must have for the trained model to be accurate. Ref. [60] compares three choices, a uniformly random unitary of the whole register, a brickwork of random two-qubit gates of finite depth, and the evolution under a physical spin-chain Hamiltonian for a fixed time.

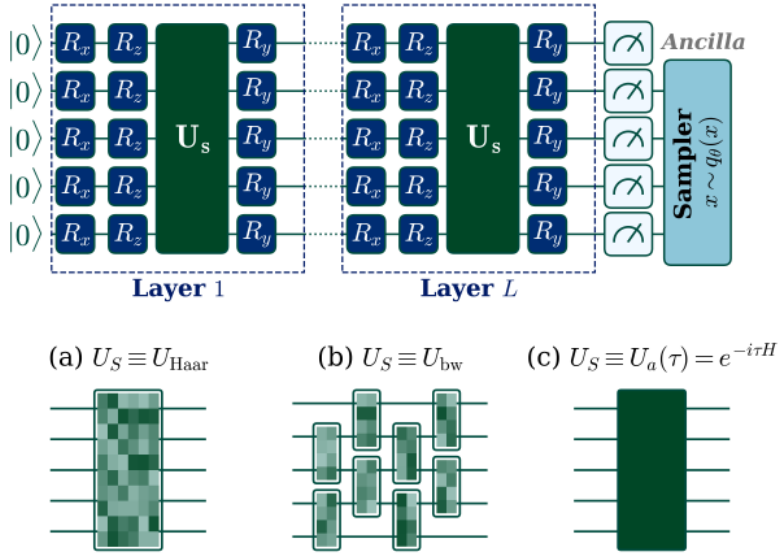


Figure 71: Architecture of the quantum scrambling Born machine (12.19), reproduced from Ref. [60]. Each layer applies trainable rotations $\hat{R}_x$, $\hat{R}_z$ and $\hat{R}_y$ to every qubit around the fixed scrambler $\hat{U}_{\mathrm{s}}$, and the measured bit strings are samples of the model, with the top qubit as an unmeasured ancilla. The scrambler is (a) a Haar-random unitary, (b) a brickwork of random two-qubit gates or (c) the evolution $e^{-i\tau\hat{\mathcal{H}}}$ under a spin-chain Hamiltonian for the time τ.

The reference scrambler is a Haar-random unitary, a unitary drawn uniformly from all unitaries of the 2^N-dimensional state space, so that no direction of this space is preferred. Applied to $|0\ldots0\rangle$ it produces a random state, which is nearly maximally entangled across every cut of the register. The average entanglement entropy of such a state is called the Page value, and for a part of $n_A \leq N/2$ qubits it lies close to the maximum of n_A bits. Section 16 derives these properties of random states. With a Haar scrambler on $N = 8$ qubits, the model learns a target of five Gaussian peaks on the bins of Section 12.3, and its KL divergence falls as the number of layers grows (Figure 72). Shallow models capture only the envelope of the target, and deeper ones resolve every peak. One or two traced-out ancillas lower the divergence most for shallow circuits, where the pure-state model is most restricted.

A Haar-random unitary of many qubits cannot be implemented efficiently, so the next question is how much scrambling a physical entangler must supply. A brickwork of random two-qubit gates acting on neighbouring pairs, repeated K times, is a circuit of depth K. Each brickwork layer extends the region influenced by one qubit by one site in each direction and, by the argument of Section 6.5,

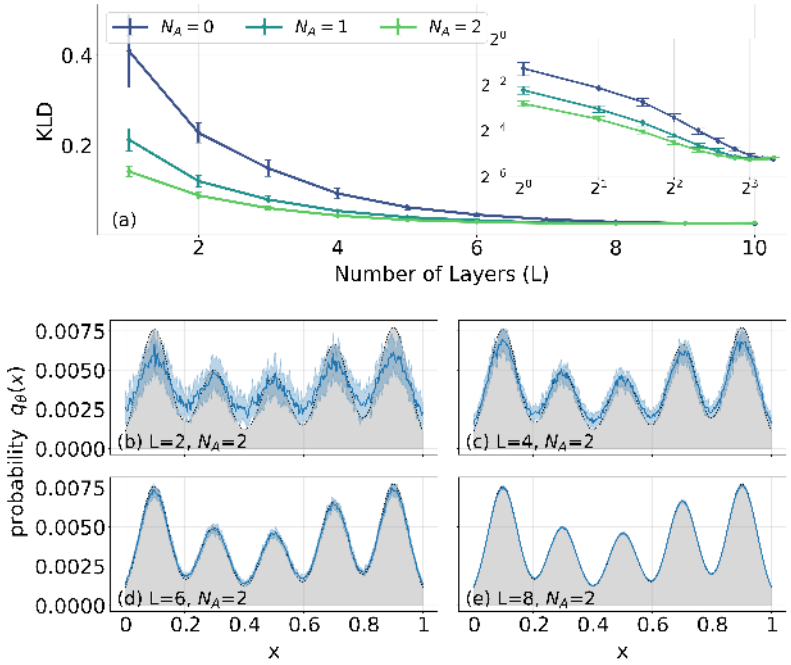


Figure 72: The scrambling Born machine with a Haar-random scrambler, $N = 8$, reproduced from Ref. [60]. (a) KL divergence between target and model against the number of layers L, without ancillas and with $N_A = 1$ or 2 ancilla qubits. (b)–(e) Learned distribution (blue) against the target of five Gaussian peaks (grey) for $L = 2$, 4, 6 and 8.

adds a bounded amount of entanglement across any cut. At $K = 1$ the scrambler entangles only neighbouring qubits, and the trained model saturates far above the divergence reached with the Haar scrambler (Figure 73). For depths of order $N/2$ every qubit is connected to every other, the entanglement of the scrambled states approaches the Page value, and the curves for all numbers of ancillas collapse onto the Haar level. The accuracy of the model is thus controlled by how close the entangler comes to the entanglement of a random state, and once it is there the microscopic origin of the scrambler hardly matters.

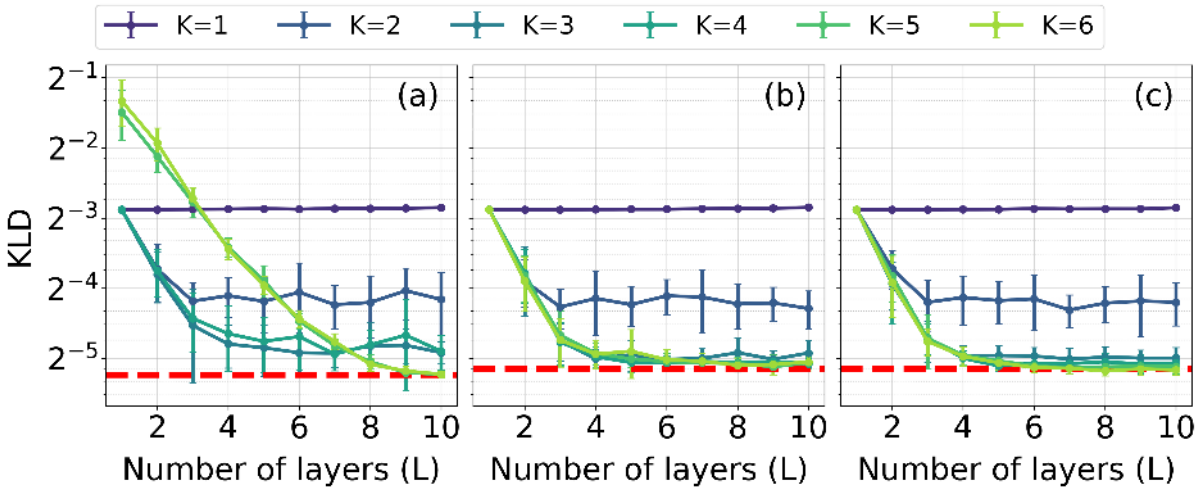


Figure 73: Brickwork scramblers of depth K, reproduced from Ref. [60]. KL divergence of the trained model against the number of layers L for $N_A = 0$, 1 and 2 ancilla qubits, with the level of the Haar scrambler as a red dashed line. Shallow brickworks saturate above this level, and a depth of order $N/2$ reaches it.

Analog quantum simulators do not apply gates but let the qubits evolve under their native interactions. The scrambler can then be the evolution $\hat{U}_\mathrm{s} = e^{-i\tau\hat{\mathcal{H}}}$ for a time τ under a nearest-neighbour Hamiltonian $\hat{\mathcal{H}}$, and Ref. [60] uses the Ising chain and the XX chain in a magnetic field. For short times the evolution barely entangles the qubits, and training fails at every depth (Figure 74).

Around $\tau \approx 0.5$ in units of the inverse coupling, the half-chain entropy of the evolved states approaches the Page value, and for $L \geq 5$ the divergence drops to the level of the Haar scrambler. At longer times the divergence of shallow models rises again, whereas deep models stay near the Haar level. Both chains are integrable, which means that their dynamics is fixed by many conserved quantities and does not explore the state space like a random unitary. The trained rotations between the evolutions vary from site to site and break this structure.

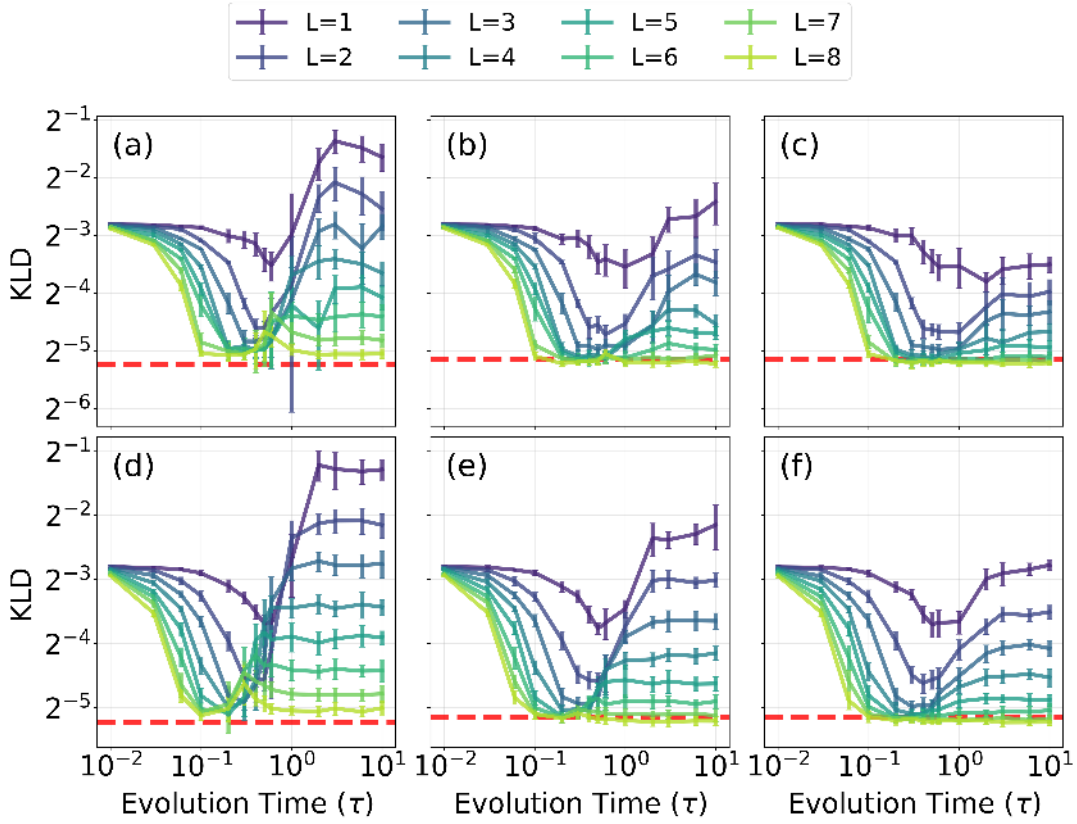


Figure 74: Analog scramblers, reproduced from Ref. [60]. KL divergence of the trained model against the evolution time τ of the scrambler for $L = 1, \ldots, 8$ layers, with the Ising chain in the top row and the XX chain in a field in the bottom row. The three columns of each row differ in the model settings specified in Ref. [60]. The red dashed lines mark the level of the Haar scrambler.

A variant replaces the fixed scrambler by the evolution under a trainable Hamiltonian. Each layer evolves the register for a fixed time τ under its own Hamiltonian

$$\hat{V}(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} e^{-i\tau\hat{\mathcal{H}}^{(\ell)}}, \qquad \hat{\mathcal{H}}^{(\ell)} = J^{(\ell)} \sum_i \hat{X}_i \hat{X}_{i+1} + \sum_i \left(h_{x,i}^{(\ell)} \hat{X}_i + h_{y,i}^{(\ell)} \hat{Y}_i + h_{z,i}^{(\ell)} \hat{Z}_i\right), \tag{12.20}$$

whose coupling $J^{(\ell)}$ and site-dependent fields $h_{a,i}^{(\ell)}$ form the $(3N+1)L$ trainable parameters. Learning a distribution then becomes the search for Hamiltonians whose dynamics produces the data, the forward counterpart of Hamiltonian learning, which infers a Hamiltonian from measured data. Figure 75 shows a target of four Gaussian peaks over two variables, encoded in two registers by the binning of Section 12.3, learned with $N = 10$ qubits including two ancillas and $L = 10$ layers, which gives 310 parameters. All four peaks are reproduced with their weights.

13 Combinatorial optimisation with QUBO and QAOA

This section is not covered in the lecture slides and accompanies Lab 073.

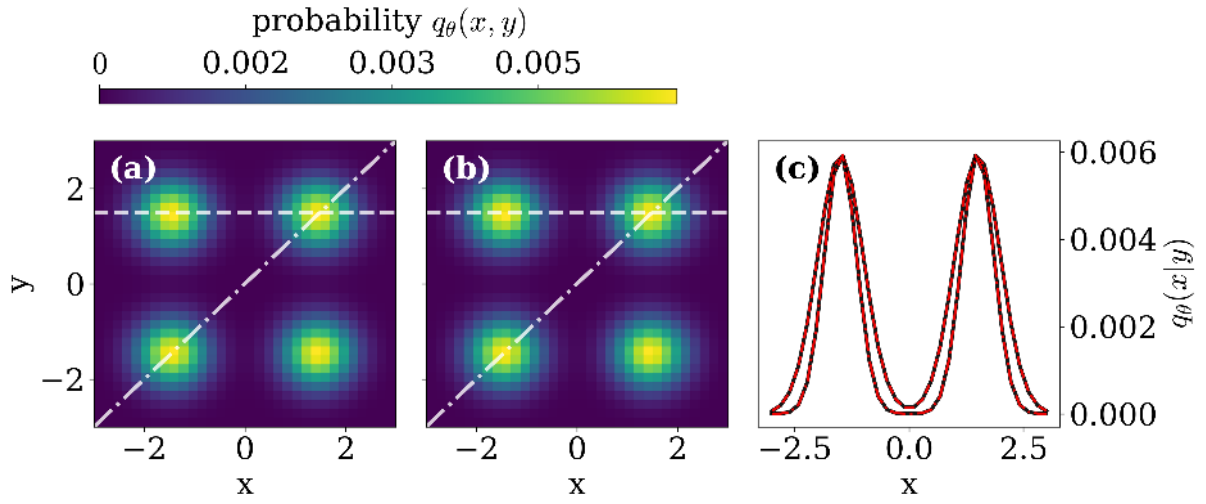


Figure 75: The scrambling Born machine with trainable Hamiltonians (12.20), reproduced from Ref. [60] and not computed for these notes. (a) Target distribution $p(x, y)$, a mixture of four Gaussian peaks on a grid of bins. (b) Learned distribution, $N = 10$ qubits with two ancillas and $L = 10$ layers. (c) Slices along the dashed and dash-dotted lines of (a) and (b), target in black and model in red.

13.1 Binary optimisation on a quantum register

A Born machine learns a distribution from data. A circuit can also be trained to concentrate its distribution on the bit strings that solve an optimisation problem, and this use proceeds in four stages (Figure 76). The problem is posed as the minimisation of a cost $C(\mathbf{x})$ over bit strings $\mathbf{x}$. The cost is rewritten as a Hamiltonian $\hat{\mathcal{H}}_C$ that is diagonal in the computational basis, so that every bit string is an eigenstate with its cost as the energy and the optimal bit string is the ground state. A parametrised circuit, the quantum approximate optimisation algorithm, prepares a state with large weight on low-cost bit strings, and its angles are trained by a classical optimiser that lowers the expected cost $\langle \hat{\mathcal{H}}_C \rangle$. Every measurement of all qubits finally returns one candidate bit string, a classical computer evaluates its cost, and the best candidate is the answer. The subsections below follow these stages.

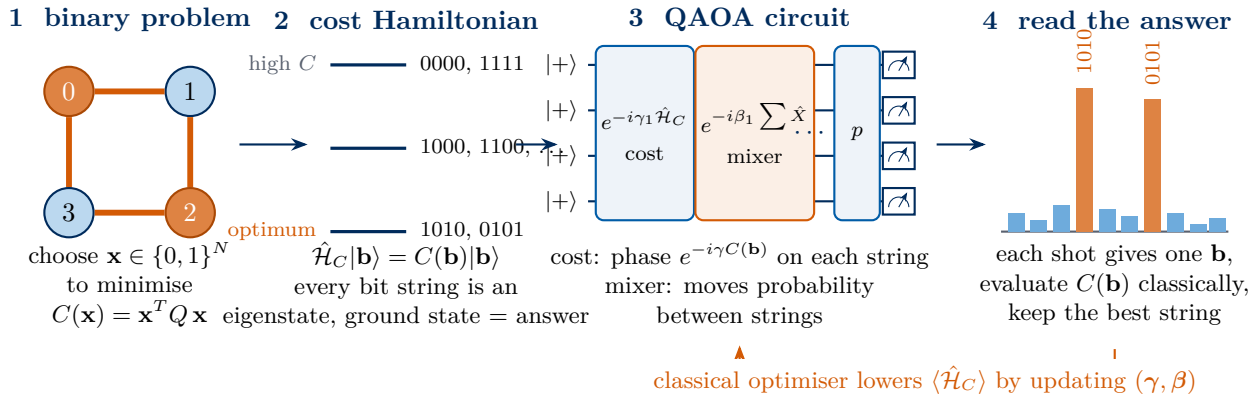


Figure 76: Binary optimisation on a quantum register. (1) A binary problem with the cost $C(\mathbf{x}) = \mathbf{x}^T Q\, \mathbf{x}$, here a cut of a square. (2) The cost Hamiltonian is diagonal, every bit string is an eigenstate and the optimum is the ground state. (3) The QAOA circuit alternates cost layers, which give each bit string the phase $e^{-i\gamma C(\mathbf{b})}$, and mixing layers, which move probability between bit strings. (4) Each shot gives one bit string, its cost is evaluated classically and the best one is kept, while a classical optimiser updates the angles to lower $\langle \hat{\mathcal{H}}_C \rangle$.

13.2 Quadratic unconstrained binary optimisation

Many decisions are choices between two options made for many items at once. A delivery is assigned to one of two vans, a component to one of two chips, a vertex of a network to one of two groups.

When the cost of a choice depends on pairs of items, the task is a quadratic unconstrained binary optimisation (QUBO) problem. It has N binary variables $x_i \in \{0, 1\}$, collected in $\mathbf{x} = (x_0, \ldots, x_{N-1})$, and a real $N \times N$ matrix Q, and it asks for the vector that minimises the quadratic cost (Figure 77)

$$C(\mathbf{x}) = \mathbf{x}^T Q\, \mathbf{x} = \sum_{i,j} Q_{ij}\, x_i x_j. \tag{13.1}$$

A binary variable satisfies $x_i^2 = x_i$, so the diagonal elements Q_{ii} act as linear costs, and every quadratic function of binary variables has the form (13.1) up to a constant. The word unconstrained means that all 2^N vectors are allowed, and constraints are added to the cost as penalty terms. Each variable is stored in one qubit, so N is also the number of qubits.

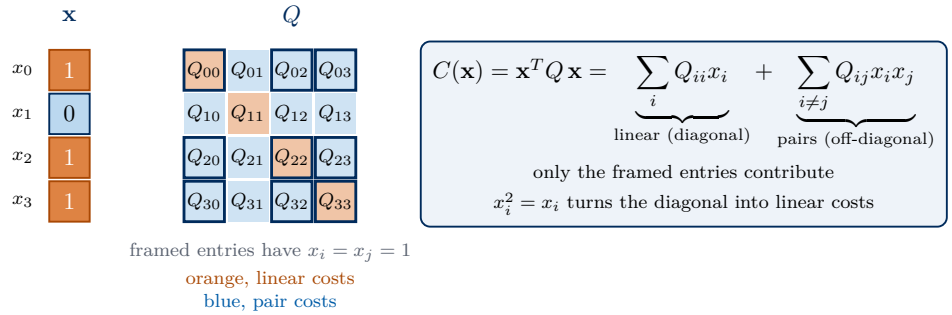


Figure 77: A QUBO problem. Each of the N binary variables is a yes-or-no choice, the matrix Q assigns linear costs on its diagonal and pair costs off the diagonal, and the answer is the bit string with the lowest cost (13.1) among 2^N candidates.

Checking every vector costs 2^N evaluations of C, about 10^{15} for $N = 50$. QUBO is NP-hard, so no known algorithm solves every instance in a time that grows polynomially with N, and practical solvers for large instances are heuristics. A greedy method flips one variable at a time as long as the cost decreases. Simulated annealing [61] also accepts some cost increases, with a probability that decreases during the run, so that it can leave local minima. For the small instances below, exhaustive search gives the exact answer against which the quantum algorithm is compared.

MaxCut, which divides the vertices of a graph into two groups so that as many edges as possible join the two groups, serves as the example of this section. Number partitioning, the maximum independent set, scheduling, graph colouring and the travelling salesman problem take the same form once their constraints are written as penalties [62].

13.3 MaxCut

A graph consists of N vertices and a set E of edges, each joining a pair of vertices (i, j). We may think of the vertices as people and the edges as conflicts between them, and we must split the people into two rooms so that as many conflicting pairs as possible end up in different rooms. A split of the vertices into two groups is called a *cut*, because a line drawn between the groups cuts every edge that crosses it. We label the groups by one binary variable per vertex, $x_i = 0$ for the first group and $x_i = 1$ for the second, so a cut is a bit string $x_0 x_1 \ldots x_{N-1}$. The *size* of the cut is the number of edges whose two ends carry different labels, and the maximum-cut problem (MaxCut) asks for the bit string with the largest cut size.

On a square, four vertices $0, 1, 2, 3$ are joined in a ring by the edges $(0, 1)$, $(1, 2)$, $(2, 3)$ and $(3, 0)$ (Figure 78). The bit string 0000 puts all vertices in one group and cuts no edge, and 1000 moves vertex 0 to the other group and cuts its two edges. The string 1010 alternates the groups around the ring and cuts all four edges, so the maximum cut of the square is four, reached by 1010 and by its complement 0101. Exchanging the two groups never changes a cut, so optimal bit strings always come in complementary pairs. Around a triangle, a cycle of three vertices, the labels cannot alternate, and at most two of its three edges can be cut. Every cycle of odd length leaves at least one edge uncut.

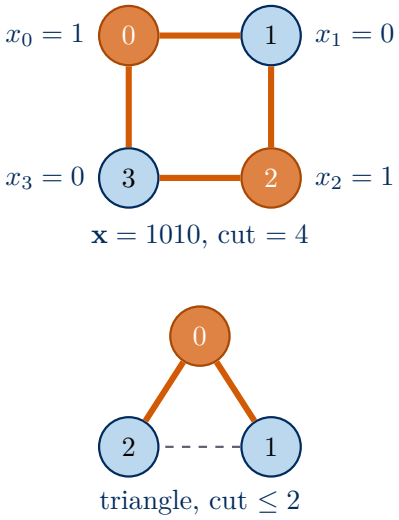


Figure 78: MaxCut on a square and on a triangle. On the square the bit string 1010 puts vertices 0 and 2 in one group and vertices 1 and 3 in the other and cuts all four edges. On the triangle at most two of the three edges can be cut.

The cube graph has eight vertices, the corners of a cube, and twelve edges (Figure 79). The split 11001001 cuts six of them. The cube contains only cycles of even length, so its vertices can be coloured alternately, and the split 10100101 cuts all twelve edges, as does its complement 01011010.

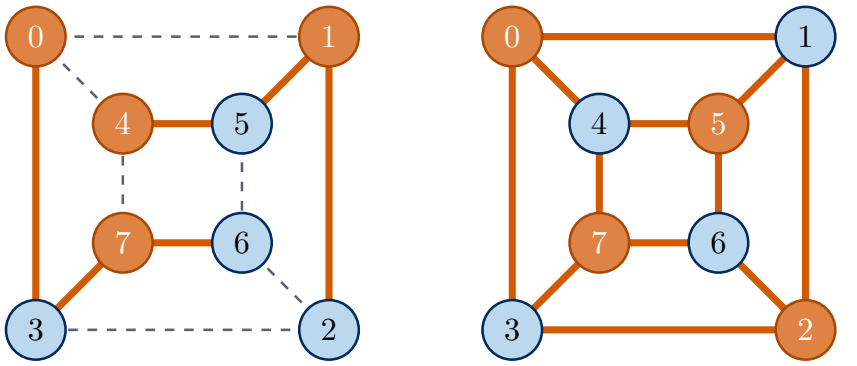


Figure 79: MaxCut on the cube graph, with the vertices of group $x_i = 0$ in blue and of group $x_i = 1$ in orange. Cut edges are solid and uncut edges dashed. (a) The split 11001001 cuts six of the twelve edges. (b) The alternating split 10100101 cuts all twelve.

For a square the answer is evident, whereas a general graph has 2^N candidates to compare. MaxCut is one of the problems shown to be NP-complete in its decision form in Ref. [63], and no algorithm is known that finds the optimum of every graph in a time that grows polynomially with N. Approximate algorithms are therefore judged by the *approximation ratio*, the cut they find divided by the maximum cut. The Goemans–Williamson algorithm, based on semidefinite programming, runs in polynomial time and guarantees an expected ratio of at least 0.878 on every graph [64]. MaxCut appears in the partitioning of networks and circuits, in clustering, where the edges carry the dissimilarities of items, and in physics as the ground-state problem of an Ising antiferromagnet, in which a cut edge is a satisfied bond.

An edge (i, j) is cut when exactly one of x_i, x_j equals one, and the expression $x_i + x_j - 2x_ix_j$ equals one in that case and zero otherwise. The cut size is therefore

$$\mathrm{cut}(\mathbf{x}) = \sum_{(i,j)\in E} (x_i + x_j - 2x_ix_j), \tag{13.2}$$

and maximising the cut is minimising $C(\mathbf{x}) = -\mathrm{cut}(\mathbf{x})$, a QUBO with the matrix

$$Q_{ii} = -d_i, \qquad Q_{ij} = Q_{ji} = \begin{cases} 1 & (i,j) \in E \\ 0 & \text{otherwise} \end{cases} \quad (i \neq j), \tag{13.3}$$

where the degree d_i is the number of edges at vertex i. Each edge contributes $Q_{ij} + Q_{ji} = 2$ to the product term, and the linear terms x_i collect on the diagonal.

On a qubit register each bit is read as the eigenvalue $z_i = 1 - 2x_i = \pm 1$ of $\hat{Z}_i$, and the cut indicator of one edge takes a compact form in these spin variables. Table 6 lists the four configurations of an edge. The edge is cut when the spins are antiparallel, $z_i z_j = -1$, and the indicator equals $(1 - z_i z_j)/2$,

$$\mathrm{cut}(\mathbf{x}) = \sum_{(i,j)\in E} \frac{1 - z_i z_j}{2}. \tag{13.4}$$

Maximising the cut thus means making as many neighbouring spins antiparallel as possible, which is the ground-state problem of an Ising antiferromagnet on the graph.

$x_i x_j$	(z_i, z_j)	$z_i z_j$	cut	$x_i + x_j - 2x_i x_j$	$\frac{1}{2}(1 - z_i z_j)$
00	$(+1,+1)$	$+1$	no	0	0
01	$(+1,-1)$	-1	yes	1	1
10	$(-1,+1)$	-1	yes	1	1
11	$(-1,-1)$	$+1$	no	0	0

Table 6: The four configurations of one edge (i,j). The cut indicator equals $x_i + x_j - 2x_i x_j$ in the binary variables and $(1 - z_i z_j)/2$ in the spin variables $z_i = 1 - 2x_i$.

13.4 The Ising Hamiltonian of a QUBO

A qubit measured in the computational basis gives the bit b_i, and its operator $\hat{Z}_i$ has the eigenvalue $z_i = 1 - 2b_i = \pm 1$. We identify each binary variable with the bit of one qubit, $x_i = b_i$, so that

$$x_i = \frac{1 - z_i}{2}. \tag{13.5}$$

Inserting (13.5) into (13.1) gives $C = \frac{1}{4}\sum_{i,j} Q_{ij}(1 - z_i - z_j + z_i z_j)$. The diagonal terms $i = j$ contain $z_i^2 = 1$ and add a constant, and collecting the constant, linear and quadratic terms gives

$$C = \underbrace{\frac{1}{4}\sum_{i,j} Q_{ij} + \frac{1}{4}\sum_i Q_{ii}}_{\text{constant } c_0} + \sum_i \underbrace{\Big(-\frac{1}{4}\sum_j (Q_{ij} + Q_{ji})\Big)}_{\text{field } h_i} z_i + \sum_{i<j} \underbrace{\frac{Q_{ij} + Q_{ji}}{4}}_{\text{coupling } J_{ij}} z_i z_j. \tag{13.6}$$

The result is the energy of classical Ising spins $z_i = \pm 1$ with couplings J_{ij}, fields h_i and a constant c_0. Replacing each spin by its Pauli operator gives the cost Hamiltonian

$$\hat{\mathcal{H}}_C = \sum_{i<j} J_{ij}\, \hat{Z}_i \hat{Z}_j + \sum_i h_i\, \hat{Z}_i + c_0\, \hat{I}. \tag{13.7}$$

It contains only $\hat{Z}$ operators and is therefore diagonal in the computational basis,

$$\hat{\mathcal{H}}_C |\mathbf{b}\rangle = C(\mathbf{b})\, |\mathbf{b}\rangle. \tag{13.8}$$

Every basis state $|\mathbf{b}\rangle$ is a candidate solution with its cost as the energy, and the ground state of $\hat{\mathcal{H}}_C$ is the optimal bit string (Figure 80). A degenerate ground level holds all optimal bit strings. For MaxCut the fields vanish, since $\sum_j (Q_{ij} + Q_{ji}) = -2d_i + 2d_i = 0$, and (13.6) becomes

$$\hat{\mathcal{H}}_C = \frac{1}{2} \sum_{(i,j)\in E} \hat{Z}_i \hat{Z}_j - \frac{|E|}{2}\, \hat{I}, \tag{13.9}$$

in agreement with the edge indicator (13.4), so a cut edge lowers the energy by one relative to an uncut edge. For the cube graph, $\hat{\mathcal{H}}_C$ has the couplings $J_{ij} = 1/2$ on the twelve edges and $c_0 = -6$, and $|10100101\rangle$ has the energy $12 \cdot (-\frac{1}{2}) - 6 = -12$, minus the maximum cut.

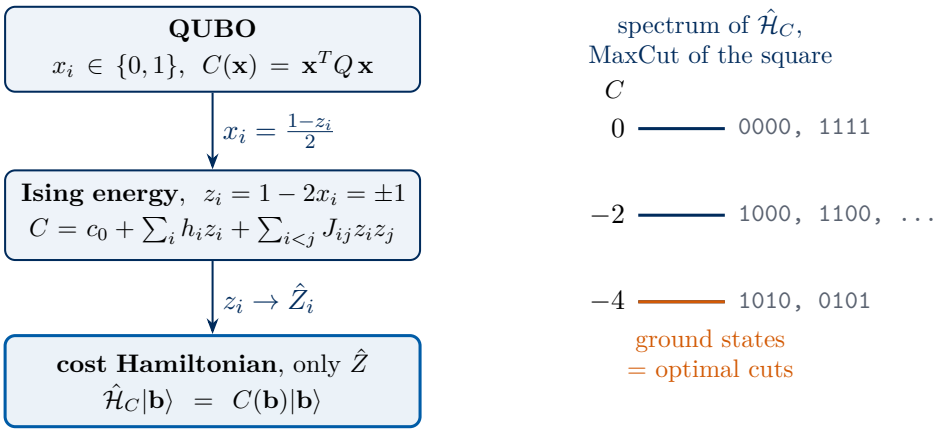


Figure 80: From a QUBO to a diagonal Ising Hamiltonian. Each binary variable becomes the bit of one qubit through $x_i = (1 - z_i)/2$, the quadratic cost becomes couplings J_{ij}, fields h_i and a constant, and every computational basis state is a candidate solution whose energy is its cost.

13.5 Answers from measured bit strings

A quantum algorithm for a QUBO prepares some state $|\psi\rangle = \sum_{\mathbf{b}} \psi_{\mathbf{b}} |\mathbf{b}\rangle$ and measures all qubits in the computational basis. Each shot returns one bit string $\mathbf{b}$ with probability $|\psi_{\mathbf{b}}|^2$, which is one candidate solution, and its cost $C(\mathbf{b})$ is evaluated on a classical computer in a time polynomial in N. From M shots we keep the best bit string. If the optimal bit strings together carry the probability P_{opt}, at least one of M independent shots is optimal with probability

$$P_{\text{hit}}(M) = 1 - (1 - P_{\text{opt}})^M, \tag{13.10}$$

and the number of shots for a chance of 95 per cent is $M_{95} = \lceil \ln 0.05 / \ln(1 - P_{\text{opt}}) \rceil$, about $3/P_{\text{opt}}$ for small P_{opt}. The state is useful when its weight on optimal or near-optimal bit strings is much larger than the uniform weight 2^{-N} of random guessing, which is what the uniform superposition $|+\rangle^{\otimes N}$ gives.

In VQE the same circuits and measurements give the ground-state energy, a number estimated as an average over many shots. Here the answer is one sampled bit string, whose cost is known exactly as soon as it is measured, and the average energy serves only as the training signal for the circuit.

As an example we take a graph of $N = 8$ vertices, a ring in which each vertex is also joined to the opposite vertex, with 12 edges and three neighbours per vertex (Figure 81). It contains cycles of odd length, such as 0–1–2–3–4–0, so no division cuts all edges. Exhaustive search over the 256 bit strings gives the maximum cut 10, reached by 8 bit strings in complementary pairs. A uniformly random bit string cuts each edge with probability one half, six edges on average, and is optimal with probability $8/256 = 1/32$, so about 95 random shots are needed for a 95 per cent chance of an optimum.

13.6 The QAOA circuit

The quantum approximate optimisation algorithm (QAOA) [65] prepares a state concentrated on low-cost bit strings with a circuit of p layers, where p plays the role of the depth L (Figure 82). It starts in $|+\rangle^{\otimes N}$, the uniform superposition of all candidates, and alternates two evolutions. The cost layer is the evolution under the cost Hamiltonian for a time γ_ℓ, and the mixing layer is the evolution under a transverse field for a time β_ℓ,

$$\hat{U}_C(\gamma_\ell) = e^{-i\gamma_\ell \hat{\mathcal{H}}_C}, \qquad \hat{U}_B(\beta_\ell) = e^{-i\beta_\ell \sum_i \hat{X}_i} = \bigotimes_{i=0}^{N-1} \hat{R}_x(2\beta_\ell). \tag{13.11}$$

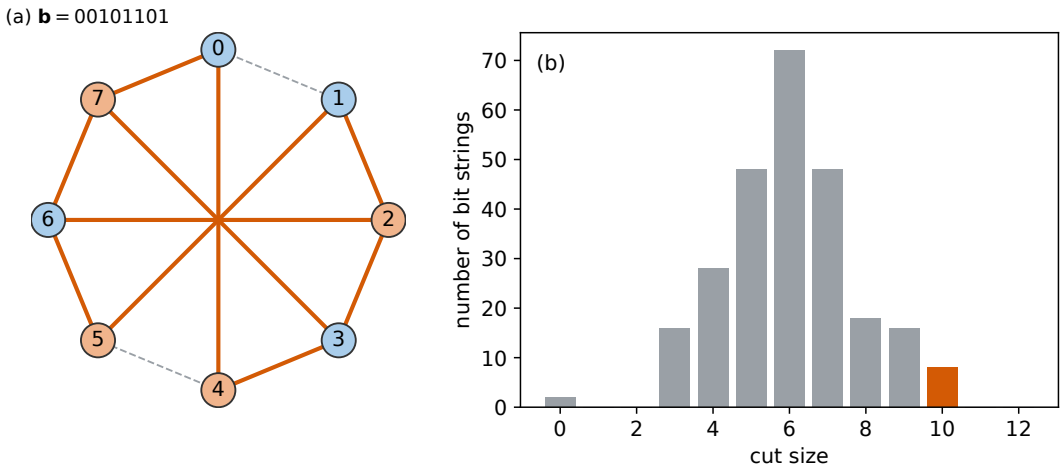


Figure 81: The MaxCut example. (a) The graph of eight vertices with the vertices coloured by the optimal bit string 00101101, blue for $b_i = 0$ and orange for $b_i = 1$, the ten cut edges solid and the two uncut edges dashed. (b) Number of bit strings for each cut size, with the eight optimal strings in orange. The optimum lies in the tail of the distribution.

The QAOA state with $2p$ angles $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$ is

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = \hat{U}_B(\beta_p)\hat{U}_C(\gamma_p) \cdots \hat{U}_B(\beta_1)\hat{U}_C(\gamma_1)\, |+\rangle^{\otimes N}. \tag{13.12}$$

The cost layer is diagonal and multiplies the amplitude of each bit string by the phase $e^{-i\gamma_\ell C(\mathbf{b})}$, so bit strings of different cost acquire different phases. The mixing layer rotates every qubit and makes the amplitudes of bit strings that differ by single bit flips interfere. The cost layer alone leaves all probabilities unchanged, and the mixing layer alone leaves $|+\rangle^{\otimes N}$ unchanged, since it is an eigenstate of $\sum_i \hat{X}_i$. Only the alternation moves probability, and the angles are chosen so that the interference collects it on low-cost bit strings. The circuit has the structure of the HVA (6.5), with the cost Hamiltonian in place of the Ising couplings of the chain.

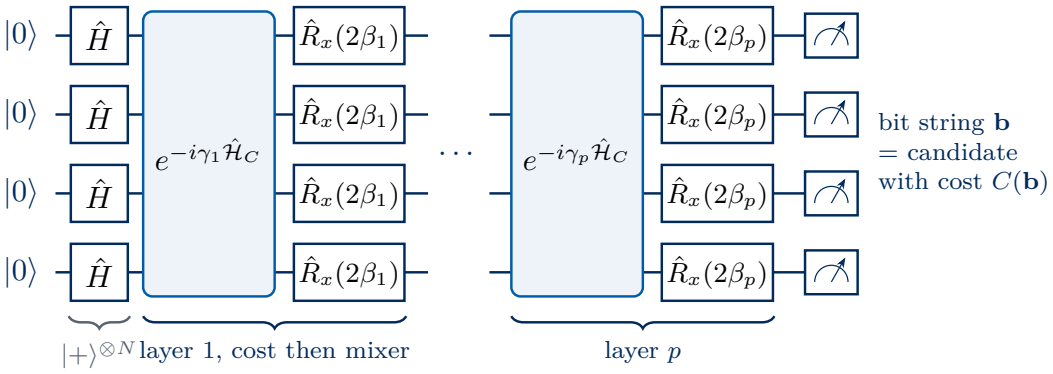


Figure 82: The QAOA circuit (13.12). Hadamard gates prepare $|+\rangle^{\otimes N}$, and each of the p layers applies the cost layer $e^{-i\gamma_\ell \hat{\mathcal{H}}_C}$, compiled on a device into $\hat{R}_{zz}$ and $\hat{R}_z$ gates, and the mixing layer of $\hat{R}_x(2\beta_\ell)$ rotations. All qubits are measured in the computational basis, and each shot is a candidate solution.

On a device the cost layer is compiled into gates, and since all terms of (13.7) commute,

$$e^{-i\gamma \hat{\mathcal{H}}_C} = e^{-i\gamma c_0} \prod_{i<j} \hat{R}_{zz}(2\gamma J_{ij}) \prod_i \hat{R}_z(2\gamma h_i), \tag{13.13}$$

with one $\hat{R}_{zz}$ gate per edge for MaxCut and a global phase that has no effect on any measurement. The trained quantity is the expected cost

$$E(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle \boldsymbol{\gamma}, \boldsymbol{\beta} | \hat{\mathcal{H}}_C | \boldsymbol{\gamma}, \boldsymbol{\beta} \rangle = \sum_{\mathbf{b}} \left| \langle \mathbf{b} | \boldsymbol{\gamma}, \boldsymbol{\beta} \rangle \right|^2 C(\mathbf{b}), \tag{13.14}$$

the average cost of the measured bit strings. Since $\hat{\mathcal{H}}_C$ is diagonal, it is estimated from the same computational-basis shots that deliver the candidates, with a single measurement setting. Each QAOA angle is shared by several gates. For MaxCut, γ_ℓ enters the 12 gates $\hat{R}_{zz}(2\gamma_\ell J_{ij})$ with $c_g = 2J_{ij} = 1$, and β_ℓ enters N gates $\hat{R}_x(2\beta_\ell)$ with $c_g = 2$. By (5.8) the two derivatives of one layer need $2 \times (12 + 8) = 40$ circuits on a device, and a full gradient $40p$ circuits.

13.7 Training and sampling

At $p = 1$ the expected cost depends on two angles and can be drawn as a map. The eigenvalues of $\hat{\mathcal{H}}_C$ are integers, so $\hat{U}_C(\gamma + 2\pi) = \hat{U}_C(\gamma)$. A shift $\beta \to \beta + \pi/2$ multiplies the mixing layer by $(-i)^N \hat{\Pi}$, which flips every bit and leaves every cut unchanged, and reversing both angles complex-conjugates the state. The rectangle $0 \le \gamma \le \pi$, $-\pi/4 \le \beta \le \pi/4$ therefore contains every value of E (Figure 83). At $\gamma = 0$ or $\beta = 0$ the state keeps the uniform distribution and the expected cut equals the random value of six. The best depth-one state cuts 8.31 edges on average.

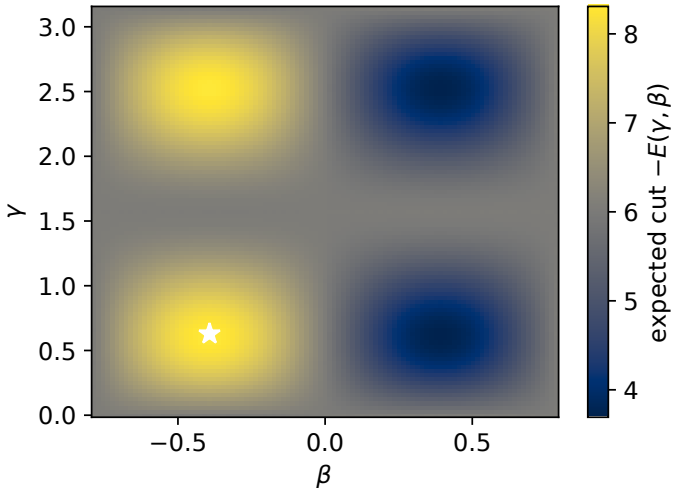


Figure 83: Expected cut $-E(\gamma, \beta)$ of the depth-one QAOA state (13.14) for the graph of Figure 81, on a grid of 101×101 angles, with the best grid point marked by a star. Exact state-vector values.

For $p > 1$ a grid is too expensive, since its size grows as the number of grid points per angle to the power $2p$. We minimise E with exact gradients and Adam from sixteen starting angle sets per depth, fifteen random and one linear ramp in which γ_ℓ grows and $|\beta_\ell|$ shrinks linearly with ℓ, and keep the best. We judge the trained state by the approximation ratio $r = -E/\mathrm{cut}_{\max}$, which compares the expected cut with the optimum and equals 0.6 for random guessing, by the probability P_{opt} of an optimal bit string and by M_{95}. Table 7 and Figure 84 show that each layer raises both measures. The probability of the optimum grows from 1/32 for random guessing to about one third at $p = 1$ and to 0.909 at $p = 4$, where the state puts almost all its weight on the maximum cut, and the number of shots needed to see an optimum falls from about a hundred to two. The landscape becomes harder to optimise with depth. Of the 16 starts, 16, 11, 3 and 1 reach the best expected cost within 10^{-3} at $p = 1$, 2, 3 and 4, so the $p = 4$ result rests on a single start.

In the deeper circuits (Figure 85) the trained cost angles γ_ℓ grow from the first layer to the last, and the mixing angles β_ℓ are negative and shrink in magnitude towards zero. The circuit starts with mostly mixing and ends with mostly cost evolution, the schedule of a slow interpolation from the mixing Hamiltonian to the cost Hamiltonian.

A device gives the distribution of Figure 84(b) only through shots, and Figure 86 shows how quickly an optimal bit string appears in a record of shots. We sample 1000 bit strings from the trained $p = 4$ state and the same number of uniformly random bit strings (Figure 87). The estimate of the expected cost from the shots, $\bar{C} = -9.870 \pm 0.016$, lies 2.5 standard errors from the exact value -9.830, a deviation that occurs in about one per cent of records, so the training signal is available

	expected cut $-E$	approximation ratio r	P_{opt}	M_{95}
random guessing	6.0	0.600	0.0312	95
$p = 1$	8.31	0.831	0.322	8
$p = 2$	9.25	0.925	0.522	5
$p = 3$	9.66	0.966	0.770	3
$p = 4$	9.83	0.983	0.909	2

Table 7: Trained QAOA states for MaxCut on the graph of Figure 81, with the maximum cut 10. Exact probabilities of the trained states. M_{95} is the number of shots for a 95 per cent chance of measuring an optimal bit string.

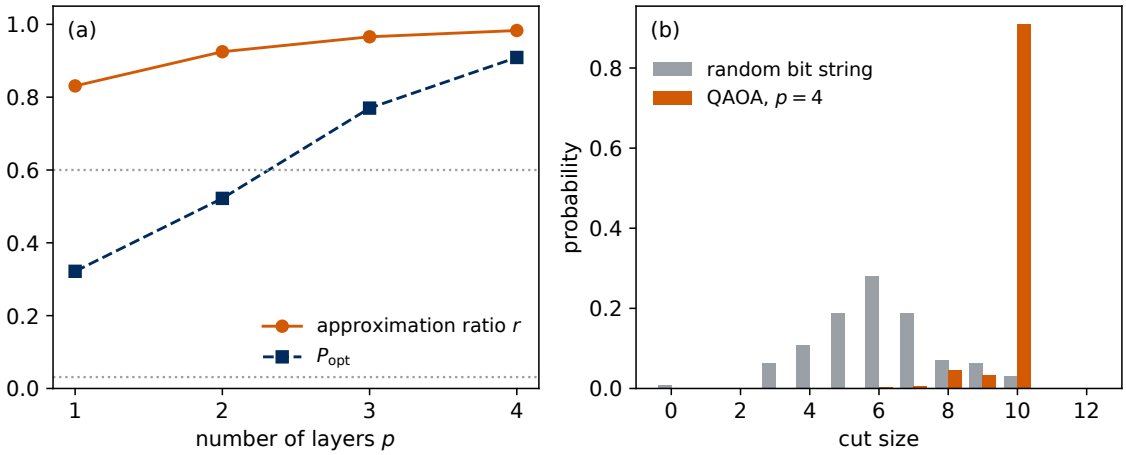


Figure 84: (a) Approximation ratio r (circles) and probability P_{opt} of an optimal bit string (squares) of the trained QAOA states against the number of layers p, with the random-guessing values 0.6 and 1/32 dotted. (b) Exact probability of each cut size for a uniformly random bit string (grey) and for the trained $p = 4$ state (orange).

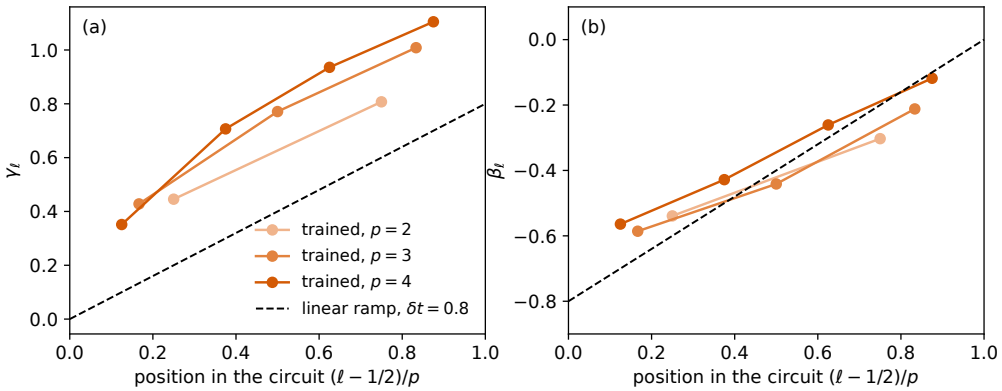


Figure 85: Trained QAOA angles for MaxCut, (a) γ_ℓ and (b) β_ℓ, against the position $(\ell - 1/2)/p$ of the layer in the circuit for $p = 2$, 3 and 4, compared with the linear annealing ramp (13.17) (black). The cost angles grow and the mixing angles shrink in magnitude along the circuit.

from the same records with the stated statistical error. An optimal bit string appears in QAOA shot 2, whereas the first optimal random bit string is shot 26. Both sources find the optimum within the 1000 shots, as expected for eight qubits. For applications the relevant quantity is the growth of the required number of shots with N, which is exponential for random guessing and depends for the QAOA on the problem class and on how the depth must grow with N.

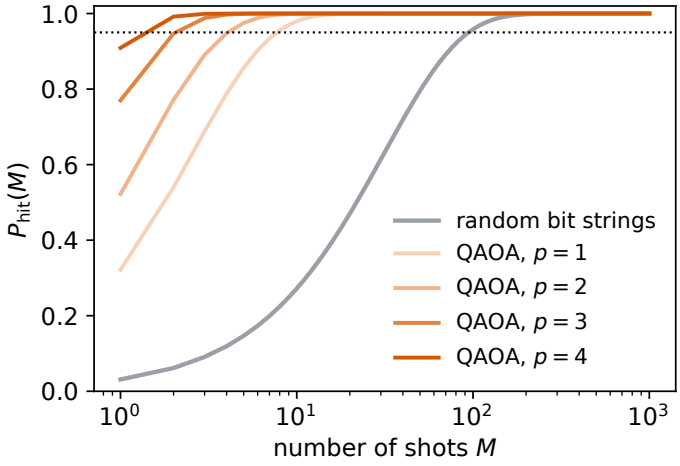


Figure 86: Probability $P_{\text{hit}}(M) = 1 - (1 - P_{\text{opt}})^M$ of measuring at least one optimal bit string in M shots (13.10), for random guessing and for the trained QAOA states, MaxCut on the graph of Figure 81. The dotted line marks 95 per cent.

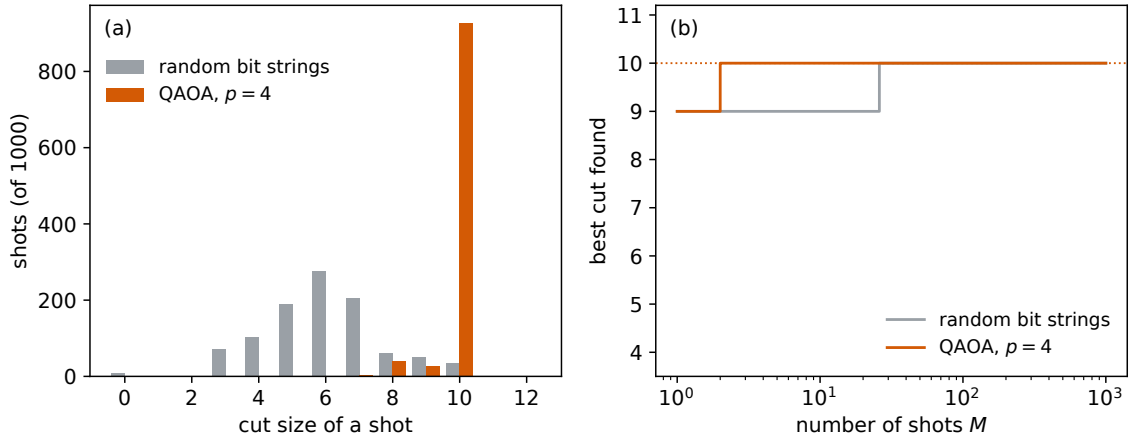


Figure 87: Sampling the trained $p = 4$ QAOA state. (a) Histogram of the cut sizes of 1000 measured bit strings for the QAOA state and for random bit strings. (b) Best cut among the first M shots, with the maximum cut dotted. Simulated measurement records.

13.8 Relation to adiabatic evolution and annealing

The structure of the QAOA follows from adiabatic evolution under the interpolating Hamiltonian

$$\hat{\mathcal{H}}(s) = -(1-s)\sum_i \hat{X}_i + s\,\hat{\mathcal{H}}_C, \qquad 0 \le s \le 1, \tag{13.15}$$

whose ground state at $s = 0$ is $|+\rangle^{\otimes N}$ and at $s = 1$ the optimal bit string. If s is increased slowly enough from 0 to 1, the adiabatic theorem keeps the state in the instantaneous ground state, and the evolution ends in the optimum [66]. Quantum annealing follows the same schedule on dedicated hardware [67]. Dividing the evolution into p steps of duration δt at the values $s_\ell = (\ell - 1/2)/p$ and splitting each step into its two groups of mutually commuting terms gives

$$e^{-i\delta t\hat{\mathcal{H}}(s_\ell)} \approx e^{+i(1-s_\ell)\delta t\sum_i \hat{X}_i}\, e^{-is_\ell\delta t\hat{\mathcal{H}}_C}, \tag{13.16}$$

with an error of order δt^2. The product is a QAOA layer (13.11) with

$$\gamma_\ell = s_\ell\,\delta t, \qquad \beta_\ell = -(1-s_\ell)\,\delta t, \tag{13.17}$$

the linear ramp used as one of the starting points of the training. As $p \to \infty$ the QAOA therefore contains the discretised adiabatic evolution, which reaches the optimum [65]. The trained angles at

small p show the same pattern, with γ_ℓ growing and $|\beta_\ell|$ shrinking along the circuit. At small depth the trained circuit need not follow the instantaneous ground state, since its angles are optimised only for the final expected cost.

14 Circuits as classifiers with quantum kernels

14.1 Classification after a feature map

In the preceding sections the circuit prepared a state whose energy or whose distribution of bit strings was the result. Parametrised circuits were soon used as general machine-learning models, in which a data point enters the circuit through its gate angles and the output is a prediction [23], and circuits that classify labelled data were proposed in Refs. [68, 69]. Classification is the standard task of this kind. A training set $\mathcal{D}$ contains $|\mathcal{D}|$ points $\mathbf{x}_i \in \mathbb{R}^2$ with labels $y_i \in \{-1, +1\}$, and the classifier must predict the label of new points drawn from the same distribution. As a running example we take two noisy concentric rings in the square $[-1, 1]^2$, the inner ring with label $+1$ and the outer one with label -1.

A linear classifier assigns the label by the side of a straight line on which a point lies, and no straight line separates two concentric rings. A new coordinate changes this (Figure 88). The map $\boldsymbol{\phi}(\mathbf{x}) = (x_1, x_2, x_1^2 + x_2^2)$ lifts every point to a height equal to its squared distance from the centre, the inner ring ends low and the outer ring high, and a horizontal plane separates them. Such a map into a space of higher dimension is called a *feature map*, and the vector $\boldsymbol{\phi}(\mathbf{x})$ is the feature vector of the point. With a suitable feature map a hyperplane is enough to classify the data.

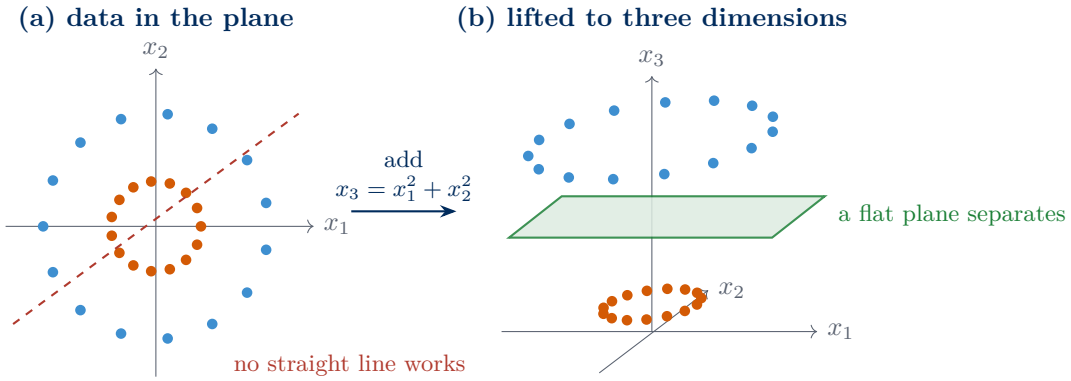


Figure 88: Two rings that no straight line separates in the plane (a) become separable by a plane after the feature map $\boldsymbol{\phi}(\mathbf{x}) = (x_1, x_2, x_1^2 + x_2^2)$ lifts the inner ring low and the outer ring high (b).

14.2 The kernel trick and the support-vector machine

A support-vector machine (SVM) [70] finds the separating hyperplane $\mathbf{w} \cdot \boldsymbol{\phi}(\mathbf{x}) + b = 0$ in feature space that keeps the largest distance, the margin, to the nearest training points of both classes. The margin equals $2/\|\mathbf{w}\|$ when the hyperplane is scaled so that $y_i(\mathbf{w} \cdot \boldsymbol{\phi}(\mathbf{x}_i) + b) \geq 1$ for all training points, and the SVM therefore minimises $\frac{1}{2}\|\mathbf{w}\|^2$ under these constraints. Overlapping classes are handled by allowing violations $\xi_i \geq 0$ of the constraints at the price $C \sum_i \xi_i$, where the regularisation constant C is chosen by cross-validation. The minimiser has the form $\mathbf{w} = \sum_i \alpha_i y_i \boldsymbol{\phi}(\mathbf{x}_i)$ with weights $0 \leq \alpha_i \leq C$, which follow from the convex quadratic problem

$$\max_{\boldsymbol{\alpha}} \; \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j \, y_i y_j \, K_{ij}, \qquad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0, \tag{14.1}$$

and the trained classifier is

$$f(\mathbf{x}) = \operatorname{sign}\Big(\sum_{i=1}^{|\mathcal{D}|} \alpha_i \, y_i \, k(\mathbf{x}_i, \mathbf{x}) + b \Big). \tag{14.2}$$

Both expressions contain the data only through inner products of feature vectors, the *kernel*

$$k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x}) \cdot \boldsymbol{\phi}(\mathbf{x}'), \qquad K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j), \tag{14.3}$$

where K is the $|\mathcal{D}| \times |\mathcal{D}|$ kernel matrix of the training points. The feature vectors themselves are never needed, which allows feature spaces of very high or infinite dimension as long as the kernel can be computed. This replacement of feature vectors by a kernel is the kernel trick [71]. The problem (14.1) is convex whenever K is positive semidefinite, so it has one global optimum that standard software finds. Most weights α_i vanish, and the training points with $\alpha_i > 0$, which lie on or inside the margin, are the support vectors. A kernel acts as a similarity measure, large for points that the feature map sends to nearby vectors, and the classifier (14.2) labels a new point by a weighted vote of its similarities to the support vectors. For the lift of Figure 88 the kernel is $k(\mathbf{x}, \mathbf{x}') = \mathbf{x} \cdot \mathbf{x}' + |\mathbf{x}|^2 |\mathbf{x}'|^2$.

14.3 Quantum states as feature vectors

A circuit $\hat{U}(\mathbf{x})$ whose gate angles depend on the data maps every point to a state $|\phi(\mathbf{x})\rangle = \hat{U}(\mathbf{x})|0 \ldots 0\rangle$ in a space of dimension 2^N [4, 5]. Its overlaps define the *quantum kernel*

$$k(\mathbf{x}, \mathbf{x}') = \left|\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle\right|^2 = \mathrm{Tr}\left[\hat{\rho}(\mathbf{x})\, \hat{\rho}(\mathbf{x}')\right], \qquad \hat{\rho}(\mathbf{x}) = |\phi(\mathbf{x})\rangle\langle\phi(\mathbf{x})|. \tag{14.4}$$

The second form shows that the feature vector is the density matrix $\hat{\rho}(\mathbf{x})$, an element of the real vector space of Hermitian matrices with the inner product $\mathrm{Tr}(\hat{A}\hat{B})$, so the kernel matrix is positive semidefinite. The squared modulus removes the global phase of the state, which has no physical meaning. The device never writes down a feature vector and only measures the similarity of two states, and a classical SVM uses the measured kernel matrix (Figure 89). Two variants are in use. In a *fixed* quantum kernel the encoding circuit $\hat{U}(\mathbf{x})$ is chosen in advance and only the convex SVM problem is solved. In a *trainable* quantum kernel the encoding $\hat{U}(\mathbf{x}; \boldsymbol{\theta})$ carries free parameters, which are first tuned so that points of the same class become similar, and the SVM is solved afterwards.

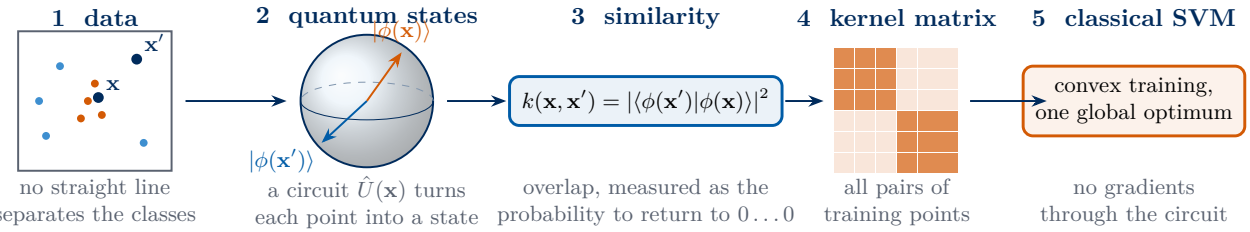


Figure 89: A quantum kernel. Every data point is mapped by a circuit to a quantum state, the device measures the overlap of two states, the overlaps form the kernel matrix, and a classical support-vector machine finds the classifier.

A kernel entry is measured by the inversion test. The register is prepared by $\hat{U}(\mathbf{x})$, the inverse circuit $\hat{U}(\mathbf{x}')^\dagger$ is applied, and all qubits are measured. The probability of the outcome $0 \ldots 0$ is

$$\left|\langle 0 \ldots 0 | \hat{U}(\mathbf{x}')^\dagger \hat{U}(\mathbf{x}) | 0 \ldots 0 \rangle\right|^2 = \left|\langle \phi(\mathbf{x}') | \phi(\mathbf{x}) \rangle\right|^2 = k(\mathbf{x}, \mathbf{x}'), \tag{14.5}$$

and its frequency $\tilde{k}$ in S shots estimates the kernel. The number of outcomes $0 \ldots 0$ follows a binomial distribution, so the statistical error of an entry is $\sqrt{k(1-k)/S}$. The diagonal entries equal one and K is symmetric, so a training set needs $|\mathcal{D}|(|\mathcal{D}| - 1)/2$ distinct circuits, a number that grows quadratically with the size of the data set.

14.4 A fixed quantum kernel

For the two rings each coordinate rotates its own qubit,

$$|\phi(\mathbf{x})\rangle = \hat{R}_y(\pi x_1)|0\rangle \otimes \hat{R}_y(\pi x_2)|0\rangle. \tag{14.6}$$

The rotations about one axis combine, $\hat{R}_y(a)^\dagger \hat{R}_y(b) = \hat{R}_y(b-a)$, and $\langle 0|\hat{R}_y(\vartheta)|0\rangle = \cos(\vartheta/2)$, so the overlap factorises into

$$k(\mathbf{x}, \mathbf{x}') = \cos^2 \frac{\pi(x_1 - x_1')}{2} \cos^2 \frac{\pi(x_2 - x_2')}{2}. \tag{14.7}$$

The kernel equals one for identical points and decreases as the points move apart in either coordinate. The data set contains two noisy rings with 200 training and 200 test points, and the constant C is chosen by five-fold cross-validation. A linear SVM reaches a test accuracy of 0.54, close to guessing, whereas the SVM with the quantum kernel (14.7) reaches 0.995 with 120 support vectors and draws a closed boundary around the inner ring (Figure 90). This kernel is a product of cosines and is computed classically just as easily, so the example shows the mechanism of a quantum kernel and makes no claim of a quantum advantage.

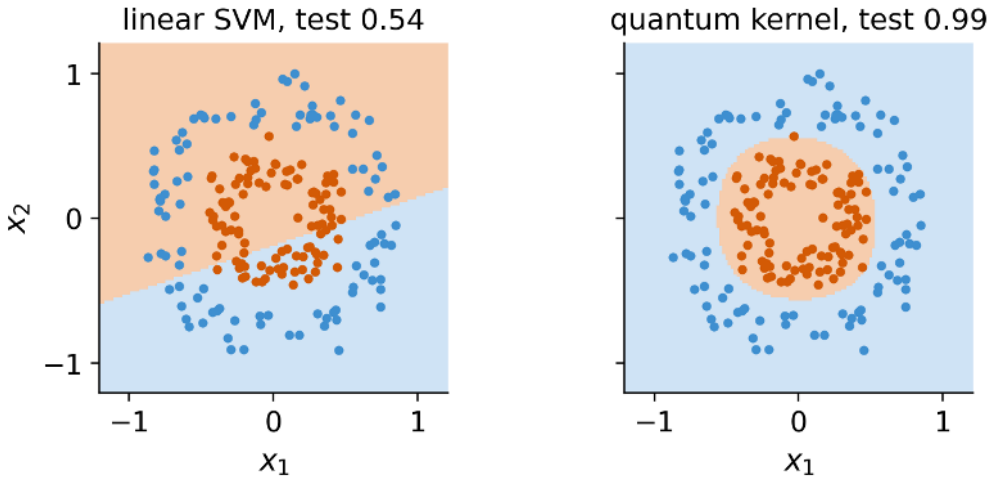


Figure 90: Classification of two noisy rings, inner ring orange and outer ring blue, by a linear SVM (left) and by an SVM with the fixed quantum kernel (14.7) (right). The shading shows the predicted class, and the titles give the test accuracies on 200 test points. Exact kernel values.

Figure 91 shows the kernel matrix of the training points sorted by class. Points of the same ring are similar and form bright blocks, and points of different rings are dissimilar. Estimating every entry by the inversion test with $S = 100$ shots gives a root-mean-square error of 0.035, close to the binomial value, and the test accuracy stays at 0.995. The estimated matrix is no longer positive semidefinite, since shot noise produces negative eigenvalues, and the SVM still finds a good classifier. The training matrix requires 19 900 circuits of 100 shots each.

14.5 Exponential concentration

A feature map that spreads the data over the whole register sends different points to nearly unrelated states. Two independent random states of N qubits have the average overlap

$$\big\langle |\langle \phi|\psi\rangle|^2 \big\rangle = \frac{1}{d}, \qquad d = 2^N. \tag{14.8}$$

To see this, fix $|\phi\rangle$ as the first vector of an orthonormal basis. A random state has no preferred direction, so its squared overlaps with all d basis vectors have the same average, and they sum to one. If the feature map behaves like a random state for every input, all off-diagonal kernel values shrink as 2^{-N}, the kernel matrix approaches the identity, and every point looks equally unlike every other. Such a kernel carries no information about the classes [72].

Concentration also makes the kernel expensive to measure. The relative statistical error of an entry estimated from S shots is $\sqrt{k(1-k)/S}/k \approx 1/\sqrt{kS}$ for small k, so a relative error ϵ needs $S \approx 1/(k\epsilon^2)$ shots, which grows as $2^N/\epsilon^2$. To test this, we encode data with a deep circuit of $L = N$ layers, each applying rotations $\hat{R}_z \hat{R}_y$ to every qubit followed by a chain of $\widehat{\mathrm{CZ}}$ gates. Every input

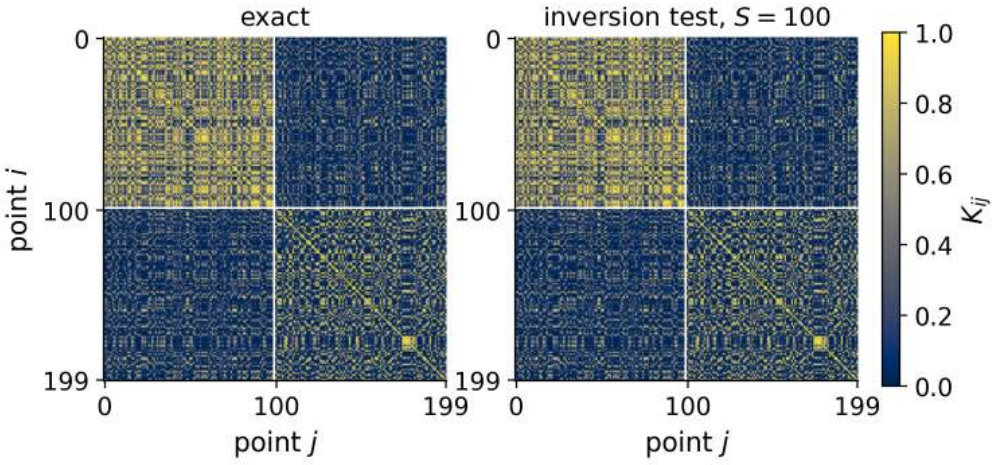


Figure 91: Kernel matrix K_{ij} of the 200 training points, sorted by class, computed exactly (left) and estimated by the inversion test (14.5) with $S = 100$ shots per entry (right). Simulated measurement records in the right panel.

point supplies all $2NL$ rotation angles, and we draw 400 pairs of input points with angles uniform in $[0, 2\pi)$ and compute their kernel values (Figure 92). The mean and the standard deviation of the kernel both follow 2^{-N}, with a fitted decay $2^{-0.98N}$, and an entry at $N = 10$ needs about 9×10^4 shots for a relative error of ten per cent. The same concentration of random states limits the training of deep circuits (Section 16). A useful feature map must keep similar points similar.

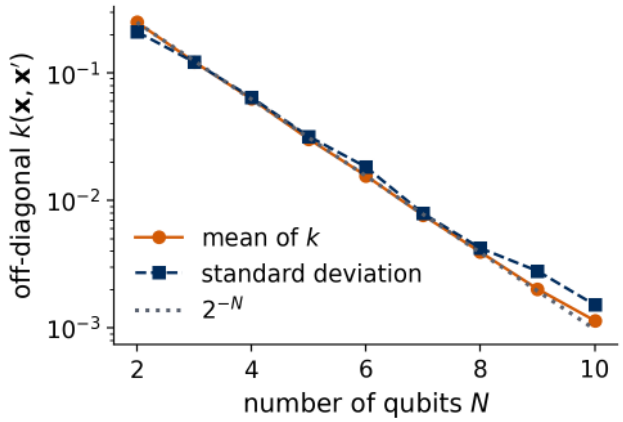


Figure 92: Exponential concentration of a deep quantum kernel. Mean (circles) and standard deviation (squares) of the off-diagonal kernel values of 400 random pairs of points, encoded by $L = N$ layers of rotations with uniformly random angles and $\widehat{\mathrm{CZ}}$ chains, against the number of qubits N, with the line 2^{-N} dotted. Exact state vectors.

14.6 A trainable quantum kernel

A trainable kernel adjusts the encoding to the task. We give the rotations of (14.6) free scales, $\hat{R}_y(w_1 x_1) \otimes \hat{R}_y(w_2 x_2)$, which turns π in (14.7) into w_1 and w_2. The scales are tuned to maximise the *kernel-target alignment* [73], which compares the kernel matrix with the ideal similarity $y_i y_j$, equal to $+1$ for points of the same class and -1 otherwise,

$$A(\mathbf{w}) = \frac{\sum_{i,j} K_{ij}(\mathbf{w})\, y_i y_j}{\|K(\mathbf{w})\|_F \, \|\mathbf{y}\mathbf{y}^T\|_F}, \qquad \|K\|_F^2 = \sum_{i,j} K_{ij}^2, \tag{14.9}$$

the cosine of the angle between the two matrices read as vectors of $|\mathcal{D}|^2$ entries. The alignment is a differentiable function of the encoding parameters and needs no SVM, so it can be maximised by

gradient ascent before the convex problem is solved [74]. For a quantum kernel, whose entries are non-negative, the alignment of balanced classes cannot exceed $1/\sqrt{2}$. The numerator is at most the sum of the entries over pairs of the same class, and by the Cauchy–Schwarz inequality this sum is at most $\|K\|_F$ times the square root of the number $|\mathcal{D}|^2/2$ of such pairs, while $\|\mathbf{y}\mathbf{y}^T\|_F = |\mathcal{D}|$.

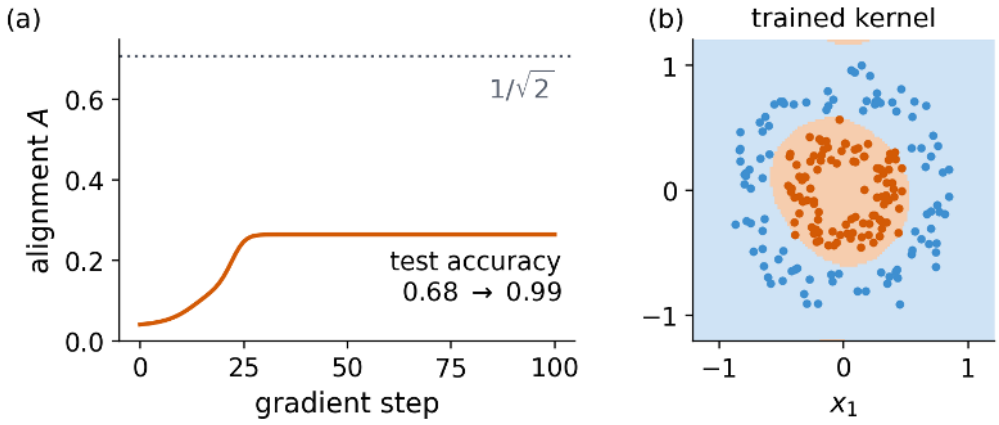


Figure 93: Training the scales of the encoding. (a) Kernel-target alignment (14.9) against the gradient step, starting from $w_1 = w_2 = 2\pi$, with the bound $1/\sqrt{2}$ for balanced classes dotted. (b) Decision regions of the SVM with the trained kernel and the training points. Exact kernel values.

We start from the poor scales $w_1 = w_2 = 2\pi$, at which the kernel oscillates twice as fast as in (14.7) and many points of the same ring look dissimilar. The alignment is then 0.04 and the SVM reaches a test accuracy of 0.68. After 100 steps of gradient ascent the scales settle at $w_1 \approx 1.11\pi$ and $w_2 \approx 1.13\pi$, close to the value π used above, the alignment rises to 0.27 and the test accuracy to 0.99 (Figure 93). Only the embedding is trained, and the SVM on the trained kernel remains a convex problem with a unique optimum.

15 Circuits as classifiers with data re-uploading

15.1 A single qubit as a classifier

A quantum kernel uses the circuit only to compute similarities and leaves the decision to a classical SVM. A circuit can also produce the label itself, when the data enter as gate angles and a measurement returns the class. We study how much the smallest possible quantum model, a single qubit, can learn in this way.

Classification task

The data are points $\mathbf{x} = (x_1, x_2)$ drawn uniformly from the square $[-1, 1]^2$. A point has the label $y = 1$ when it lies inside the circle of radius $\sqrt{2/\pi}$ and $y = 0$ otherwise. The circle covers half of the area of the square, so guessing gives an accuracy of one half. We look for a circuit of one qubit that returns the label of a new point, and success is the accuracy on new points. A second task with three classes, a disc, a ring and the outside region, follows in Section 15.9.

15.2 A qubit state as a point on the Bloch sphere

Every pure state of one qubit can be written, up to a global phase, with two angles,

$$|\psi\rangle = \cos\frac{\vartheta}{2}\,|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}\,|1\rangle, \qquad 0 \le \vartheta \le \pi, \quad 0 \le \varphi < 2\pi. \tag{15.1}$$

The angles are the polar angle ϑ, measured from the north pole, and the azimuthal angle φ of a point on the unit sphere, the Bloch sphere, with the Bloch vector $\mathbf{n} = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$

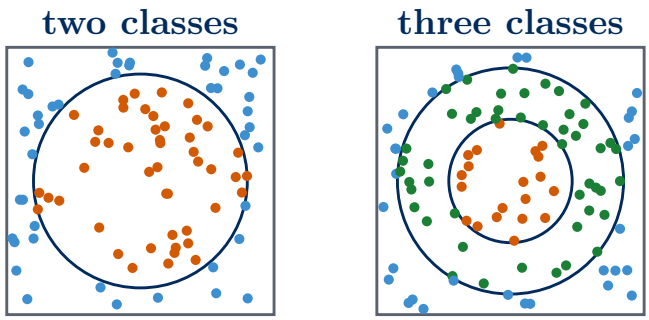


Figure 94: The two classification tasks. Left, points inside the circle form class 1 and points outside class 0. Right, three classes, a disc, a ring and the outside region.

(Figure 95). The north pole is $|0\rangle$, the south pole $|1\rangle$, and the points on the equator are equal superpositions of the two. A measurement in the computational basis returns 0 with probability $\cos^2(\vartheta/2)$, so the north pole always gives 0, the south pole always gives 1 and the equator gives both outcomes with equal probability.

Gates move the point on the sphere. The rotation $\hat{R}_y(\alpha) = e^{-i\alpha\hat{Y}/2}$ tilts the Bloch vector by the angle α about the y axis, and $\hat{R}_z(\alpha) = e^{-i\alpha\hat{Z}/2}$ turns it by α about the z axis, which changes φ and leaves the measurement probabilities unchanged. Every rotation of the sphere, and hence every single-qubit gate up to a global phase, is a product of three such rotations,

$$\hat{U}(\boldsymbol{\phi}) = \hat{R}_z(\phi_3)\,\hat{R}_y(\phi_2)\,\hat{R}_z(\phi_1), \tag{15.2}$$

with the Euler angles $\boldsymbol{\phi} = (\phi_1, \phi_2, \phi_3)$.

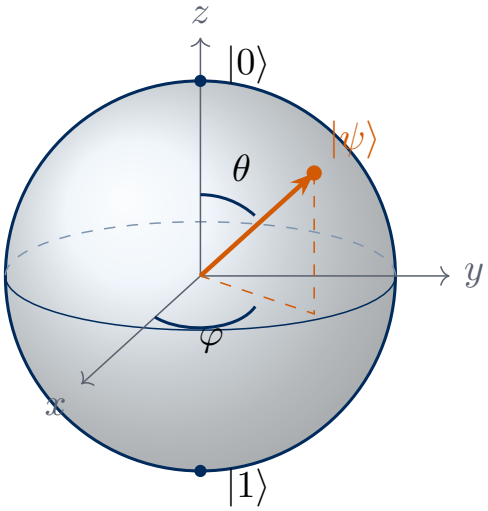


Figure 95: A qubit state (15.1) as a point on the Bloch sphere, with the polar angle ϑ (drawn as θ) from the north pole $|0\rangle$ and the azimuthal angle φ about the z axis.

15.3 Data as rotation angles

A data point enters the qubit through the angles of the gate (15.2). We add the point to trainable angles, $\boldsymbol{\phi} = \boldsymbol{\theta} + \mathbf{w} \circ \mathbf{x}$, with the three-component input $\mathbf{x} = (x_1, x_2, 0)$, trainable offsets $\boldsymbol{\theta}$ and weights $\mathbf{w}$, and $\circ$ the product of vectors taken component by component,

$$|\psi(\mathbf{x})\rangle = \hat{U}(\boldsymbol{\theta} + \mathbf{w} \circ \mathbf{x})|0\rangle. \tag{15.3}$$

Every data point thus becomes a point on the Bloch sphere, and a classifier of this kind is a map from the plane to the sphere. Training tunes $\boldsymbol{\theta}$ and $\mathbf{w}$ so that the points of the two classes land in different regions of the sphere.

15.4 Classes as label states

The class is read from the final state by comparing it with fixed *label states*, one state $|\psi_c\rangle$ for every class c. The label states should be as distinguishable as possible, i.e. as far apart on the Bloch sphere as possible, because the overlap of two states with Bloch vectors $\mathbf{n}$ and $\mathbf{n}'$ is

$$|\langle\psi|\psi'\rangle|^2 = \frac{1+\mathbf{n}\cdot\mathbf{n}'}{2}, \tag{15.4}$$

which is smallest for opposite Bloch vectors. For two classes the label states are the poles, $|0\rangle$ for class 0 and $|1\rangle$ for class 1, which are orthogonal and have fidelity zero (Figure 96). A qubit has at most two mutually orthogonal states, so for more classes the label states are spread evenly over the sphere. Three classes use three Bloch vectors 120° apart on a great circle, whose scalar product $-1/2$ gives by (15.4) the pairwise fidelity $\cos^2 60° = 1/4$. Four classes use the corners of a regular tetrahedron inscribed in the sphere, whose Bloch vectors have the pairwise scalar product $-1/3$ and therefore the pairwise fidelity $1/3$, the smallest possible for four states of a qubit.

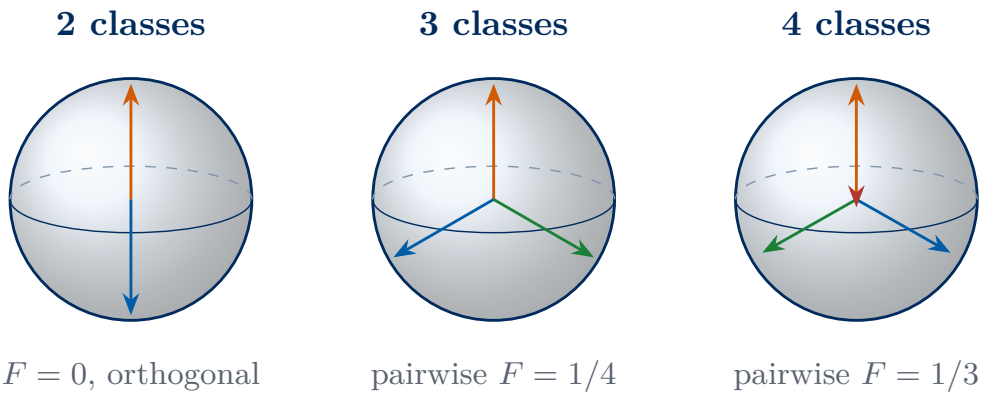


Figure 96: Label states for two, three and four classes, spread as far apart as possible on the Bloch sphere. Two classes use the poles, which are orthogonal. Three classes use Bloch vectors 120° apart on a great circle, with pairwise fidelity 1/4, and four classes the corners of a regular tetrahedron, with pairwise fidelity 1/3.

The fidelity of the final state with label state c decides the class,

$$F_c(\mathbf{x}) = \big|\langle\psi_c|\psi(\mathbf{x})\rangle\big|^2, \qquad \hat{y}(\mathbf{x}) = \arg\max_c F_c(\mathbf{x}), \tag{15.5}$$

so a point is assigned the class whose label state is closest to its final state. On a device $F_c(\mathbf{x})$ is the frequency with which a measurement in a basis containing $|\psi_c\rangle$ returns $|\psi_c\rangle$ in repeated runs at the same input. Training pulls every training point towards the label state of its own class by minimising the average infidelity over the training set $\mathcal{D}$ of points $\mathbf{x}_m$ with labels y_m [75],

$$\mathcal{L}(\boldsymbol{\theta},\mathbf{w}) = \frac{1}{|\mathcal{D}|}\sum_{m=1}^{|\mathcal{D}|}\big[1 - F_{y_m}(\mathbf{x}_m)\big]. \tag{15.6}$$

For two classes with the label states $|0\rangle$ and $|1\rangle$ the two fidelities are $F_1 = P(1\,|\,\mathbf{x})$, the probability of measuring one, and $F_0 = 1 - P(1\,|\,\mathbf{x})$. The rule (15.5) then compares a single probability with one half,

$$P(1\,|\,\mathbf{x}) = \big|\langle 1|\psi(\mathbf{x})\rangle\big|^2, \qquad \hat{y}(\mathbf{x}) = \begin{cases} 1, & P(1\,|\,\mathbf{x}) > \tfrac{1}{2}, \\ 0, & \text{otherwise}, \end{cases} \tag{15.7}$$

and the loss (15.6) becomes

$$\mathcal{L}(\boldsymbol{\theta}, \mathbf{w}) = \frac{1}{|\mathcal{D}|} \sum_{m=1}^{|\mathcal{D}|} \Big[y_m \big(1 - P(1 \,|\, \mathbf{x}_m)\big) + (1 - y_m)\, P(1 \,|\, \mathbf{x}_m) \Big], \tag{15.8}$$

where the first term asks the points of class 1 to reach $|1\rangle$ and the second asks the points of class 0 to stay near $|0\rangle$. Training sends each class towards its own pole, and the decision boundary consists of the points mapped onto the equator.

15.5 A single rotation draws a band

In a single gate (15.3), the first rotation $\hat{R}_z$ only multiplies $|0\rangle$ by a phase, the rotation $\hat{R}_y(\phi_2)$ gives $|1\rangle$ the probability $\sin^2(\phi_2/2)$, and the last $\hat{R}_z$ changes no probability. The probability of reading one,

$$P(1 \,|\, \mathbf{x}) = \sin^2 \frac{\theta_2 + w_2 x_2}{2}, \tag{15.9}$$

depends on x_2 alone, so the classifier can only cut out a horizontal band of the plane (Figure 101, left). The best band, $|x_2| < 0.58$, reaches a test accuracy of 0.76.

15.6 Loading the data repeatedly

Data re-uploading [75] loads the data again in every layer, each time with its own offsets and weights (Figure 97),

$$|\psi(\mathbf{x})\rangle = \hat{U}(\boldsymbol{\theta}_L + \mathbf{w}_L \circ \mathbf{x}) \cdots \hat{U}(\boldsymbol{\theta}_2 + \mathbf{w}_2 \circ \mathbf{x})\, \hat{U}(\boldsymbol{\theta}_1 + \mathbf{w}_1 \circ \mathbf{x}) |0\rangle. \tag{15.10}$$

The L layers carry $6L$ trainable numbers. As an analogy, each layer acts like a neuron of a network, an affine function $\boldsymbol{\theta}_\ell + \mathbf{w}_\ell \circ \mathbf{x}$ of the input followed by a nonlinear operation, here a rotation of the Bloch vector. Successive rotations do not commute, so the input enters the final state through nested trigonometric functions of increasing complexity.

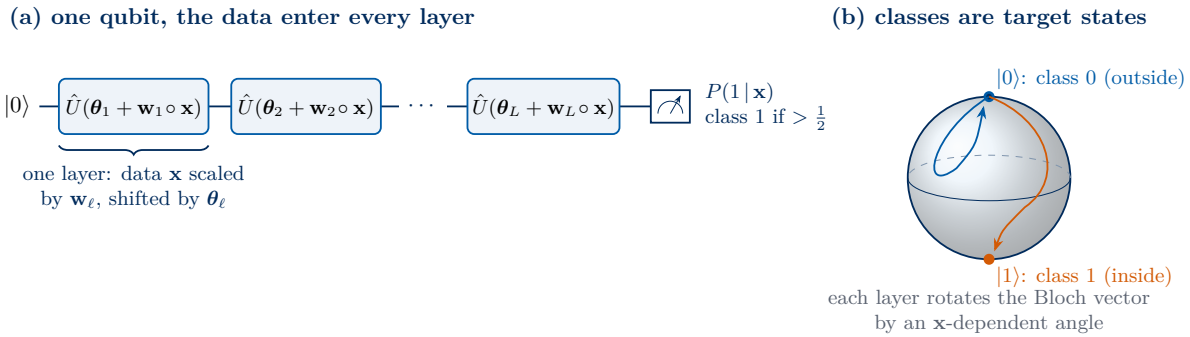


Figure 97: Data re-uploading. (a) A single qubit passes through L layers $\hat{U}(\boldsymbol{\theta}_\ell + \mathbf{w}_\ell \circ \mathbf{x})$, each loading the data point again with its own trainable offsets and weights, and is measured at the end. (b) On the Bloch sphere every data point follows its own sequence of rotations from $|0\rangle$, and training sends the points of class 0 towards the north pole and those of class 1 towards the south pole.

15.7 Re-uploading as a Fourier series

The class of a point is decided by the function $f(\mathbf{x}) = P(1 \,|\, \mathbf{x})$ through the rule $f(\mathbf{x}) > 1/2$, so the decision boundary is the level set $f(\mathbf{x}) = 1/2$. A model can therefore draw only those boundaries that are level sets of functions in its family, and the size of this family is its *expressive power*. For neural networks the universal approximation theorem states which functions the family contains. For a re-uploading circuit the family follows from Fourier analysis [76]. The frequencies of f set how fast it can change with $\mathbf{x}$. Low frequencies give smooth functions with simple boundaries, and high frequencies allow strongly curved boundaries and several separate ones.

We take a single input x loaded by the gate $\hat{S}(x) = e^{-ix\hat{Z}/2} = \mathrm{diag}(e^{-ix/2}, e^{ix/2})$, with trainable gates $\hat{W}_0, \ldots, \hat{W}_L$ between the L uploads,

$$|\psi(x)\rangle = \hat{W}_L\, \hat{S}(x)\, \hat{W}_{L-1} \cdots \hat{W}_1\, \hat{S}(x)\, \hat{W}_0 |0\rangle. \tag{15.11}$$

Inserting the identity $\hat{I} = |0\rangle\langle 0| + |1\rangle\langle 1|$ before every $\hat{S}(x)$ writes each amplitude as a sum over the 2^L sequences of basis states visited at the uploads. Every upload multiplies a term by $e^{-ix/2}$ when the qubit is in $|0\rangle$ and by $e^{+ix/2}$ when it is in $|1\rangle$. A sequence with s visits to $|1\rangle$ and $L - s$ visits to $|0\rangle$ therefore carries the factor $e^{-i(L-2s)x/2}$, and every amplitude is a combination of the exponentials $e^{-ikx/2}$ with $k \in \{-L, -L+2, \ldots, L\}$, with coefficients fixed by the trainable gates. An expectation value $f(x) = \langle\psi(x)|\hat{O}|\psi(x)\rangle$ of an observable $\hat{O}$ contains products of one amplitude and one complex-conjugate amplitude, whose exponents combine to $e^{i(k-k')x/2}$. The difference $(k - k')/2$ is an integer between $-L$ and L, so

$$f(x) = \sum_{\omega=-L}^{L} c_\omega\, e^{i\omega x}, \qquad c_{-\omega} = c_\omega^*, \tag{15.12}$$

where the second relation expresses that f is real. The circuit computes a truncated Fourier series. The encoding fixes the available frequencies, up to L for L uploads, the trainable gates set the coefficients c_ω, and a trainable weight w in $\hat{S}(wx)$ rescales every frequency to ωw. A single upload gives at most a cosine of one frequency, which is the origin of the band (15.9).

Figure 98 tests (15.12) on the target $f(x) = 0.05 + 0.35\cos x + 0.25\sin 2x - 0.3\cos 3x$, with the observable $\hat{Z}$ and general single-qubit gates (15.2) as $\hat{W}_\ell$. With one and two uploads the trained circuit reproduces only the frequencies it contains, and with three uploads it fits the target to numerical precision.

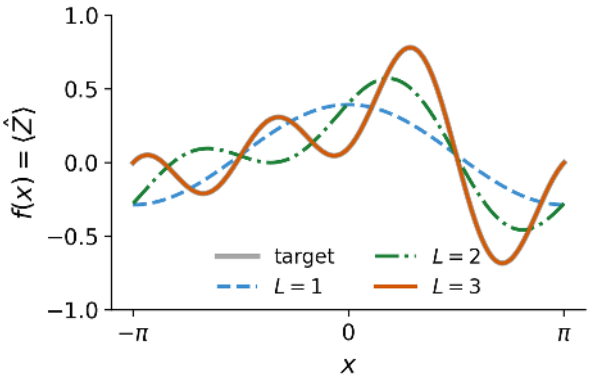


Figure 98: A single-qubit re-uploading model as a Fourier series. The target with the frequencies 1, 2 and 3 (grey) and the best fits $f(x) = \langle\hat{Z}\rangle$ with $L = 1$, 2 and 3 uploads. Only $L = 3$ contains frequency 3 and fits the target. Exact state-vector simulation.

The classifier compares the Fourier series with one half, and the boundary lies where the curve crosses this level (Figure 99). A single frequency crosses it at most twice per period and splits the input range into two intervals, whereas higher frequencies give more crossings and more class regions. The two-dimensional classifier (15.10) follows the same rule. The coordinate x_1 enters through $\hat{R}_z$ and x_2 through $\hat{R}_y$, which has the same eigenvalues $e^{\mp i\alpha/2}$ in its own eigenbasis, so $P(1\,|\,\mathbf{x})$ is a trigonometric polynomial in x_1 and x_2 whose frequencies grow with L.

15.8 A universal classifier

For a single input variable a single-qubit model can express any Fourier series of the form (15.12) and therefore approximate any square-integrable function [77]. For several input variables the number of multivariate Fourier terms grows exponentially with the number of variables, whereas a pure

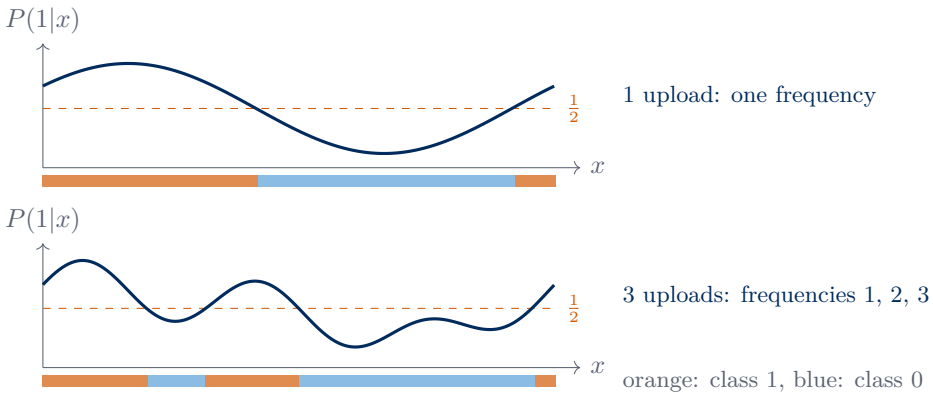


Figure 99: A Fourier series draws the decision boundary. The output $P(1\,|\,x)$ of one upload (top, one frequency) and of three uploads (bottom, frequencies 1, 2 and 3) against the input x, with the threshold $1/2$ dashed. The strips below each curve show the predicted classes, orange for class 1 and blue for class 0.

single-qubit model has a number of parameters that grows only linearly with the number of layers, so it cannot express all functions [77]. Trainable classical weights $\mathbf{w}_\ell$ that rescale the data in every layer, as in (15.10), restore universality [78], partly because these weights are themselves a classical part of the model. We train the circuit (15.10) on the circle task with 400 training points by minimising (15.8) with Adam and exact gradients, keep the best of four starts for each depth and test it on 4000 new points. The test accuracy grows from 0.73 at $L = 1$, a band slightly worse than the best one, to 0.94 at $L = 6$, and the training and test accuracies stay close to each other at every depth (Figure 100). At $L = 6$ the learned region follows the circle closely (Figure 101).

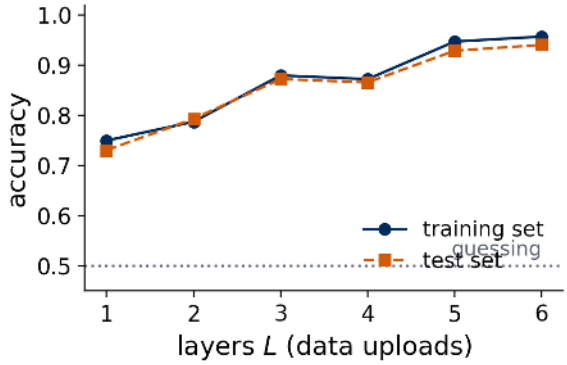


Figure 100: Accuracy of the single-qubit re-uploading classifier on the training set (circles) and on the test set (squares) against the number L of layers, for the circle task with 400 training and 4000 test points, best of four starts per depth. The dotted line marks guessing. Exact state-vector simulation with exact gradients.

15.9 Three classes

The label states of Section 15.4 extend the classifier to more classes. We divide the square into a disc $|\mathbf{x}| < 0.45$ (class 0), a ring $0.45 < |\mathbf{x}| < 0.8$ (class 1) and the outside region (class 2), with 600 training and 4000 test points (Figure 94, right). The label states $|\psi_c\rangle$ are three states whose Bloch vectors lie 120° apart on the great circle in the x–z plane (Figure 96). We predict with the largest fidelity (15.5) and train the loss (15.6) with Adam, keeping the best of four starts per depth. The classes cover the fractions 0.16, 0.34 and 0.50 of the square, and always answering the largest class gives a test accuracy of 0.49, above the value $1/3$ of random guessing.

The test accuracy rises from 0.57 at $L = 1$ and 0.70 at $L = 2$ to 0.88 at $L = 4$ and saturates near 0.89 at $L = 6$ and $L = 8$ (Figure 102). With enough layers the classifier draws both circular boundaries, and its errors lie along them and near the centre of the disc.

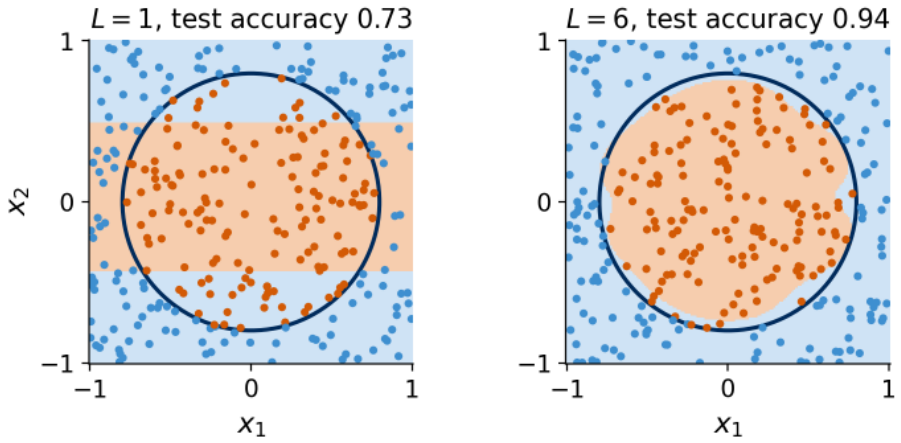


Figure 101: Predicted classes (shading, orange for class 1) of the single-qubit classifier with $L = 1$ (left) and $L = 6$ (right) layers, with training points and the true circle. One layer draws a band, six layers follow the circle.

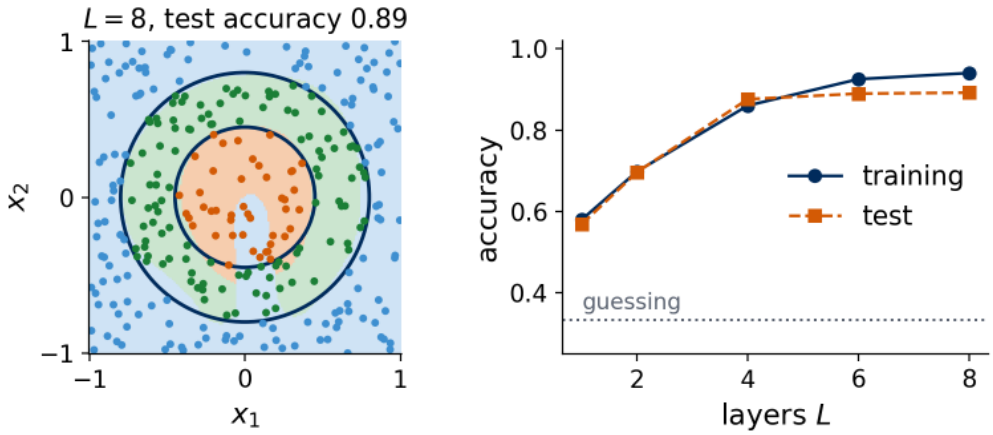


Figure 102: Single-qubit re-uploading classifier for three classes, a disc, a ring and the outside region. Left, predicted classes (shading) with $L = 8$ layers, the training points and the true boundaries. Right, training and test accuracy against the number L of layers, best of four starts per depth, with random guessing at 1/3 dotted. Exact state-vector simulation.

15.10 Classifiers with several qubits

A register of N qubits offers more label states and more places to load the data. Its 2^N computational basis states are mutually orthogonal, so up to 2^N classes receive label states with zero pairwise fidelity, whereas the single-qubit label states for more than two classes overlap (Figure 96). Four classes on two qubits use $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$. The fidelity (15.5) of the final state with a basis state is the probability of the corresponding bit string, so repeated measurements of all qubits estimate every F_c at once, and the predicted class is the most frequent bit string.

The multi-qubit classifiers of Ref. [75] load the data into every qubit in every layer, each qubit with its own offsets and weights, and entangle the qubits with $\widehat{\mathrm{CZ}}$ gates after each layer except the last (Figure 103). The $\widehat{\mathrm{CZ}}$ gate of Section 6.2 multiplies the amplitude of $|11\rangle$ by -1. On four qubits the gates act on the pairs $(1, 2)$ and $(3, 4)$ in one layer and on $(2, 3)$ and $(1, 4)$ in the next, in the qubit numbering of Ref. [75]. The last layer carries no $\widehat{\mathrm{CZ}}$, because a diagonal gate placed directly before a measurement in the computational basis changes only phases and leaves every probability unchanged. Ref. [75] also reads out the qubits separately, compares the reduced state of each qubit with the single-qubit label states of Section 15.4, and combines the fidelities of all qubits with trainable weights in the cost, which it calls the weighted-fidelity cost.

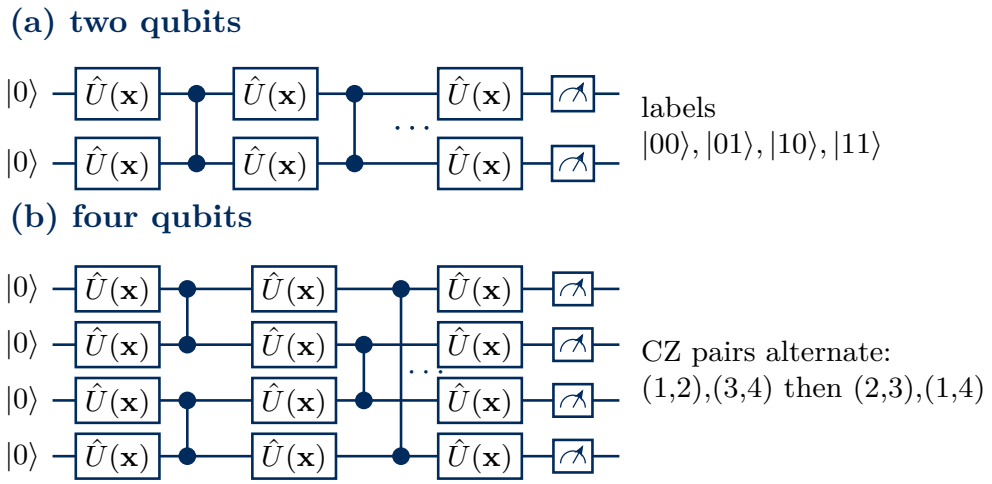


Figure 103: Multi-qubit re-uploading classifiers. (a) Two qubits, both loading the data $\mathbf{x}$ in every layer through $\hat{U}(\mathbf{x}) = \hat{U}(\boldsymbol{\theta}_{\ell,q} + \mathbf{w}_{\ell,q} \circ \mathbf{x})$, with a $\widehat{\mathrm{CZ}}$ gate after each layer except the last and the four basis states as label states. (b) Four qubits, with $\widehat{\mathrm{CZ}}$ gates on alternating pairs of qubits.

Table 8 collects the test accuracies that Ref. [75] reports with two layers and the weighted-fidelity cost for four problems, a circle, three circles and an annulus in the plane and a four-dimensional hypersphere. Every multi-qubit circuit is more accurate than the single qubit in every problem. At two layers the $\widehat{\mathrm{CZ}}$ gates help the four-qubit circuits on the annulus and on the hypersphere and lower the two-qubit accuracy on three circles. With more layers most of these differences shrink, whereas on the hypersphere the entangled circuits stay ahead, with 0.97 against 0.93 for two qubits at four layers. The class of a point of the hypersphere depends on all four coordinates together, and Section 15.11 isolates the effect of the entangling gates in this situation.

problem (classes, input dimension)	1 qubit	2 qubits		4 qubits	
		without $\widehat{\mathrm{CZ}}$	with $\widehat{\mathrm{CZ}}$	without $\widehat{\mathrm{CZ}}$	with $\widehat{\mathrm{CZ}}$
circle (2, 2)	0.94	0.96	0.96	0.96	0.96
three circles (4, 2)	0.76	0.90	0.83	0.90	0.89
annulus (3, 2)	0.76	0.79	0.82	0.87	0.96
hypersphere (2, 4)	0.87	0.92	0.91	0.90	0.98

Table 8: Test accuracies of re-uploading classifiers with $L = 2$ layers and the weighted-fidelity cost, from Tables 1–4 of Ref. [75].

15.11 Entanglement and the read-out qubit

The role of the $\widehat{\mathrm{CZ}}$ gates becomes visible when the data are split over the qubits. We take points $\mathbf{x} = (x_1, x_2, x_3, x_4)$ drawn uniformly from $[-1, 1]^4$ and assign class 1 to the points with $|\mathbf{x}|^2 < 4/\pi$. A four-dimensional ball of this radius has half the volume of the cube, but its radius $\sqrt{4/\pi} \approx 1.13$ exceeds one, so part of it lies outside the cube and class 1 holds 48% of the points. In each layer qubit 0 loads $(x_1, x_2, 0)$ and qubit 1 loads $(x_3, x_4, 0)$, and only qubit 0 is measured, with the label states $|0\rangle$ and $|1\rangle$ and the loss (15.8). We compare this circuit without $\widehat{\mathrm{CZ}}$ gates, the same circuit with a $\widehat{\mathrm{CZ}}$ gate after each layer except the last, and a single qubit that loads $(x_1, x_2, 0)$ and then $(x_3, x_4, 0)$ in every layer. Each model is trained on 1000 points with Adam, the best of four starts is kept and the accuracy is measured on 4000 test points.

Without $\widehat{\mathrm{CZ}}$ gates the register stays in a product state $|a(x_1, x_2)\rangle \otimes |b(x_3, x_4)\rangle$, and the measured probability of qubit 0 depends on x_1 and x_2 alone. The probability of class 1 at given x_1 and x_2 depends only on $x_1^2 + x_2^2$ and decreases with it, so the best rule of this kind is a threshold on $x_1^2 + x_2^2$, with a test accuracy of 0.75. A $\widehat{\mathrm{CZ}}$ gate, written as $\widehat{\mathrm{CZ}} = \hat{I} \otimes |0\rangle\langle 0| + \hat{Z} \otimes |1\rangle\langle 1|$, applies $\hat{Z}$ to qubit 0

in the branch where qubit 1 is in $|1\rangle$. The rotations of the next layer turn this relative phase into a change of the probabilities of qubit 0, which then depend on the amplitudes of qubit 1 and hence on x_3 and x_4.

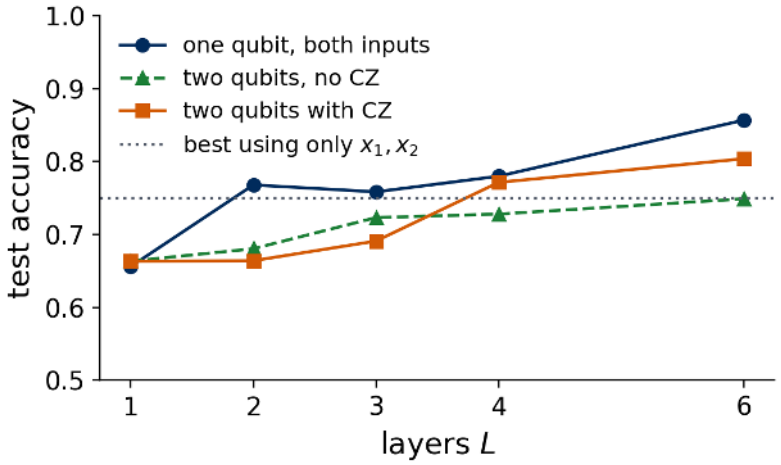


Figure 104: Test accuracy on the four-dimensional hypersphere against the number L of layers for one qubit loading all four inputs (circles), two qubits without $\widehat{\mathrm{CZ}}$ (triangles) and two qubits with $\widehat{\mathrm{CZ}}$ gates (squares), where qubit 0 loads (x_1, x_2), qubit 1 loads (x_3, x_4) and only qubit 0 is measured. The dotted line marks the best accuracy of any rule that uses x_1 and x_2 only. Best of four starts, exact state-vector simulation.

The circuit without $\widehat{\mathrm{CZ}}$ approaches the limit 0.75 of rules that use x_1 and x_2 only and stays below it at every depth (Figure 104). With $\widehat{\mathrm{CZ}}$ gates the accuracy passes the limit from four layers on and reaches 0.80 at $L = 6$, so information from qubit 1 arrives at the read-out qubit through the entangling gates. The single qubit that loads all four inputs reaches 0.86 at $L = 6$. In this experiment entanglement is needed because the data are distributed over two qubits and the class depends on all inputs together, whereas a single qubit that receives all inputs reaches a higher accuracy without a partner.

15.12 Two qubits on an annulus

When every qubit loads all inputs, the advantage of entanglement is smaller. We use the two-qubit circuit of Figure 103(a), in which both qubits load $\mathbf{x}$, with the label states $|00\rangle$ for class 0 and $|11\rangle$ for class 1, and predict class 1 when the probability of reading one, averaged over the two qubits, exceeds one half. Without the $\widehat{\mathrm{CZ}}$ gates the two qubits evolve independently and the output is the average of two single-qubit classifiers. We compare one qubit, two qubits without $\widehat{\mathrm{CZ}}$ and two qubits with $\widehat{\mathrm{CZ}}$ on a two-class version of the annulus, class 1 for $0.4 < |\mathbf{x}| < 0.8$ and class 0 elsewhere, which requires two boundaries, with 500 training and 4000 test points and the best of six starts per model and depth. About 63 per cent of the test points lie outside the annulus, so always answering class 0 gives an accuracy of 0.63.

All three models need several uploads before they draw two boundaries (Figure 105). At $L = 5$ the entangled model reaches a test accuracy of 0.94, against 0.91 for one qubit and for two qubits without $\widehat{\mathrm{CZ}}$, and its decision region reproduces the annulus. At smaller depths the order of the models changes from one depth to the next, and at $L = 3$ the single qubit is the best of the three. On this task the number of uploads matters more than the entanglement, and the lead of the entangled model at $L = 5$ rests on one best-of-six comparison per depth.

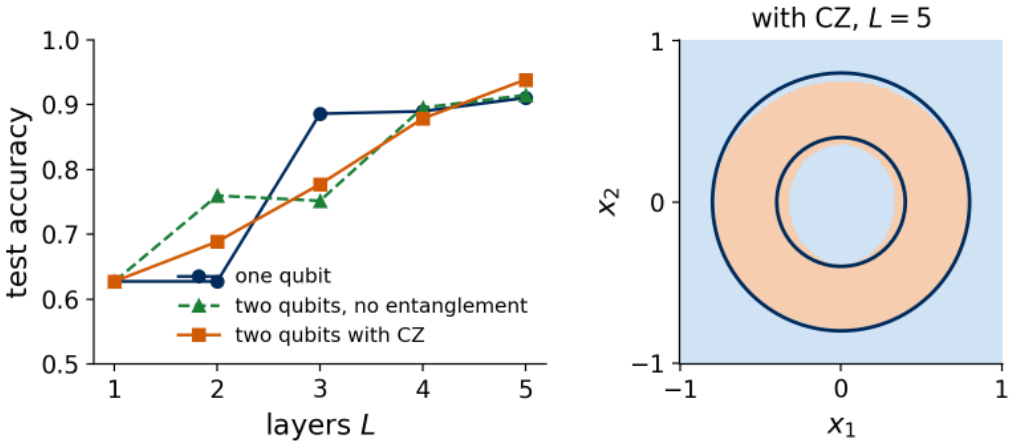


Figure 105: Re-uploading on the annulus task. Left, test accuracy against the number of layers for one qubit, two qubits without entanglement and two qubits with a $\widehat{\mathrm{CZ}}$ gate after each layer except the last, best of six starts. Right, predicted classes of the two-qubit model with $\widehat{\mathrm{CZ}}$ at $L = 5$, with the two true boundaries. Exact state-vector simulation.

16 Fundamental limitations of variational quantum circuits

Every model of these notes was trained by gradient descent on the angles of a circuit, with gradients that a device estimates from a finite number of shots. This section shows that circuits able to prepare almost any state of a large register have gradients that vanish exponentially with the number of qubits, and it traces this flatness back to the geometry of random quantum states.

16.1 Gradients at random angles

On a device every gradient component is estimated from a finite number of shots, and by (5.10) it is useful only when it exceeds the shot noise. We estimate how large a gradient component typically is at the start of a training run, when the angles are drawn at random, and how this size changes with the number of qubits. Along a single angle every cost of the form $C(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{O}|\psi(\boldsymbol{\theta})\rangle$, with a Hermitian observable $\hat{O}$, is a cosine (5.4), $a + b\cos(\theta_k - c)$, with constants that depend on the other angles. Its derivative $-b\sin(\theta_k - c)$ averages to zero over θ_k for any values of the other angles, so the mean gradient over uniformly random angles vanishes for every circuit. The typical size of a component is therefore measured by its variance,

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \big\langle (\partial_k C)^2 \big\rangle_{\boldsymbol{\theta}}, \tag{16.1}$$

where $\langle\cdot\rangle_{\boldsymbol{\theta}}$ is the average over the uniform distribution of the angles and $\partial_k C = \partial C/\partial\theta_k$. When this variance decays exponentially with N, the landscape is flat almost everywhere, and this flatness is called a *barren plateau* [79] (Figure 106).

16.2 Random states and the Haar measure

A deep circuit with randomly chosen angles prepares a state that looks random, so the typical behaviour of such circuits follows from the properties of random states. A state of N qubits is a unit vector in $\mathbb{C}^d$ with $d = 2^N$, and a *uniformly* random state is one that prefers no direction of this space. Random states are prepared by random circuits, which belong to the *unitary group* $U(d)$ of all $d \times d$ complex matrices $\hat{U}$ with $\hat{U}^\dagger\hat{U} = \hat{I}$. Every gate and every circuit on N qubits is such a matrix with $d = 2^N$, and the product of two of them and the inverse of each belong to $U(d)$ again. The *Haar measure* is the uniform probability distribution on $U(d)$, the unique probability distribution μ that no fixed unitary $\hat{V}$ can change by multiplication,

$$d\mu(\hat{U}) = d\mu(\hat{V}\hat{U}) = d\mu(\hat{U}\hat{V}). \tag{16.2}$$

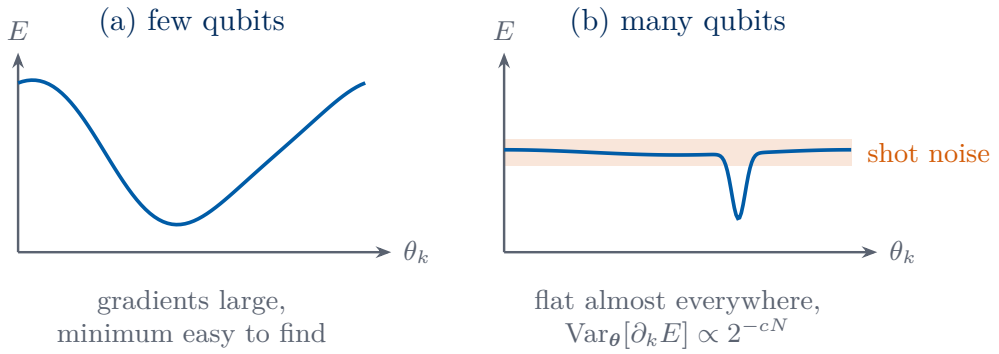


Figure 106: A barren plateau. For a small register the cost landscape has slopes that the shot noise does not hide. For a large register and a deep random circuit the landscape is flat almost everywhere, with the variance of the gradient decaying exponentially with the number of qubits, and the minimum lies in a narrow region.

A Haar-random state is $|\psi\rangle = \hat{U}|0\ldots0\rangle$ with a Haar-random $\hat{U}$, and by (16.2) its distribution does not change under any unitary rotation. Such a state is sampled by drawing d independent complex Gaussian amplitudes $g_{\mathbf{b}}$ and normalising the vector. The joint density of the Gaussian amplitudes depends only on $\sum_{\mathbf{b}} |g_{\mathbf{b}}|^2$, which every unitary leaves unchanged, so the normalised vector has the required invariance. A Haar-random unitary is sampled from a matrix of Gaussian entries by an orthogonalisation [80], and Ref. [81] gives an introduction to calculations with the Haar measure.

For one qubit the Haar-random states fill the Bloch sphere uniformly (Figure 107). The polar angle ϑ of the Bloch vector is then not uniform, whereas its cosine is, because the area of the sphere between the heights z and $z + dz$ does not depend on z. Random circuits used in practice do not sample the full Haar measure. An ensemble of unitaries that reproduces the averages of all polynomials of degree t in the entries of $\hat{U}$ and of degree t in those of $\hat{U}^\dagger$ is called a unitary t-design [82]. The averages of expectation values and of their squares, which decide the statements below, need only a 2-design, and circuits of random two-qubit gates form approximate 2-designs once their depth grows with N.

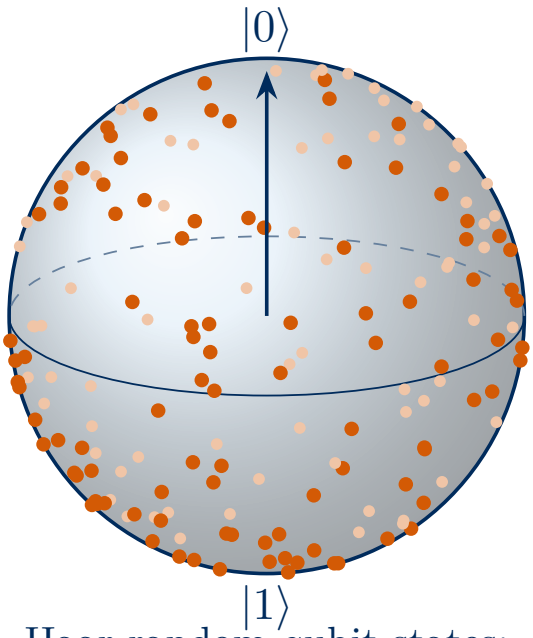


Figure 107: Haar-random states of one qubit are uniformly distributed on the Bloch sphere. Equal areas of the sphere receive equal numbers of points, so the height $z = \cos\vartheta$ of the Bloch vector, and not the polar angle ϑ, is uniform.

16.3 Concentration of measure and Lévy's lemma

For a uniformly random point of the unit sphere in $\mathbb{R}^n$, the coordinates have equal averages of their squares by symmetry, and these averages sum to one, so $\langle x_1^2\rangle = 1/n$. The density of a single coordinate is proportional to $(1-x_1^2)^{(n-3)/2}$, which for large n is a narrow Gaussian of width $1/\sqrt{n}$ around zero. Almost every point therefore lies close to every equator, and the larger the dimension, the stronger this concentration.

Lévy's lemma states the same for every smooth function of a random state. Let $f(\psi)$ be a real function of a state, for example the expectation value $f(\psi) = \langle\psi|\hat{O}|\psi\rangle$ of an observable $\hat{O}$. The number η measures how fast f can change, and f is called Lipschitz with constant η when

$$|f(\psi) - f(\phi)| \le \eta \, \big\||\psi\rangle - |\phi\rangle\big\| \tag{16.3}$$

for all pairs of states. An expectation value has $\eta \le 2\|\hat{O}\|$, where $\|\hat{O}\|$ is the largest absolute eigenvalue of $\hat{O}$. A deviation from the average by more than $\epsilon > 0$ counts as atypical, and for a Haar-random state [83, 84]

$$\Pr\Big(\big|f(\psi) - \langle f\rangle_{\text{Haar}}\big| > \epsilon\Big) \le 2\, e^{-Cd\epsilon^2/\eta^2}, \tag{16.4}$$

with a positive numerical constant C independent of d. The probability of any deviation of fixed size ϵ thus falls exponentially in the dimension $d = 2^N$, that is doubly exponentially in the number of qubits. Geometrically, the sphere of states in high dimension has almost all of its area close to any equator, and a smooth function cannot vary much over this region. Almost every random state therefore gives practically the same value of every smooth quantity, its Haar average. For expectation values the typical deviations from the Haar average are of order $1/\sqrt{d} = 2^{-N/2}$, with exponentially small probability of larger deviations.

The expectation value of $\hat{Z}_0$ shows this concentration explicitly. The Born probabilities $p_{\mathbf{b}} = |g_{\mathbf{b}}|^2 / \sum_{\mathbf{b}'} |g_{\mathbf{b}'}|^2$ of a random state are uniformly distributed over all probability vectors of length d, with the moments

$$\langle p_{\mathbf{b}}\rangle = \frac{1}{d}, \qquad \langle p_{\mathbf{b}}^2\rangle = \frac{2}{d(d+1)}, \qquad \langle p_{\mathbf{b}} p_{\mathbf{b}'}\rangle = \frac{1}{d(d+1)} \quad (\mathbf{b} \neq \mathbf{b}'). \tag{16.5}$$

The expectation value $\langle\hat{Z}_0\rangle = \sum_{\mathbf{b}} s_{\mathbf{b}} p_{\mathbf{b}}$ has the signs $s_{\mathbf{b}} = (-1)^{b_0}$, equal to $+1$ for half of the bit strings and -1 for the other half, so its mean vanishes. Its variance follows from (16.5) and from $\sum_{\mathbf{b}\neq\mathbf{b}'} s_{\mathbf{b}} s_{\mathbf{b}'} = \big(\sum_{\mathbf{b}} s_{\mathbf{b}}\big)^2 - d = -d$,

$$\mathrm{Var}\langle\hat{Z}_0\rangle = d\,\frac{2}{d(d+1)} - d\,\frac{1}{d(d+1)} = \frac{1}{d+1} \approx 2^{-N}. \tag{16.6}$$

Any Pauli string $\hat{P} \neq \hat{I}$ is obtained from $\hat{Z}_0$ by a unitary rotation, and the Haar distribution is invariant under it, so (16.6) holds for every Pauli string. Figure 108 shows the distribution of $\langle\hat{Z}_0\rangle$ in sampled random states for two, six and twelve qubits, which narrows around zero as N grows, and Figure 109 confirms the scaling of the variance. The same moments give the average weight of a fixed bit string, $\langle|\langle 0\ldots0|\psi\rangle|^2\rangle = 1/d$.

16.4 Typical states

Lévy's lemma makes almost every random state *typical*, meaning that its measurable properties equal their averages over the Haar measure up to errors exponentially small in N. A single typical state then represents the whole ensemble, in the same way as a single microstate represents the microcanonical ensemble of statistical physics. Three properties of typical states matter for variational circuits.

A typical state is locally featureless. We divide the register into a subsystem A of dimension d_A and the rest B of dimension d_B, with $d = d_A d_B$. The reduced density matrix $\hat{\rho}_A = \mathrm{Tr}_B|\psi\rangle\langle\psi|$ of a random state has the average purity [85]

$$\langle \mathrm{Tr}\hat{\rho}_A^2\rangle = \frac{d_A + d_B}{d_A d_B + 1} \approx \frac{1}{d_A} + \frac{1}{d_B}. \tag{16.7}$$

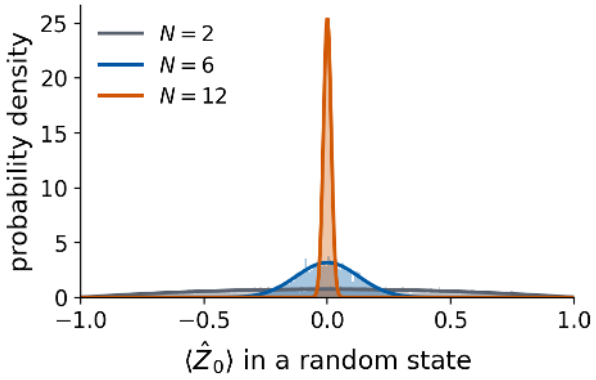


Figure 108: Distribution of $\langle \hat{Z}_0 \rangle$ in 4000 Haar-random states of $N = 2$, 6 and 12 qubits, histograms of sampled states and exact densities (lines). The distribution concentrates around zero with a width of order $2^{-N/2}$.

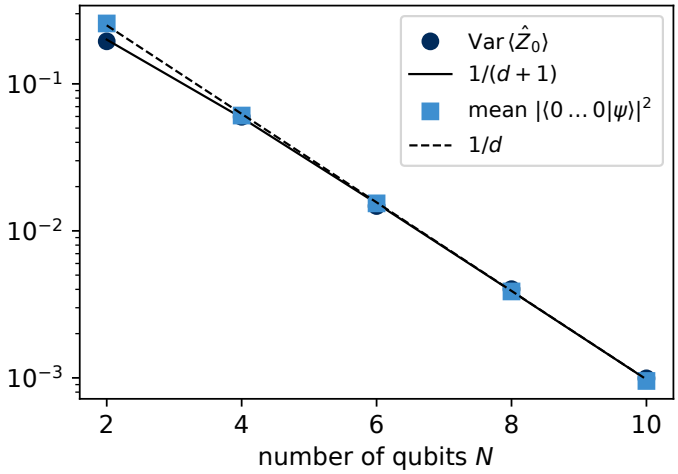


Figure 109: Concentration in random states of N qubits. The variance of $\langle \hat{Z}_0 \rangle$ and the mean overlap $|\langle 0 \dots 0 | \psi \rangle|^2$ over random states (markers) follow $1/(d+1)$ and $1/d$ with $d = 2^N$ (lines) (16.6). Sampled state vectors.

The maximally mixed state $\hat{I}_A/d_A$ has the purity $1/d_A$, and the squared distance $\mathrm{Tr}(\hat{\rho}_A - \hat{I}_A/d_A)^2 = \mathrm{Tr}\hat{\rho}_A^2 - 1/d_A$ is therefore of order $1/d_B$. A subsystem much smaller than the rest is thus almost maximally mixed, $\hat{\rho}_A \approx \hat{I}_A/d_A$, every observable of A takes the value $\mathrm{Tr}\hat{O}_A/d_A$, and every Pauli string on A has $\langle \hat{P} \rangle \approx 0$. This statement is canonical typicality [86, 87]. The state of the whole register is pure, yet all its information resides in correlations among many qubits and is invisible to local measurements. Two typical states are moreover nearly orthogonal, $|\langle \psi | \phi \rangle|^2 \approx 1/d$, by (14.8). A deep random circuit therefore produces states that all look alike locally, and changing one of its angles hardly changes any local cost.

16.5 Page entanglement

The entanglement of a typical state follows from the same reasoning. For a subsystem of $n_A \leq N/2$ qubits, with $d_A = 2^{n_A}$ and $d_B = 2^{N-n_A}$, the average entanglement entropy of a random state is [88]

$$\langle S_{\mathrm{vN}}(\hat{\rho}_A) \rangle \approx n_A - \frac{d_A}{2 d_B \ln 2} \text{ bits}, \tag{16.8}$$

only slightly below the maximum of n_A bits. This Page value grows with the number of qubits of A, a volume law, and its fluctuations between random states are small. Figure 110 compares it with sampled random states of ten qubits and with the ground state of the critical Ising chain. The ground state carries about half a bit for every size of A, because ground states of local Hamiltonians obey

an area law (Section 2.4). Such states are atypical and occupy a tiny corner of the Hilbert space. A circuit that can reach every state mostly produces typical states, far from the targets of physical interest.

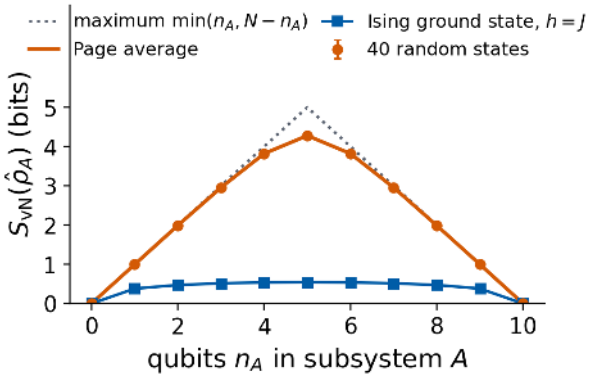


Figure 110: Entanglement entropy of the first n_A qubits of a chain of $N = 10$ qubits, averaged over 40 Haar-random states (circles, with error bars of the spread), the Page average (16.8) (orange line), the maximum $\min(n_A, N - n_A)$ (dotted) and the ground state of the critical Ising chain (squares).

16.6 Barren plateaus as a consequence of typicality

A circuit with many layers of randomly chosen gates behaves, for averages of expectation values and of their squares, like a Haar-random unitary, a 2-design. By (16.6) every Pauli expectation value in its output state then has a variance of about 2^{-N} over the random angles, and a cost that is an expectation value in a nearly random state carries almost no information about the angles. The gradient inherits this concentration when the differentiated rotation acts first on $|0\ldots0\rangle$ and a Haar-random unitary $\hat{W}$ represents all gates after it. The gradient variance of the cost $C = \langle\hat{O}\rangle$ is then [79]

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \frac{\mathrm{Tr}(\hat{O}_0^2)}{2(d^2-1)}, \tag{16.9}$$

where $\hat{O}_0 = \hat{O} - \mathrm{Tr}(\hat{O})\hat{I}/d$ is the traceless part of $\hat{O}$. The formula follows from the Haar average $\langle \mathrm{Tr}(\hat{W}^\dagger \hat{A}\hat{W}\hat{B})^2\rangle_{\hat{W}} = \mathrm{Tr}(\hat{A}^2)\mathrm{Tr}(\hat{B}^2)/(d^2-1)$ for traceless Hermitian $\hat{A}$ and $\hat{B}$, with $\hat{A} = \hat{O}_0$ and $\hat{B}$ the traceless operator produced by differentiating the rotation, which has $\mathrm{Tr}(\hat{B}^2) = 1/2$. The decay is therefore set by $\mathrm{Tr}(\hat{O}_0^2)$, which we compare for three costs,

$$C_G(\boldsymbol{\theta}) = 1 - |\langle 0\ldots0|\psi(\boldsymbol{\theta})\rangle|^2, \qquad C_L(\boldsymbol{\theta}) = \frac{1}{N}\sum_{q=0}^{N-1}\frac{1-\langle\hat{Z}_q\rangle}{2}, \qquad E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{\mathcal{H}}_{\mathrm{TFIM}}|\psi(\boldsymbol{\theta})\rangle. \tag{16.10}$$

The *global cost* C_G is the probability that a measured bit string differs from $0\ldots0$ and depends on all qubits together. The *local cost* C_L is the probability that a single qubit, averaged over the qubits, reads one. Both vanish only in $|0\ldots0\rangle$, so they have the same minimum, whereas C_L is a sum of single-qubit terms, the trash Hamming distance (8.9) with all N qubits as trash. The energy is a sum of terms on one or two neighbouring qubits, so it is a local cost as well. For the global cost $\hat{O} = \hat{I} - |0\ldots0\rangle\langle 0\ldots0|$ and $\mathrm{Tr}(\hat{O}_0^2) = 1 - 1/d$, so its variance $1/[2d(d+1)] \approx 2^{-2N-1}$ decays as 4^{-N}. A sum of Pauli strings $\sum_j c_j \hat{P}_j$ has $\mathrm{Tr}(\hat{O}_0^2) = d\sum_j c_j^2$, so the energy has Var $\approx [(N-1)J^2 + Nh^2]/2^{N+1}$ and the local cost Var $\approx 2^{-N}/(8N)$. For a circuit that acts as a 2-design every cost decays exponentially with N, the global one twice as fast.

16.7 Global and local costs

Figure 111 shows the variance of the derivative with respect to the first angle of the HEA, estimated from 300 random angle vectors for $N = 2, \ldots, 10$. The global cost decays exponentially at depth $L = N$, with a fitted decay $\mathrm{Var} \propto 2^{-cN}$ with $c = 1.96$, close to the 2-design value two. A shallow circuit gives the global cost no protection, since at $L = 2$ it decays at a similar rate, $c = 1.81$. The global cost is the probability of a single bit string out of 2^N, exponentially small for almost all angles, whatever the depth. This dependence on the cost was proved for layered circuits of local 2-designs [38], and the HEA shows the same behaviour.

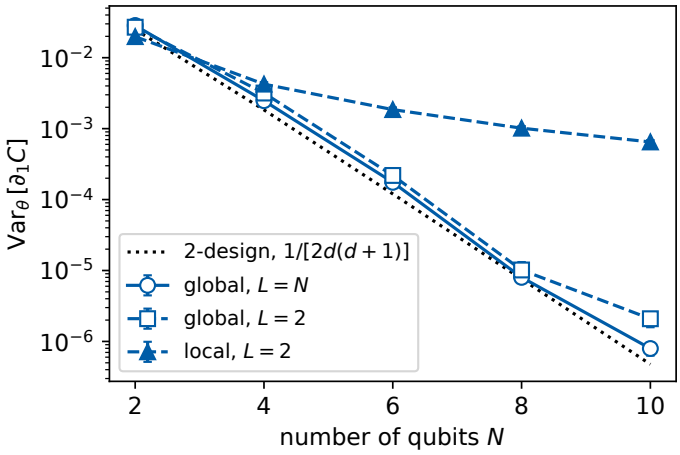


Figure 111: Variance of the derivative of the global cost C_G and the local cost C_L (16.10) with respect to the first angle of the HEA, over 300 uniformly random angle vectors, against the number of qubits, for the depths $L = N$ and $L = 2$. The dotted line is the 2-design value $1/[2d(d+1)]$ of the global cost. Exact gradients from the state vector.

The local cost at $L = 2$ decays slowly because of a light cone. The differentiated angle rotates qubit 0 in the first block, and each $\widehat{\mathrm{CZ}}$ layer passes its influence on by one qubit. After two layers it reaches only qubits 0, 1 and 2, so only $\langle \hat{Z}_0 \rangle$, $\langle \hat{Z}_1 \rangle$ and $\langle \hat{Z}_2 \rangle$ depend on it, and these three expectation values depend only on the angles of the first few qubits. For $N \geq 5$ the distribution of their derivative no longer changes with N, and the prefactor $1/N$ of C_L gives

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_1 C_L] = \frac{\mathrm{const}}{N^2} \qquad (L = 2,\ N \geq 5), \tag{16.11}$$

a power law in N. At $N = 10$ the local gradient variance is 311 times larger than the global one of the same circuit. For local costs and depths growing as $\log N$ the variance decays at most polynomially [38], whereas circuits deep enough to form 2-designs bring back the exponential decay (16.9) for local costs as well.

16.8 The energy gradient of structured ansätze

For the energy of the TFIM, the cost of VQE, the gradient variance of the HEA at depth $L = N$ decays far more slowly than the 2-design value at the sizes studied here and lies more than an order of magnitude above it at $N = 10$ (Figure 112). Each term of the energy is affected only by the gates inside its light cone, and at depth N the circuit has only just connected the two ends of the chain and is still far from a random unitary. Once the circuit is deep enough to form a 2-design, the decay approaches (16.9).

The gradient variance of the HVA grows with N. Each of its angles multiplies a sum over all bonds or all qubits, so one angle changes all $2N - 1$ terms of the energy together, and the derivative collects contributions from the whole chain. The HVA also never approaches a random unitary. Its gates commute with the parity and generate only a small family of unitaries, whose dimension grows

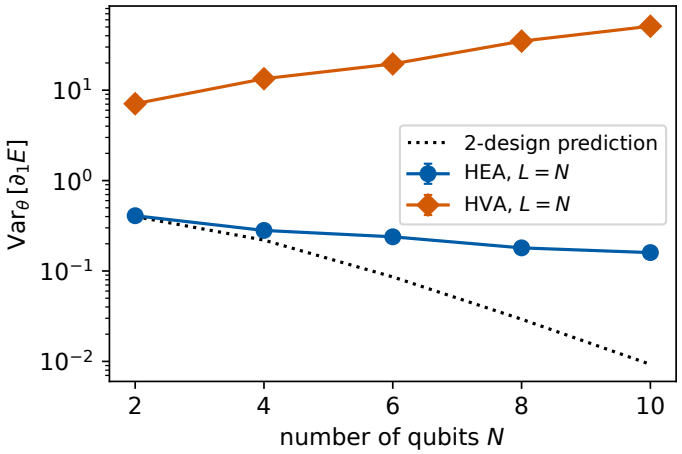


Figure 112: Variance of the derivative of the TFIM energy at $h = J$ over random angles, for the HEA with respect to its first angle and for the HVA with respect to γ_1, both at depth $L = N$, against the number of qubits. The dotted line is the 2-design value of (16.9). Exact gradients.

polynomially with N, so the states it prepares fill a small part of the 2^N dimensional space. Mild or absent barren plateaus for this ansatz were reported in Ref. [32]. In general the gradient variance is set by the dimension of the set of operators generated by the gates of a circuit, its dynamical Lie algebra [89, 90].

16.9 Shots needed to resolve a gradient

A barren plateau is a problem of measurement cost, because a component estimated by the parameter-shift rule from two energies with M shots per setting has the statistical variance $\sigma^2/(2M)$ by (5.10), where σ^2 is the single-shot variance of the energy. The typical gradient becomes visible above the shot noise when this variance equals (16.1), which requires

$$M^\star = \frac{\sigma^2}{2\,\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E]} \tag{16.12}$$

shots per setting and per shifted circuit. In a random state $\sigma^2 = d\sum_j c_j^2/(d+1) \approx \sum_j c_j^2$, and with (16.9) the 2-design value is $M^\star = d - 1 = 2^N - 1$, so every added qubit doubles the number of shots for one gradient component. For an angle shared by gates with factors c_g, the chain rule (5.8) measures each term separately and their variances add with the weights c_g^2, so the numerator of (16.12) becomes $\sigma^2 \sum_g c_g^2$.

At $N = 10$ (Figure 113) a typical energy gradient of the deep HEA is resolved with about 58 shots per setting, while a 2-design would need about 1023. For the 2-design the count reaches about 10^6 at $N = 20$ and 10^{15} at $N = 50$, for a single component of a single gradient. The HVA angle γ_1 is resolved with about 6.7 shots per setting in each of its 18 shifted circuits at $N = 10$. A better optimiser cannot remove a barren plateau, because every optimiser on a device has to resolve the gradient, or differences of energies, above the shot noise, and (16.12) sets the cost of doing so.

16.10 Noise and remedies

Noise on a device produces a second kind of plateau. Depolarising noise after each layer replaces part of the state by the maximally mixed state, on which every traceless observable has expectation value zero, so the whole cost landscape contracts towards a constant by a factor that decays exponentially with the depth [91]. For circuits whose depth grows linearly with N the gradient then vanishes exponentially in N, independently of the random-unitary argument and also for structured ansätze. The remedies within the circuit are a smaller depth and lower error rates.

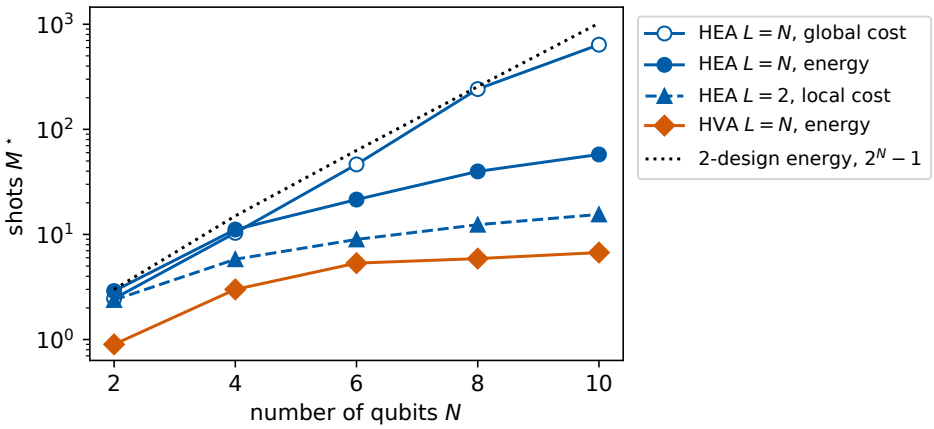


Figure 113: Shots per setting and per shifted circuit needed to resolve a typical energy gradient (16.12) at random angles, against the number of qubits, for the HEA at depth $L = N$ with the global cost and with the energy, for the HEA at $L = 2$ with the local cost, for the HVA angle γ_1 with the energy, and the 2-design value $M^\star = 2^N - 1$ (dotted). Measured gradient variances from exact state vectors.

The known ways of keeping the gradient measurable change the cost, the circuit or the starting point. Local costs decay at most polynomially for circuits of logarithmic depth [38]. Shallow circuits keep the light cone of each gate small and limit the effect of noise. Structured ansätze that respect the symmetries of the problem, such as the HVA, prepare states in a small part of the state space and keep large gradients [32, 89]. Initial angles chosen so that the circuit starts as the identity [92], and circuits trained layer by layer [93], start the training away from the random region. All of these measures restrict the set of states that the circuit explores. A circuit that can prepare almost any state has exponentially small gradients, so expressibility and trainability have to be balanced [94].

The same mechanism reaches beyond variational eigensolvers. In the scrambling Born machine (12.19) with a Haar-random scrambler, every rotation is followed by a Haar-random unitary, so (16.9) applies directly and the gradients of the trained angles decay as 2^{-N}. Fixing the entangler saves calibration on hardware but does not remove the plateau. The exponential concentration of quantum kernels in Section 14.5 and the advantage of local costs for the quantum autoencoder in Section 8.3 are further consequences of the concentration of random states.

17 Summary

1 Quantum computers as learning models

Learning tasks are classified by the source of the data and the processing device into CC, CQ, QC and QQ. These notes use a quantum device for the processing, with quantum data in the variational eigensolver and the autoencoder and classical data in the Born machine, the QAOA and the classifiers. Parametrised circuits and quantum kernels are trained by classical optimisers, quantum reservoirs and extreme learning machines train only a linear readout of a fixed quantum system, and learning from quantum data and classical learning for quantum physics form further directions.

2 The many-body problem and its classical approximations

A state of N spins needs 2^N amplitudes, and exact methods are limited to a few dozen spins. Every trial state bounds the ground-state energy from above (2.4),

$$\langle\psi|\hat{\mathcal{H}}|\psi\rangle = \sum_m |c_m|^2 E_m \geq E_0,$$

and the Rayleigh–Ritz method minimises the energy over an ansatz. Mean field takes a product ansatz. For the TFIM (2.2) the site-factorised state $|\phi(\theta)\rangle^{\otimes N}$ has $E_{\mathrm{MF}} = -J(N-1)\cos^2\theta - hN\sin\theta$ (2.6),

whose minimum obeys the self-consistent Weiss equation (2.8) and places the transition at $h_c^{\mathrm{MF}} = 2J(N-1)/N \to 2J$, twice the exact $h_c = J$. Product states carry no entanglement. Matrix product states bound the entanglement by $\log_2 \chi$ and describe area-law states, and Monte Carlo methods face the sign problem.

3 Parametrised circuits as quantum ansätze

A circuit of rotations prepares the trial state on the register (3.1),

$$|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle, \qquad \hat{R}_P(\theta) = e^{-i\theta\hat{P}/2} = \cos\tfrac{\theta}{2}\,\hat{I} - i\sin\tfrac{\theta}{2}\,\hat{P},$$

with angles that are pulse areas $\theta = \Omega t$ or interaction times $\theta = Jt$. The device prepares and measures, and a classical computer updates the angles (the VQE loop). The energy error bounds the infidelity through the gap (3.3), $E - E_0 \geq \Delta E\,(1-F)$.

4 The energy from bit strings

With $\hat{\mathcal{H}} = \sum_j c_j \hat{P}_j$, a Pauli string takes the value $(-1)^{\sum_{q\in S} b_q}$ after the rotation of its factors into $\hat{Z}$ by $\hat{H}$ or $\hat{H}\hat{S}^\dagger$. Strings sharing a basis on every qubit form one setting, two for the TFIM and three for the XXZ chain. The sample-mean estimate (4.5) has the standard error (4.7)

$$\sigma_{\bar{E}} = \sqrt{\frac{1}{M}\sum_g \sigma_g^2},$$

so each tenfold reduction of the error costs a hundred times more shots.

5 Gradients

Along one angle the energy is a shifted cosine (5.4), $E(\theta_k) = a + b\cos(\theta_k - c)$, because every generator squares to the identity. Hence the exact parameter-shift rule (5.7) and, for an angle shared by gates with $\theta_g = c_g\varphi$, the chain rule (5.8),

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\boldsymbol{\theta} + \tfrac{\pi}{2}\mathbf{e}_k) - E(\boldsymbol{\theta} - \tfrac{\pi}{2}\mathbf{e}_k)}{2}, \qquad \frac{\partial E}{\partial \varphi} = \sum_g c_g \frac{\partial E}{\partial \theta_g}.$$

On a device the shift is a change of pulse length by $\pm\pi/(2\Omega)$, and one gradient costs $2n_{\mathrm{rot}}GM$ shots (5.9). In simulation reverse-mode automatic differentiation gives the same gradient at the cost of a few energy evaluations. Finite differences amplify shot noise by $1/\epsilon$, and SPSA uses two energies per step.

6 Circuit ansätze

An ansatz should be expressive, trainable, native to the hardware and symmetric like the Hamiltonian. The HEA (6.3) alternates $\hat{R}_z\hat{R}_y$ blocks with $\widehat{\mathrm{CZ}}$ chains and has $n = 2N(L+1)$ angles. The HVA (6.5) alternates evolutions under the terms of the TFIM from $|+\rangle^{\otimes N}$, has $n = 2L$ shared angles and conserves the parity $\hat{\Pi} = \prod_i \hat{X}_i$. The SPA applies the exchange gate (6.7) to a singlet product and conserves $\hat{Z}_{\mathrm{tot}}$. A brickwork of L layers with one gate of operator Schmidt rank two per layer across a cut obeys $S_{\mathrm{vN}} \leq \min(L, N/2)$ bits (6.11).

7 The variational quantum eigensolver for spin chains

An ansatz with a built-in symmetry reaches only targets in its own symmetry sector. On the XYZ chain in two fields the shared-angle HVA is mirror-even for all angles (7.4) while the ground state is

mirror-odd, so its fidelity stays exactly zero, whereas the HVA with free angles and the HEA reach the ground state. A trial state with the reflection expectation value $\langle\hat{\mathcal{R}}\rangle$ obeys $F \le (1-\langle\hat{\mathcal{R}}\rangle)/2$. Local observables converge before the half-chain entanglement. The energy error is of second order in a small admixture ϵ of an excited state (7.6), whereas the entanglement entropy changes at first order (7.7), so the entropy error scales as $\sqrt{1-F}$ and exposes errors that the energy hides. In our runs (four restarts, 600 Adam steps) the structured ansätze reached better states with far fewer angles than the HEA at equal two-qubit gate count (Table 3). In the ordered TFIM a state within the ground doublet has (7.9)

$$E - E_0 = \frac{1-\langle\hat{\Pi}\rangle}{2}\,\Delta E,$$

so the symmetry-broken branch reached by the HEA has $F = 1/2$, no entanglement and an energy error of only $\Delta E/2$, while the parity-conserving HVA reaches the cat state. Variational quantum deflation (7.10) adds the penalty $\lambda|\langle\tilde{\psi}_0|\psi(\boldsymbol{\theta})\rangle|^2$ with $\lambda > \Delta E$ and gives the first excited state. Across the phase diagrams of the TFIM and of the XXZ chain in a transverse field, small energy errors coexist with wrong states wherever an ansatz breaks a symmetry or misses entanglement, and the fidelity and the entropy reveal it.

8 Compressing quantum data with a quantum autoencoder

An encoder $\hat{U}(\boldsymbol{\theta})$ moves an ensemble of states into $N-k$ latent qubits, and the compression is lossless if and only if $\hat{U}|\psi_i\rangle = |\psi_i^{\mathrm{c}}\rangle \otimes |0\ldots0\rangle$ for every state. The state passes through $\hat{\rho}_{\mathrm{in}} \to \hat{U}\hat{\rho}_{\mathrm{in}}\hat{U}^\dagger \to \mathrm{Tr}_{\mathrm{trash}} \to \otimes|0\ldots0\rangle\langle0\ldots0| \to \hat{U}^\dagger$, and the reconstruction fidelity $F = \langle\psi_{\mathrm{in}}|\hat{\rho}_{\mathrm{out}}|\psi_{\mathrm{in}}\rangle = \sum_t |\langle\alpha_0|\alpha_t\rangle|^2$ obeys $p_0^2 \le F \le p_0$ with the trash fidelity p_0. The trash Hamming distance $D_H = \frac{1}{k}\sum_j \langle\hat{n}_j\rangle$ is a local cost measured from trash bit strings, and

$$F \ge \max\{0,\, 1 - kD_H\}^2.$$

A part B of Schmidt rank χ fits into $\lceil\log_2\chi\rceil$ qubits by a unitary on B, a single state is always compressible by a global unitary, and a gate of operator Schmidt rank r across the cut divides χ by at most r. One encoder for a family reaches at most the sum of the 2^{N-k} largest eigenvalues of the average density matrix. A trained encoder leaves one dominant Schmidt coefficient across the cut between latent and trash qubits. Ising ground states of eight qubits compress well at every field, whereas the XXZ chain is hardest just above $\Delta = -1$. Frozen encoders flag other phases as anomalies in the Ising chain and across the XXZ chain in a field, and variational quantum anomaly detection maps phase diagrams on the device without labels or order parameters, including one with a topological Mott insulator.

9 Classifying phases with a quantum convolutional neural network

The cluster-Ising chain has a $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry and an SPT phase around the cluster state, detected by the string order parameter $\hat{S} = \hat{Z}_0\hat{X}_1\hat{X}_3\cdots\hat{X}_{2m-1}\hat{Z}_{2m}$, the telescoped product of cluster operators. A QCNN applies shared two-qubit kernels, pools by measuring half of the qubits and conditionally rotating their partners, and ends with a global unitary and one measured qubit, with $n = 21(\log_2 N - 1) + 15$ angles. Trained on 40 labelled ground states by the mean squared error of $p_{\boldsymbol{\theta}} = \langle\psi|\hat{W}^\dagger|1\rangle\langle1|_{\mathrm{out}}\hat{W}|\psi\rangle$, it classifies the 8-qubit phase diagram as accurately as the string order parameter, and its logarithmic depth keeps the gradients large.

10 Learning a Hamiltonian with a quantum graph recurrent neural network

Copies of an initial state go to an unknown system and to a circuit that repeats one trainable Trotter layer $\hat{U}_\delta(\boldsymbol{\theta}) = e^{-i\delta\hat{\mathcal{H}}_E(\boldsymbol{\theta})}e^{-i\delta\hat{\mathcal{H}}_V(\boldsymbol{\theta})}$, and a SWAP test with $P(0) = (1+F)/2$ compares the outputs. Agreement at many times and for several initial states fixes $\hat{\mathcal{H}}$ through the short-time expansion, so fifteen data pairs recover all couplings, fields and the graph, whereas a single pair does not. The Trotter step limits the accuracy of the learned parameters.

11 Compressing circuits with a quantum circuit autoencoder

The Choi state $\hat{J}_{\mathcal{E}} = (\mathcal{E} \otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$, prepared from Bell pairs, determines the channel through $\mathcal{E}(\hat{\rho}) = D\,\mathrm{Tr}_R[(\hat{I} \otimes \hat{\rho}^T)\hat{J}_{\mathcal{E}}]$, and its rank is the Kraus rank. Encoders $\hat{U}, \hat{V}$ rotate the two halves of the Choi state, the reconstruction is lossless if and only if $\hat{V} \circ \mathcal{E} \circ \hat{U} = \mathcal{F} \otimes \mathrm{id}_{\mathrm{trash}}$, and the Choi fidelity is bounded by the sum of the d^2 largest Choi eigenvalues. Training uses Bell tests on the trash pairs only, and complete depolarisation cannot be compressed.

12 Circuits as probability models

A continuous value x is mapped to the bin $k = \lfloor (x - x_{\min})/w \rfloor$ of width $w = (x_{\max} - x_{\min})/2^N$, and k in binary is the bit string, with b_0 the most significant bit. The Born machine (12.9) models $q_{\boldsymbol{\theta}}(\mathbf{b}) = |\langle \mathbf{b}|\hat{V}(\boldsymbol{\theta})|0\rangle^{\otimes N}|^2$, and each measurement draws one sample. Its loss is the KL divergence to the data, equal to the negative log-likelihood up to the entropy of the data (12.13),

$$D_{\mathrm{KL}}(\tilde{p}\,\|\,q_{\boldsymbol{\theta}}) = -\frac{1}{|\mathcal{D}|}\sum_i \ln q_{\boldsymbol{\theta}}(\mathbf{b}^{(i)}) - H(\tilde{p}).$$

A product circuit gives independent bits (12.16) and at best the product of the marginals. Entangling layers create correlations, and deeper circuits eventually fit the fluctuations of a finite data set. Sampled models are judged by the total variation distance, and ancilla qubits turn the model into a mixture of Born distributions.

13 Combinatorial optimisation with QUBO and QAOA

A QUBO $C = \mathbf{x}^T Q \mathbf{x}$ becomes, with $x_i = (1 - z_i)/2$, the diagonal Ising Hamiltonian (13.7) with

$$J_{ij} = \frac{Q_{ij} + Q_{ji}}{4}, \qquad h_i = -\frac{1}{4}\sum_j (Q_{ij} + Q_{ji}),$$

whose basis states are the candidates and whose ground state is the optimum. MaxCut gives $\hat{\mathcal{H}}_C = \frac{1}{2}\sum_{(i,j)\in E} \hat{Z}_i\hat{Z}_j - |E|/2$. The QAOA (13.12) alternates $e^{-i\gamma_\ell \hat{\mathcal{H}}_C}$ and $e^{-i\beta_\ell \sum_i \hat{X}_i}$ from $|+\rangle^{\otimes N}$. The answer is the best sampled bit string, found with probability $1 - (1 - P_{\mathrm{opt}})^M$ after M shots, and the expected cost from the same shots trains the angles. The trained angles resemble the Trotterised adiabatic schedule $\gamma_\ell = s_\ell \delta t$, $\beta_\ell = -(1 - s_\ell)\delta t$ (13.17).

14 Circuits as classifiers with quantum kernels

A feature map makes data linearly separable in a higher dimension, and an SVM needs only the kernel $k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x}) \cdot \boldsymbol{\phi}(\mathbf{x}')$. A circuit gives the quantum kernel

$$k(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x})|\phi(\mathbf{x}')\rangle|^2,$$

measured by the inversion test as the frequency of $0 \ldots 0$ after $\hat{U}(\mathbf{x}')^\dagger \hat{U}(\mathbf{x})$, with $|\mathcal{D}|(|\mathcal{D}|-1)/2$ circuits per training matrix. A product encoding separates two rings, deep encodings concentrate all kernel values at 2^{-N}, and the kernel-target alignment trains the encoding while the SVM stays convex.

15 Circuits as classifiers with data re-uploading

A qubit state is a point on the Bloch sphere, $|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle$, moved by $\hat{R}_z\hat{R}_y\hat{R}_z$, and a data point entering as rotation angles maps the plane onto the sphere. Classes are label states as far apart as possible, the poles for two classes, Bloch vectors 120° apart with pairwise fidelity 1/4 for three and a tetrahedron with pairwise fidelity 1/3 for four. A point is assigned the class of largest

fidelity. A single qubit loaded once can only cut a band of the plane. Loading the data in every layer, $\hat{U}(\boldsymbol{\theta}_\ell + \mathbf{w}_\ell \circ \mathbf{x})$, gives a classifier whose output is a Fourier series

$$f(x) = \sum_{\omega=-L}^{L} c_\omega e^{i\omega x}$$

with frequencies up to the number L of uploads, which bounds the decision boundaries $f = 1/2$ the model can draw. For one input variable a single qubit expresses any Fourier series, whereas for several variables universality needs the trainable classical weights on the data. One qubit approximates the circle boundary with growing accuracy and separates three classes, a disc, a ring and the outside region. N qubits offer 2^N orthogonal basis states as label states, and $\widehat{\mathrm{CZ}}$ gates after each layer except the last let the qubits exchange information. Entanglement matters when the class depends on inputs loaded into different qubits. On the four-dimensional hypersphere with the inputs split over two qubits, only the entangled circuit lets the read-out qubit learn the inputs of its partner. On an annulus loaded into both qubits, two entangled qubits gain little over one qubit at equal depth.

16 Fundamental limitations of variational quantum circuits

The mean gradient over random angles vanishes, so its variance measures the typical gradient. The Haar measure is the uniform distribution on the unitary group $U(d)$, and by Lévy's lemma every Lipschitz function of a Haar-random state deviates from its average by more than ϵ with a probability below $2e^{-Cd\epsilon^2/\eta^2}$. Pauli strings have $\mathrm{Var}\langle\hat{P}\rangle = 1/(d+1)$ in random states, small subsystems are almost maximally mixed and the entanglement is close to the Page value, whereas ground states are atypical. Behind a Haar-random unitary (16.9)

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \frac{\mathrm{Tr}(\hat{O}_0^2)}{2(d^2-1)}, \qquad d = 2^N,$$

which decays as 4^{-N} for the global cost and as 2^{-N} for sums of local Pauli strings. A local cost at fixed depth keeps a polynomial decay through the light cone, and the HVA keeps large gradients because its gates generate a small set of unitaries. Resolving a gradient needs $M^\star = \sigma^2/(2\,\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E])$ shots (16.12), about 2^N for a 2-design, and depolarising noise flattens the landscape exponentially in the depth.

A Gradient estimation by simultaneous perturbation

The parameter-shift rule (5.7) obtains each component of the gradient from two circuit configurations. A circuit configuration is one set of angles, prepared and measured in all G settings with M shots each, so it consumes

$$S = GM \tag{A.1}$$

shots. A full gradient needs $2n$ configurations, and the cost (5.9) of a training run grows linearly with the number of angles. Simultaneous-perturbation stochastic approximation (SPSA), introduced in Ref. [25], estimates all n components from two configurations in which every angle is shifted at once along a random direction. The estimate is noisy and slightly biased, so a training run needs more updates, while each update costs the same for any n. SPSA was used to train hardware-efficient circuits on a superconducting processor in Ref. [30].

A.1 The estimator

SPSA draws a random vector $\boldsymbol{\Delta} = (\Delta_1, \ldots, \Delta_n)$ whose entries are independent and take the values $+1$ and -1 with probability $1/2$ each, known as the Rademacher distribution. The averages over this distribution are

$$\langle\Delta_k\rangle = 0, \qquad \langle\Delta_k\Delta_j\rangle = \delta_{kj}, \qquad \Delta_k^2 = 1. \tag{A.2}$$

For a perturbation size $c > 0$ the energies of the two configurations $\boldsymbol{\theta} \pm c\boldsymbol{\Delta}$ follow from a Taylor expansion around $\boldsymbol{\theta}$,

$$E(\boldsymbol{\theta} \pm c\boldsymbol{\Delta}) = E(\boldsymbol{\theta}) \pm c \sum_j \Delta_j \, \partial_j E + \frac{c^2}{2} \sum_{j,l} \Delta_j \Delta_l \, \partial_j \partial_l E \pm \frac{c^3}{6} \sum_{j,l,m} \Delta_j \Delta_l \Delta_m \, \partial_j \partial_l \partial_m E + O(c^4), \quad \text{(A.3)}$$

where $\partial_j = \partial / \partial \theta_j$ and all derivatives are taken at $\boldsymbol{\theta}$. The even orders cancel in the difference of the two energies, and the central difference quotient is the slope of the energy along $\boldsymbol{\Delta}$ up to a correction of second order,

$$\frac{E(\boldsymbol{\theta} + c\boldsymbol{\Delta}) - E(\boldsymbol{\theta} - c\boldsymbol{\Delta})}{2c} = \sum_j \Delta_j \, \partial_j E + \frac{c^2}{6} \sum_{j,l,m} \Delta_j \Delta_l \Delta_m \, \partial_j \partial_l \partial_m E + O(c^4). \quad \text{(A.4)}$$

This single number depends on all components of the gradient, and SPSA multiplies it by $1/\Delta_k = \Delta_k$ to obtain an estimate of every component,

$$\tilde{g}_k = \frac{\tilde{E}(\boldsymbol{\theta} + c\boldsymbol{\Delta}) - \tilde{E}(\boldsymbol{\theta} - c\boldsymbol{\Delta})}{2c} \, \Delta_k, \qquad k = 1, \ldots, n, \quad \text{(A.5)}$$

where $\tilde{E}$ are the estimates (4.5) from S shots. Figure 114 contrasts the configurations of the two methods.

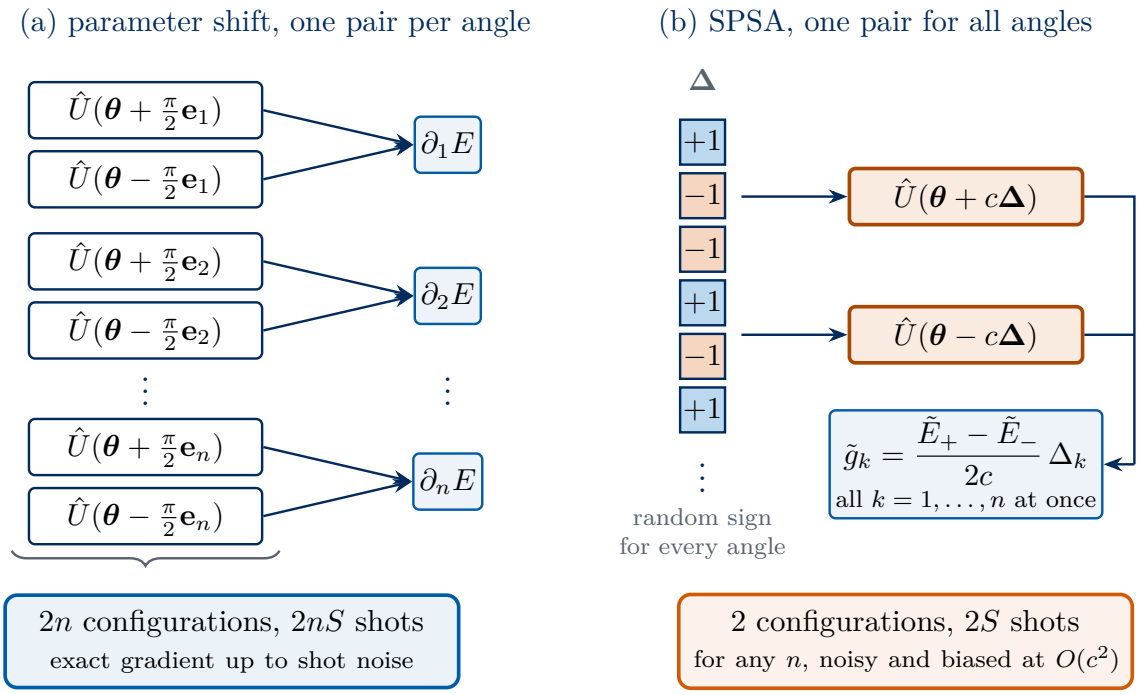


Figure 114: Circuit configurations of one gradient. (a) The parameter-shift rule shifts one angle at a time by $\pm\pi/2$ and needs a pair of configurations for each of the n angles. (b) SPSA shifts all angles at once by $\pm c$ times a random sign vector $\boldsymbol{\Delta}$ and obtains an estimate of all components from one pair of configurations. Each configuration is measured with S shots.

Inserting the expansion (A.4) for exact energies into (A.5) and averaging over $\boldsymbol{\Delta}$ with (A.2) gives

$$\langle \tilde{g}_k \rangle_{\boldsymbol{\Delta}} = \sum_j \langle \Delta_k \Delta_j \rangle \, \partial_j E + O(c^2) = \partial_k E + O(c^2). \quad \text{(A.6)}$$

The products $\Delta_k \Delta_j$ with $j \neq k$ average to zero, and only the term $j = k$ survives. The bias of order c^2 comes from the third derivatives in (A.4). The shot noise of the two estimates has zero mean and adds no bias. The expansion requires the whole displacement $c\boldsymbol{\Delta}$, of length $c\sqrt{n}$, to be small compared with the period 2π of the cosines (5.4), so the admissible perturbation size decreases with the number of angles.

A single estimate scatters around the mean (A.6). With $\Delta_k^2 = 1$ its leading term is $\tilde{g}_k = \partial_k E + \sum_{j\neq k} \Delta_k \Delta_j \, \partial_j E$, in which the products $\Delta_k \Delta_j$ are uncorrelated signs of zero mean. Together with the shot noise of the two energies, each with variance $\sigma_{\tilde{E}}^2$ from (4.7), this gives

$$\mathrm{Var}(\tilde{g}_k) = \sum_{j\neq k} \left(\partial_j E\right)^2 + \frac{\sigma_{\tilde{E}}^2}{2c^2}, \tag{A.7}$$

up to corrections of order c^2. The first term is the squared norm of the gradient without its own component. It grows with n when the components have similar size, so a single estimate has an error comparable to the full gradient. The second term is the shot noise of the parameter-shift component (5.10), $\sigma_{\tilde{E}}^2/2$, enhanced by $1/c^2$. The choice of c therefore balances the bias, which grows as c^2, against the shot noise, which grows as $1/c^2$. Averaging K independent estimates divides the variance by K at the cost of $2K$ configurations. Gradient descent needs only a direction that points downhill on average, so SPSA takes one estimate per update and lets successive small steps average the noise.

A.2 The algorithm

SPSA moves the angles against the estimate (A.5) with a step size a_t and a perturbation size c_t that both decrease with the update number t. Ref. [25] proves convergence to a local minimum when both sequences decrease to zero at suitable rates. The gain sequences recommended in Ref. [95] are

$$a_t = \frac{a}{(t+1+A)^\alpha}, \qquad c_t = \frac{c}{(t+1)^\gamma}, \qquad \alpha = 0.602, \quad \gamma = 0.101, \tag{A.8}$$

where a, c and A are chosen constants. The stability constant A is about one tenth of the planned number of updates and keeps the first steps from being too large. The constant c is chosen large enough that the energy difference of the two configurations exceeds the shot noise $\sigma_{\tilde{E}}$, and a is chosen such that the first steps change the angles by a small fraction of 2π.

SPSA with the gain sequences (A.8)

Input: initial angles $\boldsymbol{\theta}$, constants a, c, A, number of updates T, shots S per configuration.
For $t = 0, 1, \ldots, T-1$:

(i) draw $\boldsymbol{\Delta}$ with independent entries ± 1 of probability $1/2$ each,

(ii) set a_t and c_t by (A.8),

(iii) measure the two configurations $\boldsymbol{\theta} + c_t\boldsymbol{\Delta}$ and $\boldsymbol{\theta} - c_t\boldsymbol{\Delta}$ with S shots each, which gives $\tilde{E}_+$ and $\tilde{E}_-$,

(iv) form $\tilde{g}_k = (\tilde{E}_+ - \tilde{E}_-)\,\Delta_k/(2c_t)$ for all k,

(v) update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - a_t\,\tilde{\mathbf{g}}$.

Each update costs 2 configurations and $2S$ shots, independent of n.

A.3 Adam with SPSA gradients

Adam [28] can use the estimate (A.5) in place of the exact gradient, with the same two configurations per update. For each angle it keeps running averages of $\tilde{g}_k$ and of $\tilde{g}_k^2$ over past updates and moves θ_k by a fixed step size η times the ratio of the first average to the square root of the second. The estimates of one component change sign from update to update with the random Δ_k, and these flips cancel in the first average, which follows the mean gradient (A.6). The second average sets the scale of the step, so the step of every angle stays close to η even though a single estimate is large compared with the gradient, as (A.7) shows. Adam with SPSA gradients therefore runs with a fixed η and a fixed perturbation size c in place of the sequences (A.8). In the SPSA loop only the last line changes, and it becomes one Adam step with $\tilde{\mathbf{g}}$.

A.4 Cost in circuit configurations and shots

The two methods differ in the number of configurations C per update,

$$C_{\text{shift}} = 2n, \qquad C_{\text{SPSA}} = 2, \qquad \text{shots per update} = C\,S = C\,GM. \tag{A.9}$$

For the six-qubit hardware-efficient circuit with $n = 60$ angles, one parameter-shift update costs 120 configurations, and SPSA costs two. A training run that needs U updates to reach a given energy costs $U\,C$ configurations, so SPSA is cheaper whenever it needs fewer than n times as many updates as the parameter-shift rule. Table 9 lists the cost per update for the hardware-efficient circuit with $L = 1, \ldots, 6$ layers.

		parameter shift		SPSA	
L	n	configurations	shots	configurations	shots
1	24	48	96 000	2	4 000
2	36	72	144 000	2	4 000
3	48	96	192 000	2	4 000
4	60	120	240 000	2	4 000
5	72	144	288 000	2	4 000
6	84	168	336 000	2	4 000

Table 9: Circuit configurations and shots per update for the six-qubit hardware-efficient circuit with L layers and $n = 2N(L + 1)$ angles, with $G = 2$ settings and $M = 1000$ shots per setting.

A.5 Numerical comparison

We compare three optimisers on the six-qubit critical TFIM, $h = J$ in (2.2), with the hardware-efficient circuit of $L = 4$ layers and $n = 60$ angles. Every energy that an optimiser uses is an estimate (4.5) from bit strings sampled from the Born distributions of the two settings, with $M = 1000$ shots per setting, so $S = 2000$ shots per configuration. All runs start from the same small angles near $|0 \ldots 0\rangle$. Optimiser (a) is Adam with the parameter-shift gradient, (b) Adam with the SPSA estimate at fixed c, and (c) SPSA with the gain sequences (A.8). The energy error $E - E_0$ along each run is evaluated exactly from the state vector, which only a simulation can do, and is never used by the optimisers. Each optimiser runs with 5 independent seeds for the measurement records and the perturbations. The step sizes were chosen by a short scan for each method on separate seeds. The constants are $\eta = 0.05$ for (a), $\eta = 0.005$ and $c = 0.1$ for (b), and $a = 0.4$, $c = 0.3$ and A equal to one tenth of the 50 000 updates for (c). Each optimiser receives about 10^5 configurations. The bound (3.3) sets two tolerances, $E - E_0 \le \Delta E/2$, which guarantees $F \ge 1/2$, and $E - E_0 \le \Delta E/5$, which guarantees $F \ge 0.8$.

Figure 115 shows that all three optimisers reach $\Delta E/2$ within a few thousand configurations and that the two SPSA optimisers get there first. The median run of Adam with SPSA gradients crosses $\Delta E/2$ after about 2 520 configurations and Adam with the parameter-shift rule after about 4 080. The parameter-shift rule needs only 34 updates for this, against about 1 260 for Adam with SPSA gradients. SPSA is nevertheless cheaper, since each of its updates costs two configurations and each parameter-shift update costs $2n$.

The order reverses at the stricter tolerance $\Delta E/5$, which Adam with the parameter-shift rule reaches after a median of about 33 840 configurations and Adam with SPSA gradients after about 45 920. SPSA with the gain sequences (A.8) descends as fast as SPSA with Adam at first but does not reach $\Delta E/5$ within the budget in any run. Near the minimum the gradient is small while the shot-noise term of (A.7) stays, so each SPSA update makes less progress, and the low cost per update no longer compensates. The spread between seeds is large at $\Delta E/5$ for both Adam variants. SPSA is therefore the cheaper choice when the number of angles is large and a moderate accuracy suffices, whereas the parameter-shift rule reaches small energy errors with fewer circuit configurations.

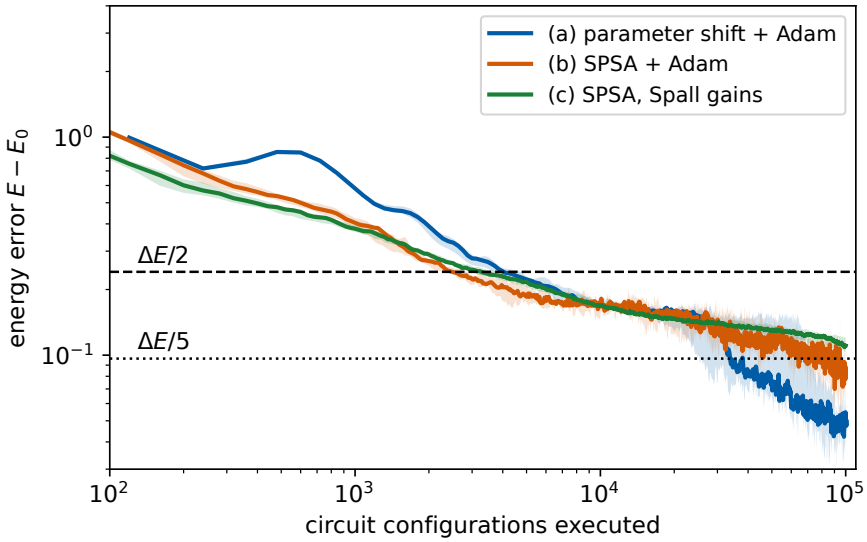


Figure 115: Exact energy error $E - E_0$ of the six-qubit trial state against the number of circuit configurations executed, for (a) Adam with parameter-shift gradients, (b) Adam with SPSA gradients and (c) SPSA with the gains (A.8). Lines show the median over 5 seeds and bands the range from the smallest to the largest value. The dashed and dotted lines mark the tolerances $\Delta E/2$ and $\Delta E/5$. Every energy used by the optimisers is estimated from $S = 2000$ shots per configuration.

References

1. E. Aïmeur, G. Brassard, S. Gambs, Machine learning in a quantum world, in Advances in Artificial Intelligence, Lect. Notes Comput. Sci. **4013**, 431 (2006), https://doi.org/10.1007/11766247_37
2. V. Dunjko, J. M. Taylor, H. J. Briegel, Quantum-enhanced machine learning, Phys. Rev. Lett. **117**, 130501 (2016), https://doi.org/10.1103/PhysRevLett.117.130501
3. J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, S. Lloyd, Quantum machine learning, Nature **549**, 195 (2017), https://doi.org/10.1038/nature23474
4. V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, J. M. Gambetta, Supervised learning with quantum-enhanced feature spaces, Nature **567**, 209 (2019), https://doi.org/10.1038/s41586-019-0980-2
5. M. Schuld, N. Killoran, Quantum machine learning in feature Hilbert spaces, Phys. Rev. Lett. **122**, 040504 (2019), https://doi.org/10.1103/PhysRevLett.122.040504
6. K. Fujii, K. Nakajima, Harnessing disordered-ensemble quantum dynamics for machine learning, Phys. Rev. Appl. **8**, 024030 (2017), https://doi.org/10.1103/PhysRevApplied.8.024030
7. P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi, M. C. Soriano, R. Zambrini, Opportunities in quantum reservoir computing and extreme learning machines, Adv. Quantum Technol. **4**, 2100027 (2021), https://doi.org/10.1002/qute.202100027
8. L. Innocenti, S. Lorenzo, I. Palmisano, A. Ferraro, M. Paternostro, G. M. Palma, Potential and limitations of quantum extreme learning machines, Commun. Phys. **6**, 118 (2023), https://doi.org/10.1038/s42005-023-01233-w
9. H.-Y. Huang, M. Broughton, J. Cotler, S. Chen, J. Li, M. Mohseni, H. Neven, R. Babbush, R. Kueng, J. Preskill, J. R. McClean, Quantum advantage in learning from experiments, Science **376**, 1182 (2022), https://doi.org/10.1126/science.abn7293
10. G. Carleo, M. Troyer, Solving the quantum many-body problem with artificial neural networks, Science **355**, 602 (2017), https://doi.org/10.1126/science.aag2302
11. E. P. Gross, Structure of a quantized vortex in boson systems, Nuovo Cimento **20**, 454 (1961), https://doi.org/10.1007/BF02731494
12. L. P. Pitaevskii, Vortex lines in an imperfect Bose gas, Sov. Phys. JETP **13**, 451 (1961), http://jetp.ras.ru/cgi-bin/e/index/e/13/2/p451?a=list

13. P. Weiss, L'hypothèse du champ moléculaire et la propriété ferromagnétique, J. Phys. Theor. Appl. **6**, 661 (1907), https://doi.org/10.1051/jphystap:019070060066100

14. P. Pfeuty, The one-dimensional Ising model with a transverse field, Ann. Phys. **57**, 79 (1970), https://doi.org/10.1016/0003-4916(70)90270-8

15. M. B. Hastings, An area law for one-dimensional quantum systems, J. Stat. Mech. (2007) P08024, https://doi.org/10.1088/1742-5468/2007/08/P08024

16. J. Eisert, M. Cramer, M. B. Plenio, Colloquium: Area laws for the entanglement entropy, Rev. Mod. Phys. **82**, 277 (2010), https://doi.org/10.1103/RevModPhys.82.277

17. S. R. White, Density matrix formulation for quantum renormalization groups, Phys. Rev. Lett. **69**, 2863 (1992), https://doi.org/10.1103/PhysRevLett.69.2863

18. U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, Ann. Phys. **326**, 96 (2011), https://doi.org/10.1016/j.aop.2010.09.012

19. M. Troyer, U.-J. Wiese, Computational complexity and fundamental limitations to fermionic quantum Monte Carlo simulations, Phys. Rev. Lett. **94**, 170201 (2005), https://doi.org/10.1103/PhysRevLett.94.170201

20. I. M. Georgescu, S. Ashhab, F. Nori, Quantum simulation, Rev. Mod. Phys. **86**, 153 (2014), https://doi.org/10.1103/RevModPhys.86.153

21. A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, J. L. O'Brien, A variational eigenvalue solver on a photonic quantum processor, Nat. Commun. **5**, 4213 (2014), https://doi.org/10.1038/ncomms5213

22. J. R. McClean, J. Romero, R. Babbush, A. Aspuru-Guzik, The theory of variational hybrid quantum-classical algorithms, New J. Phys. **18**, 023023 (2016), https://doi.org/10.1088/1367-2630/18/2/023023

23. K. Mitarai, M. Negoro, M. Kitagawa, K. Fujii, Quantum circuit learning, Phys. Rev. A **98**, 032309 (2018), https://doi.org/10.1103/PhysRevA.98.032309

24. M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, N. Killoran, Evaluating analytic gradients on quantum hardware, Phys. Rev. A **99**, 032331 (2019), https://doi.org/10.1103/PhysRevA.99.032331

25. J. C. Spall, Multivariate stochastic approximation using a simultaneous perturbation gradient approximation, IEEE Trans. Autom. Control **37**, 332 (1992), https://doi.org/10.1109/9.119632

26. S. Linnainmaa, Taylor expansion of the accumulated rounding error, BIT **16**, 146 (1976), https://doi.org/10.1007/BF01931367

27. J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, Q. Zhang, JAX: composable transformations of Python+NumPy programs (2018), https://github.com/jax-ml/jax

28. D. P. Kingma, J. Ba, Adam: a method for stochastic optimization, arXiv:1412.6980 (2014), https://arxiv.org/abs/1412.6980

29. J. Romero, R. Babbush, J. R. McClean, C. Hempel, P. J. Love, A. Aspuru-Guzik, Strategies for quantum computing molecular energies using the unitary coupled cluster ansatz, Quantum Sci. Technol. **4**, 014008 (2019), https://doi.org/10.1088/2058-9565/aad3e4

30. A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, J. M. Gambetta, Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets, Nature **549**, 242 (2017), https://doi.org/10.1038/nature23879

31. D. Wecker, M. B. Hastings, M. Troyer, Progress towards practical quantum variational algorithms, Phys. Rev. A **92**, 042303 (2015), https://doi.org/10.1103/PhysRevA.92.042303

32. R. Wiersema, C. Zhou, Y. de Sereville, J. F. Carrasquilla, Y. B. Kim, H. Yuen, Exploring entanglement and optimization within the Hamiltonian variational ansatz, PRX Quantum **1**, 020319 (2020), https://doi.org/10.1103/PRXQuantum.1.020319

33. H. F. Trotter, On the product of semi-groups of operators, Proc. Am. Math. Soc. **10**, 545 (1959), https://doi.org/10.1090/S0002-9939-1959-0108732-6

34. M. Born, V. Fock, Beweis des Adiabatensatzes, Z. Phys. **51**, 165 (1928), https://doi.org/10.1007/BF01343193

35. O. Higgott, D. Wang, S. Brierley, Variational quantum computation of excited states, Quantum **3**, 156 (2019), https://doi.org/10.22331/q-2019-07-01-156

36. C. Lanczos, An iteration method for the solution of the eigenvalue problem of linear differential and integral operators, J. Res. Natl. Bur. Stand. **45**, 255 (1950), https://doi.org/10.6028/jres.045.026

37. J. Romero, J. P. Olson, A. Aspuru-Guzik, Quantum autoencoders for efficient compression of quantum data, Quantum Sci. Technol. **2**, 045001 (2017), https://doi.org/10.1088/2058-9565/aa8072

38. M. Cerezo, A. Sone, T. Volkoff, L. Cincio, P. J. Coles, Cost function dependent barren plateaus in shallow parametrized quantum circuits, Nat. Commun. **12**, 1791 (2021), https://doi.org/10.1038/s41467-021-21728-w

39. R. Islam, R. Ma, P. M. Preiss, M. E. Tai, A. Lukin, M. Rispoli, M. Greiner, Measuring entanglement entropy in a quantum many-body system, Nature **528**, 77 (2015), https://doi.org/10.1038/nature15750

40. T. Brydges, A. Elben, P. Jurcevic, B. Vermersch, C. Maier, B. P. Lanyon, P. Zoller, R. Blatt, C. F. Roos, Probing Rényi entanglement entropy via randomized measurements, Science **364**, 260 (2019), https://doi.org/10.1126/science.aau4963

41. K. Kottmann, F. Metz, J. Fraxanet, N. Baldelli, Variational quantum anomaly detection: Unsupervised mapping of phase diagrams on a physical quantum computer, Phys. Rev. Res. **3**, 043184 (2021), https://doi.org/10.1103/PhysRevResearch.3.043184

42. I. Cong, S. Choi, M. D. Lukin, Quantum convolutional neural networks, Nat. Phys. **15**, 1273 (2019), https://doi.org/10.1038/s41567-019-0648-8

43. P. Jordan, E. Wigner, Über das Paulische Äquivalenzverbot, Z. Phys. **47**, 631 (1928), https://doi.org/10.1007/BF01331938

44. F. Pollmann, A. M. Turner, Detection of symmetry-protected topological phases in one dimension, Phys. Rev. B **86**, 125441 (2012), https://doi.org/10.1103/PhysRevB.86.125441

45. A. Pesah, M. Cerezo, S. Wang, T. Volkoff, A. T. Sornborger, P. J. Coles, Absence of barren plateaus in quantum convolutional neural networks, Phys. Rev. X **11**, 041011 (2021), https://doi.org/10.1103/PhysRevX.11.041011

46. G. Verdon, T. McCourt, E. Luzhnica, V. Singh, S. Leichenauer, J. Hidary, Quantum graph neural networks, arXiv:1909.12264 (2019), https://arxiv.org/abs/1909.12264

47. H. Buhrman, R. Cleve, J. Watrous, R. de Wolf, Quantum fingerprinting, Phys. Rev. Lett. **87**, 167902 (2001), https://doi.org/10.1103/PhysRevLett.87.167902

48. N. Wiebe, C. Granade, C. Ferrie, D. G. Cory, Hamiltonian learning and certification using quantum resources, Phys. Rev. Lett. **112**, 190501 (2014), https://doi.org/10.1103/PhysRevLett.112.190501

49. J. Wu, H. Fu, M. Zhu, H. Zhang, W. Xie, X.-Y. Li, Quantum circuit autoencoder, Phys. Rev. A **109**, 032623 (2024), https://doi.org/10.1103/PhysRevA.109.032623

50. M.-D. Choi, Completely positive linear maps on complex matrices, Linear Algebra Appl. **10**, 285 (1975), https://doi.org/10.1016/0024-3795(75)90075-0

51. A. Jamiołkowski, Linear transformations which preserve trace and positive semidefiniteness of operators, Rep. Math. Phys. **3**, 275 (1972), https://doi.org/10.1016/0034-4877(72)90011-0

52. M. Horodecki, P. Horodecki, R. Horodecki, General teleportation channel, singlet fraction, and quasidistillation, Phys. Rev. A **60**, 1888 (1999), https://doi.org/10.1103/PhysRevA.60.1888

53. M. A. Nielsen, A simple formula for the average gate fidelity of a quantum dynamical operation, Phys. Lett. A **303**, 249 (2002), https://doi.org/10.1016/S0375-9601(02)01272-0

54. Z.-Y. Han, J. Wang, H. Fan, L. Wang, P. Zhang, Unsupervised generative modeling using matrix product states, Phys. Rev. X **8**, 031012 (2018), https://doi.org/10.1103/PhysRevX.8.031012

55. S. Cheng, J. Chen, L. Wang, Information perspective to probabilistic modeling: Boltzmann machines versus Born machines, Entropy **20**, 583 (2018), https://doi.org/10.3390/e20080583

56. J.-G. Liu, L. Wang, Differentiable learning of quantum circuit Born machines, Phys. Rev. A **98**, 062324 (2018), https://doi.org/10.1103/PhysRevA.98.062324

57. M. Benedetti, D. Garcia-Pintos, O. Perdomo, V. Leyton-Ortega, Y. Nam, A. Perdomo-Ortiz, A generative modeling approach for benchmarking and training shallow quantum circuits, npj Quantum Inf. **5**, 45 (2019), https://doi.org/10.1038/s41534-019-0157-8

58. B. Coyle, D. Mills, V. Danos, E. Kashefi, The Born supremacy: quantum advantage and training of an Ising Born machine, npj Quantum Inf. **6**, 60 (2020), https://doi.org/10.1038/s41534-020-00288-9

59. S. Kullback, R. A. Leibler, On information and sufficiency, Ann. Math. Statist. **22**, 79 (1951), https://doi.org/10.1214/aoms/1177729694

60. M. Płodzień, Quantum scrambling Born machine, Phys. Rev. A **114**, 012464 (2026), https://doi.org/10.1103/qccv-flmq

61. S. Kirkpatrick, C. D. Gelatt, M. P. Vecchi, Optimization by simulated annealing, Science **220**, 671 (1983), https://doi.org/10.1126/science.220.4598.671

62. A. Lucas, Ising formulations of many NP problems, Front. Phys. **2**, 5 (2014), https://doi.org/10.3389/fphy.2014.00005

63. R. M. Karp, Reducibility among combinatorial problems, in Complexity of Computer Computations, edited by R. E. Miller, J. W. Thatcher (Plenum, New York, 1972), p. 85, https://doi.org/10.1007/978-1-4684-2001-2_9

64. M. X. Goemans, D. P. Williamson, Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming, J. ACM **42**, 1115 (1995), https://doi.org/10.1145/227683.227684

65. E. Farhi, J. Goldstone, S. Gutmann, A quantum approximate optimization algorithm, arXiv:1411.4028 (2014), https://arxiv.org/abs/1411.4028

66. E. Farhi, J. Goldstone, S. Gutmann, J. Lapan, A. Lundgren, D. Preda, A quantum adiabatic evolution algorithm applied to random instances of an NP-complete problem, Science **292**, 472 (2001), https://doi.org/10.1126/science.1057726

67. T. Kadowaki, H. Nishimori, Quantum annealing in the transverse Ising model, Phys. Rev. E **58**, 5355 (1998), https://doi.org/10.1103/PhysRevE.58.5355

68. E. Farhi, H. Neven, Classification with quantum neural networks on near term processors, arXiv:1802.06002 (2018), https://arxiv.org/abs/1802.06002

69. M. Schuld, A. Bocharov, K. M. Svore, N. Wiebe, Circuit-centric quantum classifiers, Phys. Rev. A **101**, 032308 (2020), https://doi.org/10.1103/PhysRevA.101.032308

70. C. Cortes, V. Vapnik, Support-vector networks, Mach. Learn. **20**, 273 (1995), https://doi.org/10.1007/BF00994018

71. B. E. Boser, I. M. Guyon, V. N. Vapnik, A training algorithm for optimal margin classifiers, in Proceedings of the Fifth Annual Workshop on Computational Learning Theory (ACM, 1992), p. 144, https://doi.org/10.1145/130385.130401

72. S. Thanasilp, S. Wang, M. Cerezo, Z. Holmes, Exponential concentration in quantum kernel methods, Nat. Commun. **15**, 5200 (2024), https://doi.org/10.1038/s41467-024-49287-w

73. N. Cristianini, J. Shawe-Taylor, A. Elisseeff, J. Kandola, On kernel-target alignment, in Advances in Neural Information Processing Systems **14** (2001), https://proceedings.neurips.cc/paper/2001/hash/1f71e393b3809197ed66df836fe833e5-Abstract.html

74. T. Hubregtsen, D. Wierichs, E. Gil-Fuster, P.-J. H. S. Derks, P. K. Faehrmann, J. J. Meyer, Training quantum embedding kernels on near-term quantum computers, Phys. Rev. A **106**, 042431 (2022), https://doi.org/10.1103/PhysRevA.106.042431

75. A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, J. I. Latorre, Data re-uploading for a universal quantum classifier, Quantum **4**, 226 (2020), https://doi.org/10.22331/q-2020-02-06-226

76. M. Schuld, R. Sweke, J. J. Meyer, Effect of data encoding on the expressive power of variational quantum-machine-learning models, Phys. Rev. A **103**, 032430 (2021), https://doi.org/10.1103/PhysRevA.103.032430

77. Z. Yu, H. Yao, M. Li, X. Wang, Power and limitations of single-qubit native quantum neural networks, in Advances in Neural Information Processing Systems **35**, 27810 (2022), https://arxiv.org/abs/2205.07848

78. A. Pérez-Salinas, D. López-Núñez, A. García-Sáez, P. Forn-Díaz, J. I. Latorre, One qubit as a universal approximant, Phys. Rev. A **104**, 012405 (2021), https://doi.org/10.1103/PhysRevA.104.012405

79. J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, Barren plateaus in quantum neural network training landscapes, Nat. Commun. **9**, 4812 (2018), https://doi.org/10.1038/s41467-018-07090-4

80. F. Mezzadri, How to generate random matrices from the classical compact groups, Notices Am. Math. Soc. **54**, 592 (2007), https://arxiv.org/abs/math-ph/0609050

81. A. A. Mele, Introduction to Haar measure tools in quantum information: a beginner's tutorial, Quantum **8**, 1340 (2024), https://doi.org/10.22331/q-2024-05-08-1340

82. C. Dankert, R. Cleve, J. Emerson, E. Livine, Exact and approximate unitary 2-designs and their application to fidelity estimation, Phys. Rev. A **80**, 012304 (2009), https://doi.org/10.1103/PhysRevA.80.012304

83. M. Ledoux, The Concentration of Measure Phenomenon, Mathematical Surveys and Monographs **89** (American Mathematical Society, 2001), https://doi.org/10.1090/surv/089

84. P. Hayden, D. W. Leung, A. Winter, Aspects of generic entanglement, Commun. Math. Phys. **265**, 95 (2006), https://doi.org/10.1007/s00220-006-1535-6

85. E. Lubkin, Entropy of an n-system from its correlation with a k-reservoir, J. Math. Phys. **19**, 1028 (1978), https://doi.org/10.1063/1.523763

86. S. Popescu, A. J. Short, A. Winter, Entanglement and the foundations of statistical mechanics, Nat. Phys. **2**, 754 (2006), https://doi.org/10.1038/nphys444

87. S. Goldstein, J. L. Lebowitz, R. Tumulka, N. Zanghì, Canonical typicality, Phys. Rev. Lett. **96**, 050403 (2006), https://doi.org/10.1103/PhysRevLett.96.050403

88. D. N. Page, Average entropy of a subsystem, Phys. Rev. Lett. **71**, 1291 (1993), https://doi.org/10.1103/PhysRevLett.71.1291

89. M. Ragone, B. N. Bakalov, F. Sauvage, A. F. Kemper, C. Ortiz Marrero, M. Larocca, M. Cerezo, A Lie algebraic theory of barren plateaus for deep parameterized quantum circuits, Nat. Commun. **15**, 7172 (2024), https://doi.org/10.1038/s41467-024-49909-3

90. M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, M. Cerezo, Barren plateaus in variational quantum computing, Nat. Rev. Phys. **7**, 174 (2025), https://doi.org/10.1038/s42254-025-00813-9

91. S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, P. J. Coles, Noise-induced barren plateaus in variational quantum algorithms, Nat. Commun. **12**, 6961 (2021), https://doi.org/10.1038/s41467-021-27045-6

92. E. Grant, L. Wossnig, M. Ostaszewski, M. Benedetti, An initialization strategy for addressing barren plateaus in parametrized quantum circuits, Quantum **3**, 214 (2019), https://doi.org/10.22331/q-2019-12-09-214

93. A. Skolik, J. R. McClean, M. Mohseni, P. van der Smagt, M. Leib, Layerwise learning for quantum neural networks, Quantum Mach. Intell. **3**, 5 (2021), https://doi.org/10.1007/s42484-020-00036-4

94. Z. Holmes, K. Sharma, M. Cerezo, P. J. Coles, Connecting ansatz expressibility to gradient magnitudes and barren plateaus, PRX Quantum **3**, 010313 (2022), https://doi.org/10.1103/PRXQuantum.3.010313

95. J. C. Spall, Implementation of the simultaneous perturbation algorithm for stochastic optimization, IEEE Trans. Aerosp. Electron. Syst. **34**, 817 (1998), https://doi.org/10.1109/7.705889